



# **DesignWare Cores SuperSpeed USB3.0 femtoPHY for TSMC 12-nm FFC 0.8/1.8 V**

## **Databook**

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*dwc\_usb3\_femtophy\_otg\_tsmc12ffc\_x1ns – Product Code: C523-0*

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# Revision History

| PHY Version | Date          | Databook Version | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.03a       | June 28, 2019 | 1.20             | <p>Updated:</p> <ul style="list-style-type: none"> <li>■ <a href="#">“Quick Reference”</a> on page 16</li> <li>■ Auto-extraction for <a href="#">“Signal Descriptions”</a> on page 29</li> <li>■ Table <a href="#">“Parameter Controls”</a> on page 162</li> <li>■ <a href="#">“Verilog Model Files and Simulators”</a> on page 184</li> <li>■ <a href="#">“pipe_reset_mux Modules”</a> on page 213</li> <li>■ Table <a href="#">“Scan Chains Per Scan Clock Domain”</a> on page 221</li> <li>■ Table <a href="#">“Connections Between PHY and PCS for At-Speed Testing on Entire PHY”</a> on page 225</li> <li>■ <a href="#">“Power Dissipation”</a> on page 245</li> <li>■ <a href="#">“Registers”</a> on page 253</li> <li>■ Figure <a href="#">“Timing for HIGHZ Instruction”</a> on page 472</li> <li>■ Figure <a href="#">“Timing for SAMPLE/PRELOAD Instruction”</a> on page 473</li> </ul> |
| 4.02a       | June 14, 2019 | 1.10             | <p>Updated:</p> <ul style="list-style-type: none"> <li>■ Table <a href="#">“USB 3.0 femtoPHY Power Dissipation: SuperSpeed 5-Gbps Operation Only”</a> on page 206</li> <li>■ Table <a href="#">“USB 3.0 femtoPHY Power Dissipation: HS/FS/LS Operations Only”</a> on page 207</li> <li>■ Table <a href="#">“USB 3.0 femtoPHY Power Dissipation With Voltage Regulators: SuperSpeed Plus HS/FS/LS Operations”</a> on page 209</li> <li>■ <a href="#">“Registers”</a> on page 213</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
|             | July 20, 2018 | 1.00             | <p>Updated:</p> <ul style="list-style-type: none"> <li>■ <a href="#">“Product Overview”</a> on page 15: Added <a href="#">“Quick Reference”</a> on page 16</li> <li>■ <a href="#">“Physical-Level Implementation”</a> on page 151: Updated <a href="#">“External Component Values”</a> on page 159</li> <li>■ <a href="#">“Power Dissipation”</a> on page 205</li> </ul> <p>Removed:</p> <ul style="list-style-type: none"> <li>■ References to Cadence / NCSIM / NC-Verilog throughout the databook</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                    |

| PHY Version | Date              | Databook Version | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | November 17, 2017 | -                | <ul style="list-style-type: none"> <li>■ “Product Overview” chapter: updated “Devices List” section</li> <li>■ “Deliverables” chapter: updated “Signal Integrity and Test Views” section</li> <li>■ “Functional Description” chapter: updated “Battery Charging Signals” and “Parameter Controls” sections</li> <li>■ “System-Level Implementation” chapter: updated “System Operation” and “Verilog Model Files and Simulators” sections</li> <li>■ “Physical-Level Implementation” chapter: updated “Power and Ground Connections” and “ESD Protection Strategy” sections</li> <li>■ “Production Test and Debug Implementation” chapter: updated “Scan” section</li> <li>■ “Board- and Package-Level Implementation” chapter: updated “Packaging Considerations” section</li> <li>■ “Electrical Specifications” chapter: updated “Power Dissipation” section</li> <li>■ “Signal Descriptions” chapter: updated “JTAG Port”, “Test Controls”, “Battery Charging Signals”, and “Parameter Controls” sections</li> <li>■ Updated “Control Registers” chapter</li> <li>■ “JTAG Port” appendix: updated “Using 1149.6 AC JTAG Boundary Scan” and “Simulation” sections</li> <li>■ “Scan Implementation” appendix: updated “Design for Testability Model” section</li> <li>■ “ATE Test Bench” appendix: updated “Configuring the Test Limits”, “ATE Testing Details”, and “Bench Testing Details” sections</li> </ul> |
|             | August 30, 2017   | -                | Initial version                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

# Preface

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This document describes the Synopsys DesignWare USB 3.0 femtoPHY. The USB 3.0 femtoPHY is a complete mixed-signal IP solution designed to implement USB 3.0 connectivity in a System-on-Chip (SoC) design targeted to the TSMC 12-nm FFC 0.8/1.8 V fabrication process.

## Databook Organization

This databook is organized as follows:

- [Chapter 1, “Product Overview”](#), provides an introduction to the USB 3.0 femtoPHY and its features.
- [Chapter 2, “Deliverables”](#), describes the deliverables included in the USB 3.0 femtoPHY full-package release.
- [Chapter 3, “Signal Descriptions”](#), describes the USB 3.0 femtoPHY top-level signals in detail.
- [Chapter 4, “Functional Description”](#), describes the function of each group of top-level signals in the USB 3.0 femtoPHY.
- [Chapter 5, “System-Level Implementation”](#), describes the USB 3.0 femtoPHY’s system operation.
- [Chapter 6, “Physical-Level Implementation”](#), describes the physical implementation of the USB 3.0 femtoPHY.
- [Chapter 7, “PCS Physical Implementation and Timing Model Usage”](#), provides synthesis and timing closure guidelines for the PCS layer.
- [Chapter 8, “Production Test and Debug Implementation”](#), describes the production-level test and debug implementation of the USB 3.0 femtoPHY.
- [Chapter 9, “Board- and Package-Level Implementation”](#), describes requirements and guidelines for implementing the USB 3.0 femtoPHY at the board and package level.
- [Chapter 10, “Reference Clock Implementation”](#), describes reference clock implementation requirements including jitter and layout requirements and pad/package/board connections.
- [Chapter 11, “Electrical Specifications”](#), provides USB 3.0 femtoPHY electrical specifications for the TSMC 12-nm FFC 0.8/1.8 V process.
- [Chapter 12, “Registers”](#), describes the USB 3.0 femtoPHY control registers.
- [Appendix A, “JTAG Port”](#), describes the USB 3.0 femtoPHY boundary scan support.
- [Appendix B, “Scan Implementation”](#), describes the USB 3.0 femtoPHY and PCS scan implementations.

- [Appendix C, “ATE Test Bench”](#), describes the ATE Verilog test bench and its usage.
- [Appendix D, “Application Notes”](#), provides information relevant to the USB 3.0 femtoPHY.

## Web Resources

- DesignWare IP product information: <http://www.designware.com>
- Your custom DesignWare IP page: <http://www.mydesignware.com>
- Documentation through SolvNet: <https://solvnet.synopsys.com> (Synopsys password required)
- Synopsys Common Licensing (SCL): <http://www.synopsys.com/keys>

## Reference Documentation



### Attention

Access to direct links provided is available based on availability and licensing.

- *Universal Serial Bus 3.0 Specification*, Revision 1.0, USB Implementers Forum, Inc.
- *PHY Interface For the PCI Express\* and USB 3.0 Architectures*, Version 3.0, Intel Corporation
- *Device High-speed Electrical Test Procedure*, Revision 1.2, USB Implementers Forum, Inc.
- *Host High-speed Electrical Test Procedure*, Revision 1.2, USB Implementers Forum, Inc.
- *On-The-Go Supplement to the USB 2.0 Specification*, Revision 1.0a, USB Implementers Forum, Inc.
- *USB 2.0 Link Power Management Addendum*, USB Implementers Forum, Inc.
- *USB 2.0 Transceiver Macrocell Interface (UTMI) Specification*, Version 1.05, Intel Corporation
- *UTMI+ Specification*, Revision 1.0, OTG Workgroup
- *UTMI Specification*, Revision 1.05, OTG Workgroup
- *Battery Charging Specification*, Revision 1.2, USB Implementers Forum, Inc.
- *OTG Transceiver Interface Specification*, Revision 0.72e, OTG Workgroup
- *USB Power Delivery Specification Revision 2.0*, Version 1.2, USB Implementers Forum, Inc.
- *IEEE Standard 1149.1-2001*, IEEE
- *IEEE Standard 1149.6-2003*, IEEE
- *DesignWare Cores SuperSpeed USB 3.0 Controller Databook*, Synopsys, Inc.  
[https://www.synopsys.com/dw/doc.php/iip/DWC\\_usb3/latest/doc/usb3\\_databook.pdf](https://www.synopsys.com/dw/doc.php/iip/DWC_usb3/latest/doc/usb3_databook.pdf)
- *DFT Compiler User Guide: Scan*, Synopsys, Inc.  
[https://solvnet.synopsys.com/dow\\_retrieve/B-2008.09/dftxg1/ni/dftxg1.pdf](https://solvnet.synopsys.com/dow_retrieve/B-2008.09/dftxg1/ni/dftxg1.pdf)



## Related DesignWare Cores

**Attention**

Access to direct links provided is available based on availability and licensing.

- DesignWare SuperSpeed USB 3.0 Host Controller  
[http://www.synopsys.com/dw/ipdir.php?c=dwc\\_usb\\_3\\_0\\_host](http://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_host)
- DesignWare SuperSpeed USB 3.0 Device Controller  
[http://www.synopsys.com/dw/ipdir.php?c=dwc\\_usb\\_3\\_0\\_device](http://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_device)
- DesignWare SuperSpeed USB 3.0 Hub Controller  
[https://www.synopsys.com/dw/ipdir.php?c=dwc\\_usb\\_3\\_0\\_hub](https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_hub)

## Customer Support

To obtain support for your product, contact Support Center using one of the following methods:

- *For fastest response*, use the SolvNet website. If you fill in your information as explained below, your issue is automatically routed to a support engineer who is experienced with your product. The **Sub Product** entry is critical for correct routing.

Go to <http://solvnet.synopsys.com/EnterACall> and click on the link to enter a call. Provide the requested information, including:

- ☐ **Product:** DesignWare Cores
- ☐ **Sub Product:** USB 3.0 femtoPHY
- ☐ **Tool Version:** <product\_version\_number>
- ☐ **Problem Type:**
- ☐ **Priority:**
- ☐ **Title:** Provide a short summary of the issue or error message you have encountered.
- ☐ **Description:** Specify the databook title (SuperSpeed USB3.0 femtoPHY for TSMC 12-nm FFC 0.8/1.8 V) and a description of the issue. For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
- Or, send an e-mail message to [support\\_center@synopsys.com](mailto:support_center@synopsys.com) (your e-mail is queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
  - ☐ Include the Product name, Sub Product name, process, and Tool Version number in your e-mail (as identified above) so it can be routed correctly.
  - ☐ For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
  - ☐ Attach any debug files you created.

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# 1

## Product Overview

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This chapter includes the following sections:

- [“Quick Reference”](#) on page 16
- [“General Product Description”](#) on page 17
- [“Applications”](#) on page 18
- [“Standards Compliance”](#) on page 19
- [“Features”](#) on page 20
- [“Devices List”](#) on page 23
- [“USB 3.0 femtoPHY System-Level Overview”](#) on page 24

## 1.1 Quick Reference

### USB 3.0 femtoPHY

#### Features:

For information about the features that the USB 3.0 femtoPHY supports, see [“Features”](#) on page 20.

#### Data Rates:

5.0 Gbps (USB3.0), 480 Mbps (HS), 12 Mbps (FS), 1.5 Mbps (LS - Host Mode Only)

#### EM:

88kph 100° C

#### Junction Temperature:

-40° C through +125° C

#### Process Technology:

TSMC 12-nm FFC 0.8/1.8 V

#### Power Supplies:

For information about the power supplies, see [“Power and Ground Connections”](#) on page 190

#### Metal Stacks Supported:

1P9M\_2Xa1Xd\_h\_3Xe\_vhv\_2Z,  
1P11M\_2xa1xd\_h\_3xe\_vhv\_2Y2R

#### ESD:

For information about the USB 3.0 femtoPHY's ESD requirements, see [“ESD Considerations”](#) on page 192.

#### Test Modes Supported:

Boundary Scan (ACJTAG), IDDQ, Scan (At-Speed, Stuck-At), and Diagnostics (serial loopback)

#### Protocols/Specifications:

For information about the specifications the USB 3.0 femtoPHY supports, see [“Standards Compliance”](#) on page 19.

## 1.2 General Product Description

The USB 3.0 femtoPHY is a complete mixed-signal semiconductor intellectual property (IP) solution, designed for single-chip integration into USB 3.0 SuperSpeed applications. The USB 3.0 femtoPHY, `dwc_usb3_sspxn_hspxn_ns`, includes all the necessary logical, geometric, and physical design files to implement complete USB 3.0 physical layer capability, connecting a USB OTG controller, host controller, or device controller to a USB system. The USB 3.0 femtoPHY is optimized for an SoC design targeted to the TSMC 12-nm FFC 0.8/1.8 V fabrication process.

The USB 3.0 femtoPHY supports the USB 3.0 SuperSpeed (5 Gbps) protocol and data rate and is backward-compatible with USB 2.0 high-speed (480 Mbps), full-speed (12 Mbps), and low-speed (1.5 Mbps) protocols and data rates.

This chapter provides an introduction to the USB 3.0 femtoPHY and its features.

### 1.2.1 Junction Temperature

The following table provides the USB 3.0 femtoPHY's junction temperature.

**Table 1-1 Junction Temperature**

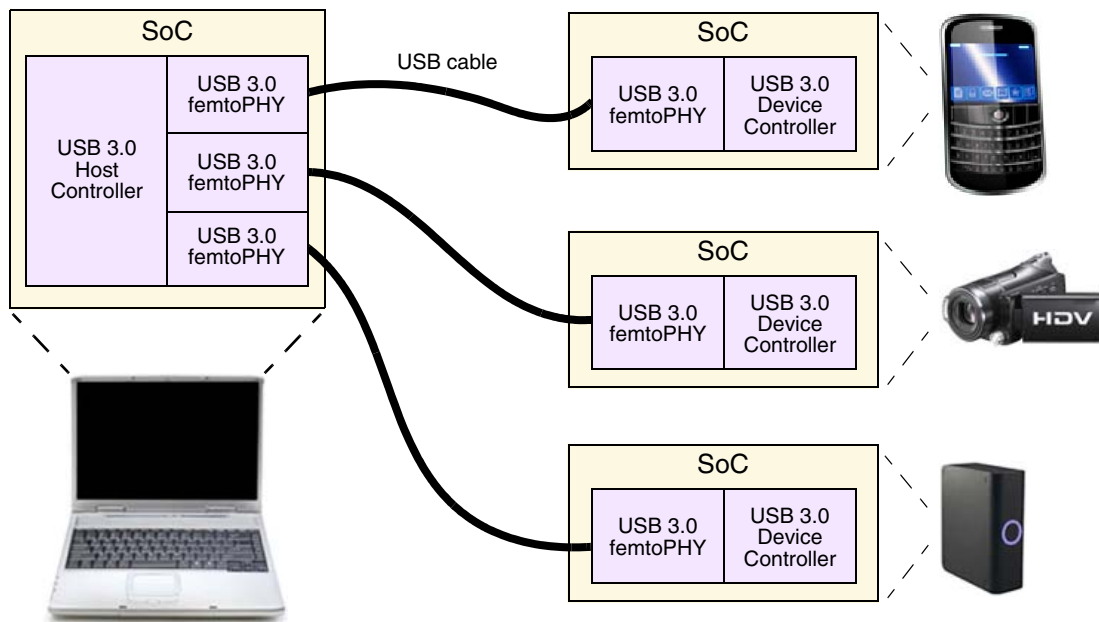
| Minimum | Maximum |
|---------|---------|
| -40° C  | +125° C |

## 1.3 Applications

The USB 3.0 femtoPHY IP targets the next generation of media storage, creation, and playback devices requiring high bandwidth for faster “sync-and-go” functionality between PCs and portable electronic devices, such as flash drives, solid state drives, camcorders, portable media players, and smartphones. Designed for low power consumption, efficient area, low test costs, process centering for optimal yield and interoperability, the USB 3.0 femtoPHY enables designers to introduce competitive products using the latest generation of the USB standard.

The following figure shows a typical application for the USB 3.0 femtoPHY.

**Figure 1-1 Typical USB 3.0 femtoPHY Application**



## 1.4 Standards Compliance

The USB 3.0 femtoPHY is fully compliant with all the required features of the following standards:

- Universal Serial Bus 3.0 Specification, Revision 1.0
  - Supports 5.0-Gbps SuperSpeed (SS) through 3-m cable, 480-Mbps high-speed (HS), 12-Mbps full-speed (FS), and 1.5-Mbps low-speed (LS) serial (Host mode only) data transmission rates, with a high-speed (HS) transmit-to-receive interpacket delay requirement of 38 bit-times as implied by the UTMI specification, version 1.05. For more information about high-speed transmit-to-receive interpacket delay, see [“High-Speed Transmit-to-Receive Interpacket Delay”](#) on page 173.
  - Supports beaconing, receiver detection, and electrical idle
  - Four-wire SuperSpeed connection (two TX and two RX unidirectional) with built-in TX and RX equalization
- PIPE 3-compliant USB 3.0 SuperSpeed interface
- On-The-Go Supplement to the USB 2.0 Specification, Revision 2.0  
Attach Detach Protocol (ADP) is not supported.
- UTMI+ Specification, Revision 1.0 (Level 3)
- Battery Charging Specification, Revision 1.2  
Accessory Charger Adapter (ACA) functions are not supported.

## 1.5 Features

This section describes general features of the USB 3.0 femtoPHY as well as features specific to the USB 3.0 and USB specifications.

### 1.5.1 General Features

The USB 3.0 femtoPHY provides the following general features:

- Optimized PHY area (for physical dimensions, see [Table 6-3](#) on page [194](#))  
Hard macro with snap-on wirebond pads and/or flip-chip bumps including all electrostatic discharge (ESD) protection
- Fully compatible with the Synopsys DesignWare USB 3.0 and USB 2.0 host and device controllers
- Supports all power management features
- Built-in VBUS threshold comparators; support for off-chip charge pump regulator to generate 5-V VBUS signals
- Single reference clock input for both USB 3.0 and USB 2.0 operating modes  
Wide range of reference clocks supported: 19.2, 20, 24, 25, 26, and 100 MHz
- Minimal external component cost: only a single external resistor required
- Built-in Self-Test (BIST) features for production, at-speed, testing on any digital tester  
SuperSpeed, high-speed, full-speed, and low-speed USB test modes supported
- Advanced, built-in diagnostics including SuperSpeed 5-Gbps on-chip sampling scope for easy debug
- Visibility and controllability of hard macro functionality through programmable registers in the design
  - Overrides on all ASIC-side inputs for easy debug
  - Access to register space via simple 16-bit parallel interface
  - Access to register space via JTAG port
- Integrated 3.3-V-to-1.8-V regulator in the PHY hard macro



## 1.5.2 USB 3.0 Features

The USB 3.0 femtoPHY provides the following USB 3.0 features:

- 5-Gbps SuperSpeed data transmission rate through 3-m USB 3.0 cable
- Spread Spectrum clock (SSC) generation and absorption  
Down-spread is programmable from -4,980 ppm through -4,003 ppm
- PIPE 3-compliant SuperSpeed USB 3.0 Transceiver interface
  - Configurable using soft PCS layer above hard macro PHY
  - Supports 8-bit interface at 500-MHz operation, 16-bit interface at 250-MHz operation, and 32-bit interface at 125-MHz operation
- Supports SuperSpeed power-down modes: U0, U1, U2, and U3
- Integrated PHY includes transmitter, receiver, SSC generation, PLL, digital core, and ESD
- Adaptive RX equalization
- Supports all USB 3.0 test modes
- Designed for excellent performance margin and receiver sensitivity
- Robust PHY architecture tolerates wide process, voltage, and temperature variations
- Low-jitter PLL technology with excellent supply isolation
- IEEE standards 1149.1 and 1149.6 (JTAG) boundary scan for internal visibility and control

### 1.5.3 USB 2.0 Features

The USB 3.0 femtoPHY provides the following USB 2.0 features:

- 480-Mbps high-speed, 12-Mbps full-speed, and 1.5-Mbps low-speed serial (Host mode only) data transmission rates
- Support for high-speed power modes: suspend, resume, and remote wakeup
- USB 2.0 test modes
- 45-Ω termination, 1.5-kΩ pull-up and 15-kΩ pull-down resistors with support for independent control of the pull-down resistors
- 8/16-bit unidirectional parallel interfaces for HS, FS, and LS (Host mode only) modes of operation in accordance with the UTMI+ specification
- Dual (HS/FS) mode host/device support (LS operation is not supported for device applications)
- Data recovery from serial data on the USB connector
- SYNC/End-of-Packet (EOP) generation and checking
- Bit stuffing and unstuffing, and bit-stuffing error detection
- Non Return to Zero Invert (NRZI) encoding and decoding
- Bit serialization and deserialization
- Holding registers for staging transmit and receive data
- Logic to support suspend, resume, and remote wakeup operations
- VBUS threshold comparators
- Parameter override bits for optimal yield and interoperability
- Provision for 24-hour, 5-V tolerance on D+ and D- lines

## 1.6 Devices List

The following table lists the primitive devices that the IP uses.

**Table 1-2 Primitive Devices**

| Device           | Description                       |
|------------------|-----------------------------------|
| ndio_mac         | N+/PW Junction DIODE              |
| pdio_gated18_mac | 2T IO 1.8V gated P+/NW DIODE      |
| ndio_gated18_mac | 2T IO 1.8V gated N+/PW DIODE      |
| pdio_hia18_mac   | 1.8V 2T ESD diode P+/NW           |
| ndio_hia18_mac   | 1.8V 2T ESD diode N+/PW           |
| nmoscap_18       | 1.8V MOSCAP Varactor              |
| nch_18_mac       | I/O 1.8V VT NMOS                  |
| nch_lvt_mac      | CORE Low VT NMOS                  |
| nch_ulvt_mac     | CORE Ultra Low VT NMOS            |
| pch_18_mac       | I/O 1.8V VT PMOS                  |
| pch_lvt_mac      | CORE Low VT PMOS                  |
| pch_ulvt_mac     | CORE Ultra Low VT PMOS            |
| pnpi2_mac        | Cascade PNP (P+/NW/Psub)          |
| rm1w             | M1 resistor                       |
| rm2w             | M2 resistor                       |
| rm3w             | M3 resistor                       |
| rm4w             | M4 resistor                       |
| rm5w             | M5 resistor                       |
| rhim_m           | P+ Poly Non-Salicided resistor 3T |
| nch_hia18_mac    | ESD 1.8V HIA nmos in PWELL        |

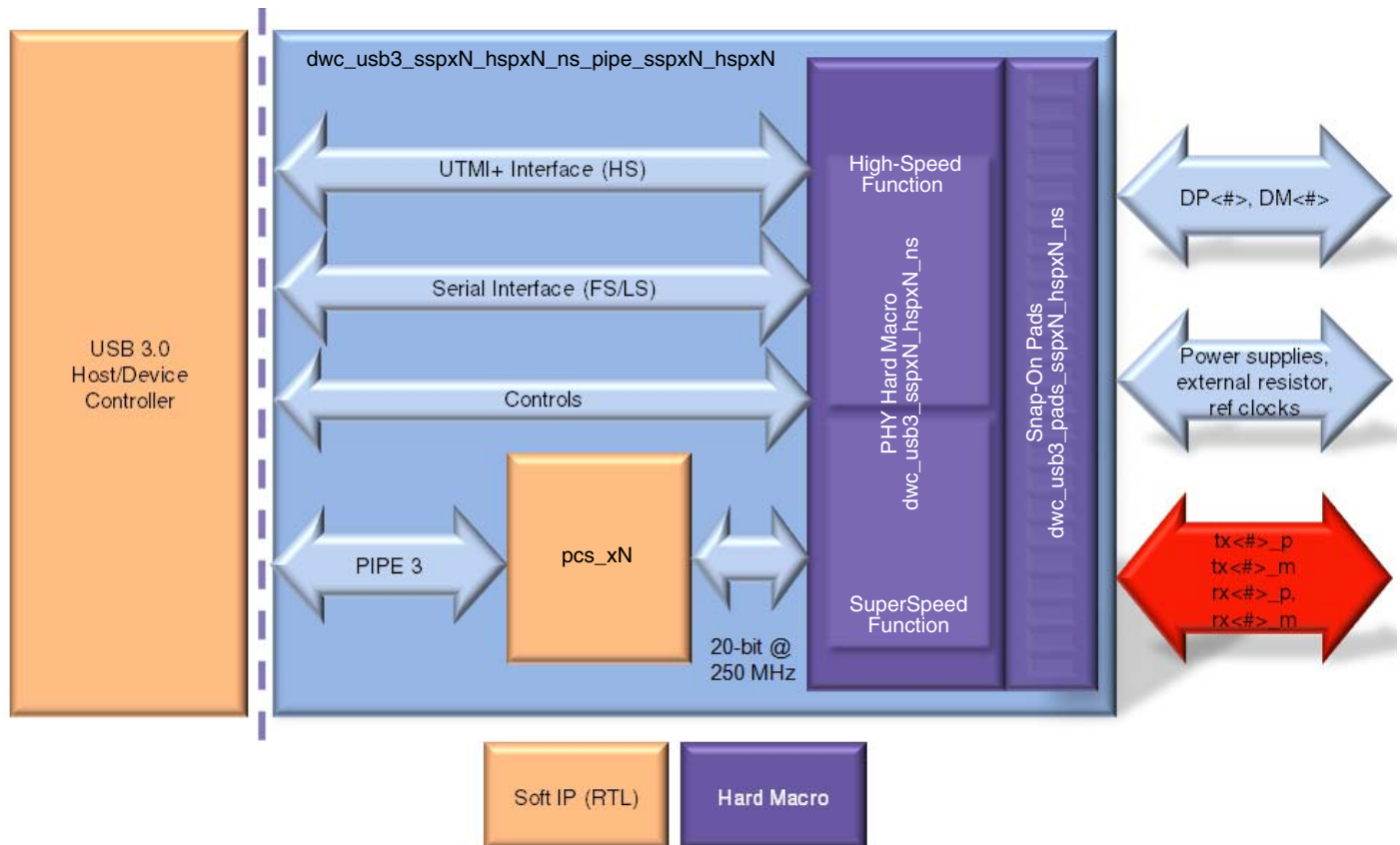
## 1.7 USB 3.0 femtoPHY System-Level Overview

This section provides a system-level overview of the USB 3.0 femtoPHY in an SoC and an overview of the USB 3.0 femtoPHY's external interfaces.

### 1.7.1 System-Level Block Diagram

The following figure shows the hierarchy of the USB 3.0 femtoPHY solution and its interface connections.

**Figure 1-2 USB 3.0 femtoPHY System-Level Block Diagram**



#### 1.7.1.1 USB 3.0 Controller

A USB 3.0 controller implements the protocol and upper layers of the USB 3.0 specification and is the interface to the application layer of any USB 3.0 solution. The USB 3.0 femtoPHY solution can be connected to any USB 3.0 host or device controller through the standard PIPE 3 and UTMI+ interfaces.

## 1.7.2 USB 3.0 femtoPHY Hierarchy

This section describes the USB 3.0 femtoPHY hierarchy.

### 1.7.2.1 Top-Level Module: `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN`

As shown in [Figure 1-2](#) on page 24, the top-level RTL wrapper for the USB 3.0 femtoPHY is `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN`. This module should be instantiated in the chip RTL, and it is through this interface that the USB 3.0 femtoPHY is both configured and connected to the controller. The `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN` module instantiates three sub-blocks as follows.

### 1.7.2.2 PHY Hard Macro: `dwc_usb3_sspaN_hspaN_ns`

`dwc_usb3_sspaN_hspaN_ns` is the hard IP that contains both analog and digital circuitry. `dwc_usb3_sspaN_hspaN_ns` contains circuitry that supports SuperSpeed (5 Gbps) operation as well as high-speed, full-speed, and low-speed legacy operations.

`dwc_usb3_sspaN_hspaN_ns` passes data to and from the controller using the following interfaces.

- **SuperSpeed (SS) function:** 20-bit TX and 20-bit RX datapaths to the PCS layer at 250 MHz
- **High-Speed (HS) function:** USB 2.0 UTMI+ (level 3) interface that supports an 8-bit interface in 60-MHz operation or a 16-bit interface in 30-MHz operation

### 1.7.2.3 Snap-On Pad Ring Hard Macro: `dwc_usb3_pads_sspaN_hspaN_ns`

`dwc_usb3_pads_sspaN_hspaN_ns` is a hard macro that is package technology and metal stack dependent. `dwc_usb3_pads_sspaN_hspaN_ns` contains wirebond pads and/or flip-chip bumps and connects to any power supply pins or any USB 3.0 femtoPHY pins that go off-chip.

For flip-chip packaging, the `dwc_usb3_pads_sspaN_hspaN_ns` block is placed at the same coordinates and orientation as the `dwc_usb3_sspaN_hspaN_ns` PHY hard macro and connects by abutment, but the `dwc_usb3_pads_sspaN_hspaN_ns` overlaps the PHY – placing the bump locations directly on top of the PHY.

This pad block connects the USB 3.0 femtoPHY pins to the USB 3.0 cable off-chip through the following pins.

- **SuperSpeed interface:** Transmit and receive differential pairs `tx<#>_(p,m)` and `rx<#>_(p,m)`. These lines are USB 3.0 specification-compliant and support SuperSpeed (5 Gbps) data transfers.
- **USB D+ (DP<#>) and D- (DM<#>) lines:** These lines are USB 2.0 and 1.1 specification-compliant and support high-speed (480 Mbps) transfers as well as full-speed (12 Mbps) and low-speed (1.5 Mbps) transfers.

#### 1.7.2.4 Physical Coding Sublayer Soft Macro: pcs\_xN

The pcs\_xN block has two sets of interface signals as follows.

- To the PHY hard macro (dwc\_usb3\_sspaN\_hspaN\_ns):
  - 20-bit, 250-MHz Transmit datapath to the PHY
  - 20-bit, 250-MHz Receive datapath from the PHY
- To the controller: A configurable PIPE 3-compliant interface whose signals are available at the top level of the USB 3.0 femtoPHY (dwc\_usb3\_sspaN\_hspaN\_ns\_pipe\_sspaN\_hspaN). The data bus width of this interface is configurable as follows:
  - 8-bit, 500-MHz interface
  - 16-bit, 250-MHz interface
  - 32-bit, 125-MHz interface

#### 1.7.3 USB Controller Integration

The connections between the dwc\_usb3\_sspaN\_hspaN\_ns\_pipe\_sspaN\_hspaN RTL and the controller depend on the controller that is used. These connections might have to be made manually; if you are using the DesignWare USB 3.0 Host/Device controller, CoreConsultant makes these connections for you.

To confirm connectivity, check your controller deliverables for test benches to run datapath simulations.

# 2

## Deliverables

The USB 3.0 femtoPHY full-package release contains all the design views and information required to instantiate the PHY in the top-level chip design and to simulate both the digital functionality of the PHY and the signal integrity of the 5-Gbps link, as well as information to enable testing the PHY in production.

For information about the USB 3.0 femtoPHY release, refer to the top-level readme file included in the product deliverables.

The deliverables are grouped into three basic categories based on usage:

- [“Front-End Views”](#) on page 27: Deliverables used in system simulation and floor planning
- [“Back-End Views”](#) on page 28: Deliverables used in physical design and finishing
- [“Signal Integrity and Test Views”](#) on page 28: Deliverables used to test and simulate link performance at the analog level

### 2.1 Front-End Views

Front-end views are provided for the hard macro portions of the USB 3.0 femtoPHY, the USB 3.0 Pads, and the PCS soft IP.

- Generic technology (GTECH) netlists and digital simulation environment
- Synthesis timing models in .db and .lib formats
- Behavioral models for analog blocks
- Library Exchange Format (LEF) file with pin sizes, pin locations, and antenna information
- PCS layer and test bench examples
- Example PCS synthesis scripts with SDC constraints
- *DesignWare Cores SuperSpeed USB3.0 femtoPHY for TSMC 12-nm FFC 0.8/1.8 V Databook*

## 2.2 Back-End Views

Back-end views are provided for the hard macro portions of the USB 3.0 femtoPHY and the USB 3.0 Pads. These deliverables tend to be metal-stack dependent, so multiple files for each hard macro are provided. The top-level cell name inside all the files is the same, but the content is different—you must select the appropriate file to match your metal stack-up.

- GDSII layout and layer map files
- Layout Versus Schematic (LVS) netlist in HSPICE format and LVS report
- Design Rule Check (DRC) report
- Parasitics in Standard Parasitics Exchange Format (SPEF)
- Gate-level netlist and Standard Delay Format (SDF) timing file
- Gate-level test bench examples

## 2.3 Signal Integrity and Test Views

The following signal integrity and test views are provided:

- Scan test environment with Automatic Test Pattern Generation (ATPG) vectors and example TetraMax scripts
- HSPICE models of RX and TX for signal integrity simulations
- Current waveforms for power delivery network simulations in the package
- IBIS-AMI/IBIS models of RX and TX for signal integrity simulations



# 3

## Signal Descriptions

---

This chapter details all possible I/O signals in the PHY. For configurable IP titles, your actual configuration might not contain all of these signals.

Inputs are on the left of the signal diagrams; outputs are on the right.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

**Active State:** Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).

**Registered:** Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of No does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the controller. A value of N/A indicates that this information is not provided for this IP title.

**Synchronous to:** Indicates which clocks in the IP sample this input (drive for an output) when considering all possible configurations. A particular configuration might not have all of the clocks listed. This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook.

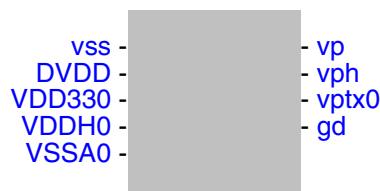
**Exists:** Names of configuration parameters that populate this signal in your configuration.

**Validated by:** Assertion or de-assertion of signals that validates the signal being described.

The I/O signals are grouped as follows:

- Power Supply Signals on [page 31](#)
- Test Controls on [page 34](#)
- UART/Autoresume/Retention Signals on [page 36](#)
- UTMI+ Controls on [page 39](#)
- Battery Charging Signals on [page 45](#)
- Clocks and Resets on [page 47](#)
- Parameter Controls on [page 54](#)
- UTMI+ Transmit Signals on [page 63](#)
- UTMI+ Receive Signals on [page 65](#)
- Signals Connected to Pads on [page 67](#)
- High-Voltage Interrupt Signals on [page 70](#)
- OTG Signals on [page 72](#)
- Serial Interface on [page 75](#)
- Serial Wire Debug Signals on [page 78](#)
- Miscellaneous Signals on [page 80](#)
- Scan Interface on [page 82](#)
- Boundary Scan on [page 111](#)
- PIPE Interface Signals on [page 115](#)
- Control Register Access on [page 123](#)
- JTAG Port on [page 125](#)
- Termination Tuning on [page 127](#)

## 3.1 Power Supply Signals



**Table 3-1 Power Supply Signals**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vp        | IO  | <p>SS Function Low Voltage Supply.</p> <p>Can be supplied from the same off-chip source as the chip VDD core supply, but must be isolated on the board through a ferrite bead and must not be connected to any other core supply in the package or on the die.</p> <p><b>Voltage Range:</b> For the vp voltage range, see "Power Supply Connections" section.</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                |
| vph       | IO  | <p>SS Function I/O Supply.</p> <p>High-speed supply for SS function.</p> <p><b>Voltage Range:</b> For the vph voltage range, see "Power Supply Connections" section.</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                |
| vptx0     | IO  | <p>Transmitter Supply Voltage.</p> <p>Similar to vp, this supply can come from the same off-chip source as the chip VDD supply, but should be isolated on the board through a ferrite bead and must not be connected to any other core supply in the package or on the die.</p> <p>Sharing vptx and vp connections adds RX asynchronous noise on the transmitter power supply. It is recommended that you isolate vptx and vp until they are out on the PCB board.</p> <p><b>Voltage Range:</b> For the vptx&lt;#&gt; voltage range, see "Power Supply Connections" section.</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p> |

**Table 3-1 Power Supply Signals (Continued)**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VSS       | I   | <p>Core Ground.</p> <p>Core vss connection to back-to-back ESD diodes in the snap-on pads.</p> <p>For robust ESD performance, connect this vss to the core ground on the chip.</p> <p><b>Note:</b> The snap-on pads might include more than one vss pin (for example, vss1,vss2).</p> <p><b>Voltage Range:</b> 0 V</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p>                                     |
| DVDD      | I   | <p>HS Low-Voltage Supply.</p> <p>Can be supplied from the same off-chip source as the chip VDD core supply, but must be isolated on the board through a ferrite bead and must not be connected to any other core supply in the package or on the die.</p> <p><b>Voltage Range:</b> For the DVDD voltage range, see "Power Supply Connections" section.</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p> |
| VDD330    | I   | <p>3.3-V Analog Power Supply.</p> <p>Analog power supply.</p> <p>For information about pad/package/board connections for this signal, see "Physical-Level Implementation" section.</p> <p><b>Voltage Range:</b> For the VDD33&lt;#&gt; voltage range, see "Power Supply Connections" section.</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                 |
| VDDH0     | I   | <p>1.8-V Analog Power Supply.</p> <p>Regulator power supply.</p> <p><b>Voltage Range:</b> For the VDDH&lt;#&gt; voltage range, see "Power Supply Connections" section.</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                 |
| VSSA0     | I   | <p>Ground.</p> <p>HS PHY ground. Tie this signal to the chip ground plane in the package, but do not connect to any core ground distribution network on the die.</p> <p><b>Voltage Range:</b> 0 V</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p>                                                                                                                                                      |

**Table 3-1 Power Supply Signals (Continued)**

| Port Name | I/O | Description                                                                                                                                                                                                                                           |
|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gd        | IO  | Ground.<br>SS function ground. Tie this signal to the chip ground plane in the package, but do not connect to any core ground distribution network on the die.<br><b>Voltage Range:</b> 0 V<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> N/A |

## 3.2 Test Controls Signals

ATERESET -  
 LOOPBACKENB0 -  
 VATESTENB -  
 test\_burnin -  
 test\_powerdown\_hsp -  
 test\_powerdown\_ssp -



**Table 3-2 Test Controls Signals**

| Port Name    | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATERESET     | I   | <p>Reset Input from Automatic Test Equipment.</p> <p>When the USB 3.0 femtoPHY is powered up (not in Suspend or Sleep mode), an automatic tester can use this test signal to disable PHYCLOCK&lt;#&gt; and FREECLK, then re-enable them with an aligned phase.</p> <ul style="list-style-type: none"> <li>1: PHYCLOCK&lt;#&gt; and FREECLK outputs are disabled.</li> <li>0: PHYCLOCK&lt;#&gt; and FREECLK outputs are available within a specific period after ATERESET is de-asserted.</li> </ul> <p>If this signal is not used, tie it low.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p> |
| LOOPBACKENB0 | I   | <p>Loopback Test Enable.</p> <p>This controller signal places the USB 3.0 femtoPHY in HS Loopback mode, which concurrently enables the HS receive and transmit logic.</p> <ul style="list-style-type: none"> <li>1: During HS data transmission, the HS receive logic is enabled.</li> <li>0: During HS data transmission, the HS receive logic is disabled.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                        |
| VATESTENB    | I   | <p>Analog Test Pin Select.</p> <p>Enables analog test voltages to be placed on the ID&lt;#&gt; pin.</p> <ul style="list-style-type: none"> <li>1: Analog test voltages can be viewed or applied on ID&lt;#&gt;.</li> <li>0: Analog test voltages cannot be viewed or applied on ID&lt;#&gt;.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                             |

**Table 3-2 Test Controls Signals (Continued)**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| test_burnin        | I   | <p>All Circuits Activator.<br/>Reserved. Tie this signal to ground (gd).<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                                                                                                 |
| test_powerdown_hsp | I   | <p>HS Function Circuits Power-Down Control.<br/>Powers down all HS function circuitry in the PHY for IDDQ testing.<br/>Before asserting this signal, ensure that VDATSRCENB&lt;#&gt;, VDATDETENB&lt;#&gt;, DCDENB&lt;#&gt;, BYPASSSEL&lt;#&gt;, and test_burnin are set to 1'b0.<br/><b>Note:</b> For more details about powering down, see "Power-Down Requirements" section.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p> |
| test_powerdown_ssp | I   | <p>SS Function Circuits Power-Down Control.<br/>Powers down all SS function circuitry in the PHY for IDDQ testing.<br/><b>Note:</b> For more details about powering down, see "Power-Down Requirements" section.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                               |

### 3.3 UART/Autoresume/Retention Signals

AUTORSMENB0 -  
 BYPASSDMDATA0 -  
 BYPASSDMEN0 -  
 BYPASSDPDATA0 -  
 BYPASSDPEN0 -  
 BYPASSSELO -  
 RETENABLEN -



**Table 3-3**     **UART/Autoresume/Retention Signals**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUTORSMENB0   | I   | <p>Autoresume Mode Enable.<br/>           Reserved. Set to 1'b'0.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| BYPASSDMDATA0 | I   | <p>Data for DM&lt;#&gt; Transmitter Digital Bypass.<br/>           In Transmitter Digital Bypass mode, this signal provides the data that is transmitted on DM&lt;#&gt;.<br/>           To use this signal, BYPASSSEL&lt;#&gt; must be set to 1'b1.<br/>           When HSXCVREXTCTL&lt;#&gt; is set to 1'b0, the FS/LS driver is used and outputs the following FS/LS states:</p> <ul style="list-style-type: none"> <li>■ 1: DM&lt;#&gt; FS/LS driver drives to a high state.</li> <li>■ 0: DM&lt;#&gt; FS/LS driver drives to a low state.</li> </ul> <p>When HSXCVREXTCTL&lt;#&gt; is set to 1'b1, set this signal to 1'b0.<br/>           If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |



**Table 3-3 UART/Autoresume/Retention Signals (Continued)**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BYPASSDMEN0   | I   | <p>DM&lt;#&gt; Transmitter Digital Bypass Enable.<br/>Enables/disables the DM&lt;#&gt; FS/LS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b0) in Transmitter Digital Bypass mode.<br/>FS mode is the standard UART mode.<br/>To use this mode, BYPASSSEL&lt;#&gt; must be set to 1'b1.</p> <ul style="list-style-type: none"> <li>1: DM&lt;#&gt; FS/LS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b0) is enabled and driven with the BYPASSDPDATA&lt;#&gt; signal.</li> <li>0: DM&lt;#&gt; FS/LS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b0) is disabled in Transmitter Digital Bypass mode.</li> </ul> <p>When HSXCVREXTCTL&lt;#&gt; is set to 1'b1, set BYPASSDMEN&lt;#&gt; to 1'b0.<br/>The PHY does not reclock the transmit data before outputting on DP&lt;#&gt; or DM&lt;#&gt;.<br/>If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.<br/><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                       |
| BYPASSDPDATA0 | I   | <p>Data for DP&lt;#&gt; Transmitter Digital Bypass.<br/>In Transmitter Digital Bypass mode, this signal provides the data that is transmitted on DP&lt;#&gt;.<br/>To use this signal, BYPASSSEL&lt;#&gt; must be set to 1'b1.<br/>When HSXCVREXTCTL&lt;#&gt; is set to 1'b0, the FS/LS driver is used and outputs the following FS/LS states:</p> <ul style="list-style-type: none"> <li>1: DP&lt;#&gt; FS/LS driver drives to a high state.</li> <li>0: DP&lt;#&gt; FS/LS driver drives to a low state.</li> </ul> <p>When HSXCVREXTCTL&lt;#&gt; is set to 1'b1, BYPASSDPDATA&lt;#&gt; controls the HS driver on both DP&lt;#&gt; and DM&lt;#&gt;.<br/>The HS driver is used and outputs the following HS states:</p> <ul style="list-style-type: none"> <li>1: HS driver transmits J.</li> <li>0: HS driver transmits K.</li> </ul> <p>If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.<br/><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-3 UART/Autoresume/Retention Signals (Continued)**

| Port Name   | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BYPASSDPEN0 | I   | <p>DP&lt;#&gt; Transmitter Digital Bypass Enable.</p> <p>Enables/disables the DP&lt;#&gt; FS/LS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b0), or the DP&lt;#&gt; and DM&lt;#&gt; HS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b1) in Transmitter Digital Bypass mode.</p> <p>FS mode is the standard UART mode. HS mode is provided for proprietary HS tests. To use either of these modes, BYPASSSEL&lt;#&gt; must be set to 1'b1.</p> <ul style="list-style-type: none"> <li>1: DP&lt;#&gt; FS/LS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b0) or DP&lt;#&gt; and DM&lt;#&gt; HS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b1) is enabled and driven with the BYPASSDPDATA&lt;#&gt; signal.</li> <li>0: DP&lt;#&gt; FS/LS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b0) or DP&lt;#&gt; and DM&lt;#&gt; HS driver (if HSXCVREXTCTL&lt;#&gt; is set to 1'b1) is disabled in Transmitter Digital Bypass mode.</li> </ul> <p>The PHY does not reclock the transmit data before outputting on DP&lt;#&gt; or DM&lt;#&gt;.</p> <p>If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p> |
| BYPASSSEL0  | I   | <p>Transmitter Digital Bypass Select.</p> <p>Enables/disables Transmitter Digital Bypass mode.</p> <p>The standard FS UART mode is selected by setting HSXCVREXTCTL&lt;#&gt; to 1'b0, and a proprietary HS test mode is selected by setting HSXCVREXTCTL&lt;#&gt; to 1'b1.</p> <ul style="list-style-type: none"> <li>1: Transmitter Digital Bypass mode is enabled.</li> <li>0: Transmitter Digital Bypass mode is disabled.</li> </ul> <p>If Transmitter Digital Bypass mode is not used, tie this input to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| RETENABLEN  | I   | <p>Lowered Digital Supply Indicator.</p> <p>Reserved. Set to 1b'1.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 3.4 UTMI+ Controls Signals

DMPULLDOWN0 -  
 DPPULLDOWN0 -  
 OPMODE0 -  
 SLEEPM0 -  
 SUSPENDM0 -  
 TERMSEL0 -  
 TXBITSTUFFEN0 -  
 TXBITSTUFFENH0 -  
 VBUSVLDEXT0 -  
 WORDINTERFACE0 -  
 XCVRSEL0 -

- AVALID0  
 - BVALID0  
 - FREECLK  
 - HOSTDISCONNECT0  
 - LINESTATE0  
 - PHYCLOCK0

**Table 3-4 UTMI+ Controls Signals**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AVALID0   | O   | <p>A-Device Session Valid Indicator.<br/>This controller signal is output from the USB 3.0 femtoPHY's Session.<br/>Valid comparator and indicates whether the session for an A-device is valid.</p> <ul style="list-style-type: none"> <li>1: Session for the A-device is valid.</li> <li>0: Session for the A-device is not valid.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p> |
| BVALID0   | O   | <p>B-Device Session Valid Indicator.<br/>This controller signal is output from the USB 3.0 femtoPHY's Session.<br/>Valid comparator and indicates whether the session for a B-device is valid.</p> <ul style="list-style-type: none"> <li>1: Session for the B-device is valid.</li> <li>0: Session for the B-device is not valid.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>  |

**Table 3-4 UTMI+ Controls Signals (Continued)**

| Port Name   | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMPULLDOWN0 | I   | <p>D- Pull-Down Resistor Enable.</p> <p>This controller signal controls the pull-down resistance on the D- line.</p> <ul style="list-style-type: none"> <li>1: Pull-down resistance on D- is enabled.</li> <li>0: Pull-down resistance on D- is disabled.</li> </ul> <p>When an A/B device is acting as a host (downstream-facing port), DPPULLDOWN&lt;#&gt; and DMPULLDOWN&lt;#&gt; are enabled.</p> <p>DMPULLDOWN&lt;#&gt; is also used for DCD in Battery Charging mode to enable RDM_DWN.</p> <p>For more information about enabling DCD, see "DCDENB&lt;#&gt;".</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                               |
| DPPULLDOWN0 | I   | <p>D+ Pull-Down Resistor Enable.</p> <p>This controller signal controls the pull-down resistance on the D+ line.</p> <ul style="list-style-type: none"> <li>1: Pull-down resistance on D+ is enabled.</li> <li>0: Pull-down resistance on D+ is disabled.</li> </ul> <p>When an A/B device is acting as a host (downstream-facing port), DPPULLDOWN&lt;#&gt; and DMPULLDOWN&lt;#&gt; are enabled.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                  |
| FREECLK     | O   | <p>Free-Running UTMI+ Clock.</p> <p>This controller signal is valid only if the USB 3.0 femtoPHY is used as a downstream-facing port. In Suspend or Sleep mode, the state of FREECLK is indeterminate.</p> <p>When this clock is valid, the output is balanced in phase and frequency to PHYCLOCK&lt;#&gt;, with a maximum skew of +/- 500 ps. This phase relationship is maintained even when a port resumes from the Suspend or Sleep state.</p> <p>This signal must not be used for latch-based logic. If FREECLK is not used, leave it disconnected.</p> <p>When this clock is not valid, leave this pin floating or ignore this signal.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p> |

**Table 3-4 UTMI+ Controls Signals (Continued)**

| Port Name       | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HOSTDISCONNECT0 | O   | <p>Peripheral Disconnect Indicator.</p> <p>This controller signal indicates to the host controller that a peripheral has been disconnected from or connected to the USB cable.</p> <ul style="list-style-type: none"> <li>1: No peripheral is connected.</li> <li>0: Peripheral is connected.</li> </ul> <p>In high-speed host operation, HOSTDISCONNECT&lt;#&gt; is updated only at the end of SOF packet transmission. In FS and LS host operations, HOSTDISCONNECT&lt;#&gt; is updated according to UTMI+ specification.</p> <p>This output is valid only when the USB 3.0 femtoPHY's DPPULLDOWN&lt;#&gt; and DMPULLDOWN&lt;#&gt; signals are both set to 1'b1. Otherwise, ignore this signal.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                |
| LINESTATE0[1:0] | O   | <p>Line State Indicator.</p> <p>This controller bus reflects the state of the single-ended receivers. In Suspend or Sleep mode, this bus is a combinatorial output (directly reflecting the current state of D- and D+, respectively).</p> <ul style="list-style-type: none"> <li>11: SE1 (D+ high, D- high)</li> <li>10: K state for high-speed and full-speed USB 2.0 traffic; J state for low-speed USB 2.0 traffic (D+ low, D- high)</li> <li>01: J state for high-speed and full-speed USB2.0 traffic; K state for low-speed USB 2.0 traffic (D+ high, D- low)</li> <li>00: SE0 (D+ low, D- low)</li> </ul> <p>During normal high-speed packet transfers, the line indicates a high-speed J state.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt; (during normal operation)</p> |
| OPMODE0[1:0]    | I   | <p>UTMI+ Operational Mode.</p> <p>This controller bus selects the UTMI+ operational mode.</p> <ul style="list-style-type: none"> <li>11: Normal operation without SYNC or EOP generation. If the XCVRSEL0 bus is not set to 2'b00 while OPMODE&lt;#&gt;[1:0] is set to 2'b11, USB 3.0 femtoPHY HS/FS/LS behavior is undefined.</li> <li>10: Disable bit stuffing and NRZI encoding.</li> <li>01: Non-Driving</li> <li>00: Normal</li> </ul> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                                           |

**Table 3-4 UTMI+ Controls Signals (Continued)**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PHYCLOCK0 | O   | <p>UTMI+ Clock.</p> <p>This controller signal clocks parallel receive and transmit data on the UTMI+.</p> <p>The PHYCLOCK&lt;#&gt; frequency is as follows:</p> <ul style="list-style-type: none"> <li>■ If WORDINTERFACE0 = 1'b1 (16-bit operation): PHYCLOCK&lt;#&gt; = 30 MHz</li> <li>■ If WORDINTERFACE0 = 1'b0 (8-bit operation): PHYCLOCK&lt;#&gt; = 60 MHz</li> </ul> <p>In Suspend or Sleep mode, PHYCLOCK&lt;#&gt; is held low. During serial interface data transmission, PHYCLOCK&lt;#&gt; must not be used. The peak output jitter of PHYCLOCK&lt;#&gt; is +/- 500 ps, which accounts for jitter added by the USB 3.0 femtoPHY and process/environment variations.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p>                        |
| SLEEPM0   | I   | <p>Sleep Assertion.</p> <p>Asserting this signal places the USB 3.0 femtoPHY in Sleep mode according to the USB 2.0 Link Power Management (LPM) addendum to the USB 2.0 specification. In Sleep mode, the transmitter is tri-stated and the USB 3.0 femtoPHY circuits are powered down.</p> <ul style="list-style-type: none"> <li>■ 1: Normal operating mode</li> <li>■ 0: Sleep mode</li> </ul> <p>If SUSPENDM&lt;#&gt; is set to 1'b0, the USB 3.0 femtoPHY remains in Suspend mode irrespective of the SLEEPM&lt;#&gt; setting.</p> <p>If the LPM function is not required, SLEEPM&lt;#&gt; must be tied to DVDD. USB 3.0 femtoPHY Sleep mode can be overridden using the test interface.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> Low<br/> <b>Synchronous To:</b> Asynchronous</p> |
| SUSPENDM0 | I   | <p>Suspend Assertion.</p> <p>Asserting this signal suspends USB 3.0 femtoPHY HS/FS and LS operation by tri-stating the HS transmitter and powering down the USB 3.0 femtoPHY HS circuits.</p> <ul style="list-style-type: none"> <li>■ 1: Normal operating mode</li> <li>■ 0: Suspend mode</li> </ul> <p>USB 3.0 femtoPHY power-down behavior can be overridden using the test interface.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> Low<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                     |

**Table 3-4 UTMI+ Controls Signals (Continued)**

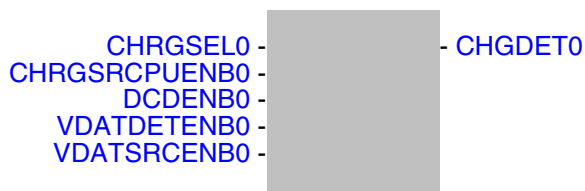
| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TERMSEL0       | I   | <p>USB Termination Select.<br/>This controller signal sets the USB 3.0 femtoPHY's terminations to FS or HS.</p> <ul style="list-style-type: none"> <li>1: Full-speed terminations are enabled.</li> <li>0: High-speed terminations are enabled.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                                                                            |
| TXBITSTUFFEN0  | I   | <p>Low-Byte Transmit Bit-Stuffing Enable.<br/>This controller signal controls bit stuffing on DATAIN&lt;#&gt;[7:0] when OPMODE&lt;#&gt;[1:0] = 2'b11.</p> <ul style="list-style-type: none"> <li>1: Bit stuffing is enabled.</li> <li>0: Bit stuffing is disabled.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                                                        |
| TXBITSTUFFENH0 | I   | <p>High-Byte Transmit Bit-Stuffing Enable.<br/>This controller signal controls bit stuffing on DATAINH&lt;#&gt;[7:0] when OPMODE&lt;#&gt;[1:0] = 2'b11.</p> <ul style="list-style-type: none"> <li>1: Bit stuffing is enabled.</li> <li>0: Bit stuffing is disabled.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                                                      |
| VBUSVLDEXT0    | I   | <p>External VBUS Valid Indicator.<br/>Valid in Device mode and only when the VBUSVLDEXTSEL0 signal is set to 1'b1. VBUSVLDEXT&lt;#&gt; indicates whether the VBUS&lt;#&gt; signal on the USB cable is valid. In addition, VBUSVLDEXT&lt;#&gt; enables the pull-up resistor on the D+ line.</p> <ul style="list-style-type: none"> <li>1: VBUS&lt;#&gt; signal is valid, and the pull-up resistor on D+ is enabled.</li> <li>0: VBUS&lt;#&gt; signal is not valid, and the pull-up resistor on D+ is disabled.</li> </ul> <p>In Host mode, this input is not used and can be tied to 1'b0.<br/><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-4 UTMI+ Controls Signals (Continued)**

| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WORDINTERFACE0 | I   | <p>UTMI+ Data Bus Width and Clock Select.</p> <p>This controller signal selects the data bus width of the UTMI+ data buses.</p> <ul style="list-style-type: none"> <li>■ 1: 16-bit data interface (PHYCLOCK&lt;#&gt; frequency is 30 MHz)</li> <li>■ 0: 8-bit data interface (PHYCLOCK&lt;#&gt; frequency is 60 MHz)</li> </ul> <p>The USB 3.0 femtoPHY supports 8/16-bit data interfaces for all valid USB 3.0 femtoPHY HS/FS/LS speed modes.</p> <p>This signal is a strapping option that must be set prior to a power-on reset and remain static during normal operation. Strapping options are not critical for STA, and any other timings or loading limits for the pin are specified in the .lib timing model included in the product deliverables.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| XCVRSEL0[1:0]  | I   | <p>Transceiver Select.</p> <p>This controller bus selects the HS, FS, or LS Transceiver.</p> <ul style="list-style-type: none"> <li>■ 11: Sends an LS packet on an FS bus or receives an LS packet.</li> <li>■ 10: LS Transceiver</li> <li>■ 01: FS Transceiver</li> <li>■ 00: HS Transceiver</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



## 3.5 Battery Charging Signals



**Table 3-5 Battery Charging Signals**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHGDET0            | O   | <p>Battery Charger Detection Output.</p> <p>Indicates whether the voltage level on DP&lt;#&gt; or DM&lt;#&gt; is greater than VDAT_REF as defined in the Battery Charger specification.</p> <ul style="list-style-type: none"> <li>1: VDP &gt; VDAT_REF (for CHRGSEL&lt;#&gt; = 1'b1) or VDM &gt; VDAT_REF (for CHRGSEL&lt;#&gt; = 1'b0)</li> <li>0: VDP &lt; VDAT_REF (for CHRGSEL&lt;#&gt; = 1'b1) or VDM &lt; VDAT_REF (for CHRGSEL&lt;#&gt; = 1'b0)</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| CHRGSEL0           | I   | <p>Battery Charging Source Select.</p> <p>Determines whether current is sourced onto or sunk from DP&lt;#&gt; or DM&lt;#&gt;.</p> <ul style="list-style-type: none"> <li>1: Data source voltage (VDAT_SRC) is sourced onto DM&lt;#&gt; and sunk from DP&lt;#&gt;.</li> <li>0: Data source voltage (VDAT_SRC) is sourced onto DP&lt;#&gt; and sunk from DM&lt;#&gt;.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                     |
| CHRGSRCPUENB0[1:0] | I   | <p>Charger Source Pull-Up Enable.</p> <p>Reserved. Set to 1b'0.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                |

**Table 3-5 Battery Charging Signals (Continued)**

| Port Name   | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCDENB0     | I   | <p>Data Contact Detection Enable.<br/>Enables current sourcing on the D+ line.<br/>For special applications, the IDP_SRC control and the RDM_DWN control are separated.<br/>For standard DCD operation in accordance with the Battery Charger v1.2 specification, simultaneously set DCDENB&lt;#&gt; = 1'b1 (to enable IDP_SRC) and set DMPULLDOWN&lt;#&gt; = 1'b1 (to enable RDM_DWN).<br/>Set both signals to 1'b0 when DCD is completed.</p> <ul style="list-style-type: none"> <li>1: IDP_SRC current is sourced onto DP&lt;#&gt;.</li> <li>0: IDP_SRC current is disabled.</li> </ul> <p><b>Note:</b> During USB 3.0 femtoPHY normal operation, set this signal low.<br/><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p> |
| VDATDETENB0 | I   | <p>Battery Charging Attach/Connect Detection Enable.<br/>Enables or disables attach/connect detection.</p> <ul style="list-style-type: none"> <li>1: Data detect voltage (CHG_DET) is enabled.</li> <li>0: Data detect voltage (CHG_DET) is disabled.</li> </ul> <p>During USB 3.0 femtoPHY normal operation, set this signal low.<br/><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                        |
| VDATSRCEB0  | I   | <p>Battery Charging Sourcing Select.<br/>Enables or disables sourcing for battery charging.</p> <ul style="list-style-type: none"> <li>1: Data source voltage (VDAT_SRC) is enabled.</li> <li>0: Data source voltage (VDAT_SRC) is disabled.</li> </ul> <p>During USB 3.0 femtoPHY normal operation, set this signal low.<br/><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                  |

## 3.6 Clocks and Resets Signals

|                    |   |                  |
|--------------------|---|------------------|
| COMMONONN          | - | CLK12MOHCI       |
| PORTRESET0         | - | CLK480M          |
| REFCLKSEL          | - | CLK48MOHCI       |
| fsel               | - | mpll_refssc_clk  |
| mpll_multiplier    | - | ref_output_clk   |
| mpll_refssc_clk_en | - | ref_repeat_clk_m |
| phy_reset          | - | ref_repeat_clk_p |
| ref_alt_clk_m      | - |                  |
| ref_alt_clk_p      | - |                  |
| ref_clkdiv2        | - |                  |
| ref_repeat_clk_en  | - |                  |
| ref_ssp_en         | - |                  |
| ref_use_pad        | - |                  |
| ref_use_xo         | - |                  |
| ref_xo_en          | - |                  |
| ssc_en             | - |                  |
| ssc_range          | - |                  |
| ssc_ref_clk_sel    | - |                  |

**Table 3-6 Clocks and Resets Signals**

| Port Name  | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLK12MOHCI | O   | <p>12-MHz Reference Clock for OHCI.</p> <p>12-MHz reference output clock signal that can be used by logic external to the USB 3.0 femtoPHY. This signal does not control any UTMI+ input or output.</p> <p>This signal starts pulsing when the HS PLL is locked and is derived directly from the high-speed 480-MHz output clock. This signal starts pulsing when the HS PLL is locked and stops pulsing when the PLL is powered down in Suspend or Sleep mode.</p> <p>Setting the COMMONONN input to 1'b0 prevents the HS PLL from being powered down in Suspend or Sleep mode; therefore, CLK12MOHCI continues to toggle.</p> <p>The peak output jitter on CLK12MOHCI is +/- 200 ps.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p> |

**Table 3-6 Clocks and Resets Signals (Continued)**

| Port Name  | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLK480M    | O   | <p>480-MHz Buffered Output Clock From HS PLL.</p> <p>This 480-MHz clock is a buffered output clock from the HS PLL that can be used by logic external to the USB 3.0 femtoPHY.</p> <p>Setting the COMMONONN input to 1'b0 prevents the HS PLL from being powered down in Suspend or Sleep mode. Therefore, CLK480M continues to toggle while COMMONONN is asserted, even in Suspend or Sleep mode.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                                                                                                                              |
| CLK48MOHCI | O   | <p>48-MHz Reference Clock for OHCI.</p> <p>48-MHz reference output clock signal that can be used by logic external to the USB 3.0 femtoPHY.</p> <p>This signal does not control any UTMI+ input or output and is derived from the USB 3.0 femtoPHY's internal PLL's 480-MHz output through an internal divide-by-10 circuit.</p> <p>This signal starts pulsing when the PLL is locked and stops pulsing when the PLL is powered down in Suspend or Sleep mode.</p> <p>Setting the COMMONONN input to 1'b0 prevents the PLL from being powered down in Suspend or Sleep mode. Therefore, CLK48MOHCI continues to toggle while COMMONONN is asserted, even in Suspend or Sleep mode.</p> <p>The peak output jitter on CLK48MOHCI is +/- 200 ps.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p> |
| COMMONONN  | I   | <p>Common Block Power-Down Control.</p> <p>Controls the power-down signals in the PLL block when the USB 3.0 femtoPHY is in Suspend or Sleep mode.</p> <ul style="list-style-type: none"> <li>■ 1: In Suspend or Sleep mode, the PLL block is powered down.</li> <li>■ 0: In Suspend or Sleep mode, the PLL block remains powered and continues to draw current.</li> </ul> <p>While the PHY is not in either Suspend mode nor Sleep state, COMMONONN can be dynamically controlled.</p> <p>For COMMONONN timing requirements when entering and exiting Suspend or Sleep mode, see "COMMONONN Control for Suspend and Sleep Modes" section.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> Low<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                          |

**Table 3-6 Clocks and Resets Signals (Continued)**

| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PORTRESET0     | I   | <p>Per-Port Reset.</p> <p>When asserted, this customer-specific signal resets the corresponding port's USB2.0 transmit and receive logic without disabling the clocks within the USB 3.0 femtoPHY.</p> <ul style="list-style-type: none"> <li>1: The transmit and receive finite state machines (FSMs) are reset, and the line_state logic combinatorially reflects the state of the single-ended receivers.</li> <li>0: The transmit and receive FSMs are operational, and the line_state logic becomes sequential after 11 PHYCLOCK&lt;#&gt; cycles.</li> </ul> <p>Asserting PORTRESET&lt;#&gt; does not override any USB 3.0 femtoPHY inputs that normally control the USB 2.0 state, nor does it cause any transient, illegal USB states.</p> <p>Within 100 ns of asserting PORTRESET&lt;#&gt;, the controller must set the inputs that control the USB 2.0 to values that cause a safe state. A safe state for Host and Device modes is defined as follows:</p> <p>Host mode: Non-driving (OPMODE&lt;#&gt;[1:0] = 2'b01) with the 15-k(Ohm) pull-down resistors enabled (DPPULLDOWN&lt;#&gt; and DMPULLDOWN&lt;#&gt; = 1'b1)</p> <p>Device mode: Non-driving (OPMODE&lt;#&gt;[1:0] = 2'b01) with the 1.5-k(Ohm)pull-up resistor disabled.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p> |
| REFCLKSEL[1:0] | I   | <p>Reference Clock Select for PLL Block</p> <p>Reserved. Set it to 2b'10.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| fsel[5:0]      | I   | <p>Frequency Select.</p> <p>Selects the reference clock frequency used for both SS and HS operations.</p> <p>The value for fsel combined with the other clock and enable signals determine the clock frequency used for SS and HS operations and if a shared or separate reference clock is used. For information about using this bus, see "Clocks and Resets" section.</p> <p><b>Voltage Range:</b> fsel[5:3]: 0-vp; fsel[2:0]: 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Table 3-6 Clocks and Resets Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mpll_multiplier[6:0] | I   | <p>MPLL Frequency Multiplier Control</p> <p>Multiplies the reference clock to a frequency suitable for intended operating speed.</p> <p>For information about the corresponding multiplier encodings, see "Timescales" section.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                        |
| mpll_refssc_clk      | O   | <p>Spread Reference Clock Output.</p> <p>Reserved</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| mpll_refssc_clk_en   | I   | <p>Double-Word Clock Enable.</p> <p>Reserved. Set this control to 1'b0.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                               |
| phy_reset            | I   | <p>Core and State Machine Reset.</p> <p>Resets the core and all state machines including the JTAG interface, with the exception of the reference clock buffer.</p> <p>Asserting phy_reset triggers the assertion of the tx&lt;#&gt;_reset and rx&lt;#&gt;_reset signals.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                              |
| ref_alt_clk_m        | I   | <p>ref_alt_clk_m is a Low-Swing Differential Input Clock Pair Internal Pin.</p> <p>Reference clock from on-chip source. Either ref_pad_clk_{p,m} or ref_alt_clk_{p,m} can be used, but never both.</p> <p>The selection is based on ref_use_pad and whether the reference clock is provided to the core via an ondie distribution network or directly to the core from external pins.</p> <p>This input can handle only full-swing CMOS levels.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p> |

**Table 3-6 Clocks and Resets Signals (Continued)**

| Port Name         | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ref_alt_clk_p     | I   | <p>ref_alt_clk_p is Low-Swing Differential Input Clock Pair Internal Pin. Reference clock from on-chip source. Either ref_pad_clk_{p,m} or ref_alt_clk_{p,m} can be used, but never both.</p> <p>The selection is based on ref_use_pad and whether the reference clock is provided to the core via an ondie distribution network or directly to the core from external pins.</p> <p>This input can handle only full-swing CMOS levels.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p> |
| ref_clkdiv2       | I   | <p>Input Reference Clock Divider Control.</p> <p>If the input reference clock frequency is greater than 100 MHz, this signal must be asserted. The reference clock frequency is then divided by 2 to keep it in the range required by the MPLL.</p> <p>When this input is asserted, the ref_ana_usb2_clk (if used) frequency is the reference clock frequency divided by 4.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                  |
| ref_output_clk    | O   | <p>Buffered Version of Input Reference Clock</p> <p>This a buffered version of either ref_pad_clk_{p,m} or ref_alt_clk_{p,m} depending on the settings of ref_use_pad.</p> <p>If ref_clkdiv2 is asserted, this clock is then the divided-by-two reference clock.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p>                                                                                                                                                                       |
| ref_repeat_clk_en | I   | <p>Clock Repeater Buffer Enable.</p> <p>Reserved</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                           |
| ref_repeat_clk_m  | O   | <p>ref_repeat_clk_m is Buffered Differential Reference Clock Output.</p> <p>This signal is enabled by ref_repeat_clk_en. The clock output is at CMOS levels.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> ref_pad_clk_m/ref_pad_clk_p or ref_alt_clk_m/ref_alt_clk_p</p>                                                                                                                                                                                                                  |

**Table 3-6 Clocks and Resets Signals (Continued)**

| Port Name        | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ref_repeat_clk_p | O   | <p>ref_repeat_clk_p is Buffered Differential Reference Clock Output. This signal is enabled by ref_repeat_clk_en. The clock output is at CMOS levels.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> ref_pad_clk_m/ref_pad_clk_p or ref_alt_clk_m/ref_alt_clk_p</p>                                                                                                           |
| ref_ssp_en       | I   | <p>Reference Clock Enable for SS function.</p> <p>Enables the reference clock to the prescaler. The ref_ssp_en signal must remain de-asserted until the reference clock is running at the appropriate frequency, at which point ref_ssp_en can be asserted. For lower power states, ref_ssp_en can also be de-asserted.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> N/A</p> |
| ref_use_pad      | I   | <p>Select Reference Clock Connected to ref_pad_clk_{p,m}.</p> <p>When asserted, selects the external ref_pad_clk_{p,m} inputs as the reference clock source. When ref_use_pad is de-asserted, ref_alt_clk_{p,m} are selected for an on-chip reference clock source.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> N/A</p>                                                     |
| ref_use_xo       | I   | <p>Use External Crystal Oscillator.</p> <p>Reserved. Set it to 1b'0.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                          |
| ref_xo_en        | I   | <p>Crystal Oscillator Always On Enable.</p> <p>Reserved. Set it to 1b'0.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                               |
| ssc_en           | I   | <p>Spread Spectrum Enable.</p> <p>Enables spread spectrum clock production (0.5% down-spread at ~ 31.5 KHz) in the USB 3.0 femtoPHY. Set this signal to 1'b1.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                  |



**Table 3-6 Clocks and Resets Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ssc_range[2:0]       | I   | <p>Spread Spectrum Clock Range.</p> <p>Selects the range of spread spectrum modulation when ssc_en is asserted and the PHY is spreading the high-speed transmit clocks. Applies a fixed offset to the phase accumulator.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                         |
| ssc_ref_clk_sel[8:0] | I   | <p>Spread Spectrum Reference Clock Shifting.</p> <p>Enables non-standard oscillator frequencies to generate targeted MPLL output rates. Input corresponds to frequency-synthesis coefficient.</p> <ul style="list-style-type: none"> <li>■ ssc_ref_clk_sel[8:6] = modulus -1</li> <li>■ ssc_ref_clk_sel[5:0] = 2's complement push amount</li> </ul> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |

### 3.7 Parameter Controls Signals

COMPDISTUNE0 -  
 OTGTUNE0 -  
 SQRXTUNE0 -  
 TXFSLSTUNE0 -  
 TXHSXVTUNE0 -  
 TXPREEMPAMPTUNE0 -  
 TXPREEMPPULSETUNE0 -  
 TXRESTUNE0 -  
 TXRISETUNE0 -  
 TXVREFTUNE0 -  
 VDATREFTUNE0 -  
 lane0\_tx\_term\_offset -  
 los\_bias -  
 los\_level -  
 pcs\_rx\_los\_mask\_val -  
 pcs\_tx\_deemph\_3p5db -  
 pcs\_tx\_deemph\_6db -  
 pcs\_tx\_swing\_full -  
 tx\_vboost\_lvl -

**Table 3-7 Parameter Controls Signals**

| Port Name         | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMPDISTUNE0[2:0] | I   | <p>Disconnect Threshold Adjustment.</p> <p>Adjusts the voltage level for the threshold used to detect a disconnect event at the host. To enable tuning at the board level, connect this bus to a register.</p> <ul style="list-style-type: none"> <li>■ 111: + 4.5%</li> <li>■ 110: + 3%</li> <li>■ 101: + 1.5%</li> <li>■ 100: 0 (default)</li> <li>■ 011: - 1.5%</li> <li>■ 010: - 3%</li> <li>■ 001: - 4.5%</li> <li>■ 000: - 6%</li> </ul> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OTGTUNE0[2:0]  | I   | <p>VBUS Valid Threshold Adjustment.</p> <p>This bus adjusts the voltage level for the VBUS&lt;#&gt; valid threshold. To enable tuning at the board level, connect this bus to a register.</p> <ul style="list-style-type: none"> <li>■ 111: + 9%</li> <li>■ 110: + 6%</li> <li>■ 101: + 3%</li> <li>■ 100: 0 (default)</li> <li>■ 011: - 3%</li> <li>■ 010: - 6%</li> <li>■ 001: - 9%</li> <li>■ 000: - 12%</li> </ul> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>         |
| SQRXTUNE0[2:0] | I   | <p>Squelch Threshold Adjustment.</p> <p>Adjusts the voltage level for the threshold used to detect valid highspeed data. To enable tuning at the board level, connect this bus to a register.</p> <ul style="list-style-type: none"> <li>■ 111: - 20%</li> <li>■ 110: - 15%</li> <li>■ 101: - 10%</li> <li>■ 100: - 5%</li> <li>■ 011: 0 (default)</li> <li>■ 010: + 5%</li> <li>■ 001: + 10%</li> <li>■ 000: + 15%</li> </ul> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name        | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXFSLSTUNE0[3:0] | I   | <p>FS/LS Source Impedance Adjustment.</p> <p>Adjusts the low- and full-speed single-ended source impedance while driving high. To enable tuning at the board level, connect this bus to a register.</p> <ul style="list-style-type: none"> <li>■ 1111: - 5%</li> <li>■ 0111: -2.5%</li> <li>■ 0011: 0 (default)</li> <li>■ 0001: + 2.5%</li> <li>■ 0000: +5%</li> </ul> <p>This parameter control is encoded in thermometer code. Any non-thermometer code setting (that is, 1001) is reserved and not supported.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |
| TXHSXVTUNE0[1:0] | I   | <p>Transmitter High-Speed Crossover Adjustment.</p> <p>This bus adjusts the voltage at which the DP&lt;#&gt; and DM&lt;#&gt; signals cross while transmitting in HS mode.</p> <ul style="list-style-type: none"> <li>■ 11: 0 (default)</li> <li>■ 10: + 15 mV</li> <li>■ 01: - 15 mV</li> <li>■ 00: Reserved</li> </ul> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                               |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name             | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXPREEMPAMPTUNE0[1:0] | I   | <p>HS Transmitter Pre-Emphasis Current Control.</p> <p>This signal controls the amount of current sourced to DP&lt;#&gt; and DM&lt;#&gt; after a J-to-K or K-to-J transition. The HS Transmitter pre-emphasis current is defined in terms of unit amounts. One unit amount is approximately 600 uA and is defined as 1X pre-emphasis current.</p> <ul style="list-style-type: none"> <li>■ 11: 3x pre-emphasis current</li> <li>■ 10: 2x pre-emphasis current</li> <li>■ 01: 1x pre-emphasis current</li> <li>■ 00: Disabled (default)</li> </ul> <p>If these signals are not used, set them to 2'b00.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                           |
| TXPREEMPULSETUNE0     | I   | <p>HS Transmitter Pre-Emphasis Duration Control.</p> <p>This signal controls the duration for which the HS pre-emphasis current is sourced onto DP&lt;#&gt; or DM&lt;#&gt;. The HS Transmitter pre-emphasis duration is defined in terms of unit amounts.</p> <p>One unit of pre-emphasis duration is approximately 580 ps and is defined as 1X pre-emphasis duration. This signal is valid only if either TXPREEMPAMPTUNE&lt;#&gt;[1] or TXPREEMPAMPTUNE&lt;#&gt;[0] is set to 1'b1.</p> <ul style="list-style-type: none"> <li>■ 1: Short</li> <li>■ 0: Long (default)</li> </ul> <p>If TXPREEMPULSETUNE&lt;#&gt; is not used, set it to 1'b0.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name        | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXRESTUNE0[1:0]  | I   | <p>USB Source Impedance Adjustment.</p> <p>Some applications require additional devices to be added on the USB, such as a series switch, which can add significant series resistance.</p> <p>This bus adjusts the driver source impedance (for both pull-up/pull-down impedances in LS/FS mode and the pull-down impedance in HS mode) to compensate for added series resistance on the USB.</p> <ul style="list-style-type: none"> <li>■ 11: - 4 Ohm.</li> <li>■ 10: - 2 Ohm.</li> <li>■ 01: 0 (default)</li> <li>■ 00: + 1.5 Ohm.</li> </ul> <p><b>Note:</b> Any setting other than the default can result in source impedance variation across process, voltage, and temperature conditions that does not meet USB 2.0 specification limits.</p> <p>Use the default setting unless the HS/FS/LS TX impedances are all skewed in the same direction away from the specification's nominal value (45 Ohm) and are near or exceed the specification's limit (40.5-49.5 Ohm).</p> <p>If this bus is not used, leave it at the default setting.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |
| TXRISETUNE0[1:0] | I   | <p>HS Transmitter Rise/Fall Time Adjustment.</p> <p>Adjusts the rise/fall times of the high-speed waveform. To enable tuning at the board level, connect this bit to a register.</p> <ul style="list-style-type: none"> <li>■ 11: - 10%</li> <li>■ 10: 0 (default)</li> <li>■ 01: + 15%</li> <li>■ 00: + 20%</li> </ul> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name         | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXVREFTUNE0[3:0]  | I   | <p>HS DC Voltage Level Adjustment.<br/>Adjusts the high-speed DC level voltage. To enable tuning at the board level, connect this bus to a register.</p> <ul style="list-style-type: none"> <li>■ 1111: + 8.75%</li> <li>■ 1110: + 7.5%</li> <li>■ 1101: + 6.25%</li> <li>■ 1100: + 5%</li> <li>■ 1011: + 3.75%</li> <li>■ 1010: + 2.5%</li> <li>■ 1001: + 1.25%</li> <li>■ 1000: 0 (default)</li> <li>■ 0111: - 1.25%</li> <li>■ 0110: - 2.5%</li> <li>■ 0101: - 3.75%</li> <li>■ 0100: - 5%</li> <li>■ 0011: - 6.25%</li> <li>■ 0010: - 7.5%</li> <li>■ 0001: - 8.75%</li> <li>■ 0000: - 10%</li> </ul> <p>For more information about the parameter controls, see "Parameter Controls" section.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| VDATREFTUNE0[1:0] | I   | <p>Data Detect Voltage Adjustment.<br/>Adjusts the threshold voltage level (Vdat_ref) used to detect data during charger-type detection.</p> <ul style="list-style-type: none"> <li>■ 11: - 20%</li> <li>■ 10: - 10%</li> <li>■ 01: 0 (default)</li> <li>■ 00: + 10%</li> </ul> <p>For more information about the parameter controls, see the "Parameter Controls" section.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                       |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name                 | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lane0_tx_term_offset[4:0] | I   | <p>Transmitter Termination Offset.<br/>Reserved. Set to 5'b00000.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                         |
| los_bias[2:0]             | I   | <p>Loss-of-Signal Detector Threshold Level Control.<br/>Sets the LOS detection threshold level. To enable tuning at the board level, connect this bus to a register.</p> <ul style="list-style-type: none"> <li>■ 111: 135 mV</li> <li>■ 110: 120 mV</li> <li>■ 101: 105 mV</li> <li>■ 100: 90 mV</li> <li>■ 011: 75 mV</li> <li>■ 010: 60 mV</li> <li>■ 001: 45 mV</li> <li>■ 000: Invalid</li> </ul> <p>Default: 3'b101</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| los_level[4:0]            | I   | <p>Loss-of-Signal Detector Sensitivity Level Control.<br/>Reserved. Set this bus to 0x09.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                 |



**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name                | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pcs_rx_los_mask_val[9:0] | I   | <p>Configurable Loss-of-Signal Mask Width.</p> <p>Sets the number of reference clock cycles to mask the incoming LFPS in U3 and U2 states.</p> <p>Masks the incoming LFPS for the number of reference clock cycles equal to the value of pcs_rx_los_mask_val[9:0]. This control filters out short, non-compliant LFPS glitches sent by a non-compliant host.</p> <p>For normal operation, set to a targeted mask interval of 10 us (value = 10 us / Tref_clk). If the ref_clkdiv2 signal is used, the value = 10 us / (2 * Tref_clk).</p> <p>Setting this bus to 10'b0 disables masking. This bus should be accessible to general configuration registers for system testing and debug. The value should be defined when in reset. Changing this value during operation might disrupt normal operation of the link.</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| pcs_tx_deemph_3p5db[5:0] | I   | <p>TX De-Emphasis at 3.5 dB.</p> <p>This static value sets the TX driver de-emphasis value when pipeX_tx_deemph[1:0] is set to 2'b01 (according to the PIPE3 specification). To enable tuning at the board level for RX eye compliance, connect this signal to a register.</p> <p>Default: 28</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| pcs_tx_deemph_6db[5:0]   | I   | <p>TX De-Emphasis at 6 dB.</p> <p>This static value sets the TX driver de-emphasis value when pipeP_tx_deemph[1:0] is set to 2'b10 (according to the PIPE3 specification).</p> <p>This bus is provided for completeness and as a second potential launch amplitude.</p> <p>Default: 28</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Table 3-7 Parameter Controls Signals (Continued)**

| Port Name              | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pcs_tx_swing_full[6:0] | I   | <p>TX Amplitude (Full Swing Mode).</p> <p>This static value sets the launch amplitude of the transmitter. To enable tuning at the board level for RX eye compliance, connect this signal to a register.</p> <p>Default: 127 (7b'1111111)</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |
| tx_vboost_lvl[2:0]     | I   | <p>TX Voltage Boost Level.</p> <p>Sets the boosted transmit launch amplitude (mVppd). To enable tuning at the board level, connect this bus to a register.</p> <p>Default: 3'b100</p> <p>For more information about the parameter controls, see "Parameter Controls" section.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                        |

3.8 UTMI+ Transmit Signals



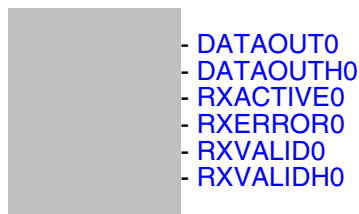
Table 3-8 UTMI+ Transmit Signals

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATAIN0[7:0]  | I   | <p>UTMI+ Data Input Bus, Low Byte.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b1, this controller bus receives the low byte of the 16- bit word of transmit data on the UTMI+.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b0, this bus receives the entire 8-bit word of transmit data on the UTMI+.</p> <p>This data is sampled on the next rising edge of PHYCLOCK&lt;#&gt; after both TXREADY&lt;#&gt; and TXVALID&lt;#&gt; are set to 1'b1.</p> <p>Voltage Range:&lt;/b&gt; 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                         |
| DATAINH0[7:0] | I   | <p>UTMI+ Data Input Bus, High Byte.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b1, this controller bus receives the high byte of the 16- bit transmit data on the UTMI+. This data is sampled on the next rising edge of PHYCLOCK&lt;#&gt; after both TXREADY&lt;#&gt; and TXVALIDH&lt;#&gt; are set to 1'b1.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b0, this bus is not valid and must be ignored.</p> <p>If this bus is not used, tie it to the digital ground supply or ignore any signaling on it.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p> |

**Table 3-8 UTMI+ Transmit Signals (Continued)**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXREADY0  | O   | <p>UTMI+ Transmit Ready Indicator.</p> <p>This controller signal indicates that the USB 3.0 femtoPHY is ready to accept data on the DATAIN&lt;#&gt; and DATAINH&lt;#&gt; buses.</p> <ul style="list-style-type: none"> <li>■ If TXVALID&lt;#&gt; and TXREADY&lt;#&gt; are set to 1'b1, the data on DATAIN&lt;#&gt; and DATAINH&lt;#&gt; is sampled at the rising edge of PHYCLOCK&lt;#&gt;. The controller's Serial Interface Engine (SIE) immediately presents data for the next transfer on DATAIN&lt;#&gt; and DATAINH&lt;#&gt;.</li> <li>■ If TXVALID&lt;#&gt; and TXREADY&lt;#&gt; are set to 1'b0, the controller must hold the previously asserted data on DATAIN&lt;#&gt; and DATAINH&lt;#&gt;. When TXVALID&lt;#&gt; is set to 1'b0, the UTMI+ controller no longer requires TXREADY&lt;#&gt; as an input.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p> |
| TXVALID0  | I   | <p>UTMI+ Transmit Valid Indicator, Low Byte.</p> <p>This controller signal indicates that data on the DATAIN&lt;#&gt; bus is valid.</p> <p>The assertion of TXVALID&lt;#&gt; initiates a SYNC pattern on the USB. The de-assertion of TXVALID&lt;#&gt; initiates an EOP on the USB. The SYNC pattern must be initiated on the USB between one and two PHYCLOCK&lt;#&gt; cycles after the assertion of TXVALID&lt;#&gt;.</p> <p>To enable the FLSRCV&lt;#&gt;, FSVPLUS&lt;#&gt;, and FSVMINUS&lt;#&gt; outputs, TXVALID&lt;#&gt; must be set to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                       |
| TXVALIDH0 | I   | <p>UTMI+ Transmit Valid Indicator, High Byte.</p> <p>This controller signal indicates that data on the DATAINH&lt;#&gt; bus (high byte) is valid.</p> <p>This signal is valid only in 16-bit Data mode (WORDINTERFACE&lt;#&gt; = 1'b1).</p> <p>If this signal is not used, tie it low.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## 3.9 UTMI+ Receive Signals



**Table 3-9 UTMI+ Receive Signals**

| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATAOUT0[7:0]  | O   | <p>UTMI+ Data Output Bus, Low Byte.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b1, this controller bus transfers the low byte of a 16-bit word of receive data on the UTMI+.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b0, this bus transfers the entire 8-bit word of receive data on the UTMI+.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                  |
| DATAOUTH0[7:0] | O   | <p>UTMI+ Data Output Bus, High Byte.</p> <p>If WORDINTERFACE&lt;#&gt; = 1'b1, this controller bus transfers the high byte of the 16-bit receive data on the UTMI+. If WORDINTERFACE&lt;#&gt; = 1'b0, this bus is not valid and must be ignored.</p> <p>If this bus is not used, leave it floating or ignore any signaling on it.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                        |
| RXACTIVE0      | O   | <p>UTMI+ Receive Active.</p> <p>This USB 2.0 controller signal indicates a SYNC pattern is detected for a received packet.</p> <p>The USB 3.0 femtoPHY de-asserts RXACTIVE&lt;#&gt; as follows:</p> <ul style="list-style-type: none"> <li>■ HS: When the USB returns to the Idle state</li> <li>■ FS: When the USB 3.0 femtoPHY detects an EOP</li> </ul> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p> |

**Table 3-9 UTMI+ Receive Signals (Continued)**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RXERROR0  | O   | <p>UTMI+ Receive Error Indicator.<br/>This USB 2.0 controller signal indicates an error is detected in the current received packet.</p> <ul style="list-style-type: none"> <li>1: A receive error was detected.</li> <li>0: No error was detected.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                            |
| RXVALID0  | O   | <p>UTMI+ Receive Data Valid Indicator, Low Byte.<br/>This USB 2.0 controller signal indicates that the DATAOUT&lt;#&gt;[7:0] bus holds valid data.<br/>The data is valid when both RXACTIVE&lt;#&gt; and RXVALID&lt;#&gt; are set to 1'b1.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p>                                                                                                                                                                                           |
| RXVALIDH0 | O   | <p>UTMI+ Receive Data Valid Indicator, High Byte.<br/>This USB 2.0 controller signal indicates that the DATAOUT&lt;#&gt;[7:0] bus holds valid data.<br/>The data is valid when RXACTIVE&lt;#&gt;, RXVALID&lt;#&gt;, and RXVALIDH&lt;#&gt; are set to 1'b1.<br/>This signal is valid only in 16-bit Data mode (WORDINTERFACE&lt;#&gt; = 1'b1).<br/>If this signal is not used, leave this pin floating or ignore any signaling on it.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> PHYCLOCK&lt;#&gt;</p> |

### 3.10 Signals Connected to Pads



**Table 3-10 Signals Connected to Pads**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DM0       | IO  | <p>USB D- Signal.</p> <p>This bidirectional pin carries USB2.0 data to and from the USB 3.0 femtoPHY.</p> <p>In HS operation, this pin receives/transmits a maximum of 800 mV or 400 mV nominally. In FS or LS operation, this pin receives/transmits 3.3 V nominally.</p> <p>For information about pad/package/board connections for this signal, see "Physical-Level Implementation" section.</p> <p><b>Voltage Range:</b> 0-VDD33&lt;#&gt;</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |
| DP0       | IO  | <p>USB D+ Signal.</p> <p>This bidirectional pin carries USB2.0 data to and from the USB 3.0 femtoPHY.</p> <p>In HS operation, this pin receives/transmits a maximum of 800 mV or 400 mV nominally. In FS or LS operation, this pin receives/transmits 3.3 V nominally.</p> <p>For information about pad/package/board connections for this signal, see "Physical-Level Implementation" section.</p> <p><b>Voltage Range:</b> 0-VDD33&lt;#&gt;</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-10 Signals Connected to Pads (Continued)**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID0           | IO  | <p>USB Mini-Receptacle Identifier and Test Point for DC Points Probes Inside USB 3.0 femtoPHY.</p> <p>Differentiates a mini-A from a mini-B plug. The ID&lt;#&gt; line is sampled only when the IDPULLUP0 signal is high. After sampling the ID&lt;#&gt; line, IDDIGO indicates whether a mini-A or mini-B cable is connected. If this signal is not used, internal resistance pulls the signal's voltage level up to the analog I/O supply level when IDPULLUP0 is high. If IDPULLUP0 is low, ID&lt;#&gt; is pulled down to the PHY ground.</p> <p>The ID&lt;#&gt; pin can also be used as an input/output analog test signal. Voltage on this node is determined by the test interface mode and the VATESTENB signal. If ID&lt;#&gt; is not used, leave it floating and tie VATESTENB low.</p> <p><b>Voltage Range:</b> 0-1.8 V<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| VBUS0         | IO  | <p>USB 5-V Power Supply Pin.</p> <p>USB power supply pin. A charge pump external to the USB 3.0 femtoPHY must provide power to this pin.</p> <p>The PHY VBUS&lt;#&gt; pin must not connect directly to the 5-V VBUS voltage on the USB 3.0 link; instead, this pin must be isolated by an external resistor (REXT1) so that the PHY VBUS&lt;#&gt; pin sources a lower voltage.</p> <p>For information about pad/package/board connections for this signal, see "Physical-Level Implementation" section.</p> <p>For information about the charge pump connection to the USB 3.0 femtoPHY, see "Charge Pump Connection" section.</p> <p>For the external resistor value, see "External Components" section.</p> <p><b>Voltage Range:</b> 0-VDD33&lt;#&gt;<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                           |
| ref_pad_clk_m | I   | <p>ref_pad_clk_m is Low-Swing Differential Input Clock Pair With Pad. Reference clock from external source: either ref_pad_clk_{p,m} or ref_alt_clk_{p,m} can be used, but never both.</p> <p>The selection is based on ref_use_pad and whether the reference clock is provided to the core via an ondie distribution network or external pins.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



**Table 3-10 Signals Connected to Pads (Continued)**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ref_pad_clk_p | I   | <p>ref_pad_clk_p is Low-Swing Differential Input Clock Pair With Pad. Reference clock from external source: either ref_pad_clk_{p,m} or ref_alt_clk_{p,m} can be used, but never both.</p> <p>The selection is based on ref_use_pad and whether the reference clock is provided to the core via an ondie distribution network or external pins.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p> |
| resref        | IO  | <p>Reference Resistor Connection.</p> <p>Attach a precision resistor-to-ground on the board, with the following specifications:</p> <ul style="list-style-type: none"> <li>Resistance: 200 Ohm</li> <li>Temperature: Coefficient +/-100 ppm/degree C</li> <li>Tolerance: +/-1%</li> </ul> <p><b>Voltage Range:</b> 0-250 mV<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                              |
| rx0_m         | I   | <p>rxX_m is High-Speed Differential Receive Pair.</p> <p>Receive differential pair.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                    |
| rx0_p         | I   | <p>rxX_p is High-Speed Differential Receive Pair.</p> <p>Receive differential pair.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                    |
| tx0_m         | O   | <p>txX_m is High-Speed Differential Transmit Pair.</p> <p>Transmit differential pair.</p> <p><b>Voltage Range:</b> 0-vptx&lt;#&gt;<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                       |
| tx0_p         | O   | <p>txX_p is High-Speed Differential Transmit Pair.</p> <p>Transmit differential pair.</p> <p><b>Voltage Range:</b> 0-vptx&lt;#&gt;<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                       |

## 3.11 High-Voltage Interrupt Signals



**Table 3-11 High-Voltage Interrupt Signals**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMSEHV0   | O   | <p>Single-Ended D- Indicator (High Voltage).<br/>Indicates the state of the D- line during normal operation.<br/>This output does not depend on the level of vp and can be used as an interrupt for power management purposes.</p> <ul style="list-style-type: none"> <li>■ 1: The voltage on D- is high.</li> <li>■ 0: The voltage on D- is low.</li> </ul> <p>If this signal is not used, leave it floating.<br/><b>Voltage Range:</b> 0-1.8 V<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| DPSEHV0   | O   | <p>Single-Ended D+ Indicator (High Voltage).<br/>Indicates the state of the D+ line during normal operation.<br/>This output does not depend on the level of vp and can be used as an interrupt for power management purposes.</p> <ul style="list-style-type: none"> <li>■ 1: The voltage on D+ is high.</li> <li>■ 0: The voltage on D+ is low.</li> </ul> <p>If this signal is not used, leave it floating.<br/><b>Voltage Range:</b> 0-1.8 V<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| IDHV0     | O   | <p>Mini A/B Plug Indicator (High Voltage).<br/>Indicates whether the connected plug is a mini-A or mini-B plug.<br/>This output does not depend on the level of vp and can be used as an interrupt for power management purposes.</p> <ul style="list-style-type: none"> <li>■ 1: The connected plug is a mini-B plug.</li> <li>■ 0: The connected plug is a mini-A plug.</li> </ul> <p><b>Voltage Range:</b> 0-1.8 V<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                            |

**Table 3-11 High-Voltage Interrupt Signals (Continued)**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OTGSESSVLDHV0 | O   | <p>OTG Device Session Valid Indicator (High Voltage).<br/>This controller signal is output from the USB 3.0 femtoPHY's Session.</p> <p>Valid comparator and indicates whether the voltage on VBUS is below OTG Device Session Valid threshold.</p> <ul style="list-style-type: none"><li>■ 1: The voltage on VBUS is above OTG Device Session Valid threshold.</li><li>■ 0: The voltage on VBUS is below OTG Device Session Valid threshold.</li></ul> <p><b>Voltage Range:</b> 0-1.8 V<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p> |

## 3.12 OTG Signals



**Table 3-12 OTG Signals**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DRVVBUS0  | I   | <p>Drive VBUS.</p> <p>This controller signal controls the VBUS Valid comparator. This signal drives 5 V on VBUS&lt;#&gt; through an external charge pump. When OTGDISABLE0 is set to 1'b0 and DRVVBUS0 is asserted, the Bandgap circuitry and VBUS Valid comparator are powered, even in Suspend or Sleep mode.</p> <ul style="list-style-type: none"> <li>1: VBUS Valid comparator is enabled.</li> <li>0: VBUS Valid comparator is disabled.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p> |
| IDDIG0    | O   | <p>Mini A/B Plug Indicator.</p> <p>This controller signal indicates whether the connected plug is a mini-A or mini-B plug. This signal must be valid within 20 ms after the IDPULLUP0 signal is set to 1'b1.</p> <ul style="list-style-type: none"> <li>1: Connected plug is a mini-B plug.</li> <li>0: Connected plug is a mini-A plug.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                        |
| IDPULLUP0 | I   | <p>Analog ID Input Sample Enable.</p> <p>This controller signal controls ID&lt;#&gt; line sampling.</p> <ul style="list-style-type: none"> <li>1: ID&lt;#&gt; pin sampling is enabled, and the IDDIG0 output is valid.</li> <li>0: ID&lt;#&gt; pin sampling is disabled, and the IDDIG0 output is not valid.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                   |

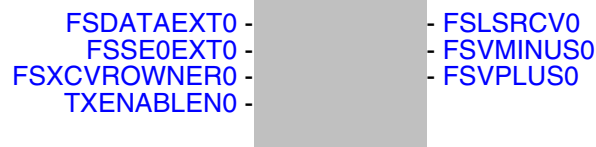
**Table 3-12 OTG Signals (Continued)**

| Port Name   | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OTGDISABLE0 | I   | <p>OTG Block Disable.</p> <p>This controller signal powers down the OTG block, which disables the VBUS Valid and Session End comparators.</p> <p>The Session Valid comparator (the output of which is used to enable the pull-up resistor on DP&lt;#&gt; in Device mode) is always on irrespective of the state of OTGDISABLE0.</p> <p>If the application does not use the OTG function, this input can be set high to save power.</p> <ul style="list-style-type: none"> <li>1: OTG block is powered down.</li> <li>0: OTG block is powered up.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p> |
| OTGSESSVLD0 | O   | <p>OTG Device Session Valid Indicator (Low Voltage).</p> <p>This controller signal is output from the USB 3.0 femtoPHY's Session Valid comparator and indicates whether the voltage on VBUS is below the OTG Device Session Valid threshold.</p> <ul style="list-style-type: none"> <li>1: The voltage on VBUS is above the OTG Device Session Valid threshold.</li> <li>0: The voltage on VBUS is below the OTG Device Session Valid threshold.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                 |
| VBUSVALID0  | O   | <p>VBUS Valid Indicator.</p> <p>This controller signal is output from the USB 3.0 femtoPHY's VBUS Valid comparator and indicates whether the VBUS0 output is at a valid level.</p> <ul style="list-style-type: none"> <li>1: VBUS&lt;#&gt; output is valid.</li> <li>0: VBUS&lt;#&gt; output is not valid.</li> </ul> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                       |

**Table 3-12 OTG Signals (Continued)**

| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VBUSVLDEXTSEL0 | I   | <p>External VBUS Valid Select.</p> <p>Selects the VBUSVLDEXT&lt;#&gt; input or the internal Session Valid comparator to indicate when the VBUS&lt;#&gt; signal on the USB cable is valid.</p> <ul style="list-style-type: none"><li>■ 1: VBUSVLDEXT&lt;#&gt; input is used.</li><li>■ 0: Internal Session Valid comparator is used.</li></ul> <p>This signal is a strapping option that must be set prior to a power-on reset and remain static during normal operation.</p> <p>Strapping options are not critical for STA, and any other timings or loading limits for the pin are specified in the .lib timing model included in the product deliverables.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p> |

### 3.13 Serial Interface Signals



**Table 3-13 Serial Interface Signals**

| Port Name  | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FSDATAEXT0 | I   | <p>USB 1.1 Transmit Data.</p> <p>This controller signal sets the USB (2.0) to either a J or K state. This signal is valid only if FSXCVROWNER&lt;#&gt; is set to 1'b1, TXENABLEN&lt;#&gt; is set to 1'b0, and FSSE0EXT&lt;#&gt; is set to 1'b0.</p> <ul style="list-style-type: none"> <li>1: D+ and D- lines are driven to a Differential 1.</li> <li>0: D+ and D- lines are driven to a Differential 0.</li> </ul> <p>If FSDATAEXT&lt;#&gt; is not used, set it to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| FLSRCV0    | O   | <p>Differential Data Receive Indicator.</p> <p>This controller signal indicates the state of differential data on the USB (2.0) during normal operation.</p> <ul style="list-style-type: none"> <li>1: Voltage on D+ is greater than the voltage on D-.</li> <li>0: Voltage on D+ is less than the voltage on D-.</li> </ul> <p>To enable FLSRCV&lt;#&gt;, the TXVALID&lt;#&gt; signal must be set to 1'b0. If FLSRCV&lt;#&gt; is not used, leave it disconnected.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>         |
| FSSE0EXT0  | I   | <p>USB 1.1 SE0 Generation.</p> <p>This controller signal sets the USB (2.0) to an SE0 state. This signal is valid only if FSXCVROWNER&lt;#&gt; is set to 1'b1 and TXENABLEN&lt;#&gt; is set to 1'b0.</p> <ul style="list-style-type: none"> <li>1: D+ and D- lines are driven to an SE0 state.</li> <li>0: D+ and D- line states are determined by FSDATAEXT&lt;#&gt;.</li> </ul> <p>If FSSE0EXT&lt;#&gt; is not used, set it to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                    |

**Table 3-13 Serial Interface Signals (Continued)**

| Port Name    | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FSVMINUS0    | O   | <p>Single-Ended D- Indicator.<br/>This controller signal indicates the state of the D- line in normal operation.</p> <ul style="list-style-type: none"> <li>1: Voltage on D- is high.</li> <li>0: Voltage on D- is low.</li> </ul> <p>To enable FSVMINUS&lt;#&gt;, the TXVALID&lt;#&gt; signal must be set to 1'b0. If FSVMINUS&lt;#&gt; is not used, leave it disconnected.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                          |
| FSVPLUS0     | O   | <p>Single-Ended D+ Indicator.<br/>This controller signal indicates the state of the D+ line in normal operation.</p> <ul style="list-style-type: none"> <li>1: Voltage on D+ is high.</li> <li>0: Voltage on D+ is low.</li> </ul> <p>To enable FSVPLUS&lt;#&gt;, the TXVALID&lt;#&gt; signal must be set to 1'b0. If FSVPLUS&lt;#&gt; is not used, leave it disconnected.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                            |
| FSXCVROWNER0 | I   | <p>UTMI+/Serial Interface Select.<br/>This controller signal enables the UTMI+ or serial interface.</p> <ul style="list-style-type: none"> <li>1: TXENABLEN&lt;#&gt;, FSDATAEXT&lt;#&gt;, and FSSE0EXT&lt;#&gt; inputs drive USB 2.0 data output onto the D+ and D- lines. Data that the USB 3.0 femtoPHY receives from the D+ and D- lines appears on the FSVMINUS&lt;#&gt; and FSVPLUS&lt;#&gt; outputs.</li> <li>0: Data on the D+ and D- lines is transmitted and received through the UTMI+.</li> </ul> <p>If FSXCVROWNER&lt;#&gt; is not used, set it to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p> |



**Table 3-13 Serial Interface Signals (Continued)**

| Port Name  | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXENABLEN0 | I   | <p>USB 1.1 Data Enable.</p> <p>This controller signal enables the FSDATAEXT&lt;#&gt; and FSSE0EXT&lt;#&gt; inputs. TXENABLEN&lt;#&gt; is valid only when the FSXCVROWNER&lt;#&gt; signal is set to 1'b1.</p> <ul style="list-style-type: none"> <li>■ 1: FSDATAEXT&lt;#&gt; and FSSE0EXT&lt;#&gt; are disabled.</li> <li>■ 0: FSDATAEXT&lt;#&gt; and FSSE0EXT&lt;#&gt; are enabled.</li> </ul> <p>If TXENABLEN&lt;#&gt; is not used, set it to 1'b1.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> Low</p> <p><b>Synchronous To:</b> Asynchronous</p> |

## 3.14 Serial Wire Debug Signals

HSXCVREXTCTL0 -  - HSRXDAT0  
HSSQUELCH0

**Table 3-14 Serial Wire Debug Signals**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HSRXDAT0  | O   | <p>HS Transceiver Asynchronous Data Output.</p> <p>This signal asynchronously outputs the state of the HS receiver and can be used with the bypass signals to perform proprietary tests on the HS RX block.</p> <p>(For information about the bypass signals, see the "UART/Autoresume/Retention Signals" section.)</p> <p>The PHY must be in an HS state with XCVRSEL&lt;#&gt;[1:0] set to 2'b00 and TERMSEL&lt;#&gt; set to 1'b0.</p> <p>To use this mode, HSXCVREXTCTL&lt;#&gt; must be set to 1'b1.</p> <p>If HSXCVREXTCTL&lt;#&gt; is set to 1'b0, HSRXDAT&lt;#&gt; is gated to 1'b0.</p> <ul style="list-style-type: none"> <li>1: If HSXCVREXTCTL&lt;#&gt; is set to 1'b1 and HSSQUELCH&lt;#&gt; is set to 1'b0, the HS differential receiver has detected an HS J.</li> <li>0: If HSXCVREXTCTL&lt;#&gt; is set to 1'b1 and HSSQUELCH&lt;#&gt; is set to 1'b0, the HS differential receiver has detected an HS K.</li> </ul> <p>HSRXDAT&lt;#&gt; is valid only when HSXCVREXTCTL&lt;#&gt; is set to 1'b1 and HSSQUELCH&lt;#&gt; is set to 1'b0.</p> <p>If HSXCVREXTCTL&lt;#&gt; is set to 1'b0 or HSSQUELCH&lt;#&gt; is set to 1'b1, HSRXDAT&lt;#&gt; is 1'b0 and does not indicate a J nor a K.</p> <p>This signal is not dependent on the state of BYPASSSEL&lt;#&gt;.</p> <p>If HSRXDAT&lt;#&gt; is not used, leave this output floating.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-14 Serial Wire Debug Signals (Continued)**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HSSQUELCH0    | O   | <p>HS Squelch Detector Asynchronous Output.</p> <p>This signal asynchronously outputs the state of the HS squelch detector and can be used with the bypass signals to perform proprietary tests on the HS RX block.</p> <p>(For information about the bypass signals, see the "UART/Autoresume/Retention Signals" section.)</p> <p>The PHY must be in an HS state with XCVRSEL&lt;#&gt;[1:0] set to 2'b00 and TERMSEL&lt;#&gt; set to 1'b0.</p> <p>To use this mode, HSXCVREXTCTL&lt;#&gt; must be set to 1'b1.</p> <p>If HSXCVREXTCTL&lt;#&gt; is set to 1'b0, HSSQUELCH&lt;#&gt; is gated to 1'b0.</p> <ul style="list-style-type: none"> <li>1: If HSXCVREXTCTL&lt;#&gt; is set to 1'b1, the HS squelch detector has not detected a valid HS VIL level. The line is squelched.</li> <li>0: If HSXCVREXTCTL&lt;#&gt; is set to 1'b1, the HS squelch detector has detected a valid HS VIL level. The line is not squelched. The HSRXDAT&lt;#&gt; signal indicates whether the detected signal is a J or a K.</li> </ul> <p>This signal is valid only when HSXCVREXTCTL&lt;#&gt; is set to 1'b1.</p> <p>If HSXCVREXTCTL&lt;#&gt; is set to 1'b0, HSSQUELCH&lt;#&gt; is 1'b0 and does not indicate the squelch condition.</p> <p>This signal is not dependent on the state of BYPASSSEL&lt;#&gt;.</p> <p>If HSSQUELCH&lt;#&gt; is not used, leave it floating.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p> |
| HSXCVREXTCTL0 | I   | <p>HS Transceiver Asynchronous Control.</p> <p>This signal asynchronously selects the HS transceiver TX/RX path and can be used with the bypass signals to perform proprietary tests on the HS TX / RX paths.</p> <p>(For information about the bypass signals, see the "UART/Autoresume/Retention Signals" section.)</p> <p>The PHY must be in an HS state with XCVRSEL&lt;#&gt;[1:0] set to 2'b00 and TERMSEL&lt;#&gt; set to 1'b0.</p> <ul style="list-style-type: none"> <li>1: HS TX/RX path is selected.</li> <li>0: FS TX/RX path is selected.</li> </ul> <p>If this signal is not used, tie this input to 1'b0.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### 3.15 Miscellaneous Signals

RX0LOSLFPSEN - 

Table 3-15 Miscellaneous Signals

| Port Name    | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX0LOSLFPSEN | I   | <p>RX LOS LFPS Filter Enable (0).</p> <p>Enables the RX LOS LFPS filter. During USB 3.0 operation, set to 1'b1.</p> <ul style="list-style-type: none"><li>1: Enables RX LOS LFPS filter.</li><li>0: Disables RX LOS LFPS filter.</li></ul> <p>Change this pin setting only during a reset.</p> <p><b>Note:</b> In Scan mode, tie this pin to 0.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> Asynchronous</p> |

### 3.16 Scan Interface Signals

|                        |                        |
|------------------------|------------------------|
| SCANCDRPHASE0CLK0      | SCANNLNSOCLK30M0       |
| SCANCDRPHASE10CLK0     | SCANNLNSOCLK48M0       |
| SCANCDRPHASE11CLK0     | SCANNLNSOCLK60M0       |
| SCANCDRPHASE12CLK0     | SCANNLNSOHSPREFCLK0    |
| SCANCDRPHASE13CLK0     | SCANNLNSOPLL480MCLK0   |
| SCANCDRPHASE14CLK0     | SCANNLNSOSERCLK0       |
| SCANCDRPHASE15CLK0     | SCANNLNSOUTMICLK0      |
| SCANCDRPHASE1CLK0      | SCANNSOCLK30M0         |
| SCANCDRPHASE2CLK0      | SCANNSOCLK48M0         |
| SCANCDRPHASE3CLK0      | SCANNSOCLK60M0         |
| SCANCDRPHASE4CLK0      | SCANNSOHSPREFCLK0      |
| SCANCDRPHASE5CLK0      | SCANNSOPLL480MCLK0     |
| SCANCDRPHASE6CLK0      | SCANNSOSERCLK0         |
| SCANCDRPHASE7CLK0      | SCANNSOUTMICLK0        |
| SCANCDRPHASE8CLK0      | SCANPSOCDRPHASE0CLK0   |
| SCANCDRPHASE9CLK0      | SCANPSOCDRPHASE10CLK0  |
| SCANCLK12MOHCI0        | SCANPSOCDRPHASE11CLK0  |
| SCANCLK30M0            | SCANPSOCDRPHASE12CLK0  |
| SCANCLK48M0            | SCANPSOCDRPHASE13CLK0  |
| SCANCLK48MOHCI0        | SCANPSOCDRPHASE14CLK0  |
| SCANCLK60M0            | SCANPSOCDRPHASE15CLK0  |
| SCANHSPREFCLK0         | SCANPSOCDRPHASE1CLK0   |
| SCANNSICLK30M0         | SCANPSOCDRPHASE2CLK0   |
| SCANNSICLK48M0         | SCANPSOCDRPHASE3CLK0   |
| SCANNSICLK60M0         | SCANPSOCDRPHASE4CLK0   |
| SCANNSIHSPREFCLK0      | SCANPSOCDRPHASE5CLK0   |
| SCANNSIPLL480MCLK0     | SCANPSOCDRPHASE6CLK0   |
| SCANNSISERCLK0         | SCANPSOCDRPHASE7CLK0   |
| SCANNSIUTMICLK0        | SCANPSOCDRPHASE8CLK0   |
| SCANPLL480MCLK0        | SCANPSOCDRPHASE9CLK0   |
| SCANPLL480MHSTXCLK0    | SCANPSOCLK48M0         |
| SCANPSICDRPHASE0CLK0   | SCANPSOHSPREFCLK0      |
| SCANPSICDRPHASE10CLK0  | SCANPSOPLL480MCLK0     |
| SCANPSICDRPHASE11CLK0  | SCANPSOPLL480MHSTXCLK0 |
| SCANPSICDRPHASE12CLK0  | SCANPSOSERCLK0         |
| SCANPSICDRPHASE13CLK0  | SCANPSOSERCLKRX0       |
| SCANPSICDRPHASE14CLK0  | SCANPSOSERCLKTX0       |
| SCANPSICDRPHASE15CLK0  | SCANPSOTESTCLK0        |
| SCANPSICDRPHASE1CLK0   | SCANPSOUTMICLK0        |
| SCANPSICDRPHASE2CLK0   | scan_bs_out            |
| SCANPSICDRPHASE3CLK0   | scan_jtag_out          |
| SCANPSICDRPHASE4CLK0   | scan_mpll_dword_out    |
| SCANPSICDRPHASE5CLK0   | scan_mpll_qword_out    |
| SCANPSICDRPHASE6CLK0   | scan_mpll_word_out     |
| SCANPSICDRPHASE7CLK0   | scan_rx0_dword_out     |
| SCANPSICDRPHASE8CLK0   | scan_rx0_scope_out     |
| SCANPSICDRPHASE9CLK0   | scan_rx0_word_out      |
| SCANPSICLK48M0         | scan_ssref_out         |
| SCANPSIHSPREFCLK0      | scan_tx0_dword_out     |
| SCANPSIPLL480MCLK0     | scan_tx0_word_out      |
| SCANPSIPLL480MHSTXCLK0 |                        |
| SCANPSISERCLK0         |                        |
| SCANPSISERCLKRX0       |                        |
| SCANPSISERCLKTX0       |                        |
| SCANPSITESTCLK0        |                        |
| SCANPSIUTMICLK0        |                        |
| SCANSERCLK0            |                        |
| SCANSERCLKRX0          |                        |

SCANSERCLKTX0 -  
 SCANTESTCLK0 -  
 SCANUTMICK0 -  
 pcs\_scan\_clk -  
 pcs\_scan\_mode -  
 pcs\_scan\_pclk -  
 pcs\_scan\_refclk -  
 pcs\_scan\_rst -  
 pcs\_scan\_rx0\_clk -  
 pcs\_scan\_shift -  
 pcs\_scan\_tx0\_clk -  
 scan\_bs\_in -  
 scan\_jtag\_in -  
 scan\_mode -  
 scan\_mpll\_dword\_clk -  
 scan\_mpll\_dword\_in -  
 scan\_mpll\_qword\_clk -  
 scan\_mpll\_qword\_in -  
 scan\_mpll\_word\_clk -  
 scan\_mpll\_word\_in -  
 scan\_rst -  
 scan\_rx0\_dword\_clk -  
 scan\_rx0\_dword\_in -  
 scan\_rx0\_scope\_clk -  
 scan\_rx0\_scope\_in -  
 scan\_rx0\_word\_clk -  
 scan\_rx0\_word\_in -  
 scan\_shift -  
 scan\_sspref\_clk -  
 scan\_sspref\_in -  
 scan\_tx0\_dword\_clk -  
 scan\_tx0\_dword\_in -  
 scan\_tx0\_out\_clk -  
 scan\_tx0\_word\_clk -  
 scan\_tx0\_word\_in -

Table 3-16 Scan Interface Signals

| Port Name         | I/O | Description                                                                                                                                                                                                                                                                                                                  |
|-------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANCDRPHASE0CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>           ASST scan clock for CDRPHASE0CLK0.<br/>           For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                           |
|--------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANCDRPHASE10CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE10CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE11CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE11CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE12CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE12CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE13CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE13CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE14CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE14CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                           |
|--------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANCDRPHASE15CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE15CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE1CLK0  | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE1CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>  |
| SCANCDRPHASE2CLK0  | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE2CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>  |
| SCANCDRPHASE3CLK0  | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE3CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>  |
| SCANCDRPHASE4CLK0  | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE4CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>  |



**Table 3-16 Scan Interface Signals (Continued)**

| Port Name         | I/O | Description                                                                                                                                                                                                                                                                                          |
|-------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANCDRPHASE5CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE5CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE6CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE6CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE7CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE7CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE8CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE8CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANCDRPHASE9CLK0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for CDRPHASE9CLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name       | I/O | Description                                                                                                                                                                                                                                                                                            |
|-----------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANCLK12MOHCI0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/> ASST scan clock for CLK12MOHCI0.<br/> For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| SCANCLK30M0     | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/> ASST scan clock for CLK30M0.<br/> For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>     |
| SCANCLK48M0     | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/> ASST scan clock for CLK48M0.<br/> For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>     |
| SCANCLK48MOHCI0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/> ASST scan clock CLK48MOHCI0.<br/> For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>     |
| SCANCLK60M0     | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/> ASST scan clock for CLK60M0.<br/> For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>     |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name        | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANHSPREFCLK0   | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for HSPREFCLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                 |
| SCANNLNSOCLK30M0 | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.<br/>ASST output of the negative-edge-triggered scan chain related to CLK30M0. This output does not contain a lock-up latch on the output.<br/>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANCLK30M0</p> |
| SCANNLNSOCLK48M0 | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.<br/>ASST output of the negative-edge-triggered scan chain related to CLK48M0. This output does not contain a lock-up latch on the output.<br/>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANCLK48M0</p> |
| SCANNLNSOCLK60M0 | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.<br/>ASST output of the negative-edge-triggered scan chain related to CLK60M0. This output does not contain a lock-up latch on the output.<br/>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANCLK60M0</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANNLNSOHSPREFCLK0  | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output of the negative-edge-triggered scan chain related to HSPREFCLK0. This output does not contain a lock-up latch on the output.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANHSPREFCLK0</p>   |
| SCANNLNSOPLL480MCLK0 | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output of the negative-edge-triggered scan chain related to PLL480MCLK0. This output does not contain a lock-up latch on the output.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANPLL480MCLK0</p> |
| SCANNLNSOSERCLK0     | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output of the negative-edge-triggered scan chain related to SERCLK0. This output does not contain a lock-up latch on the output.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANSERCLK0</p>         |
| SCANNLNSOUTMICK0     | O   | <p>ASST Non-Lock-Up Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output of the negative-edge-triggered scan chain related to UTMICK0. This output does not contain a lock-up latch on the output.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANUTMICK0</p>         |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANNSICLK30M0     | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain. ASST scan input for the negative-edge-triggered scan chain related to CLK30M0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANCLK30M0</p>         |
| SCANNSICLK48M0     | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain. ASST scan input for the negative-edge-triggered scan chain related to CLK48M0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANCLK48M0</p>         |
| SCANNSICLK60M0     | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain. ASST scan input for the negative-edge-triggered scan chain related to CLK60M0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANCLK60M0</p>         |
| SCANNSIHSPREFCLK0  | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain. ASST scan input for the negative-edge-triggered scan chain related to HSPREFCLK0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANHSPREFCLK0</p>   |
| SCANNSIPLL480MCLK0 | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain. ASST scan input for the negative-edge-triggered scan chain related to PLL480MCLK0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANPLL480MCLK0</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name      | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANNSISERCLK0 | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain.<br/>ASST scan input for the negative-edge-triggered scan chain related to SERCLK0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANSERCLK0</p>                                       |
| SCANNSIUTMICK0 | I   | <p>ASST Scan Input Data for negative-edge-triggered scan chain.<br/>ASST scan input for the negative-edge-triggered scan chain related to UTMICK0.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANUTMICK0</p>                                       |
| SCANNSOCLK30M0 | O   | <p>SCANNSOCLK30M0 is ASST Scan Data Out for negative-edge-triggered scan chain.<br/>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANCLK30M0 (output on rising edge due to lock-up latch)</p> |
| SCANNSOCLK48M0 | O   | <p>SCANNSOCLK48M0 is ASST Scan Data Out for negative-edge-triggered scan chain.<br/>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Negative edge of SCANCLK48M0 (output on rising edge due to lock-up latch)</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANNSOCLK60M0     | O   | <p>SCANNSOCLK60M0 is ASST Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANCLK60M0 (output on rising edge due to lock-up latch)</p>         |
| SCANNSOHSPREFCLK0  | O   | <p>SCANNSOHSPREFCLK0 is ASST Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANHSPREFCLK0 (output on rising edge due to lock-up latch)</p>   |
| SCANNSOPLL480MCLK0 | O   | <p>SCANNSOPLL480MCLK0 is ASST Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANPLL480MCLK0 (output on rising edge due to lock-up latch)</p> |
| SCANNSOSERCLK0     | O   | <p>SCANNSOSERCLK0 is ASST Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANSERCLK0 (output on rising edge due to lock-up latch)</p>         |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name             | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANNSOUTMICK0        | O   | <p>SCANNSOUTMICK0 is ASST Scan Data Out for negative-edge-triggered scan chain.</p> <p>ASST output for the negative-edge-triggered scan chain.</p> <p>For a list of clock domains that contain negative-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Negative edge of SCANUTMICK0 (output on rising edge due to lock-up latch)</p> |
| SCANPLL480MCLK0       | I   | <p>ASST Scan Clock for high-speed portion of PHY.</p> <p>ASST scan clock for PLL480MCLK0.</p> <p>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                     |
| SCANPLL480MHSTXCLK0   | I   | <p>ASST Scan Clock for high-speed portion of PHY.</p> <p>ASST scan clock for PLL480MHSTXCLK0.</p> <p>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                 |
| SCANPSICDRPHASE0CLK0  | I   | <p>ASST Scan Input Data.</p> <p>ASST scan input for the scan chain related to CDRPHASE0CLK0.</p> <p>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> SCANCDRPHASE0CLK0</p>                                                                                                           |
| SCANPSICDRPHASE10CLK0 | I   | <p>ASST Scan Input Data.</p> <p>ASST scan input for the scan chain related to CDRPHASE10CLK0.</p> <p>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD</p> <p><b>Active State:</b> N/A</p> <p><b>Synchronous To:</b> SCANCDRPHASE10CLK0</p>                                                                                                         |



**Table 3-16 Scan Interface Signals (Continued)**

| Port Name             | I/O | Description                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSICDRPHASE11CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE11CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE11CLK0</p> |
| SCANPSICDRPHASE12CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE12CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE12CLK0</p> |
| SCANPSICDRPHASE13CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE13CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE13CLK0</p> |
| SCANPSICDRPHASE14CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE14CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE14CLK0</p> |
| SCANPSICDRPHASE15CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE15CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE15CLK0</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                    |
|----------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSICDRPHASE1CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE1CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE1CLK0</p> |
| SCANPSICDRPHASE2CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE2CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE2CLK0</p> |
| SCANPSICDRPHASE3CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE3CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE3CLK0</p> |
| SCANPSICDRPHASE4CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE4CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE4CLK0</p> |
| SCANPSICDRPHASE5CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE5CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE5CLK0</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                    |
|----------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSICDRPHASE6CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE6CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE6CLK0</p> |
| SCANPSICDRPHASE7CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE7CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE7CLK0</p> |
| SCANPSICDRPHASE8CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE8CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE8CLK0</p> |
| SCANPSICDRPHASE9CLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CDRPHASE9CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE9CLK0</p> |
| SCANPSICLK48M0       | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to CLK48M0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCLK48M0</p>             |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name              | I/O | Description                                                                                                                                                                                                                                                                                                                                        |
|------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSIHSPREFCLK0      | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to HSPREFCLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANHSPREFCLK0</p>           |
| SCANPSIPLL480MCLK0     | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to PLL480MCLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANPLL480MCLK0</p>         |
| SCANPSIPLL480MHSTXCLK0 | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to PLL480MHSTXCLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANPLL480MHSTXCLK0</p> |
| SCANPSISERCLK0         | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to SERCLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANSERCLK0</p>                 |
| SCANPSISERCLKRX0[3:0]  | I   | <p>ASST Scan Input Data.<br/> ASST scan input for the scan chain related to SERCLKRX0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANSERCLKRX0</p>             |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name             | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSISERCLKTX0[1:0] | I   | <p>ASST Scan Input Data.<br/>ASST scan input for the scan chain related to SERCLKTX0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANSERCLKTX0</p>                                                           |
| SCANPSITESTCLK0[1:0]  | I   | <p>ASST Scan Input Data.<br/>ASST scan input for the scan chain related to TESTCLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANTESTCLK0</p>                                                             |
| SCANPSIUTMICK0[2:0]   | I   | <p>ASST Scan Input Data.<br/>ASST scan input for the scan chain related to UTMICK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANUTMICK0</p>                                                               |
| SCANPSOCDRPHASE0CLK0  | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE0CLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE0CLK0 (output on falling edge due to lock-up latch)</p>   |
| SCANPSOCDRPHASE10CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE10CLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE10CLK0 (output on falling edge due to lock-up latch)</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name             | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSOCDRPHASE11CLK0 | O   | <p>ASST Scan Output Data.<br/> ASST scan output for the scan chain related to CDRPHASE11CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE11CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE12CLK0 | O   | <p>ASST Scan Output Data.<br/> ASST scan output for the scan chain related to CDRPHASE12CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE12CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE13CLK0 | O   | <p>ASST Scan Output Data.<br/> ASST scan output for the scan chain related to CDRPHASE13CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE13CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE14CLK0 | O   | <p>ASST Scan Output Data.<br/> ASST scan output for the scan chain related to CDRPHASE14CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE14CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE15CLK0 | O   | <p>ASST Scan Output Data.<br/> ASST scan output for the scan chain related to CDRPHASE15CLK0.<br/> For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.<br/> <b>Voltage Range:</b> 0-DVDD<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> SCANCDRPHASE15CLK0 (output on falling edge due to lock-up latch)</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSOCDRPHASE1CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE1CLK0. For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE1CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE2CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE2CLK0. For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE2CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE3CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE3CLK0. For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE3CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE4CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE4CLK0. For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE4CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE5CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE5CLK0. For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE5CLK0 (output on falling edge due to lock-up latch)</p> |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSOCDRPHASE6CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE6CLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE6CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE7CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE7CLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE7CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE8CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE8CLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE8CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCDRPHASE9CLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CDRPHASE9CLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCDRPHASE9CLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOCLK48M0       | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to CLK48M0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANCLK48M0 (output on falling edge due to lock-up latch)</p>             |



**Table 3-16 Scan Interface Signals (Continued)**

| Port Name              | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSOHSPREFCLK0      | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to HSPREFCLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANHSPREFCLK0 (output on falling edge due to lock-up latch)</p>           |
| SCANPSOPLL480MCLK0     | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to PLL480MCLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANPLL480MCLK0 (output on falling edge due to lock-up latch)</p>         |
| SCANPSOPLL480MHSTXCLK0 | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to PLL480MHSTXCLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANPLL480MHSTXCLK0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOSERCLK0         | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to PSOSERCLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANPSOSERCLK0 (output on falling edge due to lock-up latch)</p>           |
| SCANPSOSERCLKRX0[3:0]  | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to PSOSERCLKRX0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANPSOSERCLKRX0 (output on falling edge due to lock-up latch)</p>       |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name             | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANPSOSERCLKTX0[1:0] | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to PSOSERCLKTX0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANPSOSERCLKTX0 (output on falling edge due to lock-up latch)</p> |
| SCANPSOTESTCLK0[1:0]  | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to TESTCLK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANTESTCLK0 (output on falling edge due to lock-up latch)</p>         |
| SCANPSOUTMICK0[2:0]   | O   | <p>ASST Scan Output Data.<br/>ASST scan output for the scan chain related to UTMICK0.<br/>For a list of clock domains that contain positive-edge-triggered scan chains, see "Scan Chains Per Scan Clock Domain" table.</p> <p><b>Voltage Range:</b> 0-DVDD<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> SCANUTMICK0 (output on falling edge due to lock-up latch)</p>           |
| SCANSERCLK0           | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for SERCLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                             |
| SCANSERCLKRX0         | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for SERCLKRX0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                           |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name     | I/O | Description                                                                                                                                                                                                                                                                                      |
|---------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCANSERCLKTX0 | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for SERCLKTX0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| SCANTESTCLK0  | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for TESTCLK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>  |
| SCANUTMICK0   | I   | <p>ASST Scan Clock for high-speed portion of PHY.<br/>ASST scan clock for UTMICK0.<br/>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>   |
| pcs_scan_clk  | I   | <p>PCS Scan Clock.<br/>PCS scan clock signal for the word clk domain in the PCS.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                      |
| pcs_scan_mode | I   | <p>PCS Scan Mode.<br/>PCS Scan mode for adding scan to PCS layer.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                     |
| pcs_scan_pclk | I   | <p>PCS Scan PCLK.<br/>PCS scan clock signal for the pclk domain in the PCS.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                           |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name        | I/O | Description                                                                                                                                                                                               |
|------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pcs_scan_refclk  | I   | PCS Scan Reference Clock.<br>PCS scan clock signal for the reference clock domain in the PCS.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous            |
| pcs_scan_rst     | I   | PCS Scan Reset.<br>PCS scan reset signal.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous                                                                |
| pcs_scan_rx0_clk | I   | PCS Scan RX Clock for Lane N.<br>PCS scan clock signal for the RX clock for laneN in the PCS.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous            |
| pcs_scan_shift   | I   | PCS Scan Shift.<br>PCS scan shift for adding scan to PCS layer.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> All PCS scan clocks                                   |
| pcs_scan_tx0_clk | I   | PCS Scan TX Clock.<br>PCS scan clock signal for the TX clock for laneN in the PCS.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous                       |
| scan_bs_in       | I   | ASST Scan Input Data.<br>ASST scan input data for bs_in.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_bs_clk                                                  |
| scan_bs_out      | O   | ASST Scan Output Data.<br>ASST scan output data for bs_out.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_bs_clk (output on falling edge due to lock-up latch) |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name                | I/O | Description                                                                                                                                                                                                                                                                             |
|--------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_jtag_in[1:0]        | I   | ASST Scan Input Data.<br>ASST scan input data for jtag_in.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_jtag_clk                                                                                                                            |
| scan_jtag_out[1:0]       | O   | ASST Scan Output Data.<br>ASST scan output data for jtag_out.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_jtag_clk (output on falling edge due to lock-up latch)                                                                           |
| scan_mode                | I   | Scan Mode.<br>Enables access to serial scan chains.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                                                                                                                   |
| scan_mpll_dword_clk      | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for mppll_dword.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous |
| scan_mpll_dword_in[1:0]  | I   | ASST Scan Input Data.<br>ASST scan input data for mppll_dword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_mpll_dword_clk                                                                                                                  |
| scan_mpll_dword_out[1:0] | O   | ASST Scan Output Data.<br>ASST scan output data for mppll_dword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_mpll_dword_clk (output on falling edge due to lock-up latch)                                                                  |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name           | I/O | Description                                                                                                                                                                                                                                                                             |
|---------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_mpll_qword_clk | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for mppll_qword.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous |
| scan_mpll_qword_in  | I   | ASST Scan Input Data.<br>ASST scan input data for mppll_qword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_mpll_qword_clk                                                                                                                  |
| scan_mpll_qword_out | O   | ASST Scan Output Data.<br>ASST scan output data for mppll_qword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_mpll_qword_clk (output on falling edge due to lock-up latch)                                                                  |
| scan_mpll_word_clk  | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for mppll_word.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous  |
| scan_mpll_word_in   | I   | ASST Scan Input Data.<br>ASST scan input data for mppll_word.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_mpll_word_clk                                                                                                                    |
| scan_mpll_word_out  | O   | ASST Scan Output Data.<br>ASST scan output data for mppll_word.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_mpll_word_clk (output on falling edge due to lock-up latch)                                                                    |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name               | I/O | Description                                                                                                                                                                                                                                                                           |
|-------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_rst                | I   | PHY Scan Reset.<br>Top-level asynchronous scan reset pin.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                                                                                                           |
| scan_rx0_dword_clk      | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for rxX_dword.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous |
| scan_rx0_dword_in[6:0]  | I   | ASST Scan Input Data.<br>ASST scan input data for rxX_dword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_rxX_dword_clk                                                                                                                   |
| scan_rx0_dword_out[6:0] | O   | ASST Scan Output Data.<br>ASST scan output data for rxX_dword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_rxX_dword_clk (output on falling edge due to lock-up latch)                                                                   |
| scan_rx0_scope_clk      | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for rxX_scope.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous |
| scan_rx0_scope_in       | I   | ASST Scan Input Data.<br>ASST scan input data for rxX_scope.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_rxX_scope_clk                                                                                                                   |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name              | I/O | Description                                                                                                                                                                                                                                                                          |
|------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_rx0_scope_out     | O   | ASST Scan Output Data.<br>ASST scan output data for rxX_scope.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_rxX_scope_clk (output on falling edge due to lock-up latch)                                                                  |
| scan_rx0_word_clk      | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for rxX_word.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous |
| scan_rx0_word_in[4:0]  | I   | ASST Scan Input Data.<br>ASST scan input data for rxX_word.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_rxX_word_clk                                                                                                                    |
| scan_rx0_word_out[4:0] | O   | ASST Scan Output Data.<br>ASST scan output data for rxX_word.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_rxX_word_clk (output on falling edge due to lock-up latch)                                                                    |
| scan_shift             | I   | Scan Shift.<br>Enables shift path for the serial scan chains.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> All PHY scan clocks                                                                                                               |
| scan_ssprof_clk        | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for ssprof.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous   |



**Table 3-16 Scan Interface Signals (Continued)**

| Port Name               | I/O | Description                                                                                                                                                                                                                                                                           |
|-------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_ssprof_in[9:0]     | I   | ASST Scan Input Data.<br>ASST scan input data for ssprof.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_ssprof_clk                                                                                                                         |
| scan_ssprof_out[9:0]    | O   | ASST Scan Output Data.<br>ASST scan output data for ssprof.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_ssprof_clk (output on falling edge due to lock-up latch)                                                                         |
| scan_tx0_dword_clk      | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for txX_dword.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous |
| scan_tx0_dword_in[1:0]  | I   | ASST Scan Input Data.<br>ASST scan input data for txX_dword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_txX_dword_clk                                                                                                                   |
| scan_tx0_dword_out[1:0] | O   | ASST Scan Output Data.<br>ASST scan output data for txX_dword.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> scan_txX_dword_clk (output on falling edge due to lock-up latch)                                                                   |
| scan_tx0_out_clk        | I   | ASST Scan Clock for SuperSpeed portion of the PHY.<br>ASST scan clock for txX_out.<br>For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous   |

**Table 3-16 Scan Interface Signals (Continued)**

| Port Name         | I/O | Description                                                                                                                                                                                                                                                                                             |
|-------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scan_tx0_word_clk | I   | <p>ASST Scan Clock for SuperSpeed portion of the PHY.<br/> ASST scan clock for txX_word.<br/> For a list of ASST clock domains and associated frequencies, see "PHY Scan Details" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| scan_tx0_word_in  | I   | <p>ASST Scan Input Data.<br/> ASST scan input data for txX_word.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> scan_txX_word_clk</p>                                                                                                                      |
| scan_tx0_word_out | O   | <p>ASST Scan Output Data.<br/> ASST scan output data for txX_word.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> scan_txX_word_clk (output on falling edge due to lock-up latch)</p>                                                                      |

## 3.17 Boundary Scan Signals

Diagram showing boundary scan signals connected to a central block. The signals are:

- acjt\_level
- bs\_capture\_dr
- bs\_clamp
- bs\_clk
- bs\_extest
- bs\_extest\_ac
- bs\_highz
- bs\_in
- bs\_init
- bs\_invert
- bs\_preload
- bs\_shift\_dr
- bs\_update\_dr
- bs\_out

**Table 3-17** Boundary Scan Signals

| Port Name       | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acjt_level[4:0] | I   | <p>1149.6 Receiver Sensitivity Level Control.</p> <p>Selects the sensitivity level for the 1149.6 receiver for AC JTAG. Set the level based on the amplitude of the incoming Step during AC JTAG boundary scan testing. For a list of acjt_level[4:0] settings, see Settings for "Signal Levels to Detect AC JTAG Pulse" table.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |
| bs_capture_dr   | I   | <p>Boundary Scan CaptureDR Signal.</p> <p>Corresponds to the CaptureDR state in the host TAP controller. This signal is sampled on the rising edge of bs_clk.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> bs_clk</p>                                                                                                                                                                        |
| bs_clamp        | I   | <p>Boundary Scan Clamp Signal.</p> <p>To be asserted when the CLAMP instruction is active in the host TAP controller.</p> <p>When bs_clamp is asserted, tx&lt;#&gt;_cm_en and tx&lt;#&gt;_data_en are internally overridden to match the transmit enable boundary scan cell.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                   |

**Table 3-17 Boundary Scan Signals (Continued)**

| Port Name    | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bs_clk       | I   | Boundary Scan Clock.<br>Boundary scan clock signal.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> N/A                                                                                                                                                                                                                                                                                                                                           |
| bs_extest    | I   | Boundary Scan EXTEST Signal.<br>To be asserted when an EXTEST instruction is active in the host TAP controller.<br>When bs_extest is asserted, tx<#>_cm_en and tx<#>_data_en are internally overridden to match the transmit enable boundary scan cell.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                                                                                                             |
| bs_extest_ac | I   | Boundary Scan AC EXTEST Signal.<br>To be asserted when an EXTEST_PULSE or EXTEST_TRAIN instruction is active in the host TAP controller.<br>If these instructions are not implemented (no 1149.6 compliance), this pin must be tied low.<br>When bs_extest_ac is asserted, tx<#>_cm_en and tx<#>_data_en are internally overridden to match the transmit enable boundary scan cell.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous |
| bs_highz     | I   | Boundary Scan High-Impedance Signal.<br>To be asserted when the HIGHZ, INTEST, or RUNBIST instruction is active in the host TAP controller.<br>When bs_highz is asserted, tx<#>_cm_en and all transmit driver controls are internally disabled.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                                                                                                                     |
| bs_in        | I   | Boundary Scan Chain Input.<br>Input to the first register of the boundary scan chain.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> bs_clk                                                                                                                                                                                                                                                                                                      |

**Table 3-17 Boundary Scan Signals (Continued)**

| Port Name  | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bs_init    | I   | <p>Boundary Scan Initialize Signal.</p> <p>This signal is similar to the "Init Clk" referenced in test receiver diagrams in the AC JTAG specification.</p> <p>While this signal is high, the hysteresis in the receiver comparator is reset. The falling edge of this signal must occur at appropriate times, as specified in the AC JTAG specification.</p> <p>Control logic is required at the chip level to create this clock from tck. The bs_init signal asynchronously latches the data value into the receiver hysteresis.</p> <p>Therefore, this signal must not glitch and must be active only at times specified in the AC JTAG specification. The minimum pulse width of bs_init is a single bs_clk phase.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> bs_clk</p> |
| bs_invert  | I   | <p>Boundary Scan Invert Signal.</p> <p>This signal is similar to the "AC Test Signal" referenced in the IEEE 1149.6 specification.</p> <p>Assert bs_invert when the BS driver is required to drive the invert of the value in the BS register.</p> <p>This signal asynchronously affects the BS transmitter and must transition only on the rising edge of jtag_clk, as specified in the 1149.6 specification.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> bs_clk</p>                                                                                                                                                                                                                                                                                                        |
| bs_out     | O   | <p>Boundary Scan Chain Output.</p> <p>Output from the last register of the boundary scan chain.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> bs_clk</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| bs_preload | I   | <p>Boundary Scan Preload Signal.</p> <p>To be asserted when PRELOAD or SAMPLE instructions are active in the host TAP controller.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Table 3-17 Boundary Scan Signals (Continued)**

| Port Name    | I/O | Description                                                                                                                                                                                                                                       |
|--------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bs_shift_dr  | I   | Boundary Scan ShiftDR Signal.<br>Corresponds to the ShiftDR state in the host TAP controller. This signal is sampled on the rising edge of bs_clk.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> bs_clk    |
| bs_update_dr | I   | Boundary Scan UpdateDR Signal.<br>Corresponds to the UpdateDR state in the host TAP controller. This signal is sampled on the falling edge of bs_clk.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> bs_clk |

## 3.18 PIPE Interface Signals

|                           |   |                      |
|---------------------------|---|----------------------|
| alt_clk_en                | - | alt_clk_req          |
| alt_clk_sel               | - | pipe0_data_bus_width |
| alt_pcs_clk               | - | pipe0_pclk           |
| alt_pipe_pclk             | - | pipe0_phystatus      |
| lane0_ext_pclk_req        | - | pipe0_power_present  |
| lane0_tx2rx_loopbk        | - | pipe0_rx_data        |
| pipe0_elasticity_buf_mode | - | pipe0_rx_datak       |
| pipe0_powerdown           | - | pipe0_rx_elecidle    |
| pipe0_reset_n             | - | pipe0_rx_status      |
| pipe0_rx_eq_training      | - | pipe0_rx_valid       |
| pipe0_rx_polarity         | - | ref_clkreq_n         |
| pipe0_rx_termination      | - | ref_pcs_clk          |
| pipe0_tx_data             | - | ref_pipe_pclk        |
| pipe0_tx_datak            | - |                      |
| pipe0_tx_deemph           | - |                      |
| pipe0_tx_detectrx         | - |                      |
| pipe0_tx_elecidle         | - |                      |
| pipe0_tx_ones_zeros       | - |                      |

**Table 3-18 PIPE Interface Signals**

| Port Name   | I/O | Description                                                                                                                                                                                                                                                                   |
|-------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alt_clk_en  | I   | Alternate Clock Enable.<br>Enables the ref_pcs_clk and ref_pipe_pclk output clocks (if necessary, powers up the MPLL).<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                                      |
| alt_clk_req | O   | Alternate Clock Source Request.<br>Indicates that the alternate clocks are needed by the slave PCS (that is, to boot the master MPLL). Connect to alt_clk_en on the master.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous |

**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name          | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alt_clk_sel        | I   | <p>Alternate Clock Source Select.<br/>Selects the alternate clock sources instead of the internal MPLL outputs for the PCS clocks.</p> <ul style="list-style-type: none"> <li>1: Uses alternate clocks.</li> <li>0: Uses internal MPLL clocks.</li> </ul> <p>Change only during a reset.<br/><b>Note:</b> In Scan mode, tie this pin to 0.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                   |
| alt_pcs_clk        | I   | <p>Alternate PCS Clock Source.<br/>Alternate input clock for pcs_clk domain.<br/>To select this clock, set alt_clk_sel to 1'b1.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                                                                                        |
| alt_pipe_pclk      | I   | <p>Alternate PIPE Clock Source.<br/>Alternate input clock for pipe_pclk domain.<br/>To select this clock, set alt_clk_sel to 1'b1.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> N/A</p>                                                                                                                                                                                                                                                                                                     |
| lane0_ext_pclk_req | I   | <p>External PIPE Clock Enable Request.<br/>When asserted, this signal enables the pipeX_pclk output regardless of power state (along with the associated increase in power consumption).<br/>You can use this input to enable pipeX_pclk in the P3 state without going through a complete boot sequence. Theoretically, this signal enables you to disable the RX terminations and use the PHY as a high-speed clock source.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p> |
| lane0_tx2rx_loopbk | I   | <p>Loopback.<br/>When this signal is asserted, data from the transmit predriver is looped back to the receive slicers. LOS is bypassed and based on the tx0_en input so that rx0_los = !tx0_data_en.<br/><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                         |



**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name                 | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipe0_data_bus_width[1:0] | O   | <p>PIPE Data Bus Width.</p> <p>This signal reports the width of the data bus for which the PHY is configured.</p> <ul style="list-style-type: none"> <li>11: Reserved</li> <li>10: 8-bit mode</li> <li>01: 16-bit mode</li> <li>00: 32-bit mode</li> </ul> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                                                                 |
| pipe0_elasticity_buf_mode | I   | <p>PIPE Elasticity Buffer Mode.</p> <p>Selects Elasticity Buffer operating mode.</p> <ul style="list-style-type: none"> <li>1: Nominal Empty Buffer mode.</li> <li>0: Nominal Half-Full Buffer mode.</li> </ul> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                                                                                                            |
| pipe0_pclk                | O   | <p>PIPE P-Clock.</p> <p>Parallel interface differential data clock. All data movement across the parallel interface is synchronized to this clock.</p> <p>This clock operates at 125 MHz, 250 MHz, or 500 MHz depending on the rate control input and the data interface width.</p> <p>The rising edge of the clock is the reference for all signals. Spread spectrum modulation on this clock is permitted.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> N/A</p>                  |
| pipe0_phystatus           | O   | <p>PIPE PHY Status.</p> <p>Communicates completion of several PHY functions including power management state transitions, rate change, and receiver detection. When this signal transitions during entry and exit from P3 states and PCLK is not running, the signaling is asynchronous.</p> <p>In error situations (where the PHY fails to assert PhyStatus), the MAC can take MAC-specific error recovery actions.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p> |

**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipe0_power_present  | O   | <p>PIPE Power Present.</p> <p>This PIPE output indicates the presence of VBUS&lt;#&gt; and is valid only when the VBUS&lt;#&gt; input to the PHY is connected.</p> <p>The signal's level is derived from the internal VBUS Level comparator integrated in the PHY's OTG block.</p> <p>If VBUS&lt;#&gt; is not connected to the PHY, this signal must not be used and must be left floating.</p> <p>For information about PHY configuration if the VBUS&lt;#&gt; pin is not bonded out nor connected, see "Non-OTG Configuration" section.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> High<br/> <b>Synchronous To:</b> pipeX_pclk</p> |
| pipe0_powerdown[1:0] | I   | <p>PIPE Power Up/Down.</p> <p>Powers up or powers down the transceiver.</p> <p>Power states for USB SS mode:</p> <ul style="list-style-type: none"> <li>■ 11: P3, lowest power state.</li> <li>■ 10: P2, longer recovery time latency, lower power state</li> <li>■ 01: P1, low recovery time latency, power saving state</li> <li>■ 00: P0, normal operation</li> </ul> <p>When transitioning from P3 to P0 states, the signaling is asynchronous, because PCLK is not running.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> Asynchronous</p>                                                         |
| pipe0_reset_n        | I   | <p>PIPE Reset.</p> <p>Resets the transmitter and receiver.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> Low<br/> <b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| pipe0_rx_data[31:0]  | O   | <p>PIPE Receive Data Bus.</p> <p>Parallel SuperSpeed data output bus.</p> <p>For a 16-bit interface, 16 bits represents two symbols of receive data. Bits [7:0] are the first symbol received, and bits [15:8] are the second symbol.</p> <p>For a 32-bit interface, 32 bits represent the four symbols of receive data. Bits [23:16] are the third symbol received, and bits [31:24] are the fourth symbol received.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                      |

**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipe0_rx_data[3:0]   | O   | <p>PIPE Receive Data Control.<br/>Data/control bit for the symbols of receive data.<br/>For a 32-bit interface, bit 0 corresponds to the low byte of RxData, bit 3 corresponds to the upper byte.<br/>For a 16-bit interface, bit 0 corresponds to the low byte of RxData[15:0], bit 1 to the upper byte.<br/>A value of 0 indicates a data byte; a value of 1 indicates a control byte.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> pipeX_pclk</p> |
| pipe0_rx_elecidle    | O   | <p>PIPE Receiver Electrical Idle.<br/>Indicates receiver detection of an electrical idle.<br/>Signal de-assertion while the PHY is in a P0, P1, P2, or P3 state indicates the detection of Low Frequency Periodic Signaling (LFPS).</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                   |
| pipe0_rx_eq_training | I   | <p>PIPE Receiver Equalization Training.<br/>In USB SuperSpeed mode, instructs the receiver to bypass normal operation to perform equalization training.<br/>While performing the training, the state of the RxData interface is undefined.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                                              |
| pipe0_rx_polarity    | I   | <p>PIPE Receive Polarity Inversion.<br/>Instructs the PHY to perform a polarity inversion on the received data.</p> <ul style="list-style-type: none"> <li>■ 1: PHY performs polarity inversion.</li> <li>■ 0: PHY does not perform polarity inversion.</li> </ul> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                          |

**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipe0_rx_status[2:0] | O   | <p>PIPE Receiver Status.<br/>Encodes receiver status and error codes for the received datastream when receiving data.</p> <ul style="list-style-type: none"> <li>111: Receive disparity error (reserved if Receive Disparity error is reported with code 0b100)</li> <li>110: Elastic Buffer underflow. This error code is not used if the Elasticity Buffer is operating in the nominal Buffer Empty mode.</li> <li>101: Elastic Buffer overflow</li> <li>100: Both 8B/10B decode error and (optionally) Receive Disparity error</li> <li>011: Receiver detected</li> <li>010: 1 SKP Ordered Set removed</li> <li>001: 1 SKP Ordered Set added</li> <li>000: Received data valid</li> </ul> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> pipeX_pclk</p> |
| pipe0_rx_termination | I   | <p>PIPE Receiver Termination.<br/>Controls presence of receiver terminations.</p> <ul style="list-style-type: none"> <li>1: Terminations present.</li> <li>0: Terminations removed.</li> </ul> <p>Implementation of this signal is required only for PHYs that support USB SS mode.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                   |
| pipe0_rx_valid       | O   | <p>PIPE Receive Data Valid.<br/>Indicates symbol lock and valid data on RxData and RxDataK.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> High<br/><b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name            | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipe0_tx_data[31:0]  | I   | <p>PIPE Transmit Data Bus.</p> <p>Parallel SuperSpeed data input bus. For a 16-bit interface, 16 bits represent two symbols of transmit data.</p> <p>Bits [7:0] are the first symbol to be transmitted, and bits [15:8] are the second symbol.</p> <p>For a 32-bit interface, 32 bits represent the four symbols of transmit data.</p> <p>Bits [23:16] are the third symbol to be transmitted, and bits [31:24] are the fourth symbol.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p> |
| pipe0_tx_datak[3:0]  | I   | <p>PIPE Transmit Data Control.</p> <p>Data/control for the symbols of transmit data.</p> <p>For a 32-bit interface, bit 0 corresponds to the low byte of TxData, bit 3 corresponds to the upper byte.</p> <p>For a 16-bit interface, bit 0 corresponds to the low byte of TxData, bit 1 corresponds to the upper byte.</p> <p>A value of zero indicates a data byte, a value of 1 indicates a control byte.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p>                            |
| pipe0_tx_deemph[1:0] | I   | <p>PIPE Transmitter De-emphasis.</p> <p>Selects transmitter de-emphasis.</p> <ul style="list-style-type: none"> <li>■ 11: Reserved</li> <li>■ 10: No de-emphasis</li> <li>■ 01: 3.5 dB de-emphasis</li> <li>■ 00: 6 dB de-emphasis</li> </ul> <p>PIPE implementations that support only a single de-emphasis level do not implement this signal.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p>                                                                                       |
| pipe0_tx_detectrx    | I   | <p>PIPE Receive Detection.</p> <p>Instructs the PHY to begin a receiver detection operation or to begin loopback.</p> <p><b>Voltage Range:</b> 0-vp<br/> <b>Active State:</b> N/A<br/> <b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                                                                                                                                                                                                                      |

**Table 3-18 PIPE Interface Signals (Continued)**

| Port Name           | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipe0_tx_elecidle   | I   | <p>PIPE Transmitter Electrical Idle.<br/>Forces TX output to the Electrical Idle state.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> pipeX_pclk</p>                                                                                                                                                                                            |
| pipe0_tx_ones_zeros | I   | <p>PIPE Transmit Compliance Patterns.<br/>Used only when transmitting USB SuperSpeed compliance patterns, CP7 or CP8.<br/>This signal causes the transmitter to transmit an alternating sequence of 50-250 1's and 50-250 0's - regardless of the state of the TxData interface.</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> Asynchronous</p> |
| ref_clkreq_n        | O   | <p>Reference Clock Removal Acknowledge.<br/>Reserved</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> Low<br/><b>Synchronous To:</b> Asynchronous</p>                                                                                                                                                                                                                             |
| ref_pcs_clk         | O   | <p>Reference Clock for alt_pcs_clk.<br/>Free-running clock to be connected to the alt_pcs_clk input on another PIPE wrapper (the slave).</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> N/A</p>                                                                                                                                                  |
| ref_pipe_pclk       | O   | <p>Reference Clock for alt_pipe_pclk.<br/>Free-running clock to be connected to the alt_pipe_pclk input on another PIPE wrapper (the slave).</p> <p><b>Voltage Range:</b> 0-vp<br/><b>Active State:</b> N/A<br/><b>Synchronous To:</b> N/A</p>                                                                                                                                              |

## 3.19 Control Register Access Signals



cr\_cap\_addr -  
 cr\_cap\_data -  
 cr\_data\_in -  
 cr\_read -  
 cr\_write -

- cr\_ack  
 - cr\_data\_out

**Table 3-19 Control Register Access Signals**

| Port Name         | I/O | Description                                                                                                                                                                                                     |
|-------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cr_ack            | O   | CR Acknowledgment.<br>Acknowledgment for the cr_cap_addr, cr_cap_data, cr_write, and cr_read control signals.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous |
| cr_cap_addr       | I   | CR Capture Address.<br>Captures cr_data_in[15:0] into the Address register.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                   |
| cr_cap_data       | I   | CR Capture Data.<br>Captures cr_data_in[15:0] into the Write Data register.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                                   |
| cr_data_in[15:0]  | I   | CR Address and Write Data Input Bus.<br>Supplies and captures address and write data.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous                          |
| cr_data_out[15:0] | O   | CR Data Output Bus.<br>Always presents last read data.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> Asynchronous                                                         |

**Table 3-19 Control Register Access Signals (Continued)**

| Port Name | I/O | Description                                                                                                                                                                       |
|-----------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cr_read   | I   | CR Read.<br>Reads from the referenced Address register.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous                         |
| cr_write  | I   | CR Write.<br>Writes the Write Data register to the referenced Address register.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous |



## 3.20 JTAG Port Signals



**Table 3-20 JTAG Port Signals**

| Port Name   | I/O | Description                                                                                                                                                                                    |
|-------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| jtag_tck    | I   | JTAG Clock.<br>JTAG state machine clock signal for use in ATE testing and debug.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> N/A                       |
| jtag_tdi    | I   | JTAG Input Port.<br>JTAG test data input signal for use in ATE testing and debug.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> jtag_tck                 |
| jtag_tdo    | O   | JTAG Output Port.<br>JTAG test data output signal for use in ATE testing and debug.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> N/A<br><b>Synchronous To:</b> jtag_tck               |
| jtag_tdo_en | O   | JTAG Output Enable.<br>JTAG test data output enable signal for use in ATE testing and debug.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> Asynchronous |
| jtag_tms    | I   | JTAG State Machine Control.<br>JTAG test mode select signal for use in ATE testing and debug.<br><b>Voltage Range:</b> 0-vp<br><b>Active State:</b> High<br><b>Synchronous To:</b> jtag_tck    |

**Table 3-20 JTAG Port Signals (Continued)**

| Port Name   | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| jtag_trst_n | I   | <p>PHY JTAG Reset.</p> <p>Top-level asynchronous JTAG reset pin.</p> <p>According to JTAG specifications, a TRST reset is optional. In the USB 3.0 femtoPHY, the JTAG interface is reset by the phy_reset signal and optionally by the jtag_trst_n signal.</p> <p>If jtag_trst_n is not used, tie it to 1'b1.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> Low</p> <p><b>Synchronous To:</b> Asynchronous</p> |

## 3.21 Termination Tuning Signals

rtune\_req -  - rtune\_ack

**Table 3-21 Termination Tuning Signals**

| Port Name | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rtune_ack | O   | <p>Resistor Tune Acknowledge.</p> <p>While asserted, indicates that a resistor tune is still in progress.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> ref_output_clk</p>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| rtune_req | I   | <p>Resistor Tune Request.</p> <p>Assertion triggers a resistor tune request (if one is not already in progress). When this signal is asserted, rtune_ack goes high until calibration of the termination impedances is complete.</p> <p>Note that termination resistance tuning disrupts the normal flow of data; therefore, assert rtune_req only when the PHY is inactive.</p> <p>Because the PHY automatically performs a tune after coming out of a reset, rtune_req can be tied low unless you want to be able to manually force a tuning.</p> <p><b>Voltage Range:</b> 0-vp</p> <p><b>Active State:</b> High</p> <p><b>Synchronous To:</b> ref_output_clk</p> |



## 4

## Functional Description

The top-level module, `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN`, is instantiated in the ASIC RTL, and control of the USB 3.0 femtoPHY is made through `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN` pins. Some of these signals connect directly to the USB 3.0 controller, while other signals are application-specific and are controlled by the ASIC. The `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN` interface signals are categorized into the following functional groups:

- “PIPE Interface” on page 130
- “Clocks and Resets” on page 138
- “Power Supply Connections” on page 143
- “Signals That Connect to External Pads” on page 143
- “Serial Test Ports” on page 144
- “Termination Tuning” on page 145
- “Test Controls” on page 145
- “OTG Signals” on page 146
- “Battery Charging Signals” on page 147
- “UART/Autoresume Signals” on page 155
- “High-Voltage Interrupt Signals” on page 159
- “Control Register Access” on page 159
- “Serial Interface” on page 160
- “UTMI+ Controls” on page 160
- “UTMI+ Receive Signals” on page 161
- “UTMI+ Transmit Signals” on page 161
- “Parameter Controls” on page 162

## 4.1 PIPE Interface

The datapath of the USB 3.0 femtoPHY is through the standard PIPE 3 interface. This interface can connect directly to any USB 3.0 controller. For more information about the PIPE 3 interface signals, refer to *PHY Interface For the PCI Express\* and USB 3.0 Architectures*.



### Note

To enable the PIPE interface to control the PHY, the following conditions must be met:

- The `ref_ssp_en` signal must be asserted, and the reference clock must be running and stable.
- The MPLL multiplier settings, `ref_use_pad`, and the parameter control signals must be set. In addition, `test_burnin`, `test_powerdown`, and `lane<#>_tx2rx_loopbk` must be de-asserted.
- The PHY must be taken out of reset (that is, `phy_reset` de-asserted).

### 4.1.1 Detection of Electrical Idle Using PHY Loss of Signal Circuitry

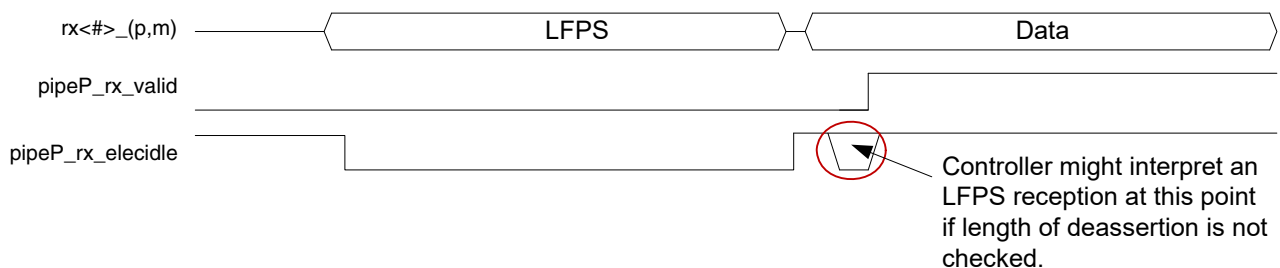
The USB 3.0 femtoPHY contains an LOS detector that informs the controller when signaling is present on the high-speed input lines. The controller then breaks the Electrical Idle condition (condition where no signal is present on the line during low-power modes or startup). The PHY de-asserts the `pipeP_rx_elecidle` signal to inform the PIPE interface when an electrical idle condition has been broken.

Due to the LOS detector's bandwidth limitations and the fact that it supports multiple protocols, the detector can detect signals up to a certain frequency. At high transition density during normal data transmission, the detector might not detect signals on the line. With low transition density data during normal data transmission, the detector might assert `pipeP_rx_elecidle` even when high-speed data is on the line. Therefore, with respect to the controller, take care to prevent false detection of either electrical idle or LFPS signaling.

#### 4.1.1.1 De-assertion of `pipeP_rx_elecidle` When LFPS is not Present at the Beginning of Data Reception

The `pipeP_rx_elecidle` signal is a qualified version of the LOS circuitry. While valid data is received, `pipeP_rx_elecidle` is held high despite possible toggling of the LOS signal. As shown in the following figure, a period of time exists at the beginning of an incoming datastream, prior to the point where `pipeP_rx_valid` is asserted, that `pipeP_rx_elecidle` might toggle. The controller must prevent false detection of electrical idle by ignoring any short de-assertions of `pipeP_rx_elecidle` and qualifying any de-assertions of `pipeP_rx_elecidle` to ensure that the pulse meets the minimum requirements for LFPS.

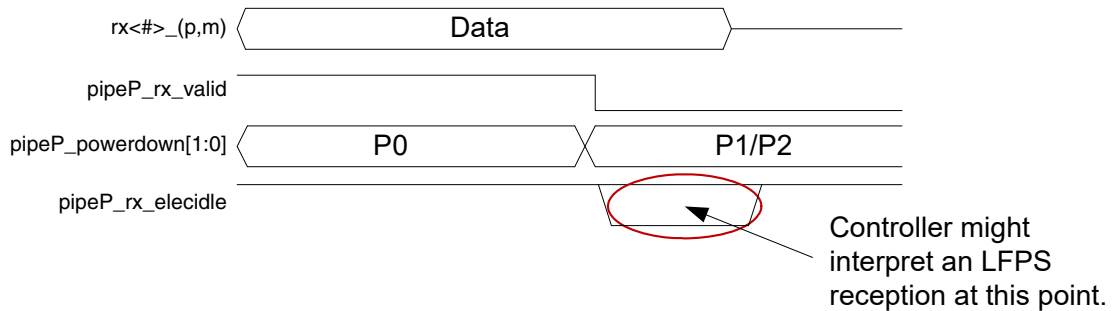
**Figure 4-1** `pipeP_rx_elecidle` Behavior at Beginning of Data Reception



#### 4.1.1.2 De-assertion of pipeP\_rx\_elecidle When LFPS is not Present During U0 Exit

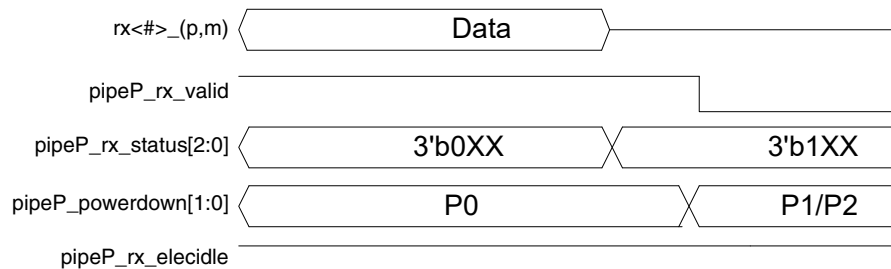
When the controller initiates exit of a U0 state, valid data is received for some period after a power-state transition, as shown in the following figure. To prevent the controller from interpreting any de-assertion of pipeP\_rx\_elecidle during this period of continued data transmission, it is recommended that you delay the exit of U0 until the valid data transmission from the far end has terminated.

**Figure 4-2 pipeP\_rx\_elecidle Behavior During U0 Exit if U0 is not Extended**



To prevent false LFPS reception during an U0 exit, the controller must delay the exit of U0 until decode/disparity errors are received on pipeP\_rx\_status, as shown in the following figure. To properly infer the electrical idle condition, eight clock cycles of decode/disparity errors are recommended.

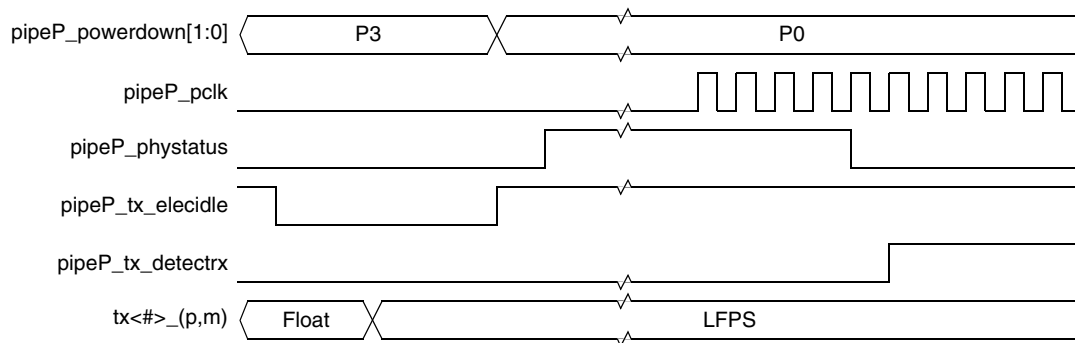
**Figure 4-3 Extension of U0 Exit to Prevent False LFPS Detection**



## 4.1.2 LFPS Enable Extension

The PIPE 3 specification addresses how to transition power states and maintain continuous, uninterrupted, Low Frequency Periodic Signaling (LFPS) transmission prior to sending data. The pcs\_xN implements a clean transition from sending LFPS to sending data using simultaneous transitions on pipeP\_powerdown[1:0] and pipeP\_tx\_elecidle. Figure 4-4 through Figure 4-5 on page 132 show timings for the relevant power state transitions.

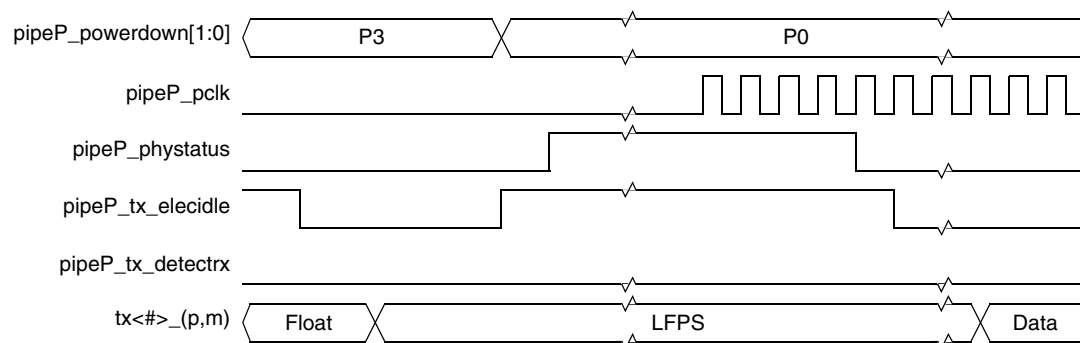
**Figure 4-4 LFPS in P3 to LFPS in P0**



As illustrated in Figure 4-4:

- In P3, pipeP\_tx\_elecidle must be low for at least five reference clock cycles before the transition to P0.
- To prevent glitches in the LFPS, the pipeP\_powerdown[1:0] and pipeP\_tx\_elecidle signals must be transitioned “together” (within three reference clock cycles of each other).
- The pipeP\_tx\_detectrx signal must be asserted in the clock cycle immediately following the falling edge of pipeP\_phystatus.

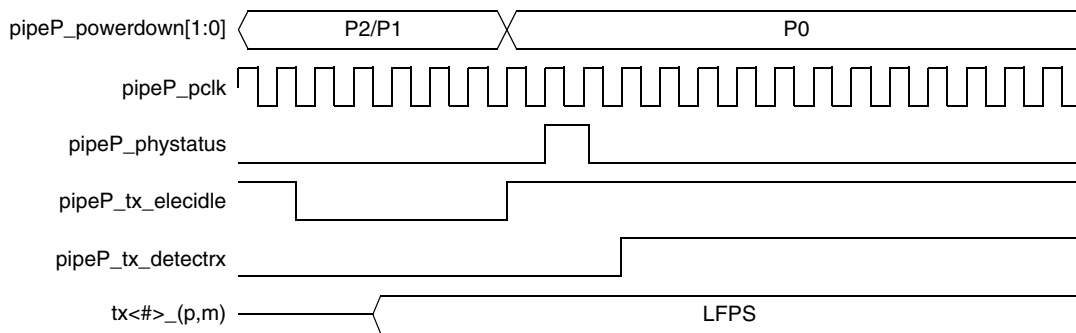
**Figure 4-5 LFPS in P3 to Data in P0**



As illustrated in Figure 4-5:

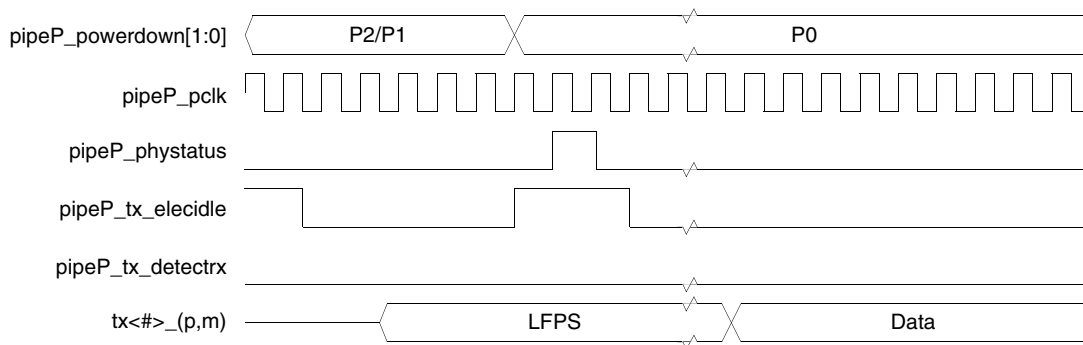
- In P3, pipeP\_tx\_elecidle must be low for at least five reference clock cycles before the transition to P0.
- To prevent glitches in the LFPS, the pipeP\_powerdown[1:0] and pipeP\_tx\_elecidle signals must be transitioned together.
- The pipeP\_tx\_elecidle signal must be de-asserted in the clock cycle immediately following the falling edge of pipeP\_phystatus.



**Figure 4-6 LFPS in P2 or P1 to LFPS in P0**

As illustrated in [Figure 4-6](#):

- To prevent glitches in the LFPS, the `pipeP_powerdown[1:0]` and `pipeP_tx_elecidle` signals must be transitioned together.
- The `pipeP_tx_detectrx` signal must be asserted in the clock cycle immediately following the falling edge of `pipeP_phystatus`.

**Figure 4-7 LFPS in P2 or P1 to Data in P0**

As illustrated in [Figure 4-7](#):

- To prevent glitches in the LFPS, the `pipeP_powerdown` and `pipe<#>_tx_elecidle` signals must be transitioned together.
- The `pipeP_tx_elecidle` signal must be de-asserted in the clock cycle immediately following the falling edge of `pipeP_phystatus`.

### 4.1.3 Low-Power States Entry and Exit

Whenever the controller is commanded to exit from a low-power state (or abort entry into a low-power state), the controller actually waits for the PIPE 3 interface to fully complete the entry into the specific low-power state, then executes the exit from this low-power state.

Specifically, the case where the receiver high-speed serial lines go to the Electrical Idle state and then resume, and the PHY is not transitioned out of the P0 state, is not permitted. If you are not using both the USB 3.0 femtoPHY and USB 3.0 Controller from Synopsys (that is, you are using a third-party controller with the Synopsys USB 3.0 femtoPHY), you must ensure that this case does not occur.

[Figure 4-8](#) on page [135](#) through [Figure 4-12](#) on page [137](#) detail this requirement.

Figure 4-8 Scenario 1: Low Power Exit Initiated by Link – Failure

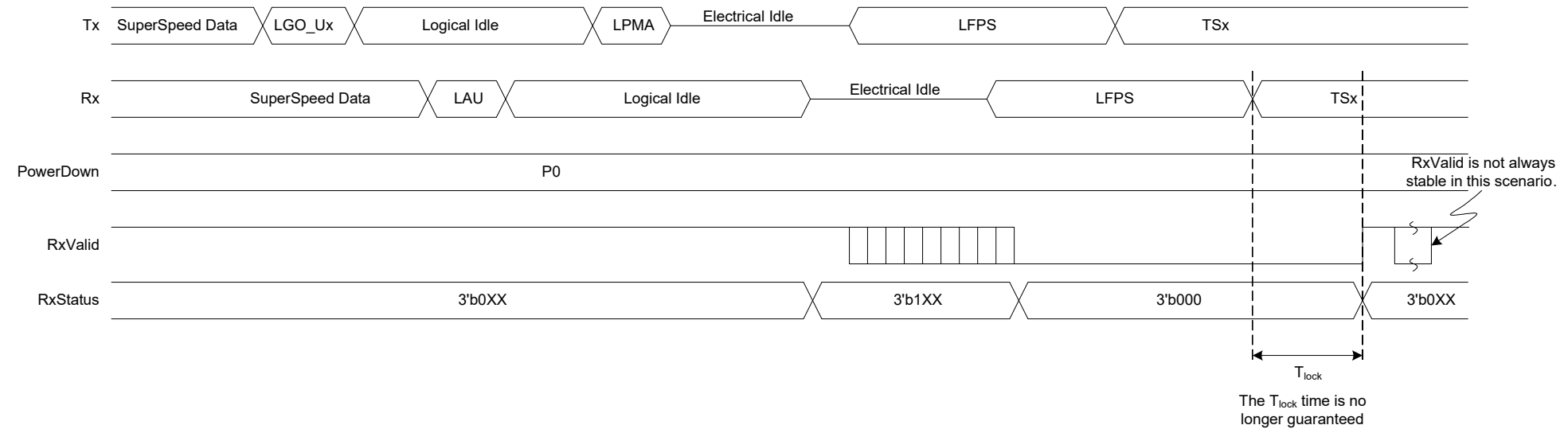


Figure 4-9 Scenario 1: Low Power Exit Initiated by Link – With P1/P2 State Transition

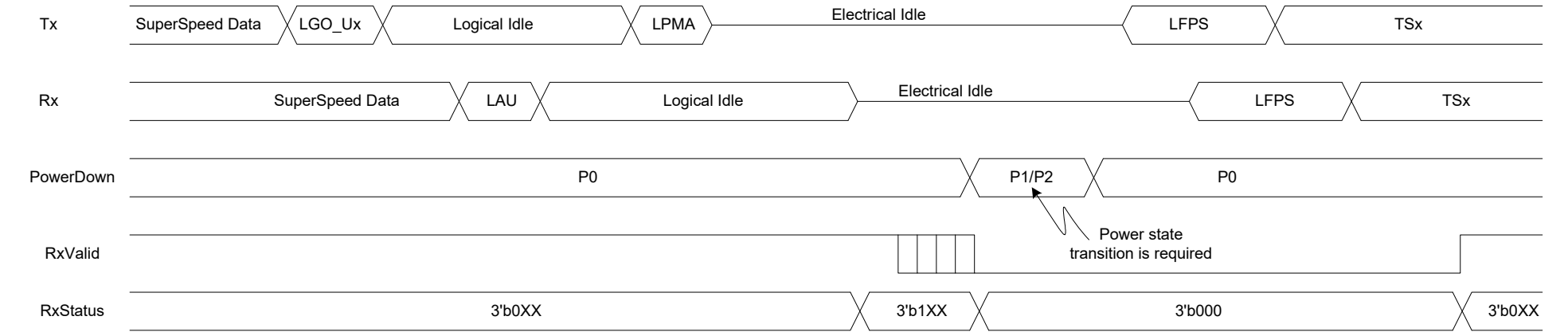


Figure 4-10 Scenario 2: Low Power Exit Initiated by Link Partner – Failure

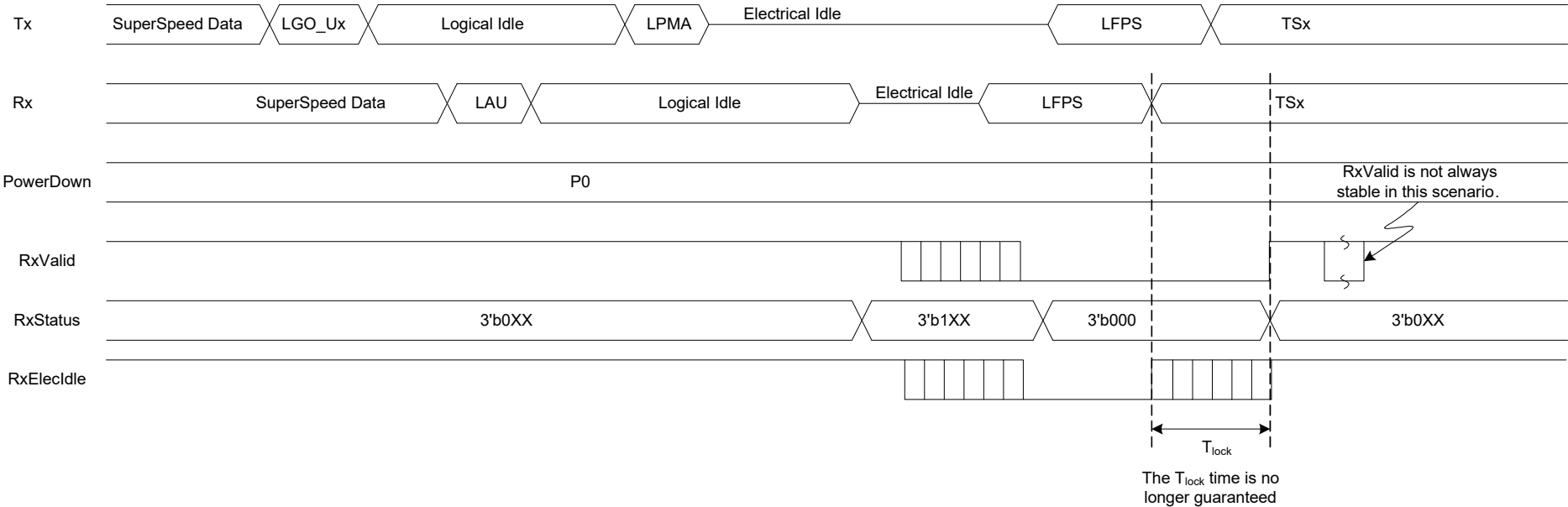


Figure 4-11 Scenario 2: Low Power Exit Initiated by Link Partner – With P1/P2 State Transition

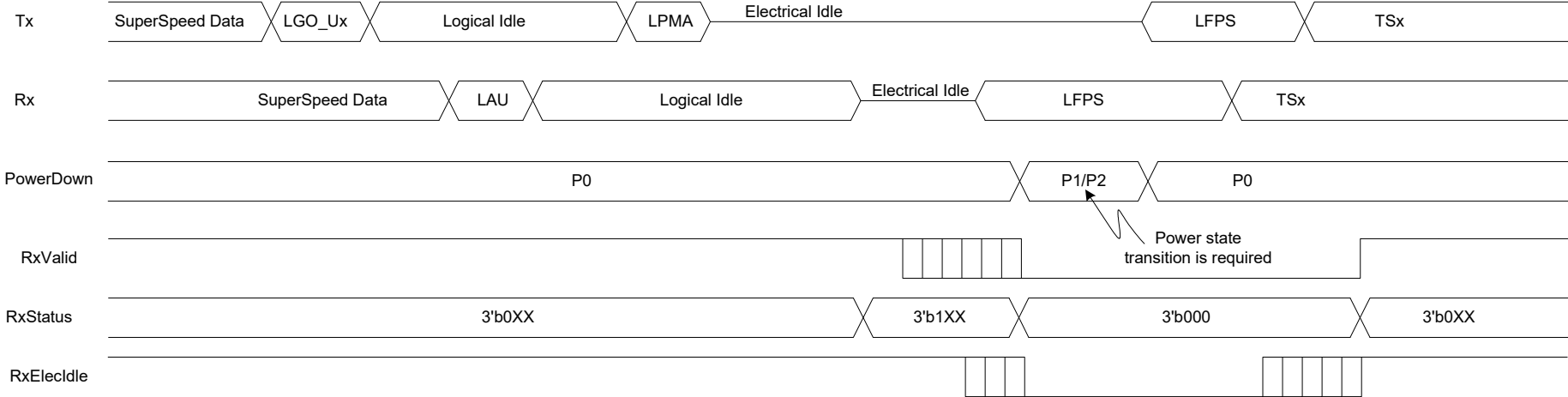
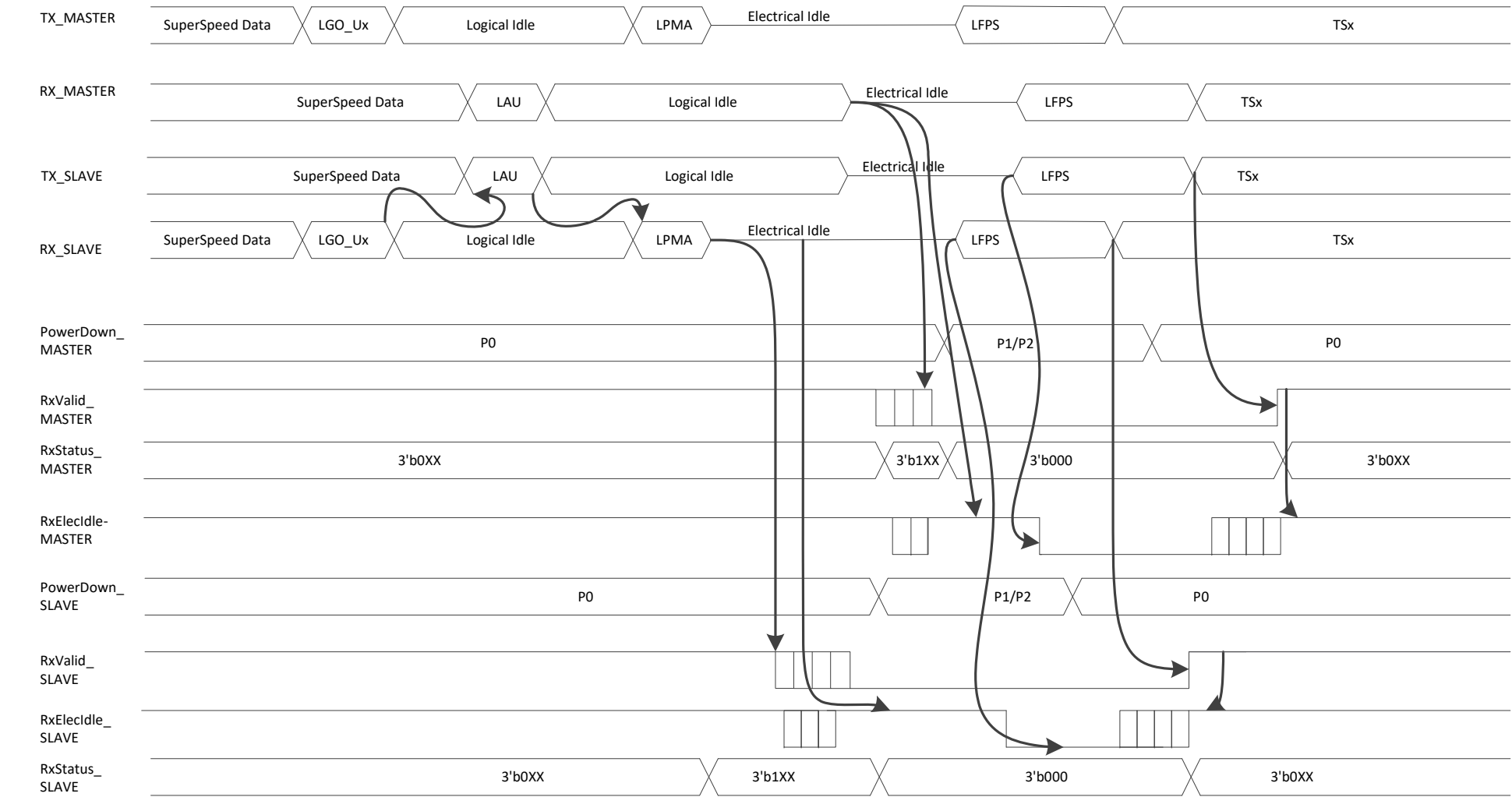


Figure 4-12 Scenario 3: Synopsys PHY on Both Sides of Link



## 4.2 Clocks and Resets

The USB 3.0 femtoPHY supports a wide range of input reference clocks from both external and on-chip clock sources.

### 4.2.1 Reference Clock

This section describes the reference clock source selection, enables, outputs, and controls for a shared reference clock.

#### 4.2.1.1 Reference Clock Source Selection

The input reference clock selection for SuperSpeed and high-speed operations is made as follows.

The `ref_use_pad` signal is used to select the input reference clock source.

- When `ref_use_pad` is set to 1'b1, the reference clock is derived off-chip through the `dwc_usb3_pads_sspN_hspN_ns` snap-on pad signals, `ref_pad_clk_m` and `ref_pad_clk_p`.
- When `ref_use_pad` is set to 1'b0, the reference clock is derived from on-chip signals, `ref_alt_clk_m` and `ref_alt_clk_p`.



#### Note

The reference clock inputs are differential. If you meet the reference clock requirements with a single end-end input clock, as described in [“Reference Clock Requirements”](#) on page 243, tie the unused reference input low or high—do not leave it unconnected.

The following table provides the input clock settings required for 5-Gbps and 480-MHz operations based on the input reference clock source frequency.

**Table 4-1 Reference Clock Frequency Selection for SuperSpeed and High-Speed Operations**

| Reference Clock (MHz) | Required Multiplication | ref_clkdiv2 | fsel[5:0] | mpll_multiplier[6:0] | ssc_ref_clk_sel[8:0] |
|-----------------------|-------------------------|-------------|-----------|----------------------|----------------------|
| 19.2                  | 130                     | 0           | 111000    | 000_0000             | 0_0000_0000          |
| 20                    | 125                     | 0           | 110001    | 000_0000             | 0_0000_0000          |
| 24                    | 104                     | 0           | 101010    | 000_0000             | 0_0000_0000          |
| 25                    | 100                     | 0           | 000010    | 110_0100             | 0_0000_0000          |
| 26                    | 96                      | 0           | 000010    | 110_0000             | 1_0000_1000          |
| 38.4                  | 130                     | 1           | 111000    | 000_0000             | 0_0000_0000          |
| 40                    | 125                     | 1           | 110001    | 000_0000             | 0_0000_0000          |
| 48                    | 104                     | 1           | 101010    | 000_0000             | 0_0000_0000          |
| 50                    | 50                      | 1           | 000010    | 110_0100             | 0_0000_0000          |
| 52                    | 96                      | 1           | 000010    | 110_0000             | 1_0000_1000          |

**Table 4-1 Reference Clock Frequency Selection for SuperSpeed and High-Speed Operations (Continued)**

| Reference Clock (MHz) | Required Multiplication | ref_clkdiv2 | fsel[5:0] | mpll_multiplier[6:0] | ssc_ref_clk_sel[8:0] |
|-----------------------|-------------------------|-------------|-----------|----------------------|----------------------|
| 100                   | 25                      | 0           | 100111    | 000_0000             | 0_0000_0000          |
| 200                   | 25                      | 1           | 100111    | 000_0000             | 0_0000_0000          |

**Note**

The reference clock source cannot be spread.

#### 4.2.1.2 Reference Clock Controls

This section describes the USB 3.0 femtoPHY reference clock enables and outputs.

##### Reference Clock Enables

To enable the reference clock buffer in the USB 3.0 femtoPHY:

1. Ensure that the reference clock is stable.
2. Assert ref\_ssp\_en, and keep it asserted for at least two reference clock cycles.
3. De-assert phy\_reset.

To enable the reference clock outputs for high-speed operation, the SUSPENDM<#> signal must be set to 1'b1.

##### Reference Clock Outputs

For convenience, ref\_output\_clk, a single-ended copy of the reference input clock is provided.

For electrical requirements of the input reference clock supplied to the USB 3.0 femtoPHY, see [“Reference Clock Implementation”](#) on page 243.

For high-speed operation, the following clock outputs are provided:

- CLK480M
- CLK12MOHCI
- CLK48MOHCI

**Note**

The reference clock inputs are differential. If you meet the reference clock requirements with a single end-end input clock, as described in [“Reference Clock Requirements”](#) on page 243, tie the unused reference input low or high—do not leave it unconnected.

### 4.2.1.3 MPLL Controls for SuperSpeed Reference Clock

The USB 3.0 femtoPHY requires a Spread Spectrum clock for transmitting 5-Gbps SuperSpeed data. The USB 3.0 femtoPHY can derive this Spread Spectrum clock from a fixed-frequency reference clock and add the required down-spread modulation, using the `ssc_en` (Spread Spectrum clock enable) signal.

The `ssc_range[2:0]` bus selects the amount of clock spreading to be added, as described in the following table.

**Table 4-2 Spread Spectrum Reference Clock Frequency Selection**

| <code>ssc_range[2:0]</code> | Down Spread of Clock (ppm) |
|-----------------------------|----------------------------|
| 3'b000                      | –4,980                     |
| 3'b001                      | –4,492                     |
| 3'b010                      | –4,003                     |
| 3'b011                      | Reserved                   |
| 3'b100                      | Reserved                   |
| 3'b101                      | Reserved                   |
| 3'b110                      | Reserved                   |
| 3'b111                      | Reserved                   |

All of these settings are within the USB 3.0 specification. The amount of EMI emission reduction might decrease as the `ssc_range` increases; therefore, the `ssc_range` settings can be registered to enable the amount of spreading to be adjusted on a per-application basis.

The USB 3.0 femtoPHY is designed to handle a wide range of input clock frequencies to support both host and device applications and to use multiplying PLLs to generate the high-speed clocks required for data transmission and reception from an accurate, low-speed reference clock source. To achieve the proper data transmission rate, the MPLL must be set based on the incoming reference clock frequency.



#### Note

Before `phy_reset` is de-asserted and `ref_ssp_en` is asserted, the input reference clock must have a single, stable value.



## 4.2.2 Clock Availability

The following table details the availability of clocks output from the PHY while in various power states.

**Table 4-3 Available Clocks From PHY in Various Power States**

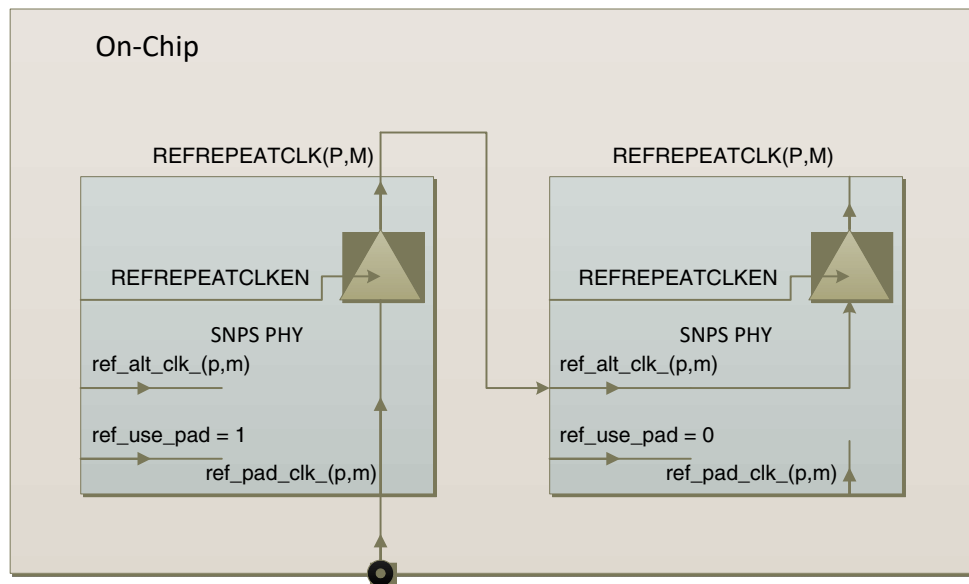
| Clock                | PIPE Power State |    |    |                  | Enable Required   |
|----------------------|------------------|----|----|------------------|-------------------|
|                      | P0               | P1 | P2 | P3               |                   |
| ref_output_clk       | Y                | Y  | Y  | Y                | ref_ssp_en        |
| ref_repeat_clk_(p,m) | Y                | Y  | Y  | Y/N <sup>a</sup> | ref_repeat_clk_en |
| pcs_clk              | Y                | Y  | Y  | N                | -                 |
| pipeP_pclk           | Y                | Y  | Y  | N                | -                 |
| rx_clk               | Y                | N  | N  | N                | -                 |

a. If all downstream PHYs that are using ref\_repeat\_clk\_(p,m) do not require a reference clock, in the P3 state, ref\_repeat\_clk\_en can be de-asserted (done at the system level).

## 4.2.3 Reference Clock Repeater

As shown in the following figure, the reference clock repeater feature enables an input reference clock to be buffered for use by another Synopsys PHY, enabling a single, off-chip clock source to supply multiple Synopsys PHYs. The ref\_repeat\_clk\_(p,m) outputs are at CMOS level. For more information about using these outputs, see [“Reference Clock Routing Guidelines for ref\\_alt\\_clk\\_m and ref\\_alt\\_clk\\_p”](#) on page 198 and [“Reference Clock Implementation”](#) on page 243.

**Figure 4-13 Reference Clock Repeater**



## 4.2.4 Resets

The `phy_reset` signal is the global reset that resets the `dwc_usb3_sspxn_hspxn_ns` and `pcs_xN`. For information about how to control `phy_reset` during power-up, see [“Powering Up and Powering Down”](#) on page 165.

## High-Speed Controls

The USB 3.0 femtoPHY provides the following high-speed control signals.

- **COMMONONN**: This signal controls whether the PLL block remains powered — consuming additional current during Suspend and Sleep modes.
- **SLEEPM<#>**: Asserting this signal places the USB 3.0 femtoPHY in Sleep mode according to the USB 2.0 Link Power Management (LPM) addendum to the USB 2.0 specification.
- **SUSPENDM<#>**: Asserting this signal suspends the USB 3.0 femtoPHY by tri-stating the transmitter and powering down the HS function’s circuits.
- **PORTRESET<#>**: When asserted, this signal resets the corresponding port’s HS transmit and receive logic without disabling the HS clocks within the USB 3.0 femtoPHY.

## 4.3 Power Supply Connections

For information about voltage levels for the USB 3.0 femtoPHY's analog and digital power supplies, see [“Power and Ground Connections”](#) on page 190.

For packaging guidelines for each supply, see [“Power and Ground Connections”](#) on page 230.

## 4.4 Signals That Connect to External Pads

The USB 3.0 femtoPHY signals that go to the outside world through the `dwc_usb3_pads_sspaN_hspaN_ns` sample snap-on pads are pinned out to the `dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN` top-level module. These pins must not be connected to anything above the top-level module except pins of the chip.

- Optional external reference clock inputs: `ref_pad_clk_m`, `ref_pad_clk_p`

If an on-chip reference clock is used (by de-asserting `ref_use_pad`), tie the `ref_pad_clk_m` and `ref_pad_clk_p` pins to `gd` in the package.

- External reference resistor: `resref`
- Receive pins: `rx<#>_m/rx<#>_p`
- Transmit pins: `tx<#>_m/tx<#>_p`
- USB 2.0 D+/D- lines: `DM<#>/DP<#>`
- USB mini-receptacle identifier: `ID<#>`
- USB 5-V power supply pin: `VBUS<#>`

## 4.5 Serial Test Ports

The USB 3.0 femtoPHY supports the following serial test ports. These test interfaces are independent and serve distinct functions.

- **Scan interface:** Automated testing of the PHY's digital functionality
- **JTAG port:** Automated testing and diagnostic port for analog test and debug of the PHY
- **Boundary Scan:** Implements AC JTAG 1149.6 on high-speed AC-coupled pins

### 4.5.1 Scan Interface

For information about the Scan interface, see [“Scan”](#) on page 218.

### 4.5.2 JTAG Port

The USB 3.0 femtoPHY provides a JTAG TAP controller with an 18-bit CRSEL register that enables access to the internal control registers. While access to these registers is not required for normal PHY operation, this interface can be used to access the PHY's BIST and diagnostic features during ATE testing in production.

For information about how this JTAG chain can be combined with other JTAG chains on the chip to provide access to diagnostics and ATE testing, see [“Scan Implementation”](#) on page 475. [“Registers”](#) on page 253 provides details about the control registers used to implement analog testing of the PHY.

### 4.5.3 Boundary Scan Port

For information about the USB 3.0 femtoPHY Boundary Scan port, see [“JTAG Port”](#) on page 445.

## 4.6 Termination Tuning

The PHY uses an external resistor to calibrate the termination impedances of the PHY's high-speed inputs and outputs. The resistor is shared between the USB 2.0 high-speed outputs and the SuperSpeed I/O. Each time the PHY is taken out of a reset, a termination calibration is performed. The termination calibration is continuous for USB 2.0 operation except when the USB 2.0 port is suspended. The termination calibration is performed only when coming out of a reset in SuperSpeed operation. Calibration of the SuperSpeed link can be requested externally by asserting the `rtune_req` (request) signal. When the calibration is complete, `rtune_ack` transitions low.

A resistor calibration on the SuperSpeed link cannot be performed while the link is operational. At no point is a request for calibrating the SuperSpeed link using `rtune_req` required; this feature is provided only for completeness.

## 4.7 Test Controls

The following input signals can be used for test purposes.

- **test\_burnin**: This input is reserved. Tie this signal low.
- **test\_powerdown\_ssp**: Asserting `test_powerdown_ssp` forces the SS function into its lowest power state by turning off all SuperSpeed blocks. (This assertion does not impact high-speed operation.) The only way to wake up the SS function from this mode is by de-asserting `test_powerdown_ssp`. For more information about using `test_powerdown_ssp`, see [“Power-Down Requirements”](#) on page 168. During Power-down test mode, the `test_burnin` input must be set to 1'b0.
- **test\_powerdown\_hsp**: Asserting `test_powerdown_hsp` forces the HS function into its lowest power state by turning off all high-speed blocks. (This assertion does not impact SuperSpeed operation.) The only way to wake up the HS function from this mode is by de-asserting `test_powerdown_hsp`. For more information about using `test_powerdown_hsp`, see [“Power-Down Requirements”](#) on page 168.
- **ATERESET**: This signal can be used to disable the high-speed controller clocks, PHYCLOCK and FREECLK, then re-enable them with an aligned phase.
- **LOOPBACKENB**: This signal places the high-speed blocks into Loopback mode and enables the high-speed receive and transmit logic concurrently. Enable this signal for test purpose only.
- **VATESTENB**: This signal places the analog test signals onto the optional ID<#> pin. If VATESTENB is not used, tie it low.

## 4.8 OTG Signals

The USB 3.0 femtoPHY uses the following OTG 1.1 signals:

- **OTGDISABLE0**: This controller signal powers down the OTG block, which disables the VBUS Valid comparator.
- **VBUSVLDEXTSEL0**: This signal selects the VBUSVLDEXT<#> input or the internal Session Valid comparator to indicate when the VBUS signal on the USB cable is valid.
- **DRVVBUS0**: This signal controls the VBUS Valid comparator and drives 5 V on VBUS through an external charge pump.
- **VBUSVALID0**: This signal is output from the USB 3.0 femtoPHY's VBUS Valid comparator and indicates whether the VBUS output is at a valid level.
- **IDPULLUP0**: This signal controls ID line sampling.
- **IDDIG0**: This signal indicates whether the connected plug is a mini-A or mini-B plug.
- **OTGSESSVLD0**: This signal is output from the USB 3.0 femtoPHY's Session Valid comparator and indicates whether the voltage on VBUS is below the OTG Device Session Valid threshold.

## 4.9 Battery Charging Signals

The USB 3.0 femtoPHY supports battery charger detection to enable devices to draw increased current from VBUS for charging and/or powering up from dedicated charging ports or charging downstream ports.

The USB 3.0 femtoPHY provides the following battery-charger-related signals:

- **OTGSESSVLD0**: This signal indicates VBUS Session Valid detection and can be used by a portable device to detect the VBUS voltage from a host or charger port.
- **CHRGSEL<#>**: This signal selects whether the Battery Charger circuit sets voltage sourcing on the D+ line and sensing on the D- line or voltage sourcing on D- and sensing on D+.
- **VDATSRCENB<#>**: This signal activates the voltage sourcing on the D+ or D- line.
- **VDATDETENB<#>**: This signal activates the Battery Charger comparator and enables detection of the D+ and D- lines according to the Battery Charging specification.
- **DCDENB<#>**: This signal enables the DCD IDP\_SRC current source in a portable device only.
- **DMPULLDOWN<#>**: This signal enables the DCD RDM\_DWN resistance in a portable device only.
- **CHGDET<#>**: This signal is the Battery Charge Detection ( $V_{DAT\_REF}$ ) comparator output.
- **FSVPLUS<#>**: This signal is the DP<#> FS/LS single-ended receiver output, which is also the  $V_{LGC}$  output.
- **FSVMINUS<#>**: This signal is the DM<#> FS/LS single-ended receiver output, which is also the  $V_{LGC}$  output.

For more information about the battery charger signals, see “Battery Charging Signals” section in chapter [“Signal Descriptions”](#) on page 29

## 4.9.1 Performing Battery Charging

The following table describes the USB 3.0 femtoPHY battery charging operations and corresponding signal settings.



### Note

- This table applies to battery charging operations, not to normal operation.
- It is highly recommended that you keep the USB 3.0 femtoPHY in Suspend mode (SUSPENDM<#> = 1'b0) while performing battery charging operations, such as those in the following table. Some customized chargers and battery charging procedures might cause an SE1 and/or noise on DP<#> / DM<#>, which can lead to false SYNC detection and RXACTIVE<#> assertion. This false detection is prevented by placing the PHY in Suspend mode.

**Table 4-4 Battery Charging Operations**

| Mode of Operation <sup>a</sup>                                                                                                                                 | Inputs     |               |              |           |               |               |                |             | Outputs     |                |            |                | DP/DM Port States        |                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|--------------|-----------|---------------|---------------|----------------|-------------|-------------|----------------|------------|----------------|--------------------------|--------------------------------------------|
|                                                                                                                                                                | CHRGSEL<#> | VDATDETENB<#> | VDATSRCEB<#> | DCDENB<#> | DPPULLDOWN<#> | DMPULLDOWN<#> | OPMODE<#>[1:0] | SUSPENDM<#> | OTGSESSVLD0 | CHGDET<#>      | FSVPLUS<#> | FSVMINUS<#>    | V <sub>DP&lt;#&gt;</sub> | V <sub>DM&lt;#&gt;</sub>                   |
| Battery charging is disabled                                                                                                                                   | X          | 0             | 0            | 0         | X             | X             | XX             | X           | X           | 0              | X          | X              | X                        | X                                          |
| PHY is operating as a portable device (PD) and is not attached to host nor charger port.                                                                       | X          | 0             | 0            | 0         | 0             | 0             | 01             | 0           | 0           | X              | X          | X              | V <sub>LGC_LOW</sub>     | V <sub>LGC_LOW</sub>                       |
| PHY is operating as a PD using the Dead Battery Provision (DBP) - Unconfigured Clause, while attached to a standard downstream port (SDP).                     | 0          | 1             | 1            | 0         | 0             | 0             | 01             | 0           | 1           | 0 <sup>b</sup> | X          | 0 <sup>b</sup> | V <sub>DP_SRC</sub>      | < V <sub>DAT_REF</sub> <sup>b</sup>        |
| PHY is operating as a PD using the DBP - Unconfigured Clause, while attached to a special port (such as a PS2 or proprietary charger) that pulls D+ / D- high. | 0          | 1             | 1            | 0         | 0             | 0             | 01             | 0           | 1           | 1 <sup>b</sup> | X          | 1 <sup>b</sup> | V <sub>DP_SRC</sub>      | Optionally > V <sub>LGC</sub> <sup>b</sup> |



**Table 4-4 Battery Charging Operations (Continued)**

| Mode of Operation <sup>a</sup>                                                                                                                                 | Inputs     |              |              |           |               |               |                |             | Outputs     |                |                |                | DP/DM Port States                 |                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|--------------|-----------|---------------|---------------|----------------|-------------|-------------|----------------|----------------|----------------|-----------------------------------|-----------------------------------------------------------------------|
|                                                                                                                                                                | CHRGSEL<#> | VDATETENB<#> | VDATSRCEB<#> | DCDENB<#> | DPPULLDOWN<#> | DMPULLDOWN<#> | OPMODE<#>[1:0] | SUSPENDM<#> | OTGSESSVLD0 | CHGDET<#>      | FSVPLUS<#>     | FSVMINUS<#>    | V <sub>DP</sub> <#>               | V <sub>DM</sub> <#>                                                   |
| PHY is operating as a PD using the DBP - Unconfigured Clause, while attached to a charging downstream port (CDP) or a dedicated charging port (DCP).           | 0          | 1            | 1            | 0         | 0             | 0             | 01             | 0           | 1           | 1 <sup>b</sup> | X              | 0 <sup>b</sup> | V <sub>DP_SRC</sub>               | > V <sub>DAT_REF</sub> and optionally < V <sub>LGC</sub> <sup>b</sup> |
| PHY is operating as a PD in Data Contact Detection (DCD) mode. No data pin contact with remote charger.                                                        | X          | 0            | 0            | 1         | 0             | 1             | 01             | 0           | 1           | X              | 1              | X              | V <sub>LGC_HI</sub>               | V <sub>LGC_LOW</sub>                                                  |
| PHY is operating as a PD in DCD mode. Data pin contact to remote charger is detected.                                                                          | X          | 0            | 0            | 1         | 0             | 1             | 01             | 0           | 1           | X              | 0 <sup>c</sup> | X              | V <sub>LGC_LOW</sub> <sup>c</sup> | V <sub>LGC_LOW</sub>                                                  |
| PHY is operating as a PD that is performing Primary Detection while attached to an SDP.                                                                        | 0          | 1            | 1            | 0         | 0             | 0             | 01             | 0           | 1           | 0 <sup>b</sup> | X              | 0 <sup>b</sup> | V <sub>DP_SRC</sub>               | < V <sub>DAT_REF</sub> <sup>b</sup>                                   |
| PHY is operating as a PD that is performing Primary Detection while attached to a special port (such as a PS2 or proprietary charger) that pulls D+ / D- high. | 0          | 1            | 1            | 0         | 0             | 0             | 01             | 0           | 1           | 1 <sup>b</sup> | X              | 1 <sup>b</sup> | V <sub>DP_SRC</sub>               | Optionally > V <sub>LGC</sub> <sup>b</sup>                            |
| PHY is operating as a PD that is performing Primary Detection while attached to a CDP or DCP.                                                                  | 0          | 1            | 1            | 0         | 0             | 0             | 01             | 0           | 1           | 1 <sup>b</sup> | X              | 0 <sup>b</sup> | V <sub>DP_SRC</sub>               | > V <sub>DAT_REF</sub> and optionally < V <sub>LGC</sub> <sup>b</sup> |

**Table 4-4 Battery Charging Operations (Continued)**

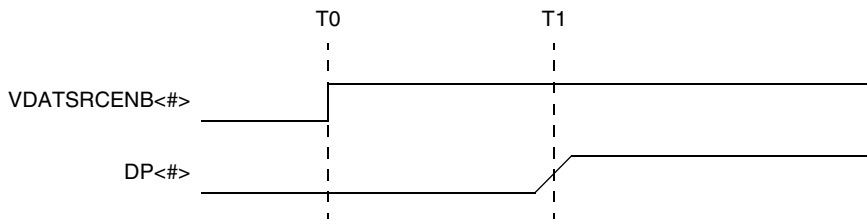
| Mode of Operation <sup>a</sup>                                                                   | Inputs     |               |               |           |               |               |                |             | Outputs     |                |                |                | DP/DM Port States                                              |                                   |
|--------------------------------------------------------------------------------------------------|------------|---------------|---------------|-----------|---------------|---------------|----------------|-------------|-------------|----------------|----------------|----------------|----------------------------------------------------------------|-----------------------------------|
|                                                                                                  | CHRGSEL<#> | VDATDETENB<#> | VDATSRCENB<#> | DCDENB<#> | DPPULLDOWN<#> | DMPULLDOWN<#> | OPMODE<#>[1:0] | SUSPENDM<#> | OTGSESSVLD0 | CHGDET<#>      | FSVPLUS<#>     | FSVMINUS<#>    | V <sub>DP</sub> <#>                                            | V <sub>DM</sub> <#>               |
| PHY is operating as a PD that is performing Secondary Detection while attached to a CDP.         | 1          | 1             | 1             | 0         | 0             | 0             | 01             | 0           | 1           | 0              | X              | X              | <V <sub>DAT_REF</sub>                                          | V <sub>DM_SRC</sub>               |
| PHY is operating as a PD that is performing Secondary Detection while attached to a DCP.         | 1          | 1             | 1             | 0         | 0             | 0             | 01             | 0           | 1           | 1              | X              | X              | >V <sub>DAT_REF</sub>                                          | V <sub>DM_SRC</sub>               |
| PHY is operating as a host CDP with no PD detected.                                              | 1          | 1             | 0             | 0         | 1             | 1             | 01             | 0           | X           | 0 <sup>d</sup> | 0 <sup>d</sup> | X              | <V <sub>DAT_REF</sub> <sup>d</sup>                             | V <sub>LGC_LOW</sub>              |
| PHY is operating as a host CDP that detects an attached PD, which supports the BC specification. | 1          | 1             | 1             | 0         | 1             | 1             | 01             | 0           | X           | 1 <sup>d</sup> | 0 <sup>d</sup> | X              | >V <sub>DAT_REF</sub><br>and<br><V <sub>LGC</sub> <sup>d</sup> | V <sub>DM_SRC</sub>               |
| PHY is operating as a host CDP that detects a connect from the PD D+ pullup <sup>e</sup> .       | 1          | 1             | 0             | 0         | 1             | 1             | 01             | 0           | X           | X              | 1 <sup>e</sup> | 0 <sup>e</sup> | >V <sub>LGC</sub> <sup>e</sup>                                 | V <sub>LGC_LOW</sub> <sup>e</sup> |

- a. This table demonstrates the PHY settings for some of the main operations in the Battery Charging specification, Revision 1.2. The specification contains important additional information, options, and sequences for these and other Battery Charging operations; refer to the full specification.
- b. The specification includes an optional DM<#> V<sub>LGC</sub> comparison during the Weak Battery algorithm and the Primary Detection. The PHY supports this option by keeping the DM<#> V<sub>LGC</sub> detection signal separate from the PHY's CHGDET signal to enable you to decide whether to include the DM<#> V<sub>LGC</sub> detection. The PHY's CHGDET signal is from the V<sub>DAT\_REF</sub> detector only, and the PHY's FSVMINUS signal is from the DM<#> V<sub>LGC</sub> detector. The PHY does not include the logical combination of these two signals. You can optionally incorporate this logic externally by AND'ing the PHY's CHGDET signal with an inverted FSVMINUS signal.
- c. The specification identifies scenarios where DCD is not expected to work, such as when a special port is pulling D+ / D– high. A PD must include a DCD timeout after which time the PD proceeds to Primary Detection.
- d. For the CDP role, the specification provides two different methodologies, one of which is represented in this table. The PHY's CHGDET signal is from the V<sub>DAT\_REF</sub> detector only, and the PHY's FSVPLUS signal is from the DP<#> V<sub>LGC</sub> detector. The PHY does not include the PRTBL\_DET logical combination of these two signals. You can incorporate this logic externally by AND'ing the PHY's CHGDET signal with an inverted FSVPLUS signal.
- e. This CDP role example assumes that the PD is a full-speed (or high-speed capable) device. If the PD is a low-speed device, the connect pull-up is on D– and is indicated by FSVMINUS.

4.9.2 Dead Battery Provision

The dead battery provision enables a device to draw increased current (up to 100 mA) from VBUS in certain conditions. This function can be used only if all power supplies are stable and at their respective, normal, valid operating levels. While the increased current is being drawn, the device must source a nominal 0.6 V on DP<#>. This source is enabled with the VDATSRCENB<#> signal, as shown in the following figure.

Figure 4-14 Dead Battery Provision



The following table describes the corresponding timing parameters.

Table 4-5 Dead Battery Provision Timing Values

| Timing Parameter | Description                               | Value       |
|------------------|-------------------------------------------|-------------|
| T0               | VDATSRCENB<#> is asserted.                | Reference   |
| T1               | DP<#> is driven to V <sub>DAT_SRC</sub> . | T1 > 100 μs |

4.9.3 Data Contact Detect

Prior to a device performing battery charger detection, DCD is performed to determine if the D+ and D- pins in the device have made contact with those in the host or charger.

When the DCDENB<#> signal is asserted, the weak pull-up current on DP<#> is enabled. For standard DCD in accordance with the Battery Charger v1.2 specification, the pull-down resistor on DM<#> (RDM\_DWN) must also be enabled by setting DMPULLDOWN<#> = 1'b1. After asserting DCDENB<#> and DMPULLDOWN<#>, the FSVPLUS<#> output must be monitored, and a positive DCD is determined when FSVPLUS<#> has been low for 10 ms.

Note that if the USB device is plugged in quickly, the FSVPLUS<#> signal might remain low and not transition to a high state. The following figure shows timings for the dead battery provision, and [Table 4-6](#) describes the corresponding timing parameters.

Figure 4-15 Data Contact Detection

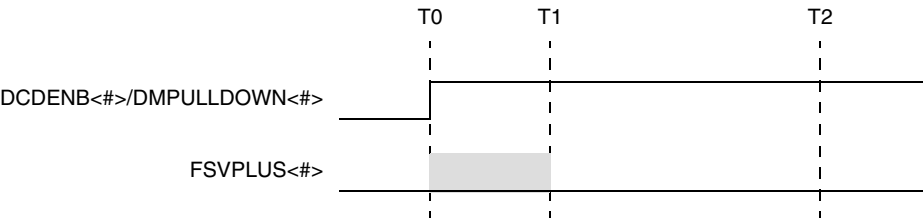


Table 4-6 Data Contact Detection

| Timing Parameter | Description                               | Value           |
|------------------|-------------------------------------------|-----------------|
| T0               | DCDENB<#> and DMPULLDOWN<#> are asserted. | Reference       |
| T1               | FSVPLUS<#> output is low.                 | T1 ≥ T0         |
| T2               | DCD is positive.                          | T2 = T1 + 10 ms |

### 4.9.4 Battery Charger Detection

This section describes battery charger detection in Device mode and Host mode.

#### Device Mode

When a device performs battery charger detection (CHRGSEL<#> = 1'b0), the VDATSRCENB<#> and VDATDETENB<#> signals are asserted, and the CHGDET<#> signal is monitored. The following figure shows timings for battery charger detection, and Table 4-7 describes the corresponding timing parameters.

 **Note**

The following timings represent a case where the CHGDET<#> output becomes valid.

Figure 4-16 Battery Charger Detection: Device Mode

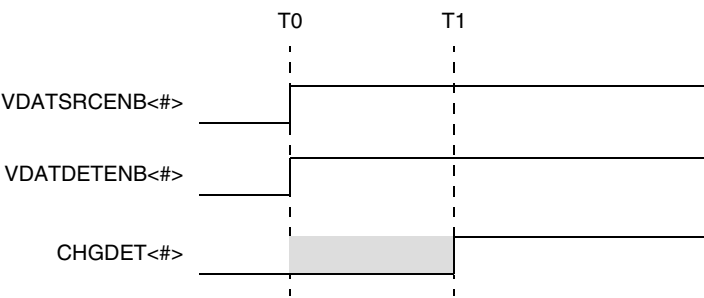


Table 4-7 Battery Charger Detection: Device Mode

| Timing Parameter | Description                                   | Value            |
|------------------|-----------------------------------------------|------------------|
| T0               | VDATSRCENB<#> and VDATDETENB<#> are asserted. | Reference        |
| T1               | CHGDET<#> output becomes valid.               | T1 > T0 + 200 μs |

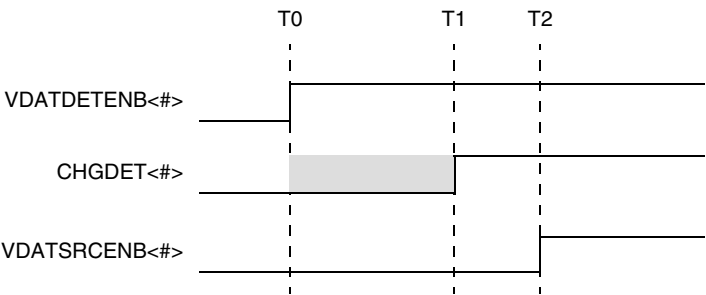
#### Host Mode

When a charging downstream port performs battery charger detection (CHRGSEL<#> = 1'b1), the VDATDETENB<#> signal is asserted, and the CHGDET<#> signal is monitored. After CHGDET<#> is determined to be valid, the VDATSRCENB<#> signal is asserted.

The following figure shows timings for battery charger detection, and [Table 4-8](#) describes the corresponding timing parameters.

 **Note** The following timings represent a case where the CHGDET<#> output becomes valid.

**Figure 4-17 Battery Charger Detection: Host Mode**



**Table 4-8 Battery Charger Detection: Host Mode**

| Timing Parameter | Description                     | Value                                                                                                         |
|------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------|
| T0               | VDATDETENB<#> is asserted.      | Reference                                                                                                     |
| T1               | CHGDET<#> output becomes valid. | $T1 > T0 + 200 \mu s$                                                                                         |
| T2               | VDATTRCENB<#> is asserted.      | For VDATTRCENB<#> timing values, refer to the TVDMSRC_EN timing values in the Battery Charging specification. |

## 4.10 UART/Autoresume Signals

The USB 3.0 femtoPHY uses the following signals for UART and Autoresume support.

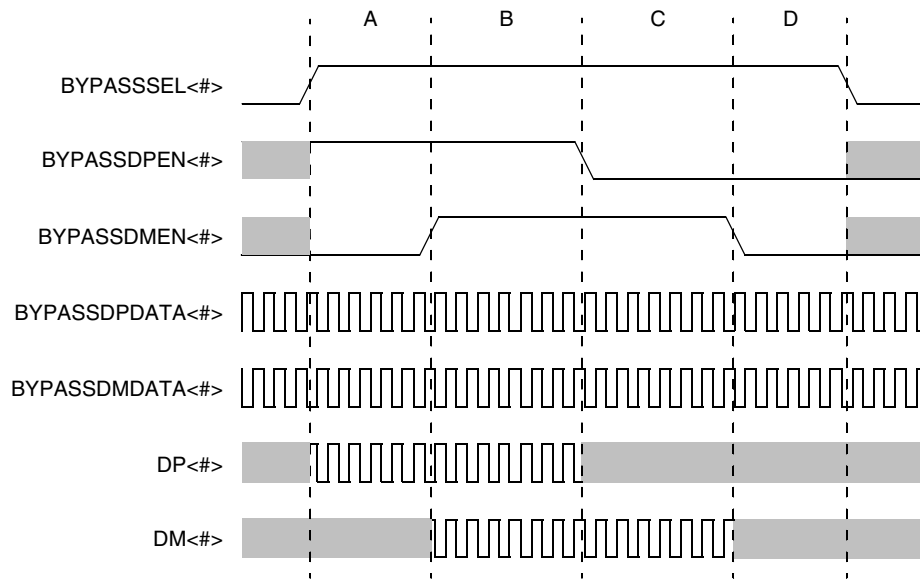
- **BYPASSSEL<#>**: This signal enables UART/Autoresume mode.
- **BYPASSDPEN<#>**: This signal enables UART/Autoresume data transmission on D+ line.
- **BYPASSDMEN<#>**: This signal enables UART/Autoresume data transmission on D- line.
- **BYPASSDPDATA<#>**: This signal provides UART/Autoresume input data for D+ line.
- **BYPASSDMDATA<#>**: This signal provides UART/Autoresume input data for D- line.
- **FSVPLUS<#>**: This signal indicates the state of the D+ line during UART/Autoresume data reception.
- **FSVMINUS<#>**: This signal indicates the state of the D- line during UART/Autoresume data reception.

When using the UART and Autoresume signals, the USB 3.0 femtoPHY's Transmitter block is active, while the PLL, high-speed devices, and OTG block are powered down, and the UTMI+ control and status signals are not active.

The USB 3.0 femtoPHY's single-ended receivers are used for UART and Autoresume data reception, while the full-speed drivers are used for data transmission. UART and Autoresume signaling levels are the same as signaling levels for USB 2.0 FS mode.

UART and Autoresume functions are accessed by setting the BYPASSSEL<#> signal to 1'b1. While in this mode, the USB 3.0 femtoPHY's FS driver is enabled via the BYPASSDPEN<#> and BYPASSDMEN<#> signals, respectively. The data driven on the D+ and D- lines is set by the BYPASSDPDATA<#> and BYPASSDMDATA<#> signals, respectively. The following figure shows the timing sequence for transmitter digital logic bypass operation.

In this mode of operation, the driver source impedance is not calibrated; however, the driver retains the settings of the last calibration. In addition, the rise/fall time control is disabled, so changes in the state of DP<#> and DM<#> driven by this mode are not expected to meet the rise and fall times as defined in the USB 2.0 specification.

**Figure 4-18 Transmitter Digital Logic Bypass Operation**

#### 4.10.1 Using UART and Autoresume Functions

To use UART and Autoresume functions:

1. Disable the OTG block by setting OTGDISABLE0 to 1'b1.
2. Disable the pull-up resistance on the D+ line by setting OPMODE<#>[1:0] to 2'b01.
3. To ensure that the PLL block is powered down in Suspend mode, set COMMONONN to 1'b1.
4. Place the USB 3.0 femtoPHY in Suspend mode by setting SUSPENDM<#> to 1'b0.
5. It is recommended that you ensure the USB 3.0 femtoPHY is in normal operating mode prior to switching to Suspend mode; otherwise, the Squelch/Receiver blocks might continue to consume power using the UART and Autoresume functions.
6. Set BYPASSSEL<#> to 1'b1.
7. To transmit data onto DP<#> and DM<#>, control BYPASSDMEN<#>, BYPASSDPEN<#>, BYPASSDMDATA<#>, and BYPASSDPDATA<#>.
8. To receive data, monitor FSVPLUS<#> and FSVMINUS<#>.

For UART/Autoresume mode timing specifications, see ["Timing Specification"](#) on page 157.



## 4.10.2 Returning to Normal Operating Mode

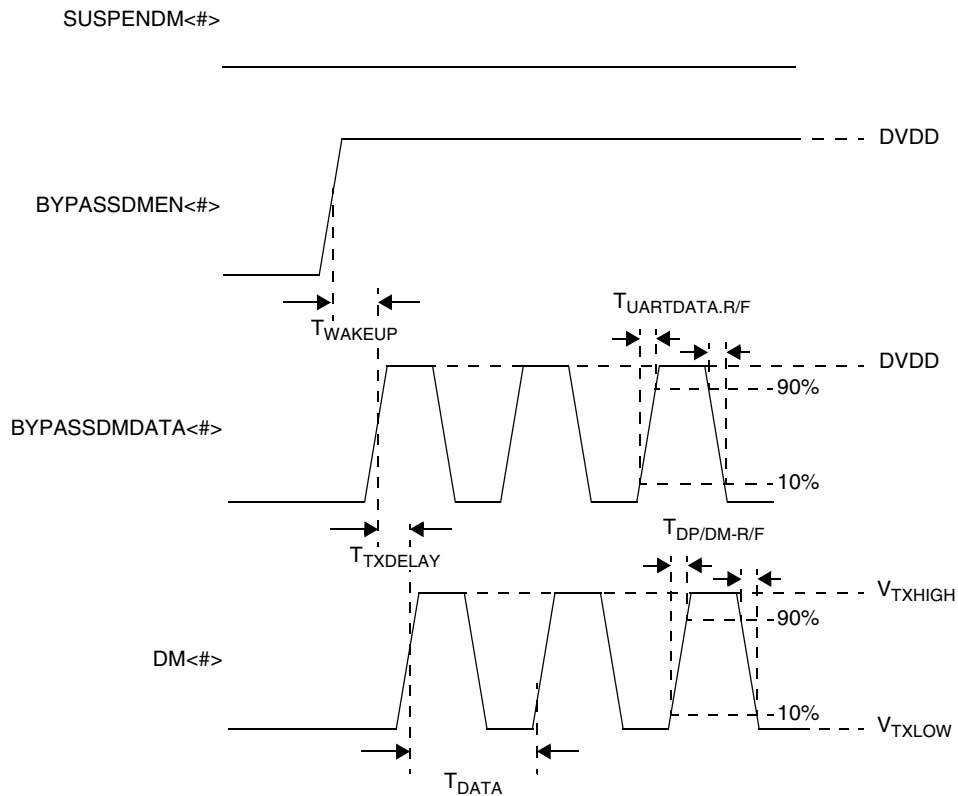
To return to normal operating mode:

1. Ensure that no activity is on the USB.
2. Set BYPASSSEL<#> to 1'b0.
3. Resume the USB 3.0 femtoPHY by following the instructions in [“Resume Operation: High-Speed Function”](#) on page 170.
4. If OTG functions are required, set OTGDISABLE0 to 1'b0.

## 4.10.3 Timing Specification

The following figure shows timing for UART and Autoresume data transmission. Only transmission on DM<#> is shown. A similar waveform is expected for transmission on DP<#>.

**Figure 4-19 UART Mode Data Transmission Timing**



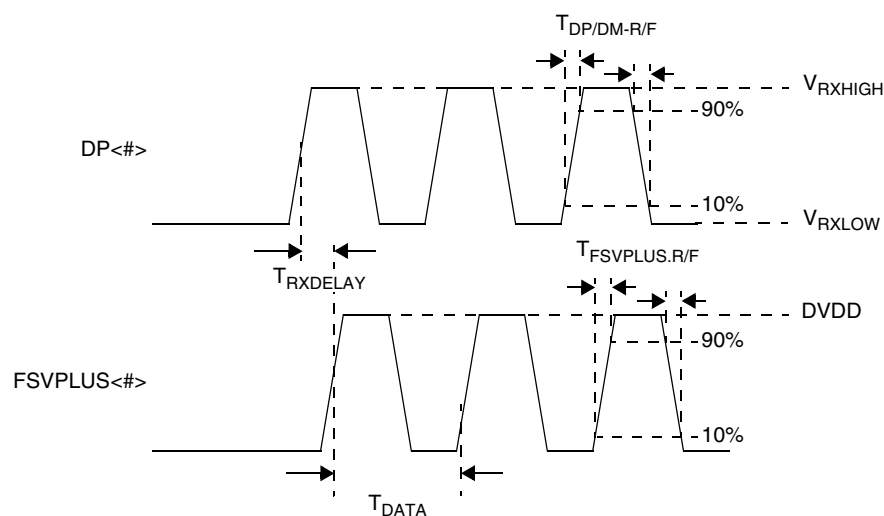
The following table describes the data transmission timing parameters.

**Table 4-9 UART Mode Data Transmission Timing Specification**

| Parameter          | Description                                     | Minimum | Typical | Maximum | Unit    |
|--------------------|-------------------------------------------------|---------|---------|---------|---------|
| $T_{WAKEUP}$       | Wakeup time after BYPASSDMEN<#> signal asserted | 20      | -       | -       | $\mu$ s |
| $T_{UARTDATA.R/F}$ | BYPASSDMDATA<#> rise/fall time                  | -       | -       | 50      | ns      |
| $T_{DATA}$         | Transfer/ receive data rate                     | -       | -       | 1       | Mbps    |

The following figure shows timing for UART and Autoresume data reception. Only transmission on DP<#> is shown. A similar waveform is expected for transmission on DM<#>.

**Figure 4-20 UART Mode Data Reception Timing**



The following table describes the data reception timing specification.

**Table 4-10 UART Mode Data Reception Timing Specification**

| Parameter       | Description                          | Minimum | Typical | Maximum | Unit |
|-----------------|--------------------------------------|---------|---------|---------|------|
| $T_{DP/DM-R/F}$ | DP<#> rise/fall time                 | -       | -       | 200     | ns   |
| $T_{RXDELAY}$   | Received data's propagation delay    | -       | -       | 50      | ns   |
| $T_{TXDELAY}$   | Transferred data's propagation delay | -       | -       | 100     | ns   |

## 4.11 High-Voltage Interrupt Signals

The high-voltage interrupt signals are analog-level outputs of the USB 3.0 femtoPHY that are used to signal interrupts on the DM<#>, DP<#>, VBUS<#>, and ID<#> pins.

These signals must not be connected to any digital logic devices, because they are at I/O signal level.

If these output signals are not used, leave them floating.

## 4.12 Control Register Access

The CR port is a simple 16-bit data/address parallel port that is provided for on-chip access to the control registers inside the USB 3.0 femtoPHY. While access to these registers is not required for normal PHY operation, this interface enables you to access some of the PHY's diagnostic features during normal operation or to override some basic PHY control signals.

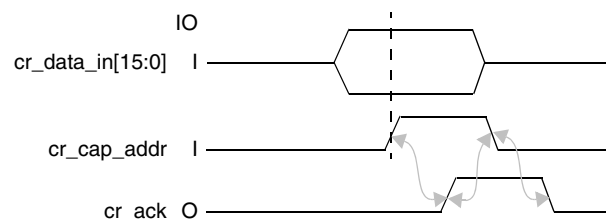
This interface is completely asynchronous – using a handshake between the cr\_cap\_addr, cr\_cap\_data, cr\_read, and cr\_write input commands with cr\_ack acknowledgments and cd\_data\_out outputs from the PHY. CR port access is divided into address, read, and write transactions, as shown in [Figure 4-21](#) through [Figure 4-23](#) on page 160.



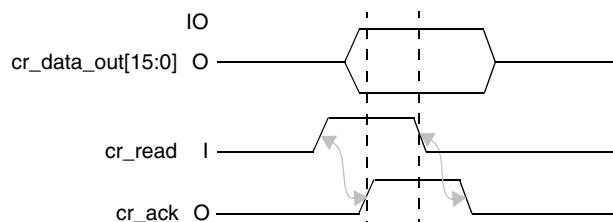
### Note

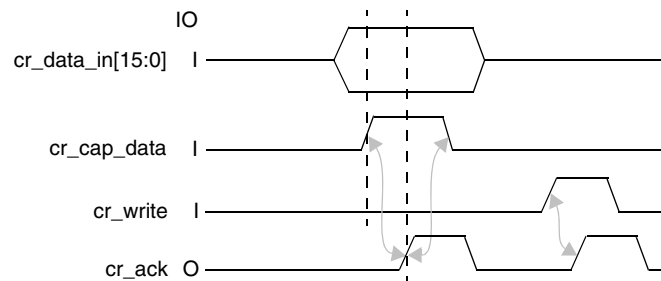
If you plan to access the CR registers without using the handshake function (for example, to run ATE vectors that consist of predetermined sequences), then you must use the minimum timing shown in “[CREG Access Through the CR Parallel Port](#)” on page 488.

**Figure 4-21 CR Bus Address Capture Transaction**



**Figure 4-22 CR Bus Read Transaction**



**Figure 4-23 CR Bus Write Transaction**

If the Control Register port is not used, tie the input signals as follows:

- `cr_cap_addr = 1'b0`
- `cr_cap_data = 1'b0`
- `cr_data_in = 16'b0`
- `cr_read = 1'b0`
- `cr_write = 1'b0`
- No connect:
  - `cr_data_out[15:0]`
  - `cr_ack`

For details about the control register fields, see [“Registers”](#) on page 253.

## 4.13 Serial Interface

The serial interface supports full-speed and low-speed data transmission rates to and from a USB 1.1 controller. To enable the serial interface, set `FSXCVROWNER<#>` to `1'b1`. To enable data transmission through the serial interface, set `TXENABLEN<#>` to `1'b0`.

For serial interface timing requirements, see [“Serial Interface Timing Requirements”](#) on page 177.

## 4.14 UTMI+ Controls

The UTMI+ control and status signals are designed to interface with USB 2.0 OTG, host, and device controllers.

For data reception and transmission via the UTMI+, the USB 3.0 femtoPHY supports 8-bit or 16-bit operation and `PHYCLOCK<#>` frequency of 30 MHz or 60 MHz. For information about USB 3.0 femtoPHY speed modes and the corresponding `PHYCLOCK<#>` frequencies when using the UTMI+, see [“High-Speed Function Speed Modes”](#) on page 174. For details about `PHYCLOCK<#>` timing behavior for power-on reset and resume operations, see [“Suspend and Sleep Operations: High-Speed Function”](#) on page 169 and [“Resume Operation: High-Speed Function”](#) on page 170. In addition, check the host controller's UTMI+ clock synchronization requirements with respect to `PHYCLOCK<#>`.

For information about operating in a non-OTG, UTMI+ Level 3 speed mode, see [“Operating the High-Speed Function in Non-OTG, UTMI+ Level 3 Speed Modes”](#) on page 175.

## 4.15 UTMI+ Receive Signals

The UTMI+ parallel receive port can be configured for 8-bit or 16-bit operation via the WORDINTERFACE<#> signal.

Data is output on the UTMI+ via the DATAOUT<#>[7:0]/DATAOUTH<#>[7:0] buses. The controller signals, RXVALID<#>/RXVALIDH<#>, indicate when these buses hold valid data.

SYNC patterns are indicated via the RXACTIVE<#> signal, and errors detected in the current received packet are indicated via the RXERROR<#> signal.

## 4.16 UTMI+ Transmit Signals

The UTMI+ parallel transmit port can be configured for 8-bit or 16-bit operation via the WORDINTERFACE<#> signal.

Data is input from the UTMI+ via the DATAIN<#>[7:0]/DATAINH<#>[7:0] buses. The controller signal, TXREADY<#>, indicates when the PHY is ready to accept data on these buses, while the TXVALID<#>/TXVALIDH<#> signals indicate when data on these buses is valid.

## 4.17 Parameter Controls

The USB 3.0 femtoPHY provides several signals that can be statically set or programmed. Some of these parameters are package-dependent; therefore, their programmability enables you to drive these parameters from a register.



### Note

To enable tuning at the board level, connect these controls to a register.

**Table 4-11 Parameter Controls**

| Parameter Control Signal              | Details                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMPDISTUNE<#>[2:0]                   | Adjusts the voltage level for the threshold used to detect a disconnect event at the host.                                                                                                                                                                                                |
| OTGTUNE0[2:0]                         | Adjusts the voltage level for the VBUS Valid threshold.                                                                                                                                                                                                                                   |
| SQRXTUNE<#>[2:0]                      | Adjusts the voltage level for the threshold used to detect valid high-speed data.                                                                                                                                                                                                         |
| TXFSLSTUNE<#>[3:0]                    | Adjusts the low- and full-speed single-ended source impedance while driving high.                                                                                                                                                                                                         |
| TXHSXVTUNE<#>[1:0]                    | The default bit setting is intended to set the crossover voltage as close as possible to 0 V.                                                                                                                                                                                             |
| TXPREEMPAMPTUNE<#>[1:0]               | Controls the amount of current sourced to DP<#> and DM<#> after a J-to-K or K-to-J transition.                                                                                                                                                                                            |
| TXPREEMPPULSETUNE<#>                  | Controls the duration for which the HS pre-emphasis current is sourced onto DP<#> or DM<#>.                                                                                                                                                                                               |
| TXRESTUNE<#>[1:0]                     | This bus adjusts the driver source impedance (for both pull-up/pull-down impedances in LS/FS mode and the pull-down impedance in HS mode) to compensate for added series resistance on the USB.                                                                                           |
| TXRISETUNE<#>[1:0]                    | The default bit setting is intended to set the HS rise/fall times to meet the USB specification. Might need to be changed due to board, package and on-chip parasitics.                                                                                                                   |
| TXVREFTUNE<#>[3:0]                    | The default bit setting is intended to create an HS transmit DC level of approximately 400 mV.                                                                                                                                                                                            |
| VDATREFTUNE<#>[1:0]                   | Adjusts the threshold voltage level ( $V_{\text{dat\_ref}}$ ) used to detect data during charger-type detection.                                                                                                                                                                          |
| los_bias[2:0]                         | Sets the LOS detection threshold level.                                                                                                                                                                                                                                                   |
| los_level[4:0]                        | Reserved                                                                                                                                                                                                                                                                                  |
| pcs_rx_los_mask_val[9:0]              | Sets the number of reference clock cycles to mask the incoming LFPS. For normal operation, set to a targeted mask interval of 10 $\mu\text{s}$ (value = $10 \mu\text{s} / \text{Tref\_clk}$ ). If the ref_clkdiv2 signal is used, the value = $10 \mu\text{s} / (2 * \text{Tref\_clk})$ . |
| pcs_tx_deemph_3p5db[5:0] <sup>a</sup> | Package-dependent                                                                                                                                                                                                                                                                         |

**Table 4-11 Parameter Controls (Continued)**

| Parameter Control Signal  | Details                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| pcs_tx_deemph_6db[5:0]a   | Package-dependent                                                                                                  |
| pcs_tx_swing_full[6:0]a   | For the default 1.0-V launch amplitude; package dependent                                                          |
| lane0_tx_term_offset[4:0] | Reserved                                                                                                           |
| tx_vboost_lvl[2:0]        | Controls the launch amplitude only when VPTX is less than the launch amplitude corresponding to the tx_vboost_lvl. |

- a. For information about changing these parameter control values dynamically during active operation, see [“Changing Parameter Controls”](#) on page 164.

The values for transmit de-emphasis and amplitude are derived from the following equations:

$$\text{Tx de-emphasis (db)} = 20 \times \log_{10} \left[ \frac{128 - 2 \times \text{pcs\_tx\_deemph}}{128} \right]$$

$$\text{Tx amplitude (V)} = (\text{corresponding launch amplitude based on tx\_vboost\_lvl[2:0]}) \times \left[ \frac{\text{pcs\_tx\_swing\_full} + 1}{128} \right]$$

Using these relationships, the default settings for a 1.0-V launch amplitude and a -3.5 dB TX equalization were calculated, and settings for the associated parameter control bits were selected as specified in [Table 4-11](#).

### 4.17.1 Changing Parameter Controls

In general, the parameter controls are static signals to be set prior to taking the PHY out of reset. However, you can dynamically change these values on-the-fly for test purposes.

In this case, changes to the transmitter to reflect the current values of the following controls:

- pcs\_tx\_deemph\_3p5db[5:0]
- pcs\_tx\_deemph\_6db[5:0]
- pcs\_tx\_swing\_full[6:0]

occur only after the pipeP\_tx\_deemph[1:0] input changes.

### 4.17.2 Fixed RX Equalization

The USB 3.0 femtoPHY's adaptive RX equalization function does not always provide an optimal setting for equalization; the result can be marginal jitter tolerance curves. Synopsys recommends using fixed equalization instead of adaptive equalization. System-level simulations and hardware testing have proven that fixed equalization is sufficient for both short and long channels.

The PHY can be configured for fixed equalization by programming relevant control registers in the USB 3.0 femtoPHY as follows.

1. Write 1'b0 to RX\_OVRD\_IN\_HI.RX\_EQ\_EN [address 16'h1006: bit 6].
2. Write 1'b1 to RX\_OVRD\_IN\_HI.RX\_EQ\_EN\_OVRD [address 16'h1006: bit 7].
3. Write a fixed value to RX\_OVRD\_IN\_HI.RX\_EQ [address 16'h1006: bits 8–10] [equalization setting] (generally 2–4, based on testing in customer environment).
4. Write 1'b1 to RX\_OVRD\_IN\_HI.RX\_EQ\_OVRD [address 16'h1006: bit 11].



# 5

## System-Level Implementation

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This chapter includes the following sections:

- [“System Operation”](#) on page 165
- [“Test Environment Overview”](#) on page 181
- [“Verilog Model Files and Simulators”](#) on page 184

### 5.1 System Operation

This section describes the USB 3.0 femtoPHY’s system operation.

#### 5.1.1 Powering Up and Powering Down

This section describes USB 3.0 femtoPHY power-up and power-down requirements.

##### 5.1.1.1 Power-Up Requirements

The USB 3.0 femtoPHY has multiple power supplies; for details, see [“Power Supply Connections”](#) on page 143.

For information about the required packaging connections for all of the PHY power and ground supplies, see [“Power and Ground Connections”](#) on page 230.

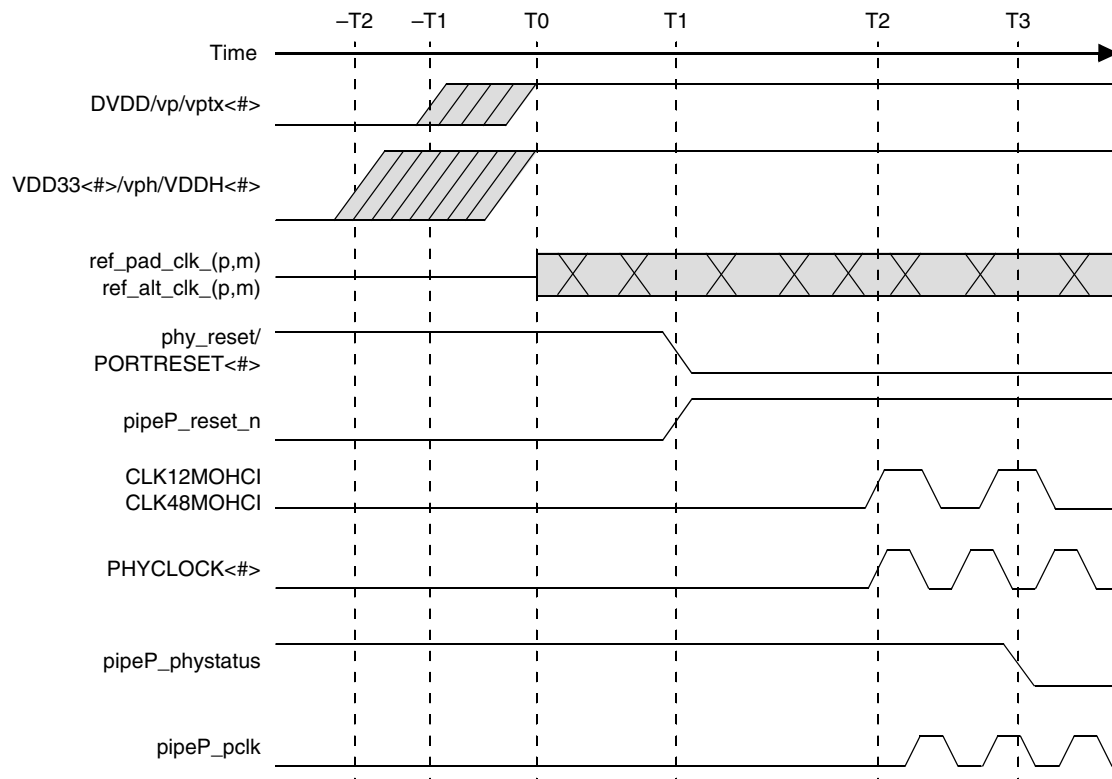
The USB 3.0 femtoPHY can have the 3.3-V and 0.8-V supplies power up in any sequence provided the low supply comes up soon after the high supply comes up.

### 5.1.1.2 Power-On Reset

While the power supplies can be brought up in any sequence provided they come up at relatively the same time, the ASIC must hold the phy\_reset pin high until the supplies are within the range specified in [Table 5-1](#) on page 167 and the reference clock is up and stable.

The following figure shows power-on reset timing for the USB 3.0 femtoPHY when the HS function is not in Suspend or Sleep mode.

**Figure 5-1 PoR Timing**



**Table 5-1 PoR Timing Values**

| Timing Parameter | Description                                                                                                                                                                           | Value         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| RefClk           | Differential reference clock input into the PHY. Equal to pins ref_alt_clk_p/ref_alt_clk_m or ref_pad_clk_m/ref_pad_clk_p depending on the setting of ref_use_pad and REFCLKSEL[1:0]. |               |
| –T2 to –T1       | Maximum time VDD33<#>/vph/VDDH<#> is applied before DVDD/vp/vptx<#>                                                                                                                   | < 10's of ms  |
| T0               | Time when supplies are within specified limits and the reference clock is up and stable                                                                                               |               |
| T1               | Time after T0 when ASIC can remove PoR                                                                                                                                                | > 10 $\mu$ s  |
| T2–T1            | Time after T1 when PHYCLOCK, CLK12MOHCI, and CLK48MOHCI are guaranteed to be available for use by ASIC                                                                                | < 75 $\mu$ s  |
| T3–T1            | Time after T1 when pipeP_phystatus de-asserts to indicate that pipeP_pclk is guaranteed to be up and stable                                                                           | < 225 $\mu$ s |

### 5.1.1.3 Power Supply Sequencing If the USB 3.0 femtoPHY is Not Used

If the USB 3.0 femtoPHY is not being used, you can save power by using one of these methods.

- Method 1: Power up all power supplies and set the test\_powerdown\_ssp and test\_powerdown\_hsp signals to 1'b1 to place the PHY into its lowest power state. It is also recommended that you assert phy\_reset to 1'b1.
- Method 2: Tie all power supplies to ground.
- Method 3: Float all power supplies.



For methods 2 and 3, you must use isolation gates on PHY outputs to ensure that the outputs do not float high and drive any active logic.

### 5.1.1.4 Power-Down Requirements

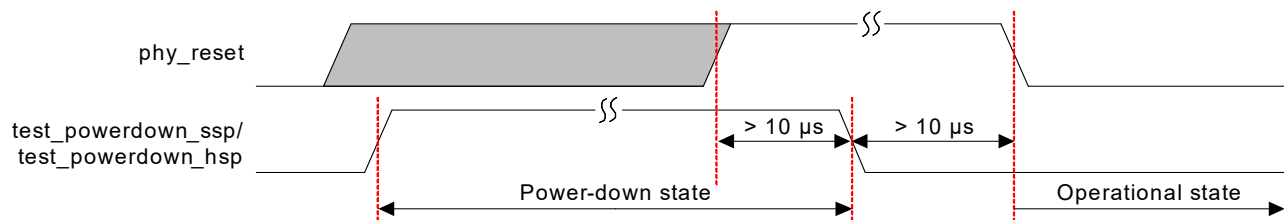
You can place the USB 3.0 femtoPHY into its lowest power state by asserting the `test_powerdown_ssp` and `test_powerdown_hsp` inputs. For certain applications, the SS function can be placed in its lowest power state while enabling the HS function to continue to operate. Similarly, the HS function can be placed in its lowest power state while enabling the SS function to continue to operate. If `test_powerdown_hsp` and `test_powerdown_ssp` are asserted to 1'b1 for IDDQ testing or for obtaining a low-power, non-functional state, then it is also recommended that you assert `phy_reset` to 1'b1.

When either one or both of the `test_powerdown_hsp` or `test_powerdown_ssp` signals are at 1'b1, these signals must not be de-asserted to 1'b0 unless the `phy_reset` signal is at 1'b1. In other words, if the PHY is configured with any of the two functions (HS or SS) powered down, then the function(s) can only be restored during a POR.

Use the following timing (as shown in [Figure 5-2](#)) when de-asserting `test_powerdown_hsp` or `test_powerdown_ssp` to 1'b0, in addition to the POR timing in “[Power-On Reset](#)” on page 166.

- Assert `phy_reset` to 1'b1 (if not already asserted) at least 10  $\mu$ s before de-asserting `test_powerdown_hsp` or `test_powerdown_ssp` to 1'b0.
- After `test_powerdown_hsp` or `test_powerdown_ssp` are de-asserted to 1'b0, wait at least another 10  $\mu$ s before de-asserting `phy_reset` to 1'b0.

**Figure 5-2 PHY Power-Down Sequence**



5.1.2 Suspend and Sleep Operations: High-Speed Function

This section describes the HS function’s suspend and sleep operations.

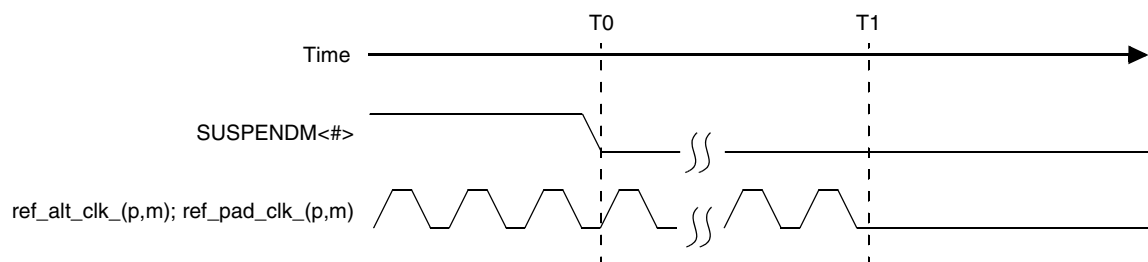
5.1.2.1 Suspend Operation

To place the HS function in Suspend mode, set the SUSPENDM<#> signal to 1'b0.

To ensure that all power-down signals are set, keep the reference clock active for a minimum of 2.5  $\mu$ s after setting SUSPENDM<#> to 1'b0.

The following figure shows the HS function’s timing behavior for suspend operation.

Figure 5-3 Suspend Operation



The following table describes the timing values for suspend operation.

Table 5-2 Suspend Operation

| Timing Parameter | Description                                                   | Value                 |
|------------------|---------------------------------------------------------------|-----------------------|
| T0               | The SUSPENDM<#> signal is set to 1'b0.                        | 0 (reference)         |
| T1               | T1 indicates when it is safe to turn off the reference clock. | 2.5 $\mu$ s (minimum) |

5.1.2.2 Sleep Operation

To place the HS function in Sleep mode, set the SLEEPM<#> signal to 1'b0. If the HS function is in Suspend mode and SLEEPM<#> is asserted or de-asserted, the HS function remains in Suspend mode. While the HS function is in Sleep mode, the input reference clock must remain active.

### 5.1.3 Resume Operation: High-Speed Function

This section describes resume operation from Suspend mode and Sleep mode.



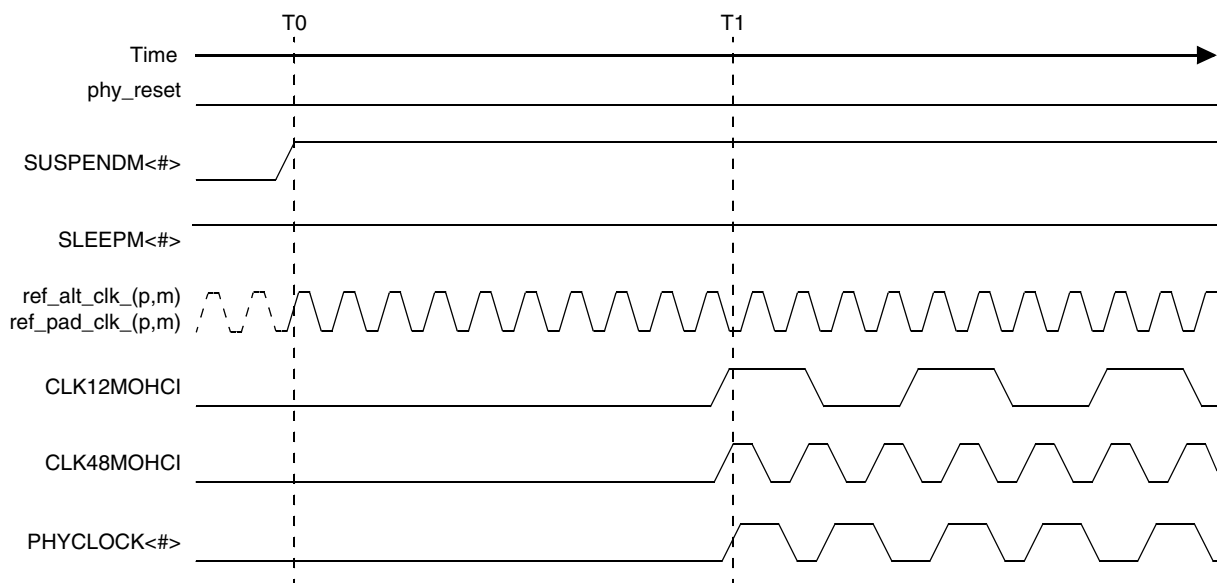
#### Note

To enable the HS function to resume, it must first be placed in either Suspend mode (set SUSPENDM<#> to 1'b0) or Sleep mode (set SLEEPM<#> to 1'b0).

#### 5.1.3.1 Resume Operation From Suspend Mode

To exit Suspend mode, set SUSPENDM<#> to 1'b1. The following figure shows the HS function's timing behavior for resume operation from Suspend mode.

**Figure 5-4 Resume Operation From Suspend Mode**



The following table describes the timing values for resume operation from Suspend mode.

**Table 5-3 Resume Operation From Suspend Mode**

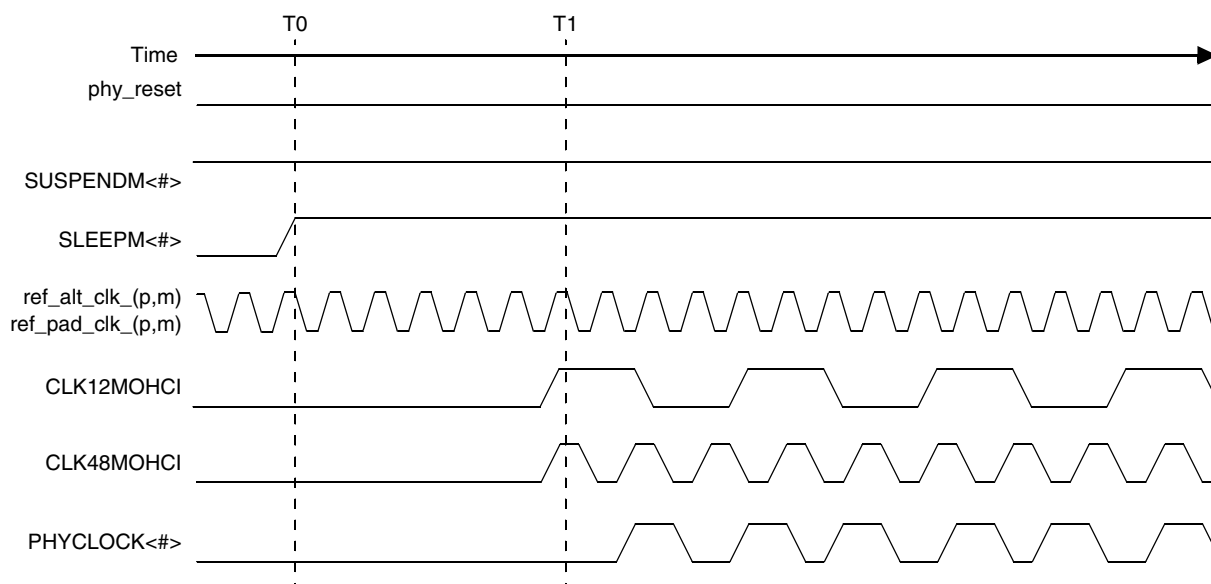
| Timing Parameter | Description                                                                                                                                                                                                                                               | Value                                                                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| T0               | The SUSPENDM<#> signal is set to 1'b1.                                                                                                                                                                                                                    | 0 (reference)                                                                          |
| T1               | T1 indicates when PHYCLOCK<#>, CLK12MOHCI, and CLK48MOHCI are available at the macro output, based on the reference clock source. If SLEEPM<#> is 1'b1, the HS function continues to resume. If SLEEPM<#> is 1'b0, the HS function remains in Sleep mode. | If COMMONONN = 1'b0: T1 < T0 + 16 $\mu$ s<br>If COMMONONN = 1'b1: T1 < T0 + 75 $\mu$ s |

### 5.1.3.2 Resume Operation From Sleep Mode

To exit Sleep mode, set SLEEPM<#> to 1'b1. However, if the HS function is currently in Suspend mode, the HS function remains in Suspend mode even if SLEEPM<#> is de-asserted.

The following figure shows the HS function's timing behavior for resume operation from Sleep mode.

**Figure 5-5 Resume Operation From Sleep Mode**



The following table describes the timing values for resume operation from Sleep mode.

**Table 5-4 Resume Operation From Sleep Mode**

| Timing Parameter | Description                                                                                         | Value                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| T0               | The SLEEPM<#> signal is set to 1'b1.                                                                | 0 (reference)                                                                         |
| T1               | T1 indicates when CLK12MOHCI is available at the macro output, based on the reference clock source. | If COMMONONN = 1'b0: T1 < T0 + 5 $\mu$ s<br>If COMMONONN = 1'b1: T1 < T0 + 75 $\mu$ s |

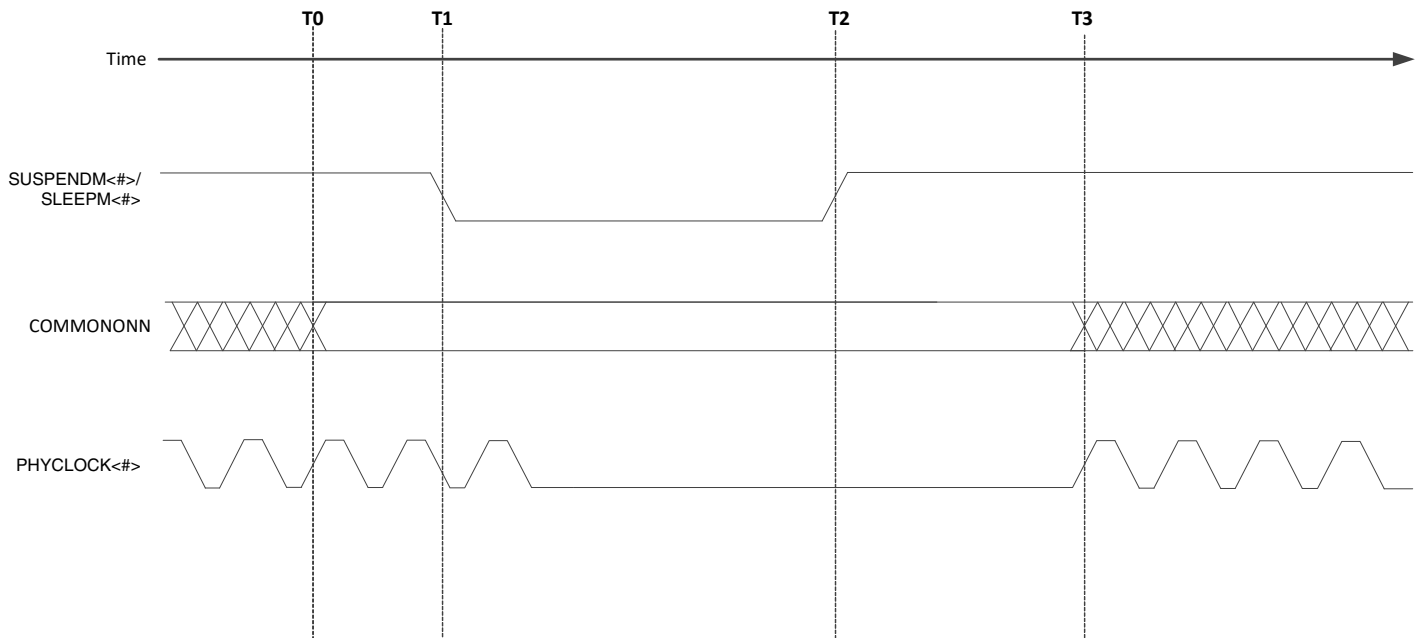
In [Figure 5-5](#) and [Table 5-4](#), the reference clock can be an external board clock or on-chip clock. The reference clock is toggling prior to or at T0 if the resume occurs after exiting Sleep mode. During Sleep mode, the reference clock runs continuously.

### 5.1.4 COMMONONN Control for Suspend and Sleep Modes

The COMMONONN signal controls the power-down signals to many of the analog blocks in the PHY during the PHY's Suspend and Sleep modes. Setting COMMONONN to 1'b0 keeps the analog blocks powered and the clocks (with the exception of PHYCLOCK, which is gated by SUSPENDM<#>/SLEEPM<#>) toggling during Suspend and Sleep modes. Setting COMMONONN to 1'b1 causes the analog blocks to power down and the clocks to be stopped during Suspend and Sleep modes.

The time required for PHYCLOCK to restart toggling after exiting Suspend and Sleep modes depends on the state of the COMMONONN signal and the reference clock being used. In order for the PHY to correctly sample COMMONONN, this signal must be stable for at least six 12-MHz clock cycles prior to asserting the SUSPENDM<#> and SLEEPM<#> signals. The following figure shows the timing requirements for COMMONONN when entering and exiting Suspend or Sleep mode.

**Figure 5-6 Timing Requirements for COMMONONN When Entering and Exiting Suspend or Sleep Mode**





The following table describes the timing requirements for COMMONONN when entering and exiting Suspend or Sleep mode.

**Table 5-5 Timing Requirements for COMMONONN When Entering and Exiting Suspend or Sleep Mode**

| Timing Parameter | Description                                                                                                                                                                                                                                                | Value                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| T0               | COMMONONN is set to the target value and must remain stable during Suspend or Sleep mode.                                                                                                                                                                  | 0 (reference 1)                                                                                    |
| T1               | SUSPENDM<#>/SLEEPM<#> is set to 1'b0.                                                                                                                                                                                                                      | $T1 \geq T0 + 500 \text{ ns}$                                                                      |
| T2               | SUSPENDM<#>/SLEEPM<#> is set to 1'b1.                                                                                                                                                                                                                      | 0 (reference 2)                                                                                    |
| T3               | Indicates when PHYCLOCK and the remaining output clocks are available at the macro output, based on the USB 3.0 femtoPHY reference clock source.<br>T3 also indicates when COMMONONN can safely be toggled for subsequent low-power Suspend or Sleep mode. | If COMMONONN = 1'b0: $T3 < T2 + 16 \mu\text{s}$<br>If COMMONONN = 1'b1: $T3 < T2 + 75 \mu\text{s}$ |

#### 5.1.4.1 High-Speed Transmit-to-Receive Interpacket Delay

The UTMI specification version 1.05, section 6.4.1 indicates that all UTMI-based PHYs must have a minimum receive-to-transmit (RX-to-TX) HS turnaround time (interpacket delay) of 38 bit-times. Based on this requirement, the USB 3.0 femtoPHY supports a minimum TX-to-RX turnaround time of 38 bit-times, as shown in the following table. Synopsys has not received reports of PHYs in the field that are non-UTMI based nor that provide an RX-to-TX turnaround time below 38 bit-times. The maximum turnaround time of 192 bit-times is a property of the PHY and USB 3.0 controller combined.

**Table 5-6 High-Speed Transmit-to-Receive Interpacket Delay**

| Parameter                     | Minimum | Maximum | Unit         |
|-------------------------------|---------|---------|--------------|
| HS TX-to-RX interpacket delay | 38      | 192     | HS bit-times |

When simulating the PHY in an HS mode USB link, set the VIP or test bench model to provide the UTMI minimum RX-to-TX turnaround time of 38 HS bit-times. If the TX-to-RX turnaround time is less than 38 bit times, the Synopsys USB 3.0 femtoPHY is not expected to receive packets reliably.

## 5.1.5 High-Speed Function Speed Modes

The following table lists the speed modes that the HS function supports, the applicable signal settings, and the corresponding PHYCLOCK<#> frequencies.



### Note

- For more information about operating the HS function in a non-OTG, UTMI+ Level 3 speed mode, see ["Operating the High-Speed Function in Non-OTG, UTMI+ Level 3 Speed Modes"](#) on page 175.
- The modes of operation indicated in [Table 5-7](#) indicate speed modes for normal data transfers only; these modes do not reflect the settings for high-speed handshakes.

**Table 5-7 High-Speed Function Speed Modes**

| Mode                     | DPPULLDOWN<#> | DMPULLDOWN<#> | TERMSEL<#> | XCVRSEL<#>[1:0] | OPMODE<#>[1:0] | WORDINTERFACE<#> | PHYCLOCK<#> |
|--------------------------|---------------|---------------|------------|-----------------|----------------|------------------|-------------|
| <b>Peripheral</b>        |               |               |            |                 |                |                  |             |
| HS OTG Peripheral        | 0             | 1             | 0          | 00              | 00/11          | 0/1              | 60/30 MHz   |
| FS OTG Peripheral        | 0             | 1             | 1          | 01              | 00             | 0/1              | 60/30 MHz   |
| FS-Serial OTG Peripheral | 0             | 1             | 1          | 01              | 00/10          | 0/1              | 60/30 MHz   |
| HS Peripheral            | 0             | 0             | 0          | 00              | 00/11          | 0/1              | 60/30 MHz   |
| FS Peripheral            | 0             | 0             | 1          | 01              | 00             | 0/1              | 60/30 MHz   |
| FS-Serial Peripheral     | 0             | 0             | 1          | 01              | 00/10          | 0/1              | 60/30 MHz   |
| <b>Host</b>              |               |               |            |                 |                |                  |             |
| HS OTG Host              | 1             | 1             | 0          | 00              | 00/11          | 0/1              | 60/30 MHz   |
| FS OTG Host              | 1             | 1             | 1          | 01              | 00             | 0/1              | 60/30 MHz   |
| LS OTG Host              | 1             | 1             | 1          | 10              | 00             | 0/1              | 60/30 MHz   |
| FS-Serial OTG Host       | 1             | 1             | 1          | 01              | 00/10          | 0/1              | 60/30 MHz   |
| LS-Serial OTG Host       | 1             | 1             | 1          | 10              | 00/10          | 0/1              | 60/30 MHz   |
| HS Host                  | 1             | 1             | 0          | 00              | 00/11          | 0/1              | 60/30 MHz   |
| FS Host                  | 1             | 1             | 1          | 01              | 00             | 0/1              | 60/30 MHz   |
| LS Host                  | 1             | 1             | 1          | 10              | 00             | 0/1              | 60/30 MHz   |

**Table 5-7 High-Speed Function Speed Modes (Continued)**

| Mode             | DPPULLDOWN<#> | DMPULLDOWN<#> | TERMSEL<#> | XCVRSEL<#>[1:0] | OPMODE<#>[1:0] | WORDINTERFACE<#> | PHYCLOCK<#> |
|------------------|---------------|---------------|------------|-----------------|----------------|------------------|-------------|
| LS Host Preamble | 1             | 1             | 1          | 11              | 00             | 0/1              | 60/30 MHz   |
| FS-Serial Host   | 1             | 1             | 1          | 01              | 00/10          | 0/1              | 60/30 MHz   |
| LS-Serial Host   | 1             | 1             | 1          | 10              | 00/10          | 0/1              | 60/30 MHz   |

### 5.1.6 Operating the High-Speed Function in Non-OTG, UTMI+ Level 3 Speed Modes

To operate the HS function in a non-OTG, UTMI+ Level 3 speed mode, as described in [Table 5-7](#):

1. Leave the following top-level pins disconnected:  
ID<#>, IDDIG0, AVALID<#>, BVALID<#>, VBUSVALID0.
2. Connect the following top-level inputs to the digital ground supply (gd):  
IDPULLUP0, and DRVVBUS0.
3. Power down the OTG block (set the OTGDISABLE0 signal to 1'b1).

#### Note

When operating in a non-OTG, UTMI+ speed mode, the OTG block's inputs and outputs are ignored. Powering down the OTG block is an optional step; however, not performing this step results in added power consumption, because the OTG block remains powered up.

### 5.1.6.1 Driving the VBUSVLDEXT<#> and VBUSVLDEXTSEL0 Input Signals

When operating in a non-OTG, UTMI+ speed mode, you are constrained to using the VBUSVLDEXT<#> signal to indicate when the VBUS signal on the USB cable is valid.

In Device mode, VBUSVLDEXT<#> and VBUSVLDEXTSEL0 enable the pull-up resistor on DP. When operating the HS function in Non-OTG Host mode or OTG mode, VBUSVLDEXT<#> and VBUSVLDEXTSEL0 can be set to 1'b0 and 1'b0, respectively. Three methods are available to drive the HS function's VBUSVLDEXT<#> and VBUSVLDEXTSEL0 input signals as follows.

#### External Comparator Method

Set VBUSVLDEXTSEL0 to 1'b1. Sample the voltage on VBUS using a comparator — external to the HS function — whose threshold is set to a predefined value. Subject the comparator's output to a level-shift down and provide the output as the VBUSVLDEXT<#> input to the HS function. In this case, the VBUSVLDEXT<#> input acts as a qualifier signal to enable the pull-up resistor on DP.

#### Internal Comparator Method

Set VBUSVLDEXTSEL0 to 1'b0. VBUSVLDEXT<#> can be set to either 1'b0 or 1'b1. In this case, VBUS is connected to the HS function's VBUS pin. The voltage on VBUS is then sampled by the HS function's internal comparator. The internal comparator samples the voltage on VBUS based on a predefined threshold, and the output is used as a qualifier signal to enable the pull-up resistor on DP.

#### External Comparator/Programmable Register Method

Set VBUSVLDEXTSEL0 to 1'b1. Make the HS function's VBUSVLDEXT<#> input register-programmable, enabling the VBUSVLDEXT<#> input to be set to the appropriate value when needed. In this case, the value observed on the HS function's VBUSVLDEXT<#> input is used as a qualifier signal to enable the pull-up resistor on DP. This method enables you to control when the pull-up resistor on DP is enabled.

## 5.1.7 Serial Interface Timing Requirements

A USB 1.1 host controller's interface exercises the TXENABLEN<#>, FSDATAEXT<#>, and FSSE0EXT<#> pins. The relative input skew between these inputs affects the output signaling on the USB. If the external skew between these signals is too large, illegal signaling might be produced on the USB.

To avoid illegal signaling on the USB, you must determine how much relative skew between these input signals is acceptable.

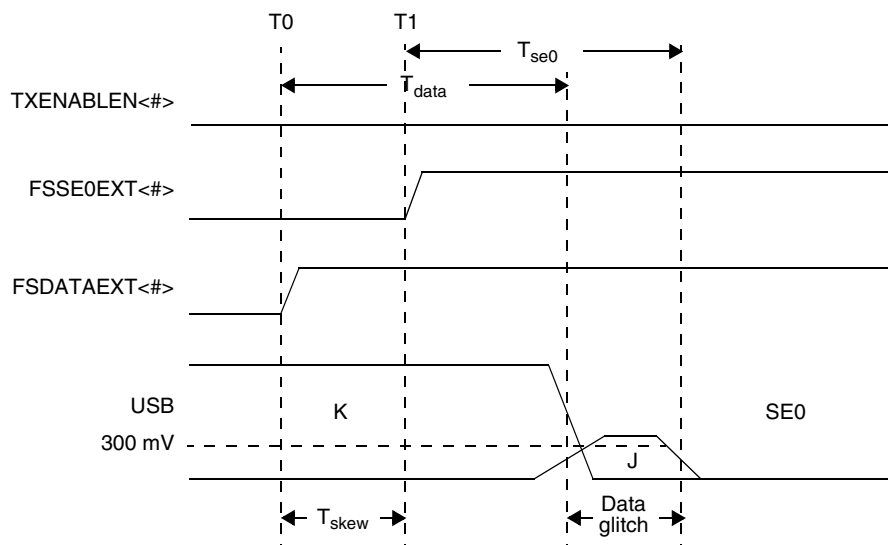
You must account for three relative skews across operating conditions as follows:

- **Skew 1:** The skew between FSSE0EXT<#> and FSDATAEXT<#> when FSSE0EXT<#> transitions from 1'b0 to 1'b1 and FSDATAEXT<#> also changes state
- **Skew 2:** The skew between FSSE0EXT<#> and FSDATAEXT<#> when FSSE0EXT<#> transitions from 1'b1 to 1'b0 and FSDATAEXT<#> transitions from 1'b0 to 1'b1
- **Skew 3:** The skew between TXENABLEN<#> and FSDATAEXT<#> when TXENABLEN<#> transitions from 1'b1 to 1'b0 and FSDATAEXT<#> transitions from 1'b1 to 1'b0

Skew 1 and Skew 2 are critical in the EOP phase of an FS/LS packet. If these skews are not considered, the transitions into or out of an EOP in FS or LS signaling can cause have illegal USB states immediately before or after the EOP.

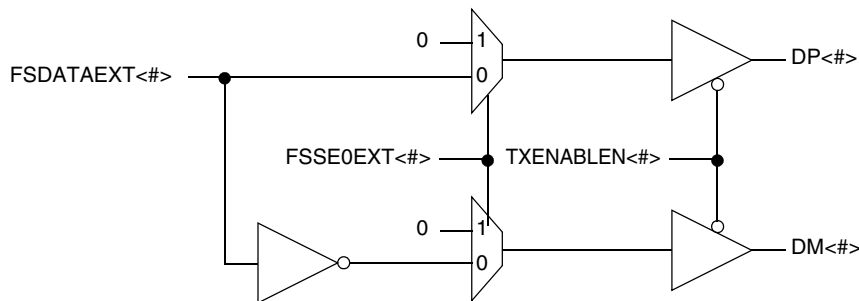
A USB data glitch occurs when signaling causes one of the USB lines to rise above 300 mV while the other line remains low, then to fall below 300 mV while the other line remains low. The following figure shows a USB data glitch.

**Figure 5-7 USB Data Glitch**



The following figure shows a functional diagram of the serial interface transmit signals.

**Figure 5-8 Functional Diagram of Serial Interface Transmit Signals**



The TXENABLEN<#> signal enables and disables the USB 1.1 transmit paths. If TXENABLEN<#> is set to 1'b0, the USB 1.1 transmit paths are enabled. If TXENABLEN<#> is set to 1'b1, the USB 1.1 transmit paths are disabled.

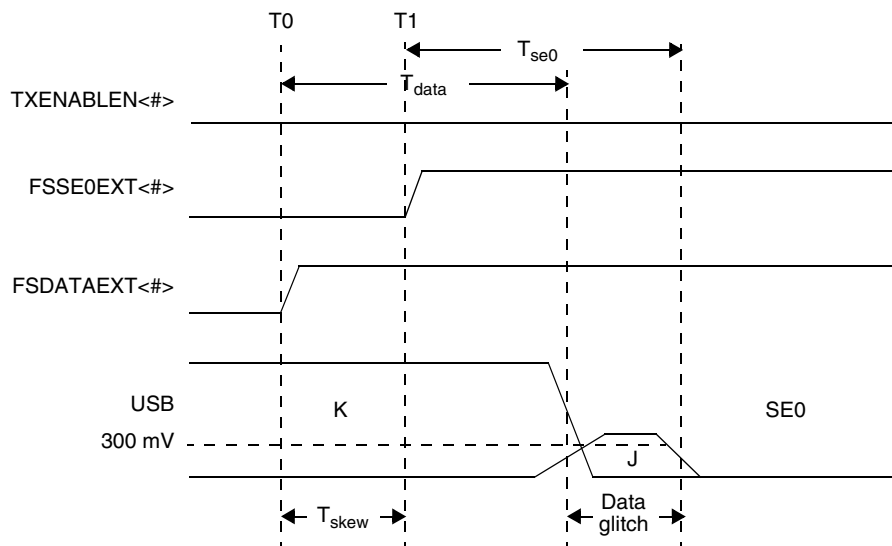
The FSSE0EXT<#> signal enables and disables SE0 generation on the USB only if TXENABLEN<#> is set to 1'b0. If TXENABLEN<#> is set to 1'b1, the value of FSSE0EXT<#> is ignored. Setting FSSE0EXT<#> to 1'b1 triggers the assertion of an SE0 onto the USB. Setting FSSE0EXT<#> to 1'b0 gives signaling control to FSDATAEXT<#>.

If FSSE0EXT<#> is set to 1'b0 and TXENABLEN<#> is set to 1'b0, FSDATAEXT<#> generates either a J state or K state onto the USB. If TXENABLEN<#> is set to 1'b1 or FSSE0EXT<#> is set to 1'b1, the value of FSDATAEXT<#> is ignored.

### 5.1.7.1 Skew 1 Description

Skew 1 relates to the transition from data to SE0 when generating an EOP in FS or LS signaling. This transition consists of FSSE0EXT<#> transitioning from 1'b0 to 1'b1. As shown in the following figure, if FSDATAEXT<#> also transitions near the FSSE0EXT<#> transition, a data glitch might occur on the USB before the SE0.

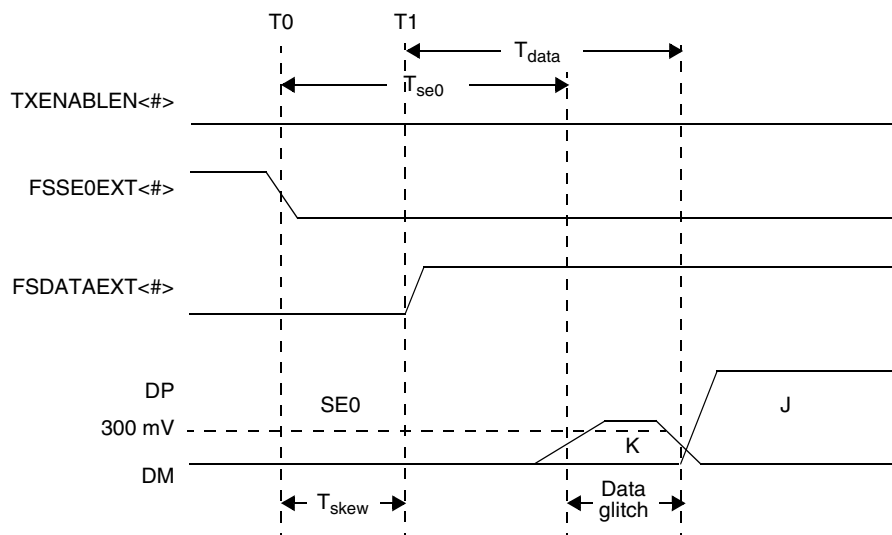
**Figure 5-9 Skew 1 Test Case**



### 5.1.7.2 Skew 2 Description

Skew 2 relates to the transition from the SE0 to J state at the end of the EOP interval in an FS/LS packet transmission. This transition consists of FSSE0EXT<#> transitioning from 1'b1 to 1'b0. As shown in the following figure, if FSDATAEXT<#> also transitions near the FSSE0EXT<#> transition, a data glitch might occur on the USB after the SE0.

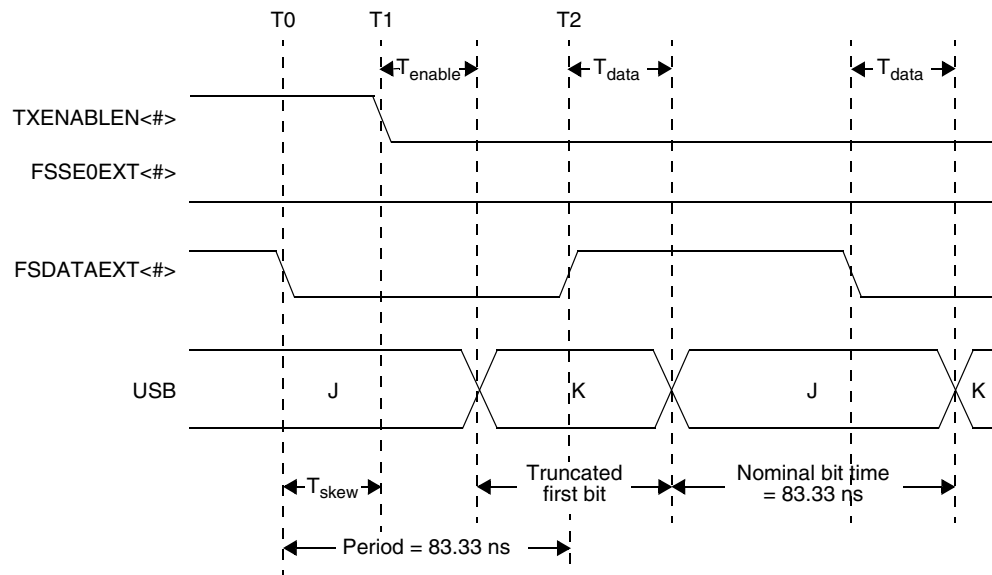
**Figure 5-10 Skew 2 Test Case**



### 5.1.7.3 Skew 3 Description

Skew 3 relates to the transition from the USB Idle state to transmitting USB data. This transition consists of TXENABLEN<#> transitioning from 1'b1 to 1'b0. As shown in the following figure, if FSSE0EXT<#> also transitions near the TXENABLEN<#> transition, the first bit of the SYNC pattern might be truncated. This truncation can be significant when transmitting through a five-tier USB topology.

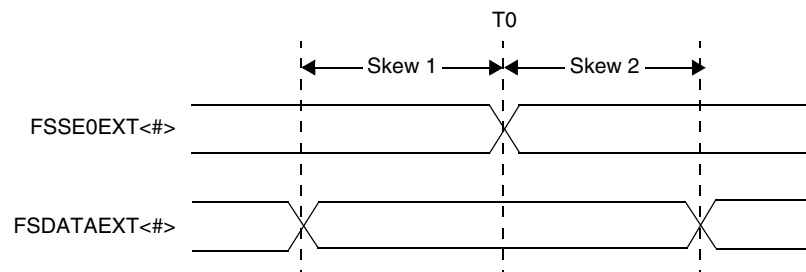
**Figure 5-11 Skew 3 Test Case**



### 5.1.7.4 Skew Timing Values

Skew 1 and Skew 2 can be combined to generate a skew range between FSSE0EXT<#> and FSDATAEXT<#>, so if the input skew between FSSE0EXT<#> and FSDATAEXT<#> is held within this range, no data glitches occur around the SE0. The following figure shows the skew range between FSSE0EXT<#> and FSDATAEXT<#>.

**Figure 5-12 Skew Range Between FSSE0EXT<#> and FSDATAEXT<#>**



Therefore, Skew 1 = 1.5 ns and Skew 2 = 1.5 ns. To avoid truncating the first bit of SYNC, ensure that Skew 3 is  $\leq 4$  ns.



## 5.2 Test Environment Overview

This section describes the USB 3.0 femtoPHY test environment.

### 5.2.1 Verilog Function Limitations

The Verilog model simulates using a four-state (0, 1, X, Z) simulator. Therefore, signals with real-world voltage and current values over time, such as those that exist on the high-speed serial data line, cannot be fully simulated with this four-state methodology.

These simulator limitations include:

- Serial output signal transmit level, pre-emphasis, and edge rate in the transmitter
- Serial input receiver equalization
- Receive and transmit I/O terminations
- Termination resistor tuning and measurement
- LOS voltage-level setting
- AC signal characteristics of the AC JTAG receiver
- Voltage margining and RX eye capture

These functions are internally verified using extensions to the four-state simulator methodology, but these functions cannot be supported in standard deliverables.

### 5.2.2 Serial Line Modeling

This section describes serial line modeling on the SuperSpeed and USB 2.0 high-speed interfaces.

#### 5.2.2.1 Serial Line Modeling: SuperSpeed Interface

The tx<#>\_m and tx<#>\_p serial lines are fundamentally analog signals. Because each wire is limited to the simulator's four states, not all aspects of the analog signals can be reflected. The following table provides encodings used to represent the analog nature of the outputs.

**Table 5-8 Differential Signal Handling in Verilog Simulation**

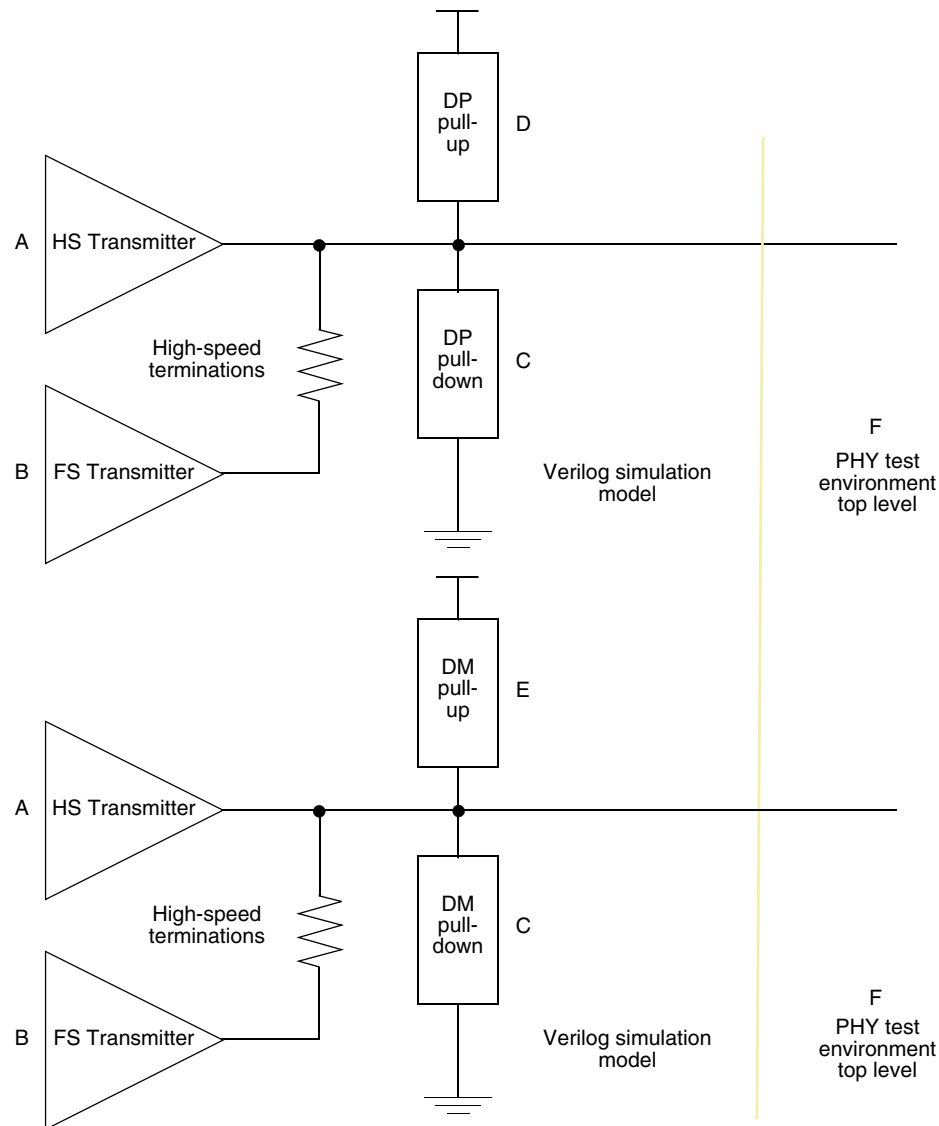
| State                              | tx<#>_p | tx<#>_m |
|------------------------------------|---------|---------|
| Differential 1                     | 1       | 0       |
| Differential 0                     | 0       | 1       |
| No signal, holding common mode     | 0       | 0       |
| No signal, not holding common mode | Z       | Z       |

The analog receiver model interprets only Differential 0 or Differential 1 encodings as valid signal levels. All other encodings result in unknown data and cause the receiver to detect a loss of signal.

### 5.2.2.2 Serial Line Modeling: USB 2.0 High-Speed Interface

The following figure shows the drive-strength properties of the DP and DM signals in the USB 3.0 femtoPHY behavioral models and test environment. When simulating with third-party BFM, pay particular attention to these drive strengths, because mismatched strengths between models that drive D+ and D- cause compatibility issues during simulations.

**Figure 5-13 Drive-Strength Properties of DP and DM**



**Drive-Strength Properties:**

- A:** HS Transmitter strength for DP/DM is "supply."
- B:** FS Transmitter strength for DP/DM is "strong."
- C:** Pull-down strength for DP/DM is "weak."
- D:** Pull-up strength for DP is "pull" (for FS operation).
- E:** Pull-up strength for DM is "pull" (for LS operation). This drive strength applies only when the PHY is connected to a host in the PHY test environment, not when the PHY is used in a higher-level test environment.
- F:** Pull-down strength for D+/D- is "weak." D+/D- are declared as wired ORs (wor).

The top-level test bench file, `top.v` (in the `simulation/system/` directory), declares the D+/D- lines as wired ORs. In any test bench that instantiates the USB 3.0 femtoPHY, you must declare the D+/D- lines as wired ORs with no strength modeling. This declaration type prevents bus contention if other models (for example, the USB device controller behavioral model or USB host BFM) are connected to the USB 3.0 femtoPHY.

### 5.2.3 Language Compatibility

Simulating the `dwc_usb3_sspN_hspN_ns_pipe_sspN_hspN` requires a Verilog-2001 compatible simulator. The source code for the `dwc_usb3_sspN_hspN_ns` and the `pcs_xN` uses ANSI-style interface declarations as well as `localparam` declarations, both of which were added to the Verilog-2001 standard.

### 5.2.4 Timescales

The USB 3.0 specification permits a maximum frequency variation of +600 through -600 ppm from nominal values. The simulation models support these frequency differences by modeling the clock and data recovery circuitry with sufficient accuracy to support reference clock frequencies and incoming serial datastream rates that differ from the nominal values. To support these frequency offsets, the simulation timescale for the PHY models is preset as:

```
`timescale 1ns / 10fs
```

Using this value during simulation enables the `pcs_xN` to manage the depth of the elasticity buffer in the receiver datapath—provided the appropriate number of SKP ordered sets are supplied by the test bench. If unexpected overflow or underflow conditions are encountered during simulation, verify that: the timescale is set correctly; the frequency differential is not greater than the amount permitted by the USB 3.0 base specification; sufficient SKP ordered sets are provided in the serial datastream to the receiver.

### 5.2.5 Simulator Options Not Supported

Due to the inclusion of analog behavioral modeling in the PHY, several simulator options cannot be supported.

The following simulation options are not supported in VCS (other simulators might use alternate option names):

- **\* +delay\_mode\_zero:** Changes all #delays to zero and kills the analog modeling in the PHY
- **\* +delay\_mode\_unit:** Changes all #delays to #1 and kills the analog modeling in the PHY
- **\* +nospecify (relevant to only gate-level simulations):** Cannot be used for any at-speed simulations and therefore, can be used for only SCAN stuck-at fault testing. This limitation prevents SDF back-annotation, which is required in any digital, gate-level simulation.

## 5.3 Verilog Model Files and Simulators

The following table describes each portion of the Verilog model and indicates the database deliverables directory where the portion is stored.

**Table 5-9 Verilog Model Directory Structure**

| Directory                             | Description                                                                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| macro/Latest//rtl/                    | Logic for the PCS section                                                                                                                                                                          |
| macro/Latest/sim/                     | Contains Bourne shell scripts for running simulations                                                                                                                                              |
| macro/Latest//testbench/              | <ul style="list-style-type: none"> <li>■ Test bench and tests for the Verilog model</li> <li>■ *.dat: Contains data files for simulations</li> <li>■ *pads: Contains pad's analog model</li> </ul> |
| macro/Latest//testbench/include/      | Contains macros included in the simulation compile and/or run stages                                                                                                                               |
| pads_<wb/fc>/Latest//behavior/        | Behavioral Verilog for the USB 3.0 Pads                                                                                                                                                            |
| phy/Latest//behavior/                 | The top-level module resides in dwc_usb3_sspaN_hspaN_ns_gtech.v, and the gate-level netlist resides in dwc_usb3_sspaN_hspaN_ns_gate_level.v. Logic for the dwc_usb3_sspaN_hspaN_ns hard macro.     |
| phy/Latest//timing/<metal stack>/sdf/ | Contains the SDF files (Standard Delay Format) used for gate simulations. Both Worst Case (WC) and Best Case (BC) SDF files are provided.                                                          |

The Verilog model has been generated for simulation and exercised on the Synopsys VCS simulator.

Simulation scripts for the USB 3.0 femtoPHY are provided for the simulator in the macro/<version>/sim/ directory. The gtech\_sim scripts include the behavioral code for the hard macro. The gates\_{sim} scripts include the behavioral code for the analog portion and gate-level code for the synthesized portions of the PHY. In general, the scripts simply add the appropriate files to the simulator command line. The top level of the Active or Pad section is hard-coded in the scripts, because the test bench supports only a single lane configuration.

“{sim}” refers to vcs for the Synopsys VCS simulator.

### 5.3.1 Gate-Level Models

For gate-level simulations, the ANI\_SIM\_GATES and ANI\_STDCELL\_\* macros must be defined. For examples of these macros, refer to the gates\_{sim} scripts in the macro/<version>/sim/ directory.

Consider the following issues when running gate-level simulations:

- Due to non-trivial clocking structures within the dwc\_usb3\_sspaN\_hspaN\_ns PHY, a zero-time or unit-delay simulation of the gate-level model would be non-functional due to race conditions in the Verilog execution caused by simulator event ordering. Therefore, SDF back-annotation of the gate delays is required. The .sdf files are provided in the phy/<version>/timing/<metal stack>/sdf/ directory and are automatically loaded at simulation startup.
- The .sdf files simply represent the hardware's timing. These files are generated at worst-case and best-case process/temperature/voltage corner. However, they are not sufficient for replacing static timing. Synopsys has validated the dwc\_usb3\_sspaN\_hspaN\_ns PHY's internal timing, and timing for the dwc\_usb3\_sspaN\_hspaN\_ns PHY interfaces is provided in the .lib files. Validate these timings using STA tools, not Verilog simulation.
- For proper operation of the gate-level model with respect to standard cell library timing checks, the following switch is required:

```
+notimingcheck
```

This switch is required due to invalid timing checks (caused by multiple clock domains, synchronizers, and so on) and the standard cell library's behavior of inserting X's into the simulation when these checks fail.

- While running gate-level simulations with the PCS in RTL, PIPE\_WIDTH must be set to 2 in the top-level test bench. This restriction does not apply in the case of gate-level simulations with synthesized RTL (with clock tree inserted).

### 5.3.2 SDF Annotation Warnings

Each simulator loads the .sdf files in a different manner. Therefore, warnings during SDF annotation vary between simulators. These messages are innocuous and can be safely ignored. However, there must be no errors during SDF annotation. The following sections contain example warnings from each supported simulator.

#### 5.3.2.1 VCS Warnings

The warnings from VCS SDF annotation are as follows. Unfortunately, VCS does not have a mechanism to suppress these warnings.

```
{path}/dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN.sdf:11382, SDF Warning: Timingchecks removed at compile-time, statement ignored
```

This warning is due to the +notimingcheck switch and is expected.

### 5.3.2.2 Test Warnings

At time zero in the simulation, the assertions for mutual exclusion fire due to the wires transitioning from “X” to their reset values, causing a warning to appear. You can ignore the warning message.

```
0 test.tb.PIPE_DUT.phy.ana.lane0.tx.p2s.serial.store_a.mux0_a Info: strong mutex ignored
```

### 5.3.3 Macros Affecting SS Mode Simulation

The following macros affect simulation.

- **‘ANI\_TIMESCALE:** This macro is at the top of each SS function file and is set to 1 ns/10 fs. For the analog modeling to function correctly, the timescale precision must be set to this value. Modern Verilog simulators are event-based; therefore, setting the timescale to this precision does not cause noticeably slower simulation times compared to a coarser precision. Verilog simulators parse timescales while modules are read, so if other logic in the simulation requires a different timescale, the appropriate timescale directive might be set in those files.
- **‘ANI\_SHORT\_RESET:** In the behavioral RTL only, some long reset and power-up/power-down timeouts are shortened by this macro. To run a simulation with a shorter reset startup time, define this macro. This macro does not affect the gate-level Verilog (always uses full timeouts).
- **‘ANI\_SIM\_GATES:** Must be defined when running gate level simulations. Note that ANI\_SHORT\_RESET has no effect if ANI\_SIM\_GATES is defined.
- **‘ANI\_STDCELL\_\***: Must be defined when running gate-level simulations.

## 5.3.4 Simulating With the Test Environment

This section provides information about simulating the USB 3.0 femtoPHY using the test environment.

### 5.3.4.1 Running Example Tests

To verify the integrity and functionality of the SS function's Verilog model, example tests are provided. These tests reside in the macro/<version>/testbench/ directory and illustrate how to drive the various interfaces to perform several basic tasks.

The tests inject data into the transmit and receive paths and verify that the same data emerges from the other side.

To run a test, modify the run scripts to work in your environment. Identify the simulator and model you want to use, and edit the associated run\_\*\_gates/gtech\_{sim} file. To reflect the path to the model and simulator in your environment, you must change several path variables at the beginning of the script. Specifically, you must set the MODEL\_PATH variable to the directory in which the PHY was unpacked.

Ensure that you have the appropriate licensing environment already set up.

To run a test, execute the associated run\_\*\_gates/gtech\_{sim} script.

The test output contains simulator messages that vary between simulators. During test execution, you will receive status messages about the PHY powering up and being enabled. If no errors are reported during and at the end of the test, and the test results indicate "0" errors, the test has passed.

### 5.3.5 Simulation Log Files

Simulation results are stored in the vrun.log file. A passing test contains no strings that include "Error" and have a successful completion message: "Test Completed with zero errors".

### 5.3.6 Test Case Descriptions

The USB 3.0 femtoPHY test environment includes the following test cases:

- **simple\_datapath:** Turns on the PHY and transmits and receives packets on the SuperSpeed and USB 2.0 high-speed interfaces.
- **control\_reg\_access:** Demonstrates how to access the PHY's internal control registers via both the JTAG port and the CR parallel port.
- **simple\_power\_modes:** Loops through a series of power-state transitions in SuperSpeed and High-Speed modes.
- **simple\_acjtag:** Loops through various boundary scan scenarios.
- **simple\_power\_aware:** Exercises top-level power-collapsing scenario.





# 6

## Physical-Level Implementation

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This chapter includes the following sections:

- [“Power and Ground Connections”](#) on page 190
- [“ESD Considerations”](#) on page 192
- [“Floorplanning Considerations”](#) on page 194
- [“Synopsys .lib and .db Files”](#) on page 199
- [“Synthesis”](#) on page 202

## 6.1 Power and Ground Connections

The following table provides the voltage levels for the USB 3.0 femtoPHY's analog and digital power supplies.

**Table 6-1 Analog and Digital Power Supplies**

| Signal                           | Description                                           | Value                                                    |
|----------------------------------|-------------------------------------------------------|----------------------------------------------------------|
| <b>Analog Power Supplies</b>     |                                                       |                                                          |
| DVDD                             | PHY analog and digital high-speed, low-voltage supply | 0.8 V (+ 10%, - 7%) at the macro pins with respect to gd |
| vp                               | PHY analog and digital SuperSpeed, low-voltage supply | 0.8 V (+ 10%, - 7%) at the macro pins with respect to gd |
| vptx<#>                          | PHY transmit supply                                   | 0.8 V (+ 10%, - 7%) at the macro pins with respect to gd |
| <b>High-Voltage I/O Supplies</b> |                                                       |                                                          |
| VDD33<#>                         | High-voltage supply for high-speed operation          | 3.3 V (+ 10%, - 7%) at the macro pins with respect to gd |
| VDDH<#>                          | High-voltage supply for high-speed operation          | 3.3 V (+ 10%, - 7%) at the macro pins with respect to gd |
| vph                              | High-voltage supply for SuperSpeed operation          | 3.3 V (+ 10%, - 7%) at the macro pins with respect to gd |
| <b>PHY Ground</b>                |                                                       |                                                          |
| gd                               | PHY analog and digital ground                         | 0 V                                                      |
| VSSA<#>                          | High-speed analog and digital ground                  | 0 V                                                      |
| <b>Core Ground</b>               |                                                       |                                                          |
| vss                              | Core vss for back-to-back ESD diode connection        | 0 V                                                      |



### Note

- The USB 3.0 femtoPHY can tolerate peak-to-peak AC noise at  $\pm 3\%$  of nominal supply.
- During PHY operation, vp and vptx<#> must be at the same voltage level.

### 6.1.1 Power and Ground Isolation

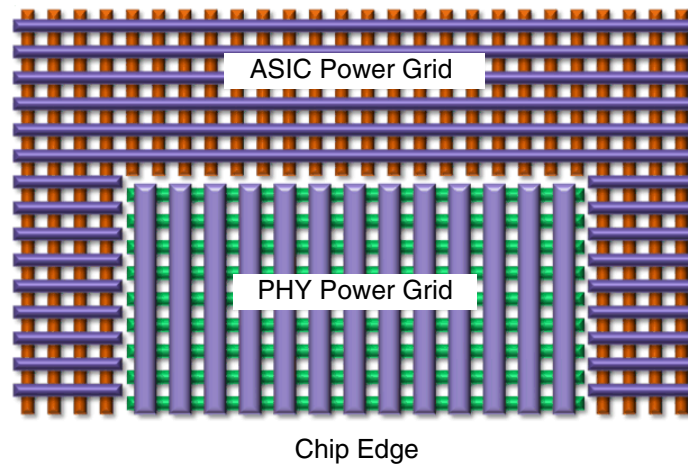
To prevent power supply noise from inducing jitter on the transmitted and received waveforms, the USB 3.0 femtoPHY's power supplies are isolated from the power supplies used on the rest of the ASIC.

The USB 3.0 femtoPHY's ground nodes, `gd` and `VSSA<#>`, are isolated on the die from the core `vss`. Back-to-back diodes between core `vss` and `gd`, core `vss` and `VSSA<#>`, and also `gd` and `VSSA<#>` are recommended and provided within the snap-on pads, but no direct metal-metal connection is permitted.

The USB 3.0 femtoPHY's ground nodes, `gd` and `VSSA<#>` should be connected to the core `vss` in the package. This connection provides a robust, low-impedance ground plane and is required by the PHY's ESD methodology.

Isolation of all power and ground on the die implies that the only connection between the hard macro and the rest of the ASIC is for signal wires. The signal integrity of this interface is maintained using a fully clock-forwarded architecture where both clock and data cross the power domains together, so that subtle differences in the supply voltages have no impact on timing.

**Figure 6-1 Power Grid Isolation**



## 6.2 ESD Considerations

This section describes ESD implementation in the PHY.

### 6.2.1 ESD Requirements

ESD performance is evaluated by the amount of voltage and current that the protection circuits can absorb during an ESD event while keeping voltages below damaging levels. The protection circuits are designed to meet the following protection levels provided in the following table.

**Table 6-2 Minimum ESD Protection Target Levels**

| ESD Category                                 | Minimum Protection Level   | JEDEC Class                      |
|----------------------------------------------|----------------------------|----------------------------------|
| Human Body Model (HBM)<br>(JS-001-2014)      | 2 kV                       | 2                                |
| Charged Device Model (CDM)<br>(JESD22-C101F) | 6 A peak discharge current | C2/C1 (500 V/250 V) <sup>a</sup> |

a. Support for either 500-V or 250-V CDM target level is dependent on maximum discharge current generated in final SoC/package implementation.

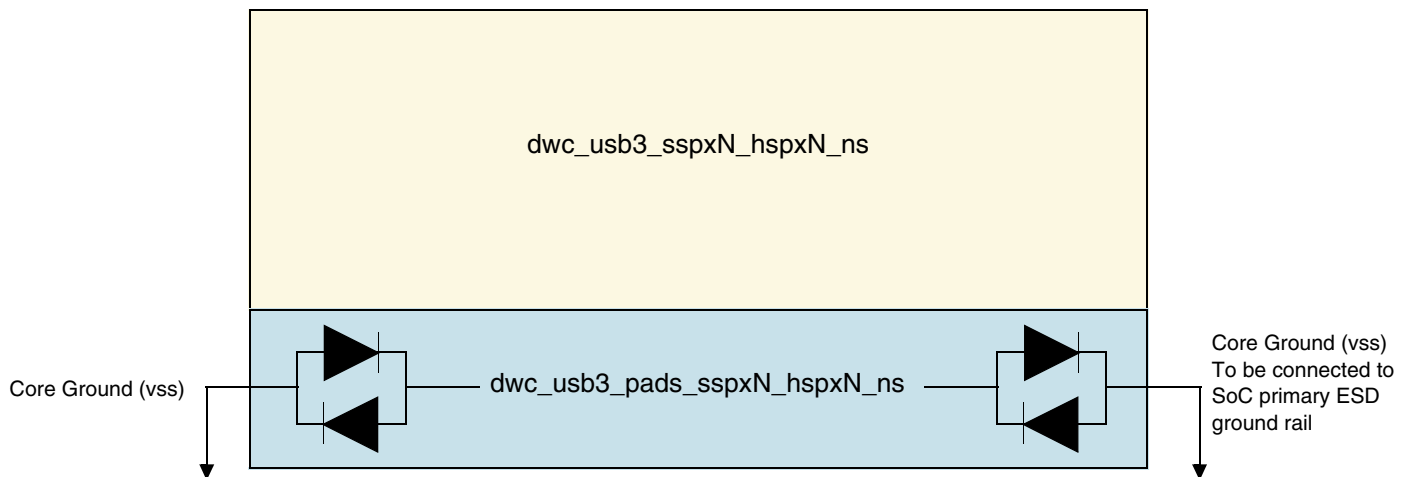
## 6.2.2 ESD Protection Strategy

The USB 3.0 femtoPHY's ESD protection strategy provides the following features:

- Sufficient metal width in all ESD paths
- Primary ESD protection devices for all signal pads and all power supplies in the PHY and analog pad strip
- Secondary ESD protection within the PHY for CDM events including series resistances and secondary diodes
- An active clamp (between power/ground): The clamp transistors are located at the primary diodes, and the clamp and primary diodes are close to each other near the input – minimizing the resistance between them ( $< 0.5 \Omega$ ) and maximizing the effectiveness of the ESD circuits.

The following figure shows the recommended ESD strategy to connect the USB 3.0 femtoPHY on the SoC.

**Figure 6-2 ESD Protection Strategy**



During ESD events, the primary ESD protection directs most of the current to the core power/ground buses and through the clamp. This path has minimal resistance, keeping the signal pad voltage and voltage across the core power buses low. Back-to-back diodes enable current flow between power domains – preventing large differences from developing between power domains. The back-to-back diodes prevent damage between interfacing circuits due to power bus differences.

## 6.2.3 ESD Considerations

The core's ESD structures cannot be relied upon to protect remote chip circuits. The back-to-back diodes soften the local, interfacing power supply differences. However, the core clamp is not effective at protecting distant chip circuits – the core protection circuits are designed to protect core circuits only.

## 6.3 Floorplanning Considerations

This section describes things to consider when floorplanning the USB 3.0 femtoPHY.

### 6.3.1 Physical Sizes

The size of the USB 3.0 femtoPHY is determined by the size of the following subcomponents:

- `dwc_usb3_sspaN_hspaN_ns`: PHY hard macro
- `dwc_usb3_pads_sspaN_hspaN_ns`: Snap-on pad strip

The following table provides block sizes for the hard IP components.

**Table 6-3 Physical Sizes of PHY Components**

| Component                                                   | Size                                                       |
|-------------------------------------------------------------|------------------------------------------------------------|
| <code>dwc_usb3_sspaN_hspaN_ns</code> PHY hard macro         | 764 x 601.92 $\mu\text{m}$ (0.432 $\text{mm}^2$ )          |
| <code>dwc_usb3_pads_sspaN_hspaN_ns</code> snap-on pad strip | Flip-chip: 764 x 150 $\mu\text{m}$ (0.0996 $\text{mm}^2$ ) |

These blocks connect by abutment, but they might overlap. If they overlap, the areas in the preceding table represent the portions that do not overlap.



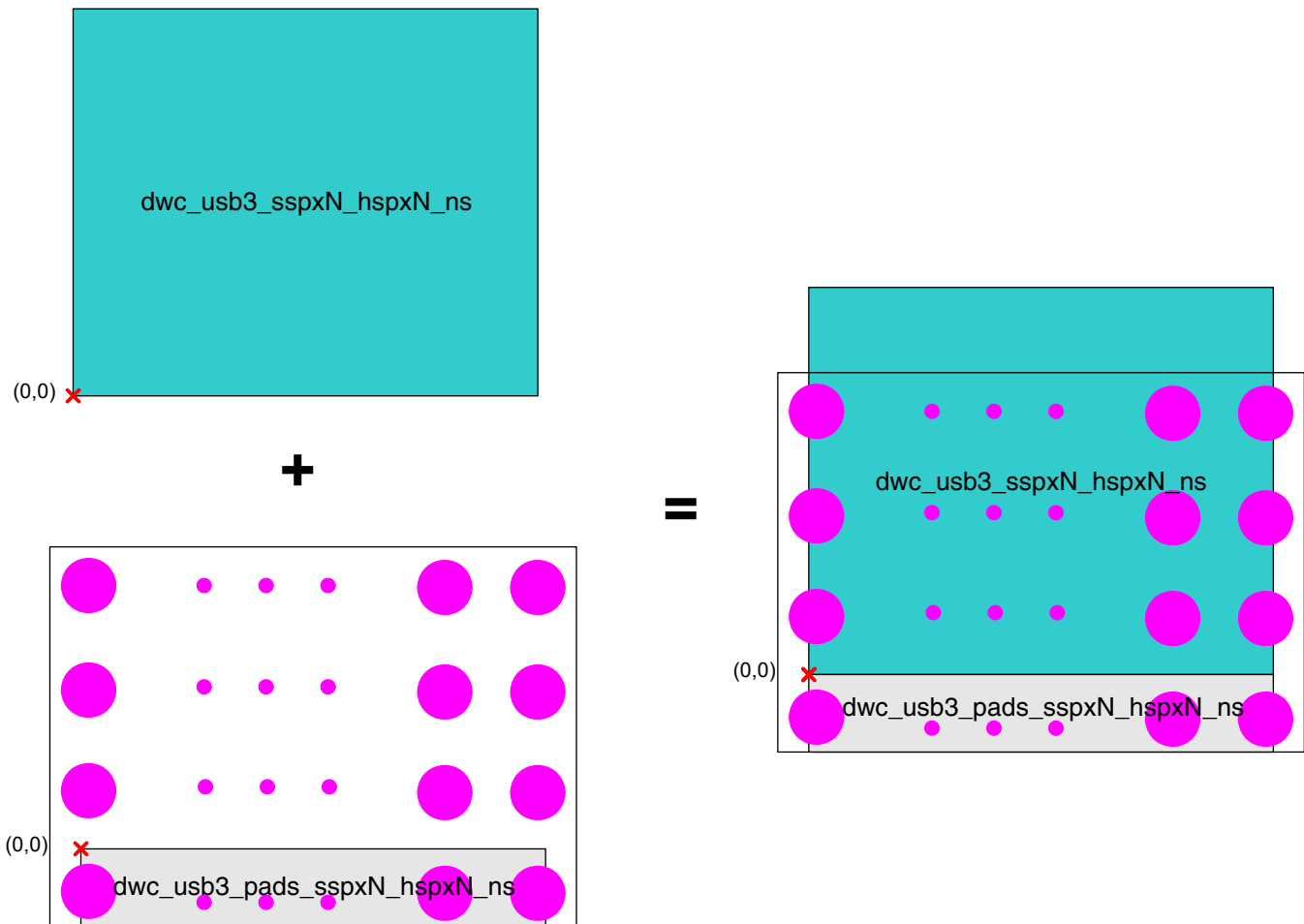
#### Note

For users who want to create a combo GDS (PHY + pads) and verify that it is LVS/DRC clean, Synopsys recommends reviewing the pipe wrapper module Verilog file for the interconnection between PHY and pads. The interconnection is by abutment in the physical GDS; however, the corresponding netlist must be created by the user.

### 6.3.2 Layout Guidelines

The PHY and the pad strip are designed to be placed at the same origin with the same rotation. As shown in the following figure(s), if these blocks are placed correctly, any overlapping routing provides the correct connections between the PHY and the associated bond pads. The recommended minimum distance the PHY can be placed from any adjacent cell is 5  $\mu\text{m}$ .

**Figure 6-3 Placement of PHY and Flip-Chip Pad Strips**



#### Note

For details about the flip-chip bump spacing, order, and locations, refer to the `usb3_pads_sspaN_hspaN` LEF file and GDS views.

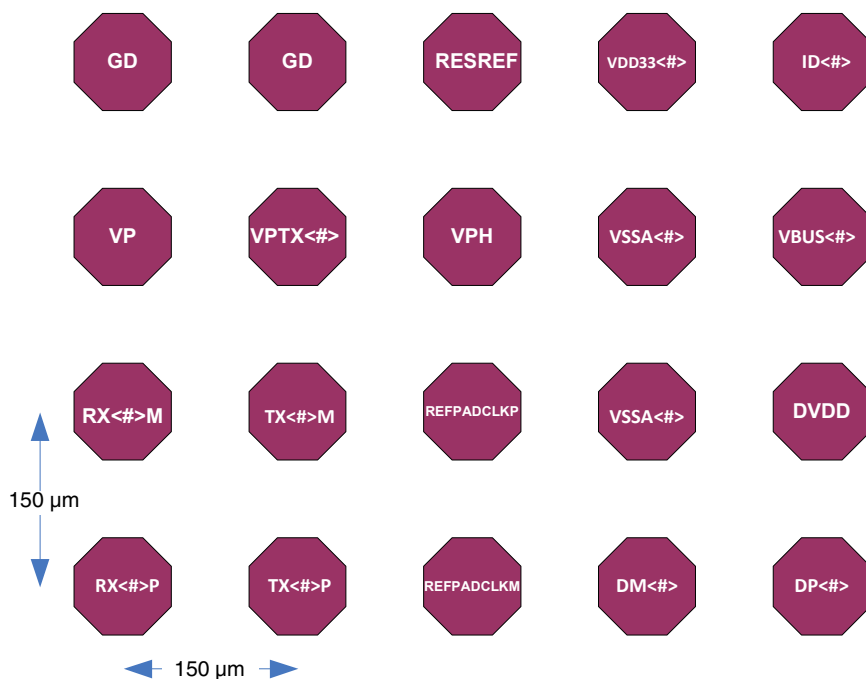
The following figure show the flip-chip bump order for dwc\_usb3\_pads\_ssp<sub>xN</sub>\_hsp<sub>xN</sub>\_ns.



### Note

The following figure is for illustration purpose only. For the exact bump order and locations, refer to the LEF deliverables.

**Figure 6-4 Flip-Chip Bump Order**





### 6.3.2.1 Rotation and Mirroring

The components do not require any rotation or mirroring before placement within the ASIC. If rotation or mirroring is required, the same rotation and/or mirroring must be done on both the PHY hard macro and the snap-on pad strip to maintain correct connectivity.



#### Note

To verify whether rotations are permitted for this technology node, refer to the `dwc_usb3_sspN_hspN_ns` LEF and `dwc_usb3_pads_sspN_hspN_ns` LEF deliverables.

### 6.3.2.2 External Components

The following table provides the value for external PHY components.

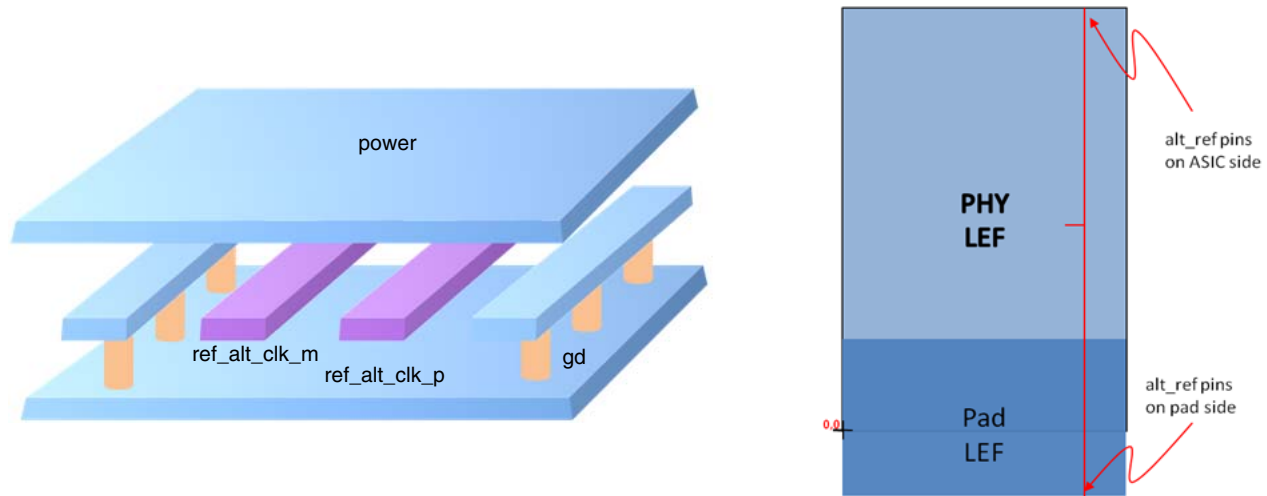
**Table 6-4 External Component Values**

| Item                                                                    | Value                                                         | Design Guidelines                                                                                                                      |
|-------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| External reference resistor resistance                                  | 200 $\Omega$ ( $\pm 1\%$ ) $\pm 100$ ppm/ $^{\circ}\text{C}$  | None                                                                                                                                   |
| External resistor $V_{\text{bus}}$                                      | 30 k $\Omega$ ( $\pm 1\%$ ) $\pm 100$ ppm/ $^{\circ}\text{C}$ | None                                                                                                                                   |
| External reference resistor maximum current                             | 2 mA                                                          | None                                                                                                                                   |
| External reference resistor parasitic capacitance ( $C_{\text{load}}$ ) | < 10 pF                                                       | It is important to meet the total capacitance at low frequency (in other words, the total capacitance of pad, package trace, and PCB). |

### 6.3.2.3 Reference Clock Routing Guidelines for ref\_alt\_clk\_m and ref\_alt\_clk\_p

In many ASICs, the ref\_use\_pad signal is de-asserted, and the reference clock is supplied from an on-chip source using ref\_alt\_clk\_m and ref\_alt\_clk\_p. In this case, the challenge is to route these clocks from the on-chip source to the PHY without adding jitter. In general, it is recommended that you use a filtered supply and route the signal through a shielded channel to the PHY, as shown in the following figure. Reference clock routing can be done either inside the chip or outside the bond pads around the peripheral of the die. For convenience, the ref\_alt\_clk\_p and ref\_alt\_clk\_m signals can be connected to pins either on the ASIC side or die-edge side of the PHY dwc\_usb3\_pads\_sspN\_hspN\_ns block.

**Figure 6-5 Alternate Ref Clock Routing and Connections**



For details on the reference clock requirements for using the ref\_alt\_clk\_p and ref\_alt\_clk\_m pins, see [“Reference Clock Requirements”](#) on page 243.

## 6.4 Synopsys .lib and .db Files

This section describes the timing models (.lib and .db files) included in the lib/ directory of the USB 3.0 femtoPHY product deliverables and provide examples of using these files for STA and synthesis.

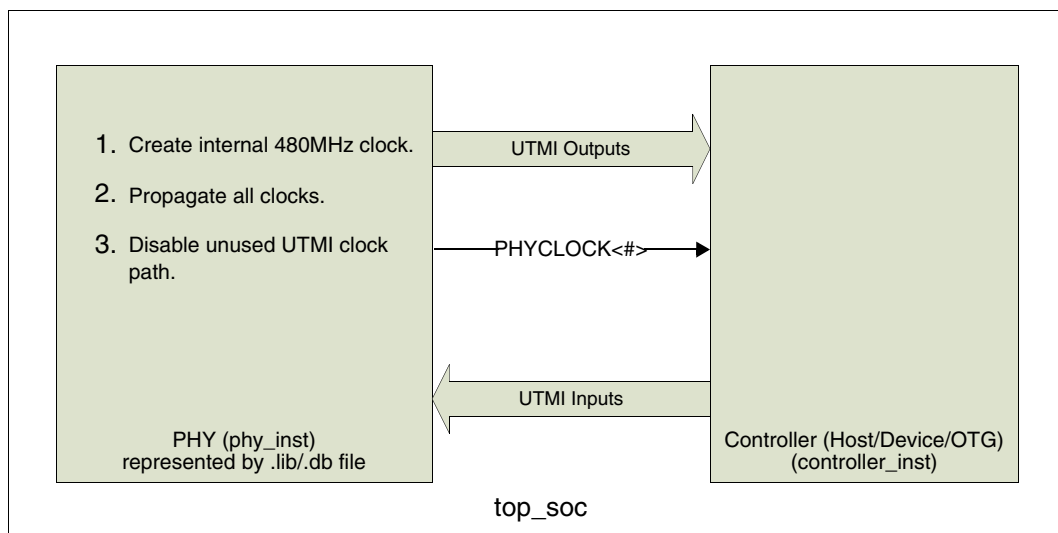
### 6.4.1 Static Timing Analysis

This section describes STA of the UTMI timing path and the SuperSpeed path.

#### 6.4.1.1 STA of the UTMI Timing Path

The following figure illustrates STA of the UTMI timing path.

**Figure 6-6 STA of the UTMI Timing Path**



As illustrated in [Figure 6-6](#), the guidelines for running STA of the DUT are as follows.

1. Create a master clock for the internal 480-MHz PLL's output. The .lib has a special internal node for this purpose. The following is an example of such a command using PrimeTime.

```
create_clock phy_inst/pll_480m_clk -name "phy_inst/pll_480m_clk_in" -period 2.083 -  
waveform "0 0.93735"
```

2. Propagate all clocks, both created and generated. The following PrimeTime commands accomplish this task.

```
set_propagated_clock [all_clocks]  
set_propagated_clock [get_generated_clocks *]
```

3. Depending on whether you are using the UTMI bus in 8-bit or 16-bit mode, disable the other (unused) clock path.

For example:

If you intend to operate the UTMI in 8-bit mode, disable the internal divide-by-16 clock path:

```
remove_generated_clock phy_inst/hsp_x1_inst_div16clk
```

If you intend to operate the UTMI in 16-bit mode, disable the internal divide-by-8 clock path:

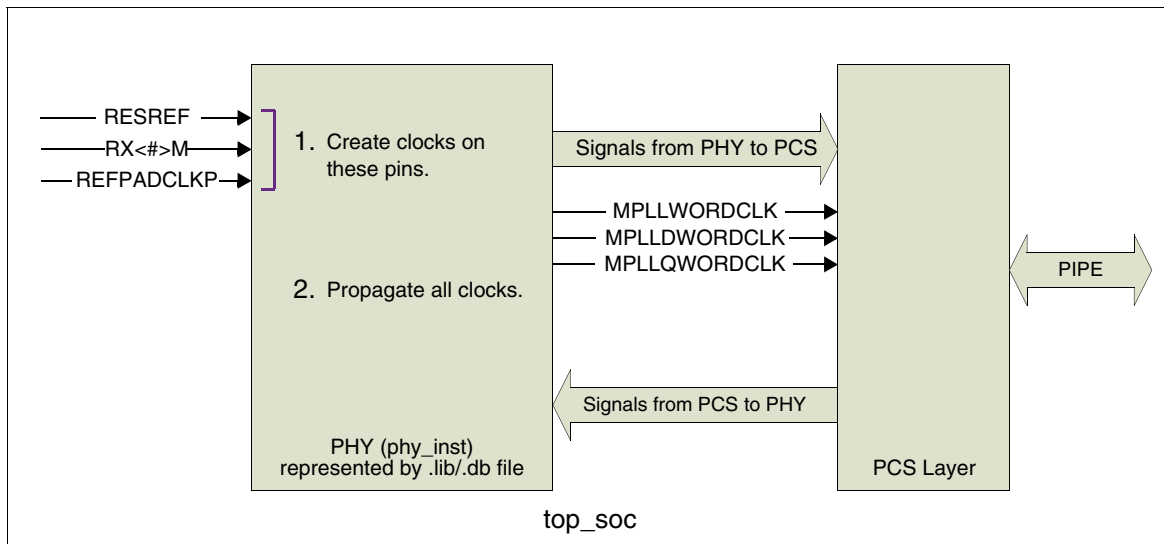
```
remove_generated_clock phy_inst /hsp_x1_inst_div8clk
```

After performing these steps, you can run timing reports on all the interface signals between the PHY and the controller.

### 6.4.1.2 STA of the SuperSpeed Path

The following figure illustrates STA of the SuperSpeed path.

**Figure 6-7 STA of the SuperSpeed Path**



As illustrated in [Figure 6-7](#), the guidelines for running STA of the SuperSpeed path in the DUT are as follows.

1. Create master clocks on the RESREF, RX<#>M, and REFPADCLKP inputs. The RESREF and RX<#>M inputs should model 500-MHz clocks, and REFPADCLKP should model a 100-MHz clock. The following is an example of such commands using PrimeTime.

```
create_clock phy_inst/RESREF -period 2
create_clock phy_inst/RX0M -period 2
create_clock phy_inst/REFPADCLKP -period 10
```

2. Propagate all clocks, both created and generated. The following PrimeTime commands accomplish this task.

```
set_propagated_clock [all_clocks]
set_propagated_clock [get_generated_clocks *]
```

## 6.5 Synthesis

For synthesis of the PCS layer (that is, the SuperSpeed path outside the PHY that leads to the PIPE interface), Synopsys provides predesigned synthesis scripts for DesignCompiler. These scripts already use the PHY's .lib/.db files to constrain the signals between the PHY and the PCS, along with the required clock definitions.

For synthesis of the HS/FS/LS link layer controller (that is, the USB 2.0 controller path outside the PHY), the same process as described in [“STA of the UTMI Timing Path”](#) on page 199 must be executed in the synthesis scripts. Doing so enables the controller's UTMI signals to inherit the timing characteristics and requirements of the PHY's UTMI signals, so that the resultant synthesized logic is timing-correct by construction.

## 7

# PCS Physical Implementation and Timing Model Usage

---

The pcs\_xN is delivered as soft RTL – meaning the pcs\_xN is hardened along with the rest of the RTL for the controller and ASIC.

Synthesizing the pcs\_xN requires a Verilog-2001 compatible synthesis tool. The source code for the pcs\_xN uses ANSI-style interface declarations as well as localparam declarations, both of which were added to the Verilog-2001 standard.

This chapter provides synthesis and timing closure guidelines for the pcs\_xN block and includes the following sections:

- [“PCS Constraints”](#) on page 204
- [“pcs\\_xN Clock Domains”](#) on page 204
- [“Synchronizers”](#) on page 211
- [“Hand-Placed Logic Constraints”](#) on page 213
- [“pipe\\_reset\\_mux Modules”](#) on page 213
- [“PCS Defines Descriptions”](#) on page 214

## 7.1 PCS Constraints

The widely used Synopsys design constraints format known as SDC describes the “design intent” and surrounding constraints for synthesis, clocking, timing, power, test, and environmental and operating conditions. Essentially all synthesized designs use SDC, and several EDA companies have translators that can read and process SDC. Typically, for synthesis and timing purposes, SDC contains the input and output delays of interface pins, clock definitions, and any timing path exceptions in the design, such as false or multi\_cycle paths.

In most cases, the pcs\_xN is synthesized in conjunction with other logic that resides on the opposite side of the PIPE interface boundary. In this situation, timing constraints can be automatically inferred from the adjoining logic and from the PHY timing model to which the pcs\_xN is connected.

When synthesizing the pcs\_xN-and-USB 3.0 femtoPHY combination as a standalone block, signals on the PIPE interface must be constrained so that sufficient time is budgeted for the adjoining logic to be connected later. As a general rule, you should be able to synthesize the pcs\_xN using an external input delay of 50% of the PIPE clock cycle and an external output delay of 75% of the PIPE clock cycle.

In addition, when synthesizing the pcs\_xN-and-USB 3.0 femtoPHY combination as a standalone block, you must consider the nature of the PIPE clock domain and the PIPE interface signals in that clock domain. Because PIPE interface signals are sampled and driven with respect to the pipeP\_pclk output, any additional clock tree delay that is added to pipeP\_pclk outside of the pcs\_xN likely causes hold violations on the PIPE outputs and setup violations on the PIPE inputs. If possible, to ensure balanced clock delays on both sides of the PIPE interface, clock tree insertion must be done at the top-most level. If this strategy is not possible, setup and hold times must be carefully managed across the PIPE interface.

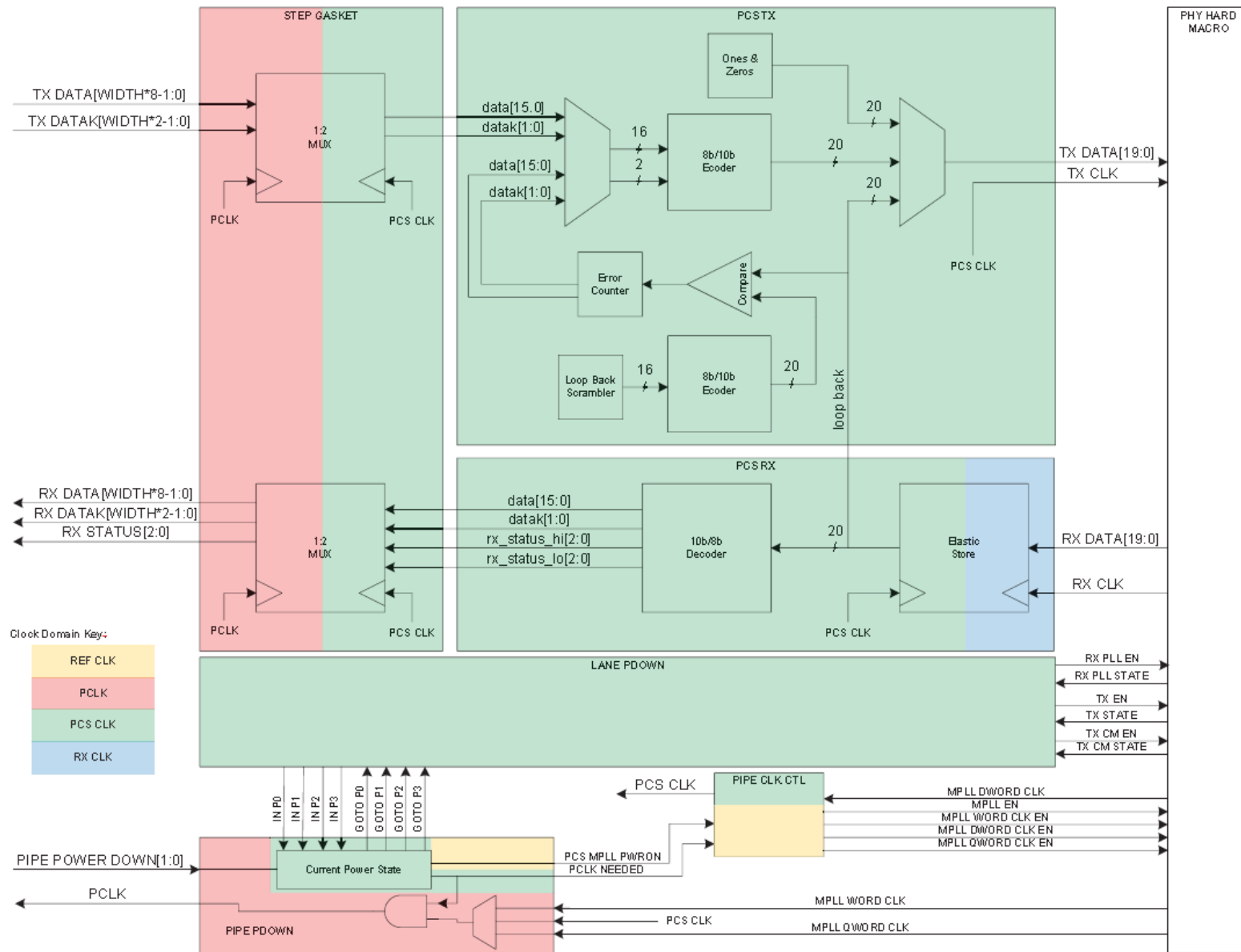
Due to the interaction between multiple clock domains within the pcs\_xN, several timing overrides are required to generate optimal results. These timing overrides are provided in the form of a TCL function that can be called from within dc\_shell or psyn\_shell to apply constraints to each pair of connected pcs\_xN/USB 3.0 femtoPHY instances.

## 7.2 pcs\_xN Clock Domains

For synthesis and static timing purposes, four clock domains exist within the pcs\_xN: the PIPE clock domains (250-MHz and 500-MHz portions), the recovered data (receiver) clock domain, and the reference clock domain. This section describes these clock domains.

[Figure 7-1](#) on page [205](#) shows the clock domains within the pcs\_xN.



**Figure 7-1 Clock Domains Within pcs\_xN**

## 7.2.1 PIPE Clock Domain

The PIPE clock domain is the primary clock domain in the pcs\_xN, and because most PIPE signals are synchronous with PCLK, this clock domain is also a primary clock domain in any higher-level logic that communicates with the PHY via the PIPE interface.

Throughout this databook, the PIPE clock is referred to as PCLK.

The PCLK frequency depends on the selected PIPE interface width: 500 MHz (1 byte), 250 MHz (2 bytes), or 125 MHz (4 bytes). This rate is not dynamically selectable and is fixed for a given PIPE width. The PCLK is sourced from the SS function's MPLL WORD, DWORD, or QWORD clocks, respectively (50% duty cycle).

In this databook, the term “pseudo timing arc” defines a timing arc that exists in the .lib file, but which does not correspond to a real path in the netlist or layout. Pseudo timing arcs are used to enable correct timing analysis in static timing tools.

The SS function's timing model contains several pseudo timing arcs from the resref\_f pin to the MPLL clock outputs. These timing arcs are used to propagate an ideal clock that must be defined at the resref\_f pin. By propagating the ideal clock through the PHY using these pseudo timing arcs, proper slopes and clock/data alignment are generated for static timing analysis on the MPLL output clock from which the PCLK tree is built.

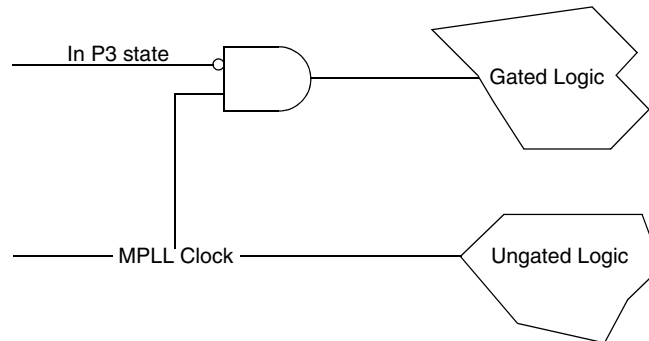
Only one 500-MHz clock must be defined on the resref\_f pin. The 250-MHz and 125-MHz clocks are automatically generated on their respective PHY outputs. The following example shows an excerpt from a synthesis script that defines the ideal clock parameters on the resref\_f pin.

```
create clock -name pipe_clk -period 2 -waveform {0 1} resref_f
set pipe_clk [get_clocks pipe_clk]
set_clock_latency -source -min $min_src_latency $pipe_clk
set_clock_latency -source -max $max_src_latency $pipe_clk
set_clock_latency -min $min_net_latency $pipe_clk
set_clock_latency -max $max_net_latency $pipe_clk
set_clock_uncertainty -setup $setup_uncertainty $pipe_clk
set_clock_uncertainty -hold $hold_uncertainty $pipe_clk
set_clock_transition -rise -fall -min $min_trans $pipe_clk
set_clock_transition -rise -fall -max $max_trans $pipe_clk
```

While applying the ideal clock to the resref\_f input enables proper timing analysis through the SS function's MPLL clock outputs, care must be taken to prevent any clock tree network from being built on the resref\_f pin itself, because this pin is a primary analog input to the PHY. Prevention is typically accomplished by applying the proper constraints to prevent the resref\_f net and PHY macro from being touched during the synthesis and place-and-route phases of the design.

As shown in the following figure, the PCLK output of the PCS is derived by gating the MPLL clock with an enable signal that enables PCLK to toggle whenever the pcs\_xN is not in the P3 state.

**Figure 7-2 Derivation of PCLK Output**



This clock gate is present in the final clock tree generated for the PIPE clock domain and feeds any logic that connects to PCLK. To ensure proper setup and hold times on the clock gate, the clock enable signal is generated on the falling edge of the MPLL clock.

To ensure proper timing between the gated and ungated portions of the PIPE clock domain, consider the following.

- To enable proper data transfer between subtrees, clock network latencies from the root to the leaves for gated and ungated portions of the clock tree must be balanced.
- To reduce the chance of clock gating setup and hold issues between the clock enable signal and the clock, clock network latencies must be kept to a minimum.

The clock enable and gating logic is implemented within an instance of `pipe_clk_gate` in the `pipe_pdown` module and can be adapted to various clock gating methodologies as required, provided the cycle-by-cycle behavior of the logic does not change.

## 7.2.2 Receiver Clock Domain

The receiver clock domain is used primarily for writing received data into the elasticity buffer inside the pcs\_xN. The only data transfer from the receiver clock domain to the PIPE clock domain occurs within the elasticity buffer, `pipe_estore`. During normal operation, multiple cycles separate the writing of data into one row of the buffer from reading the same row out of the buffer. For synthesis purposes, you can consider the transfer of data from the receiver clock domain to the PIPE clock domain to be a single-cycle path with the receiver clocks and PIPE clock being aligned. If the buffer depth approaches a single cycle, a buffer underflow condition is reported, so if the logic operates correctly as a single-cycle path, correct data transfer is guaranteed even beyond a signaled underflow condition.

The USB 3.0 femtoPHY timing model contains pseudo timing arcs from the `rx<#>_m` input to the `rx<#>_clk` and `rx<#>_data` outputs. These timing arcs are used to propagate an ideal clock that must be defined at the highest pin connected to the `rx<#>_m` input of the PHY. By propagating the ideal clock through the PHY using these pseudo timing arcs, proper slopes and skews are generated on the RX outputs from which the receiver clock tree is built.

```
create clock -name rx_clk -period 2 -waveform {0 1} rx0_m
set rx_clk [get_clocks rx_clk]
```

```

set_clock_latency -source -min $min_src_latency $rx_clk
set_clock_latency -source -max $max_src_latency $rx_clk
set_clock_latency -min $min_net_latency rx0_m
set_clock_latency -max $max_net_latency rx0_m
set_clock_uncertainty -setup $setup_uncertainty $rx_clk
set_clock_uncertainty -hold $hold_uncertainty $rx_clk
set_clock_transition -rise -fall -min $min_trans $rx_clk
set_clock_transition -rise -fall -max $max_trans $rx_clk

```

While applying the ideal clock to the rx<#>\_m input enables proper timing analysis through the PHY's rx<#>\_clk and rx<#>\_data output pins, care must be taken to prevent any clock tree network from being built on the rx<#>\_m pin, because this pin is a primary analog input to the PHY. Prevention is typically accomplished by applying the proper constraints to prevent the rx<#>\_m net and PHY macro from being touched during the synthesis and place-and-route phases of the design.

### 7.2.3 Reference Clock Domain

A small amount of sequential logic in the PCS must be able to operate when all PHY clocks other than the reference clock are disabled. This logic operates using ref\_output\_clk, which is the USB 3.0 femtoPHY's buffered version of its reference clock input. In a typical system, this clock operates nominally at 100 MHz. Some systems might need to support slower reference clocks, so it is recommended that if possible, you synthesize this clock tree using a 100-MHz clock, which is the highest frequency the USB 3.0 femtoPHY can generate on its ref\_output\_clk output pin.

The USB 3.0 femtoPHY timing model contains pseudo timing arcs from the ref\_pad\_clk\_m primary input to the ref\_output\_clk output. These timing arcs are used to propagate an ideal clock, which must be defined at the highest pin connected to the ref\_pad\_clk\_p input of the PHY. By propagating the ideal clock through the PHY using these pseudo timing arcs, proper slopes are generated on the PHY's ref\_output\_clk output from which the clock tree is built.

```

create clock -name ref_clk -period 10 \
  -waveform {0 5} ref_pad_clk_p
set_ref_clks [get_clocks ref_clks]
set_clock_latency -source -min $min_src_latency $ref_clks
set_clock_latency -source -max $max_src_latency $ref_clks
set_clock_latency -min $min_net_latency ref_pad_clk_p
set_clock_latency -max $max_net_latency ref_pad_clk_p
set_clock_uncertainty -setup $setup_uncertainty $ref_clks
set_clock_uncertainty -hold $hold_uncertainty $ref_clks
set_clock_transition -rise -fall -min $min_trans $ref_clks

```

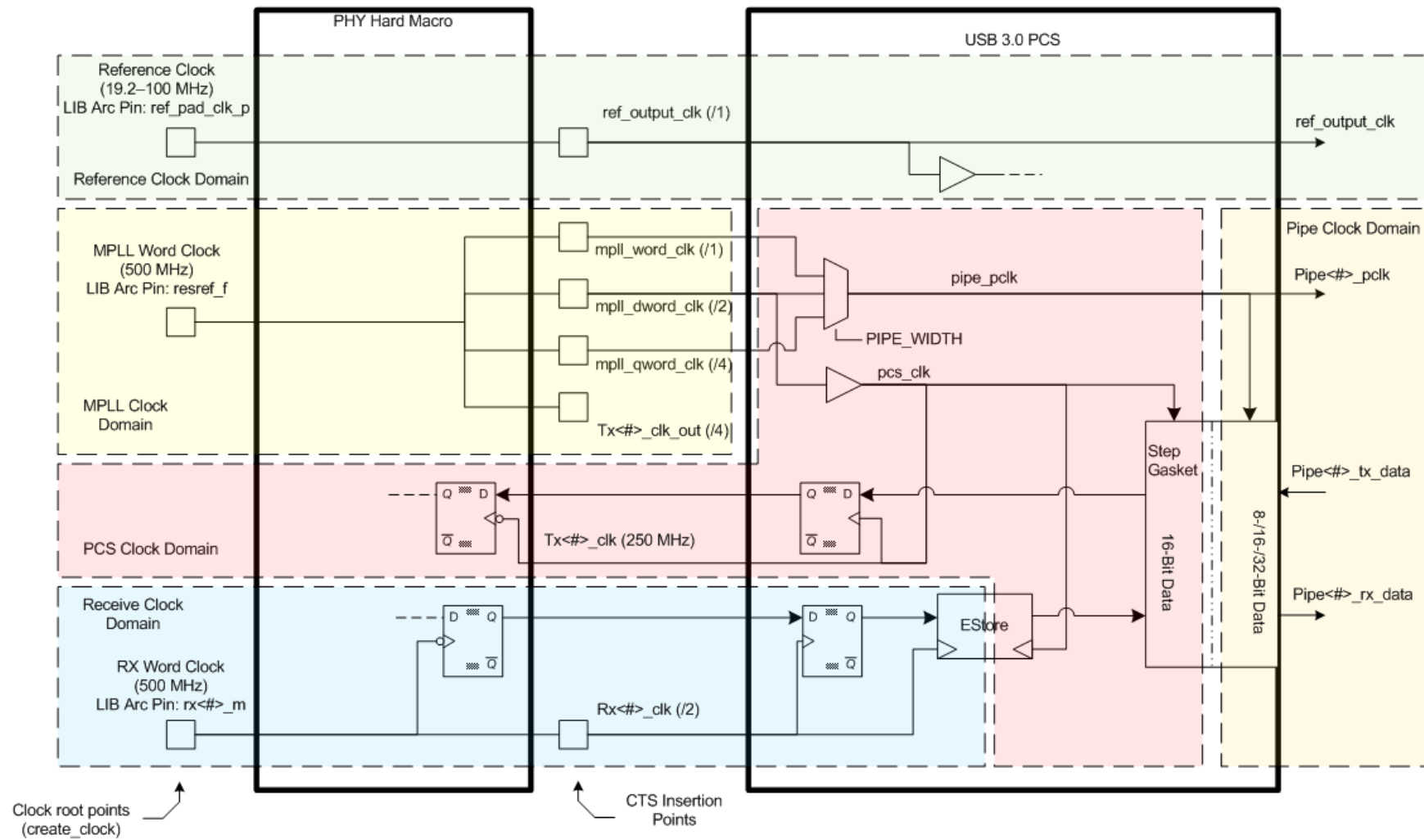
While applying the ideal clock to the ref\_pad\_clk\_p input enables proper timing analysis through the USB 3.0 femtoPHY's ref\_output\_clk output pin, care must be taken to prevent any clock tree network from being built on the ref\_pad\_clk\_p pin, because this pin is a primary analog input to the PHY. Prevention is typically accomplished by applying the proper constraints to prevent the ref\_pad\_clk\_p net and PHY macro from being touched during the synthesis and place-and-route phases of the design.

## 7.2.4 LIB Arcs and PCS Clock Tree

The following figure shows the clock tree insertion points for a single-lane PIPE wrapper. To enable synchronous data crossing, the pcs\_clk and the pipeP\_pclk domains must be balanced. The rxN\_clk and ref\_clk clocks are asynchronous to each other as well as to pcs\_clk and pipeP\_pclk.

You must provide two additional clocks at the pipe wrapper level.

- bs\_clk: Boundary scan clock
- jtag\_tck: JTAG clock; you must provide this slow clock input to the pipe wrapper for the JTAG interface.

**Figure 7-3 LIB Arcs and PCS Clock Tree**

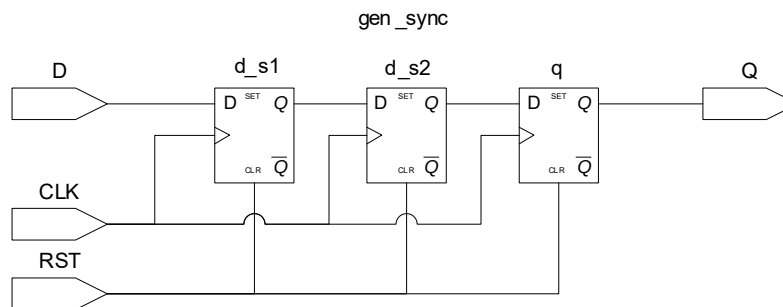
## 7.3 Synchronizers

This section describes synchronizers implemented in the pcs\_xN.

### 7.3.1 Basic Synchronizers

For single-bit signals that must be passed from one clock domain to another, an instance of the gen\_sync module is used to synchronize the source signal into the receiving clock domain, as shown in the following figure.

**Figure 7-4 gen\_sync Module**



The input to the first stage of the synchronizer can be considered completely asynchronous, so all paths to the d\_s1 register can be considered false.

The second and third registers in the gen\_sync module are d\_s2 and q. To ensure stable synchronization, constrain the maximum delay to these cells, from d\_s1/CK to d\_s2/D and from d\_s2/CK to q/D, to 20 percent of the sampling clock frequency.

For ease of timing constraint definition, this delay can be implemented as 800 ps (20 percent of 1/250 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instances. The clock-tree delay can be ignored if the paths are constrained explicitly from d\_s1/CK to d\_s2/D and from d\_s2/CK to q/D.

For multi-bit signals that must be passed from one clock domain to another, an instance of the multi\_bit\_dsync is used to synchronize the source signal into the receiving clock domain. The only difference between multi\_bit\_dsync and gen\_sync is that multi\_bit\_dsync uses two flip-flops, while gen\_sync uses three flip-flops.

The input to the first stage can be considered asynchronous, so all paths to the in\_p\_d register can be considered false.

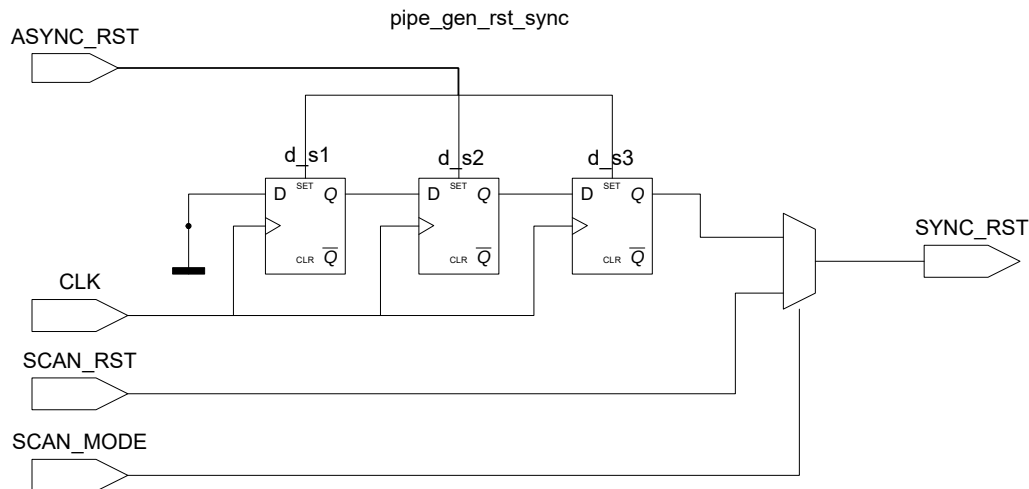
The second register in the multi\_bit\_dsync is in\_p\_dd. To ensure stable synchronization, constrain the maximum delay to these cells, from in\_p\_d/CK to in\_p\_dd/D to 20% of the sampling clock's frequency.

## 7.3.2 Reset Synchronizers

As shown in the following figure, any time an external or internal reset condition is passed from one clock domain to another domain within the pcs\_xN, the reset condition must pass through a gen\_rst\_sync module to do the following:

- Asynchronously capture the reset signal
- Stretch the reset pulse to be active for at least one full clock cycle in the receiving domain
- Synchronously de-assert the reset signal with respect to the receiving domain

**Figure 7-5** gen\_rst\_sync Module



Because assertion of a reset causes all flip-flops in the synchronizer module to capture the pulse, the de-assertion edge of the asynchronous reset input has no effect on the final flip-flop that drives the reset into the receiving domain. Therefore, all asynchronous paths through the d\_s3 register of this module can be considered false.

The second and third registers in the gen\_rst\_sync module are d\_s2 and d\_s3. To ensure stable synchronization, constrain the maximum delay to these cells, from d\_s1/CK to d\_s2/D and from d\_s2/CK to d\_s3/D, to 20 percent of the sampling clock frequency.

For ease of timing constraint definition, this delay can be implemented as 800 ps (20% of 1/250 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instance. The clock-tree delay can be ignored if the paths are constrained explicitly from d\_s1/CK to d\_s2/D and from d\_s2/CK to d\_s3/D.



## 7.4 Hand-Placed Logic Constraints

The PCS provides a single generic, digital-cell, primitive module that instantiates a specific digital cell. This cell must not be logically optimized during synthesis. The module, `pipe_clk_mux`, is typically used around asynchronous clock domain crossing (CDC) points within the design.

It is important that you process the `pipe_clk_mux` module correctly, not just synthesize it as usual with the PCS logic. Failure to do so likely creates glitch-generating and reconvergent logic that can cause problems on the PIPE interface, especially in the P3 power state when the PIPE clock is disabled and some control functions become asynchronous.

To prevent the `pipe_clk_mux` module from being optimized during synthesis, you can either replace the contents of the module with an instantiation of the appropriate standard cell from a digital library, or synthesize the module stand-alone with a restricted library, so that only the correct cell is selected by the tool. In both cases, a “`set_size_only`” attribute must be applied to prevent reoptimization of the instance during regular PCS synthesis.

## 7.5 `pipe_reset_mux` Modules

The `pipe_reset_mux` modules in the PCS include a shadow register that is useful for scan. All `rst_in_s1` registers (unique name to `pipe_reset_mux`) can be marked with a “don’t touch” attribute to preserve them through synthesis. When stitched into the scan chain, these registers provide observability into logic cones that are in bypass in Scan mode.

The current reset MUX instances are:

- `pcs_xN/clk_ctl/phy_reset_scan_mux`
- `pcs_xN/pipeP_pdown/pipe_reset_n_scan_mux`
- `pcs_xN/pipeP_pdown/mppll_state_rst`
- `pcx_xN/lane<#>/rx/phy_in_p0_reset_mux`

where P refers to the PIPE number, and N and <#> refer to the lane number.

## 7.6 PCS Defines Descriptions

The PCS RTL contains the following defines, which can be used for optional features:

- **DWC\_USB3\_BERC**: Enables optional BERC Loopback mode. BERC ordered sets are processed and replaced with BCNT ordered sets according to the USB 3.0 specification. If this define is not included, a simple loopback after the elastic store is implemented according to the USB 3.0 specification.
- **DWC\_USB3\_P3\_RESET**: Note that this macro define is for Synopsys's internal testing only. To be PIPE 3-compliant, this define must not be used. (PIPE specification states that P2 is a reset state.) Enabling this define changes the reset PIPE power state from P2 to P3 and keeps the PHY in its lower power state while enabling the controller to begin operations. Controllers must support this define.
- **DWC\_USB3\_FPGA\_TIMING\_STAGE\_PCS\_TX**: Adds register timing stages to the transmit path to aid in FPGA timing closure.
- **DWC\_USB3\_FPGA\_TIMING\_STAGE\_PCS\_RX**: Adds register timing states to the receive path to aid in FPGA timing closure.
- **DWC\_USB3\_FPGA\_TIMING\_STAGE\_ESTORE\_RD**: Adds register timing states to the elastic store to aid in FPGA timing closure.
- **DWC\_USB3\_FPGA\_PCLK\_BYPASS**: Bypasses the clock gate cell on the pipeP\_pclk signal to aid in FPGA implementation.
- **DWC\_USB3\_FPGA\_ASYNC\_TX\_RESET**: Works around a reset clock issue in the USB 3 test chip to aid in FPGA implementation.

# 8

## Production Test and Debug Implementation

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This chapter includes the following sections:

- [“Test Strategy Overview”](#) on page 215
- [“ASIC Pin Requirements to Enable Production Testing”](#) on page 216
- [“BIST Capabilities”](#) on page 216
- [“Diagnostic Features”](#) on page 218
- [“Scan”](#) on page 218

### 8.1 Test Strategy Overview

The USB 3.0 femtoPHY is designed to minimize the effort required to get the finished product chip into production with minimum test time and minimum test development. Software is provided with the IP to create test vectors that utilize the PHY’s Built-In-Self-Test (BIST) capability and JTAG port to test the PHY in production. Test development time is minimized by placing sufficient BIST hardware in the IP to compare analog values with high and low limits in the IP and simply scan out a pass or fail result for easy vector comparison on a digital tester.

## 8.2 ASIC Pin Requirements to Enable Production Testing

To enable ATE testing, the only requirements on the ASIC-side pins are access to the four JTAG pins and the following settings.

- Reset must be de-asserted.
- The reference clock must be running, properly configured using `ref_use_pad`, and enabled using `ref_ssp_en`.
- The `test_powerdown_ssp` and `test_powerdown_hsp` signals must be set to 1'b0.
- The `test_burnin` signal must be set to 1'b0.

## 8.3 BIST Capabilities

The USB 3.0 femtoPHY provides the following BIST capabilities on the SS function.

- **Pattern generation:** The Pattern Generator can generate PRBS-7, PRBS-15, or a programmable 10-bit WORD or 20-bit WORD~WORD pattern.
- **Pattern matching:** The Pattern Matcher can match the patterns of the Pattern Generator on the received clock domain so that asynchronous pattern matching is supported.
- **DAC:** A programmable DAC can be used to generate a voltage from 0-to-VDD33<#> (the analog supply).
- **ADC:** An SAR is wrapped around the DAC to turn it into an ADC. Accuracy is ~2 percent.
- **RX slicer offset:** The receiver inputs have a programmable input offset that enables the DAC and Pattern Matchers/Generator to test link integrity with a programmable eye opening. The range of programmable offset is ~+300 mV through -300 mV.
- **CDR phase offset:** The CDR can be shifted left or right from the center of the eye, so that the eye opening of #5 can be made at an arbitrary number of points within a specified eye opening.
- **Asynchronous loopback testing:** The TX can transmit a high-speed multiple of the reference clock while the RX CDR is running off a high-frequency clock with either a programmable ppm offset or with SSC enabled.
- **High-/low-limit registers:** You can set high-limit and low-limit registers in the IP using JTAG, so that the results of an ADC conversion are converted into a pass/fail bit that can be scanned out through JTAG for a simple vector comparison in the tester. This feature eliminates the need for a complex test program development.

The USB 3.0 femtoPHY provides the following BIST capabilities on the HS function:

- Ability to run BIST in low-speed, full-speed, and high-speed modes

When BIST is selected to run in all three USB 2.0 speed modes, the following patterns are transmitted in Loopback mode, as described in the following table.

- BIST controls are accessible through the USB 2.0 BIST control registers at addresses 16'H2018 and 16'H2009. For information about the USB 2.0 BIST control registers, see [“Registers”](#) on page 253.

**Table 8-1 BIST Sequence on High-Speed Function**

| Test | Description                                                        |
|------|--------------------------------------------------------------------|
| 1    | LS transmission and reception of 4 bytes of FFs using 8-bit UTMI+  |
| 2    | LS transmission and reception of 4 bytes of 00s using 8-bit UTMI+  |
| 3    | LS transmission and reception of 4 bytes of FFs using 8-bit UTMI+  |
| 4    | LS transmission and reception of 4 bytes of 00s using 8-bit UTMI+  |
| 5    | FS transmission and reception of 4 bytes of FFs using 8-bit UTMI+  |
| 6    | FS transmission and reception of 4 bytes of 00s using 8-bit UTMI+  |
| 7    | FS transmission and reception of 8 bytes of FFs using 16-bit UTMI+ |
| 8    | FS transmission and reception of 8 bytes of 00s using 16-bit UTMI+ |
| 9    | HS transmission and reception of 4 bytes of FFs using 8-bit UTMI+  |
| 10   | HS transmission and reception of 4 bytes of 00s using 8-bit UTMI+  |
| 11   | HS transmission and reception of 8 bytes of FFs using 16-bit UTMI+ |
| 12   | HS transmission and reception of 8 bytes of 00s using 16-bit UTMI+ |

ATE testing using the USB 3.0 femtoPHY's BIST function is through the JTAG port or the 16-bit parallel interface of the PHY.

### 8.3.1 Guidelines for Running BIST

To ensure that BIST operates correctly, follow these guidelines.

- Do not connect any capacitive loads or resistive terminations on the D+ and D- lines.
- Use one of the following termination options while running BIST:
  - Run BIST in two stages, so that HS BIST is run with 45-Ω termination-to-ground, while FS and LS BIST are run with no capacitive loads or resistive terminations on the D+ and D- lines.
  - Use a 130-Ω termination-to-ground for all three BIST modes. Note that this value might need to be adjusted, because it is dependent on the length of the load-board traces.

## 8.4 Diagnostic Features

In addition to the ATE capability, the USB 3.0 femtoPHY can use the JTAG port to extract an eye diagram of the signal at the input of the receiver. This feature is helpful for debug or initial system power-up.

## 8.5 Scan

This section describes the USB 3.0 femtoPHY's scan implementation.

The PHY provides a Scan interface for stuck-at fault and transition-delay fault testing (also known as at-speed-scan testing or ASST) of both the hard macro PHY and soft PCS layer.

To maximize scan coverage across the PHY/PCS layer boundary, it is recommended that you run DFT insertion on the PHY and PCS together.

An ATPG model of the digital portion of the PHY is provided (in .ctl, .spf formats) that can be used during ATPG pattern generation to stitch the scan chains together and test them during chip-level scan testing.

If scan is not being implemented in the PHY, tie all scan pins to 1'b0.

### 8.5.1 Scan Configuration

To enable scan testing, set the following signals:

- `test_burnin = 1'b0`
- `test_powerdown_ssp = 1'b0`
- `test_powerdown_hsp = 1'b1`
- `scan_mode = 1'b1`

In Scan mode, all analog blocks are powered down. All digital outputs to the analog blocks are looped back to the digital portion to provide coverage on all analog/digital interfaces. When running scan, the `vph` and `vptx<#>` supplies are not required; only the analog `vp` supply is required.

## 8.5.2 PHY Scan Details

This section describes scan functionality in the PHY.

### 8.5.2.1 Clocks and Resets

To support at-speed scan testing, the PHY provides separate scan clock domains for each functional clock domain. The following table lists the available scan clocks.

**Table 8-2 PHY Scan Clocks**

| Clock Name                                                                                           | Frequency (MHz) |
|------------------------------------------------------------------------------------------------------|-----------------|
| bs_clk (use functional input for scan)                                                               | 100             |
| jtag_tck (use functional input for scan)                                                             | 100             |
| scan_ssref_clk                                                                                       | 100             |
| scan_mpll_word_clk                                                                                   | 500             |
| scan_mpll_dword_clk                                                                                  | 250             |
| scan_mpll_qword_clk                                                                                  | 125             |
| scan_txP_word_clk                                                                                    | 500             |
| scan_txP_dword_clk                                                                                   | 250             |
| scan_txP_out_clk                                                                                     | 500             |
| scan_rxP_word_clk                                                                                    | 500             |
| scan_rxP_dword_clk                                                                                   | 250             |
| scan_rxP_scope_clk                                                                                   | 250             |
| SCANPLL480MCLK<#>                                                                                    | 480             |
| SCANPLL480MHSTXCLK<#>                                                                                | 480             |
| SCANHSPREFCLK<#>                                                                                     | 12              |
| SCANCLK12MOHCI<#>                                                                                    | 12              |
| SCANCLK30M<#>                                                                                        | 30              |
| SCANCLK48M<#>                                                                                        | 48              |
| SCANCLK48MOHCI<#>                                                                                    | 48              |
| SCANCLK60M<#>                                                                                        | 60              |
| SCANCDRPHASEXCLK<#> (where X is 0–15)                                                                | 480             |
| SCANSERCLK<#>                                                                                        | 480             |
| <b>Note:</b> P represents the SuperSpeed lane number, and <#> represents the high-speed port number. |                 |

**Table 8-2 PHY Scan Clocks (Continued)**

| Clock Name                                                                                           | Frequency (MHz) |
|------------------------------------------------------------------------------------------------------|-----------------|
| SCANSERCLKRX<#>                                                                                      | 480             |
| SCANSERCLKTX<#>                                                                                      | 480             |
| SCANTESTCLK<#>                                                                                       | 60              |
| SCANUTMICKL<#>                                                                                       | 60              |
| <b>Note:</b> P represents the SuperSpeed lane number, and <#> represents the high-speed port number. |                 |

As noted in [Table 8-2](#) on page 219, with the exception of bs\_clk and jtag\_tck, all other primary input clocks are internally multiplexed with their respective at-speed scan clock.

All output clocks from the PHY have the appropriate at-speed scan clock multiplexed onto them and do not require an additional scan MUX on the output.

All resets used in the PHY (either top-level inputs or internally generated) are controlled in Scan mode by scan\_rst and phy\_reset.

**Note**

In Scan mode, the analog portion of the PHY is powered down; therefore, all at-speed scan clocks must be provided to the PHY from external clock sources.



### 8.5.2.2 Scan Chains

The following table provides the number of scan chains per scan clock domain.

**Table 8-3 Scan Chains Per Scan Clock Domain**

| Clock Name                                                                                           | Number of Positive-Edge-Triggered Scan Chains | Number of Negative-Edge-Triggered Scan Chains |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| bs_clk (use functional input for scan)                                                               | 1                                             | N/A                                           |
| jtag_tck (use functional input for scan)                                                             | 2                                             | N/A                                           |
| scan_ssprof_clk                                                                                      | 10 (for x1)<br>15 (for x2)<br>24 (for x4)     | N/A                                           |
| scan_mpll_word_clk                                                                                   | 1                                             | N/A                                           |
| scan_mpll_dword_clk                                                                                  | 2                                             | N/A                                           |
| scan_mpll_qword_clk                                                                                  | 1                                             | N/A                                           |
| scan_txP_word_clk                                                                                    | 1                                             | N/A                                           |
| scan_txP_dword_clk                                                                                   | 2                                             | N/A                                           |
| scan_txP_out_clk                                                                                     | 0 <sup>a</sup>                                | N/A                                           |
| scan_rxP_word_clk                                                                                    | 5                                             | N/A                                           |
| scan_rxP_dword_clk                                                                                   | 7/2 <sup>b</sup>                              | N/A                                           |
| scan_rxP_scope_clk                                                                                   | 1                                             | N/A                                           |
| SCANPLL480MCLK<#>                                                                                    | 1                                             | 1                                             |
| SCANPLL480MHSTXCLK<#>                                                                                | 1                                             | N/A                                           |
| SCANHSPREFCLK<#>                                                                                     | 1                                             | 1                                             |
| SCANCLK12MOHCI<#>                                                                                    | N/A                                           | N/A                                           |
| SCANCLK30M<#>                                                                                        | N/A                                           | 1                                             |
| SCANCLK48M<#>                                                                                        | 1                                             | 1                                             |
| SCANCLK48MOHCI<#>                                                                                    | N/A                                           | N/A                                           |
| SCANCLK60M<#>                                                                                        | N/A                                           | 1                                             |
| SCANCDRPHASEXCLK<#> (where X is 0–15)                                                                | 16                                            | N/A                                           |
| SCANSERCLK<#>                                                                                        | 1                                             | 1                                             |
| SCANSERCLKRX<#>                                                                                      | 4                                             | N/A                                           |
| <b>Note:</b> P represents the SuperSpeed lane number, and <#> represents the high-speed port number. |                                               |                                               |

**Table 8-3 Scan Chains Per Scan Clock Domain (Continued)**

| Clock Name                                                                                           | Number of Positive-Edge-Triggered Scan Chains | Number of Negative-Edge-Triggered Scan Chains |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| SCANSERCLKTX<#>                                                                                      | 2                                             | N/A                                           |
| SCANTESTCLK<#>                                                                                       | 2                                             | N/A                                           |
| SCANUTMICKL<#>                                                                                       | 3                                             | 1                                             |
| <b>Note:</b> P represents the SuperSpeed lane number, and <#> represents the high-speed port number. |                                               |                                               |

- a. This is an output clock that does not contain any flops within the PHY.
- b. Lane0 for this clock contains seven scan chains. All subsequent lanes contain two scan chains.

All scan chains have a maximum of 100 flip-flops.

With one exception, all scan chains terminate with lock-up latches. The negative-edge-triggered scan chains include an additional non-lock-up latch output that does not contain an output lock-up latch.

The PHY .lib files include all appropriate timing arcs to support Shift and Stuck-at Capture modes.

### 8.5.2.3 Scan Shift and Stuck-At Capture Modes

In Scan Shift mode, connect all scan clock inputs to a single 100-MHz source. To reduce tester time in Scan Shift mode, all scan chains are shifted in and out simultaneously. When stitching the scan chains together, ensure that you select the correct latch output (either lock-up or non-lock-up) for each scan chain. In Stuck-at Capture mode, perform the capture in a one-hot fashion, so that each clock domain is tested in isolation. The maximum scan shift and stuck-at-fault testing frequency is 100 MHz.

### 8.5.2.4 At-Speed Capture Mode

In ASST Capture mode, externally drive all PHY scan clock inputs by the appropriate frequency, as described in [Table 8-2](#) on page 219.

ASST capture is performed using a pulse-pulse methodology in a one-hot fashion, so each clock domain is tested in isolation.

All synchronous inputs and output are at-speed scannable, and the PHY .lib files include appropriate timing arcs to support at-speed capture.

### 8.5.3 At-Speed Scanning of RX and TX Interfaces

The RX and TX interfaces between the PHY and PCS layer are half-cycle paths in functional mode. These interfaces are source-synchronous, where the clocks are inverted to maximize hold margin. However, to support a pulse-pulse at-speed scan capture methodology, the paths can be timed only as full-cycle paths for at-speed scan.

The following signals are affected.

RX interface:

- rx<#>\_valid
- rx<#>\_data[19:0]

TX interface:

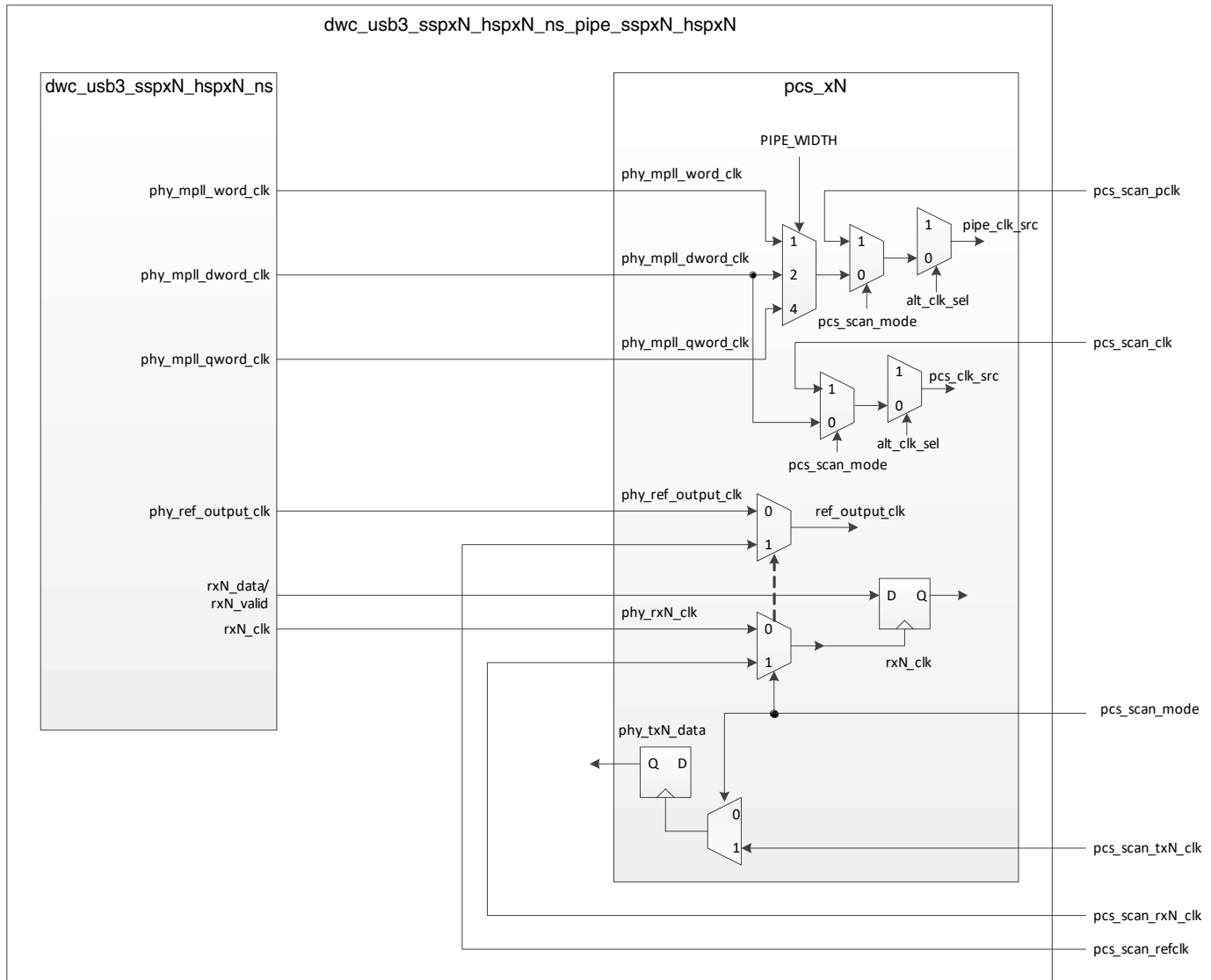
- tx<#>\_data\_en
- tx<#>\_data[19:0]
- tx<#>\_invert

## 8.5.4 PCS Layer Scan Details

The PHY includes separate scan clock and reset inputs for each PCS layer clock domain to support at-speed-scan testing of the PCS layer.

The following figure outlines the PCS scan clocking architecture.

**Figure 8-1 PCS Scan Clocking Architecture**



To provide the flexibility of running at-speed scan insertion at different hierarchical levels, the at-speed scan clocks of the PCS layer and PHY are both brought up to the PHY's top level. This design permits each scan input to be controlled independently to support DFT insertion flows, which do contain the PHY. For example, a controller and PCS layer can be synthesized and go through DFT insertion together without the PHY.

To support DFT insertion flows that do not include the PHY, all primary input clocks to the PCS contain scan MUXes despite the fact that all output clocks in the PHY already have scan multiplexed onto them.



To maximize at-speed test coverage, it is recommended that you tie the PCS layer RTL at-speed scan signals to the appropriate PHY at-speed scan signals.

The following table provides the necessary connections between the PHY and PCS layer RTL to implement at-speed scan testing on the entire PHY.

**Table 8-4 Connections Between PHY and PCS for At-Speed Testing on Entire PHY**

| PCS Layer Input                                       | PHY Input                             |
|-------------------------------------------------------|---------------------------------------|
| pcs_scan_mode                                         | scan_mode                             |
| pcs_scan_shift                                        | scan_shift                            |
| pcs_scan_rst                                          | scan_rst                              |
| pcs_scan_refclk                                       | scan_ssref_clk                        |
| pcs_scan_clk                                          | scan_mpll_dword_clk                   |
| pcs_scan_pclk                                         | scan_mpll_word_clk                    |
| pcs_scan_rxP_clk                                      | scan_rxP_dword_clk                    |
| pcs_scan_txP_clk                                      | scan_txP_dword_clk/scan_tx1_dword_clk |
| <b>Note:</b> P represents the SuperSpeed lane number. |                                       |

### 8.5.5 ATPG Vectors

When running scan vector simulations, use the full gate-level netlist. The scan netlist provided in the deliverables includes a scan model of the analog portion of the design only. When running scan vector simulations, use the full behavioral analog model within the gate-level netlist.

In addition, to prevent errors in the analog model when running scan vector simulations, define the `DWC_TETRAMAX` define.

## 8.5.6 ATPG Deliverables

The following table outlines the scan deliverables.

**Table 8-5 Scan Deliverables**

| File                                                   | Description                                                      |
|--------------------------------------------------------|------------------------------------------------------------------|
| macro/Latest/sim/run_scan_vcs                          | Run ATPG vector simulations                                      |
| macro/Latest/sim/run_scan_sim                          |                                                                  |
| phy/Latest/atpg/tetramax/<design>_tmax_atpg.tcl        | TetraMAX ATPG run script                                         |
| phy/Latest/atpg/ctl/<design>_dft.ctl                   | DFT models that can be used in hierarchical scan synthesis       |
| phy/Latest/atpg/ctl/<design>_dft.spf                   |                                                                  |
| phy/Latest/atpg/models/<design>_scan.v                 | Scan netlist output from TetraMAX                                |
| phy/Latest/atpg/vectors/<design>_test.para.v           | Test vectors and test bench for simulating with parallel loading |
| phy/Latest/atpg/vectors/<design>_test.para.dat         |                                                                  |
| phy/Latest/atpg/vectors/<design>_test.para.stil        |                                                                  |
| phy/Latest/atpg/vectors/<design>_test.para_stildpv.v   |                                                                  |
| phy/Latest/atpg/vectors/<design>_test.all.serial.v     | Test vectors and test bench for simulating with serial loading   |
| phy/Latest/atpg/vectors/<design>_test.serial.dat       |                                                                  |
| phy/Latest/atpg/vectors/<design>_test.serial.stil      |                                                                  |
| phy/Latest/atpg/vectors/<design>_test.serial_stildpv.v |                                                                  |
| phy/Latest/atpg/vectors/<design>_test.wgl              | Test vectors in WGL format                                       |





# 9

## Board- and Package-Level Implementation

---

This chapter describes requirements and guidelines for implementing the USB 3.0 femtoPHY at the board and package levels.

This chapter includes the following sections:

- [“Packaging Considerations”](#) on page 230
- [“Charge Pump Connection”](#) on page 235
- [“Non-OTG Configuration”](#) on page 236
- [“PHY Link Modeling”](#) on page 236

## 9.1 Packaging Considerations

The USB 3.0 femtoPHY supports the following packaging:

- Flip-chip

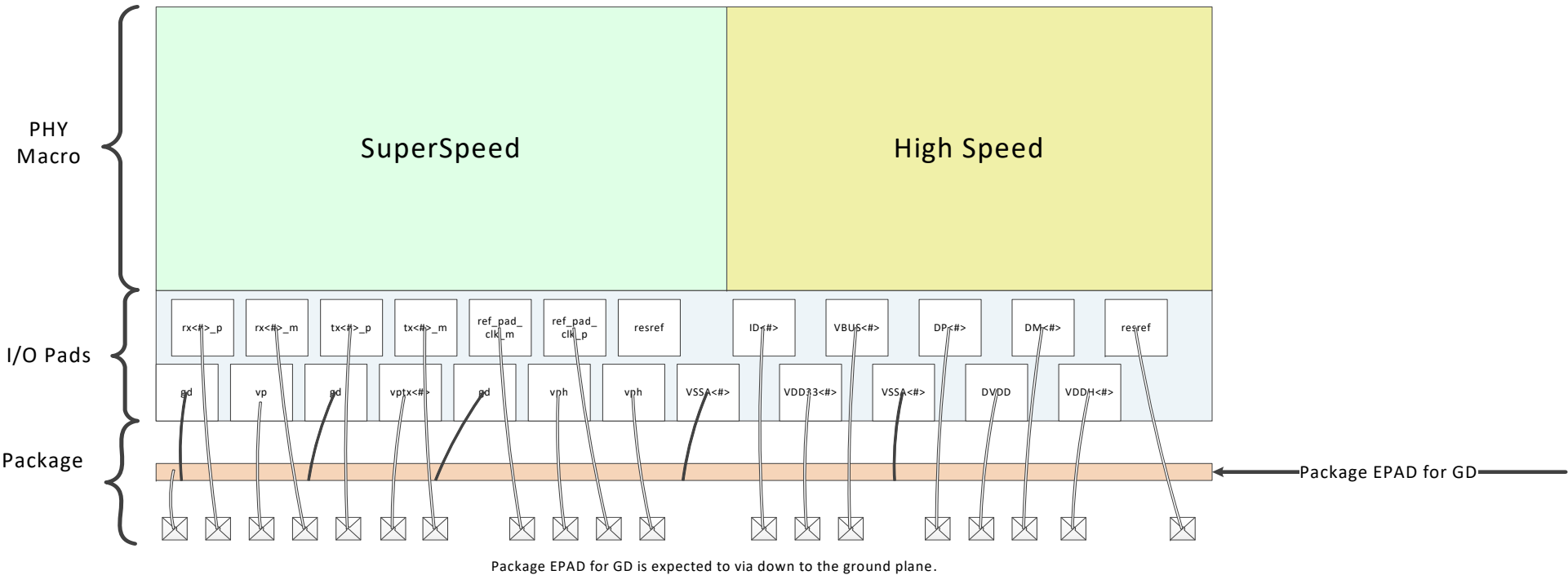
### 9.1.1 Power and Ground Connections

When implementing the USB 3.0 femtoPHY in an SoC, use the following guidelines:

- Connect the `vp`, `vptx<#>`, and `DVDD` supplies together through a ferrite bead on-board. These core supplies can also be connected to the ASIC's core supply through a ferrite bead on-board, but they must be isolated on-die and in-package.
- Keep `gd`, `VSSA<#>`, and `vss` separate on die, but tied together as a common ground in the package with the rest of the ASIC. `vph`, `VDDH<#>`, and `VDD33<#>` can be tied together in the package or through a ferrite bead on-board. PI analysis with customer-specific package/board implementation must be performed to guarantee that the PHY noise ripple specification is met. If such application is intended, it is recommended that you analyze USB 3.0-SSP and USB 3.0-HSP PHY operational modes simultaneously as well.
- In applications using multiple USB 3.0 femtoPHY macros, the core supplies can be shorted together on-board, the 3.3-V supplies can be shorted together on-package or on-board, and the ground supplies must be shorted together in the package.
- The `resref` resistor cannot be shared between multiple USB 3.0 femtoPHY macros.

Synopsys recommends that all the power/ground pins be bonded out as previously described. [Figure 9-1](#) on page [231](#) shows an example of the recommended bonding method.

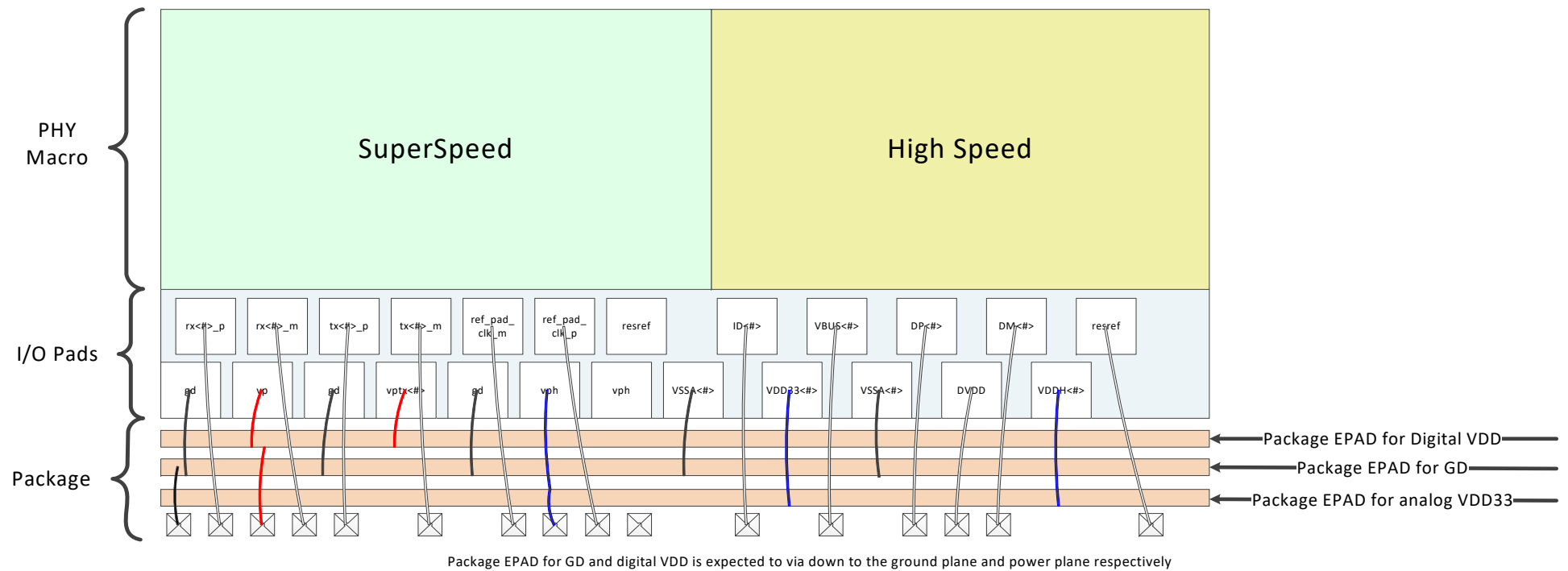
Figure 9-1 Recommended Bonding Method



However, to reduce the number of pins at the package level, you can do the following:

- Short VP and VPTX<#> at the package level.
- Short GD and VSSA<#> at the package level.
- Short VPH, VDD33<#>, and VDDH<#> at the package level.
- Bond out only one of the two VPH bond pads.

[Figure 9-2](#) on page [233](#) shows an illustration of this pin reduction method. Note that Synopsys has not tested this pin reduction method on silicon.

**Figure 9-2 Pin Reduction Method**

**Note:** Bonding out REFPADCLK(P,M), VBUS<#>, and ID<#> is optional depending on your usage model. For signal descriptions, see [“Signal Descriptions”](#) on page 29.

The following table provides the supply impedance requirements for the USB 3.0 femtoPHY.

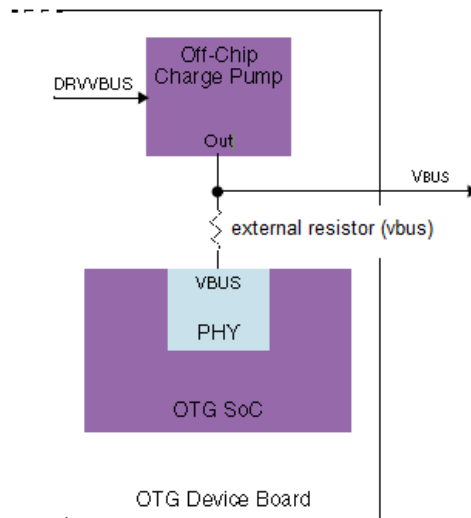
**Table 9-1 Supply Impedance Requirements**

| $L_{gd} + L_{vp}$<br>(nH) | $L_{VSSA<\#>} + L_{DVDD}$<br>(nH) | $L_{gd} + L_{vptx<\#>}$<br>(nH) | $L_{VSSA<\#>} + L_{VDD33<\#>}$<br>(nH) | $L_{gd} + L_{vph}$<br>(nH) | $L_{VSSA<\#>} + L_{VDDH<\#>}$<br>(nH) |
|---------------------------|-----------------------------------|---------------------------------|----------------------------------------|----------------------------|---------------------------------------|
| < 2.4                     | < 2.4                             | < 2.4                           | < 2.8                                  | < 2.8                      | < 2.8                                 |

## 9.2 Charge Pump Connection

An off-chip charge pump is required to provide power to the USB 3.0 femtoPHY's VBUS pin. The following figure shows the charge pump connection to the USB 3.0 femtoPHY.

**Figure 9-3 Charge Pump Connection**



The charge pump's output is connected directly to VBUS on the device board. The USB 3.0 femtoPHY's VBUS pin is also connected to VBUS. The charge pump's DRVVBUS input is an output of the OTG HNP finite state machine and an input to the USB 3.0 femtoPHY.

The VBUS pin presents a worst-case core-side load of 500 fF. This worst-case load is a small addition to the overall capacitance budget that includes the external capacitive load due to routing, pads, package, board traces, and receivers.

## 9.3 Non-OTG Configuration

When the USB 3.0 femtoPHY is configured to operate in non-OTG mode, neither routing nor I/O pad is required for the ID<#> pin. You can leave the pad strip as-is and simply leave the ID<#> bond pad floating. However, if ID<#> is left floating and not connected to a chip pin, ID<#> cannot be used as a test point for DC point probes inside the USB 3.0 femtoPHY.



### Note

The internal VBUS comparator is used to generate the PCS signal, pipeP\_power\_present, but if the PHY's VBUS<#> pin is not bonded out nor connected, this signal is not valid. Therefore, the USB 3.0 femtoPHY's pipeP\_power\_present output must be left floating, and an external VBUS comparator must be used to generate the required USB 3.0 controller power-present signal.

## 9.4 PHY Link Modeling

To assist in link modeling, detailed models of the PHY transmitter and receiver are provided along with a representative channel model.

These models include the following files:

- tx\_channel\_rx\_model\_tb.sp: Top-level SPICE test bench
- data.vec: Sample serial datastream input file for the transmitter
- tx\_model.sp: SPICE model of the transmitter
- rx\_model.sp: SPICE model of the receiver
- wb\_pkg.sp: Example wirebond package model
- bw.sp: Four bond wires modeled to comprise eight T sections
- dm.rlgc: RLGC matrix parameters
- via\_ball.sp: Example package via and ball subcircuit
- channel.s4p: 'S parameter' file representing channel
- 3MCABLE\_X1\_SE.s4p: Example S parameter file representing cable

This channel model is simply an example; you must measure the system board and connector you are using to get an accurate representation of the actual channel used in the end application.

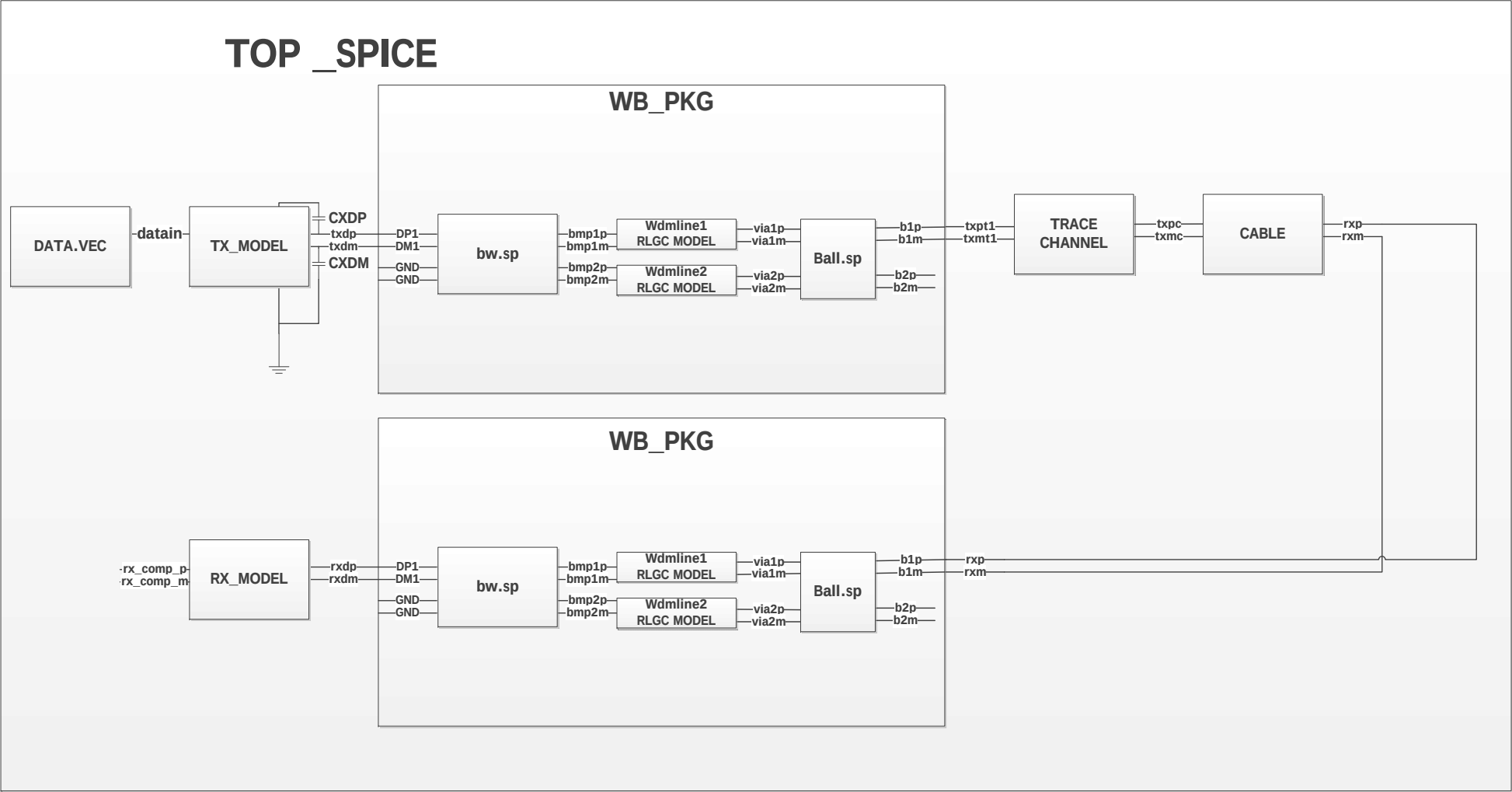
The HSPICE modeling directory in the database deliverables includes the TX and RX models. Both models are fit to detailed simulations of the PHY at both typical and worst-case conditions.

Sample test benches are provided using a representative channel model for both eye diagrams and return loss simulations.

Figure 9-4 on page 237 shows the structure of the SPICE top-level test bench.



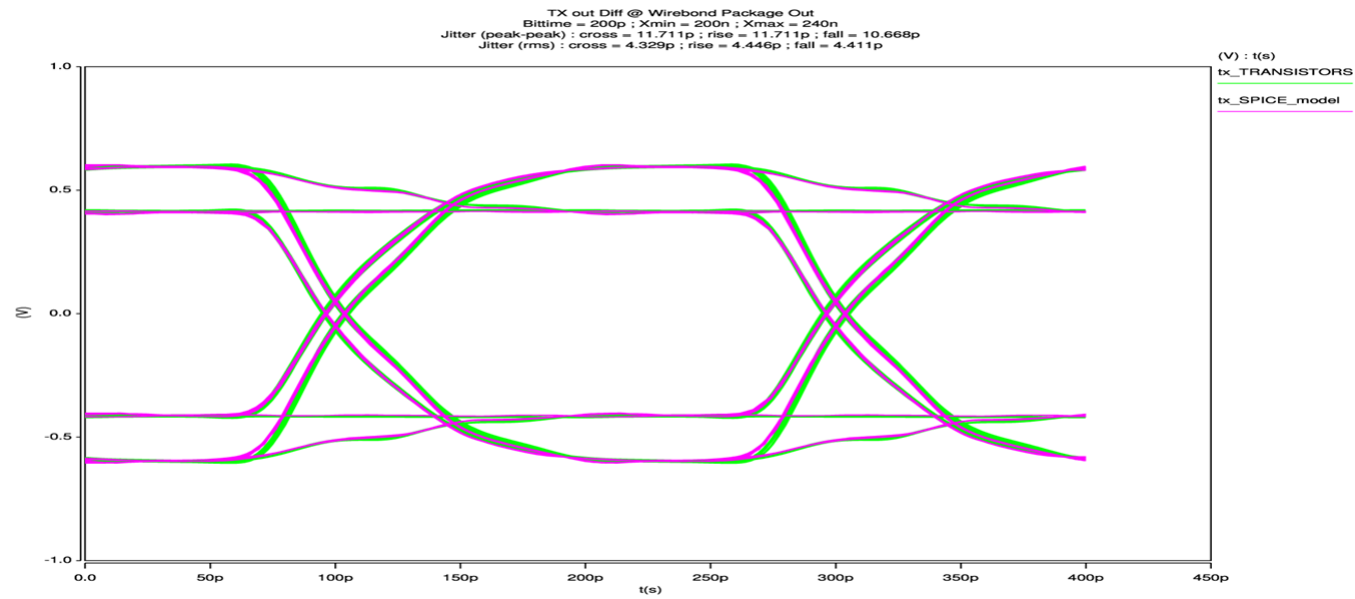
Figure 9-4 SPICE Top-Level Test Bench Structure



### 9.4.1 HSPICE Modeling of TX

The HSPICE model of the transmit driver is matched to the detailed full-transistor-level model of the TX. The following figure shows an example of the matching obtained with this modeling, where simulation is of the TX and package model into an ideal 50- $\Omega$  load.

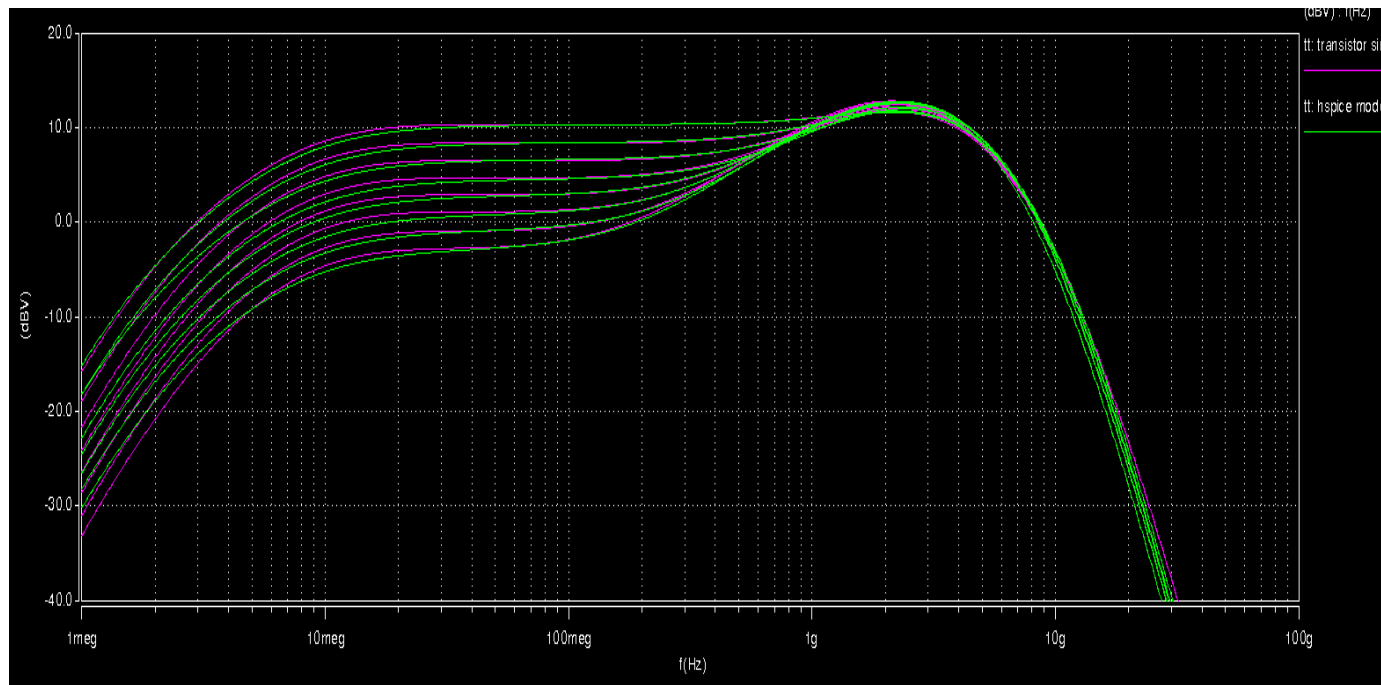
**Figure 9-5 TX Model Matching**



## 9.4.2 HSPICE Modeling of the RX

The HSPICE model of the receiver is matched to the detailed full-transistor-level model of the RX analog front-end (AFE) including the adaptive RX equalizer. The following figure shows an example of the matching obtained with this modeling of the equalizer's frequency-dependent gain.

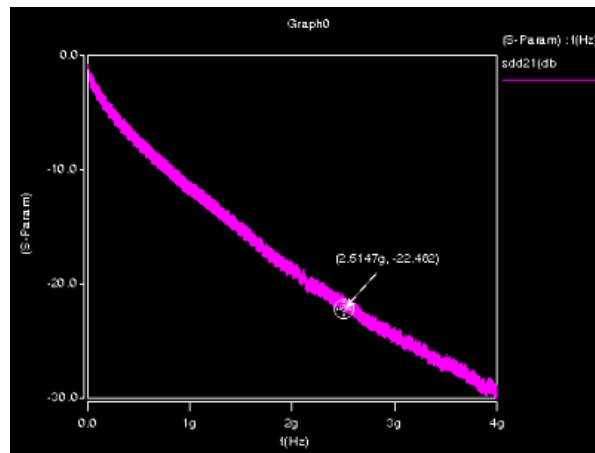
**Figure 9-6** RX Model Matching - Typical Conditions



### 9.4.3 Simulation Test Bench

A worst-case channel model is provided in the database deliverables along with a sample package model. The end-to-end channel loss of this model is  $-22$  db at  $2.5$  GHz, as shown in the following figure.

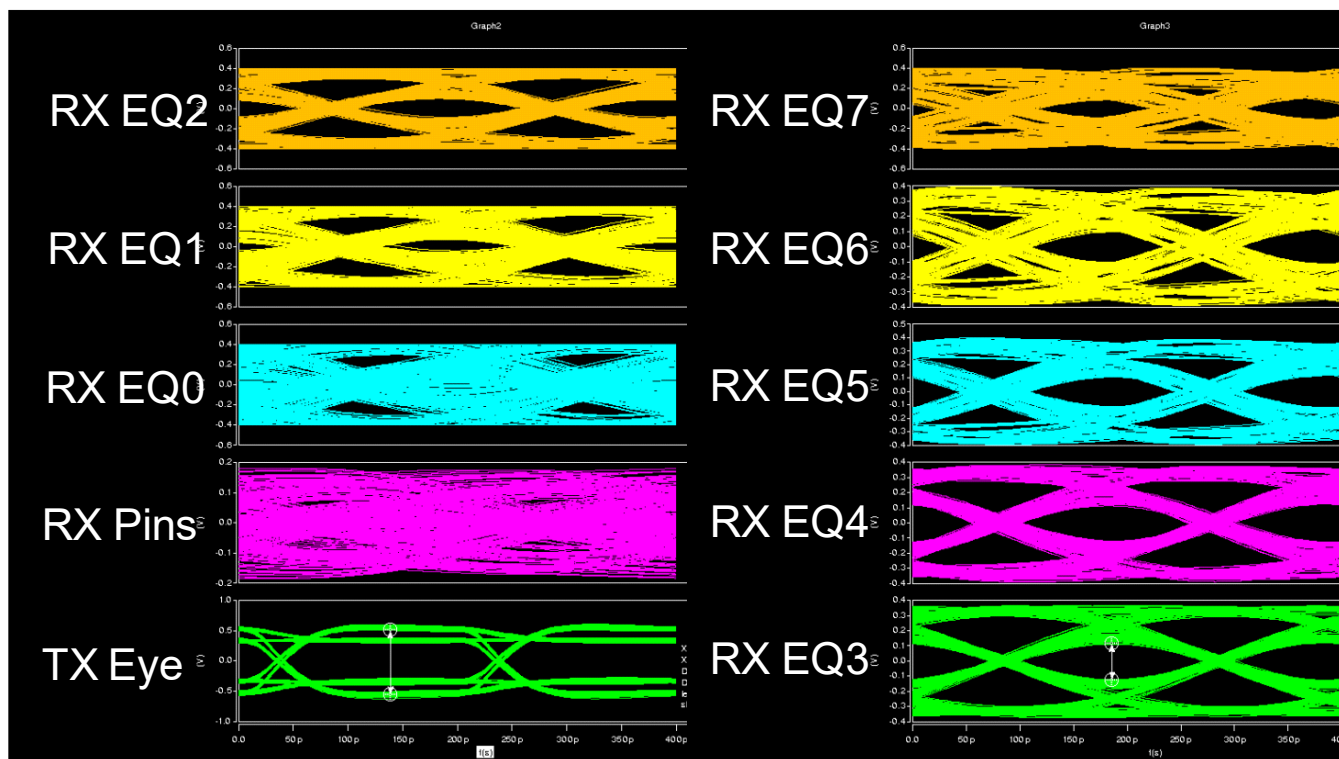
**Figure 9-7 Example Worst-Case Channel Loss**



### 9.4.4 End-to-End Simulation

The sample test bench simulates the end-to-end channel composed of TX-package-transmission line-package-RX. The RX equalizer was swept across the entire range of equalization available. With this worst-case channel model, the eye at the RX receiver pins is closed, as shown in the following figure.

**Figure 9-8 End-to-End Modeling**





## 10

## Reference Clock Implementation

This chapter includes the following section:

- “Reference Clock Requirements” on page 243

## 10.1 Reference Clock Requirements

The USB 3.0 femtoPHY is designed to handle a wide range of input clock frequencies to support both host and device applications.

The following table summarizes the requirements of the reference clock provided to the USB 3.0 femtoPHY through the `ref_pad_clk_(p,m)` and `ref_alt_clk_(p,m)` input pins to support SuperSpeed and high-speed operations.

**Table 10-1 Reference Clock Requirements**

| Symbol       | Parameter                             | Min.              | Typ. | Max.              | Unit | Conditions                                                                                               |
|--------------|---------------------------------------|-------------------|------|-------------------|------|----------------------------------------------------------------------------------------------------------|
| FREF_CLK     | Reference clock frequency             | 19.2              | -    | 200               | MHz  | -                                                                                                        |
| FREF_OFFSET  | Reference clock frequency offset      | -300              | -    | 300               | ppm  | -                                                                                                        |
| RMSJREF_CLK  | Reference clock random jitter (RMS)   | -                 | -    | 3                 | ps   | 1.5 MHz to Nyquist frequency. For example, for 100-MHz reference clock, the Nyquist frequency is 50 MHz. |
| REF_CLK_SKEW | Reference clock skew                  | -                 | -    | 200               | ps   | -                                                                                                        |
| DJREF_CLK    | Reference clock cycle-to-cycle jitter | -                 | -    | 150               | ps   | DJ over all frequency                                                                                    |
| DCREF_CLK    | Duty cycle                            | 40                | -    | 60                | %    | -                                                                                                        |
| VCMREF_CLK   | Common mode input level               | $(vp - 10\%) / 2$ | -    | $(vp + 10\%) / 2$ | V    | Differential inputs <sup>a</sup>                                                                         |

**Table 10-1 Reference Clock Requirements (Continued)**

| Symbol     | Parameter                              | Min.          | Typ. | Max.           | Unit | Conditions                       |
|------------|----------------------------------------|---------------|------|----------------|------|----------------------------------|
| VDREF_CLK  | Differential input swing               | 2 x (vp –10%) | -    | 2 x (vp + 10%) | Vpp  | Differential inputs <sup>a</sup> |
| VOLREF_CLK | Single-ended input logic: Low          | 0             | -    | 0.3            | V    | If single-ended input is used    |
| VOHREF_CLK | Single-ended input logic: High         | vp–0.3        | -    | vp             | V    | If single-ended input is used    |
| SWREF_CLK  | Input edge rate                        | 0.6           | -    | 4              | V/ns | -                                |
| R_resref   | Required external reference resistance | -             | 200  | -              | ohm  | ± 1% accuracy                    |

a.  $VDREF\_CLK / 4 + VCMREF\_CLK \leq vp + \text{diode forward biasing voltage}$  and  $VCMREF\_CLK - VDREF\_CLK / 4 \geq -\text{diode forward biasing voltage}$ .



# 11

## Electrical Specifications

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This chapter includes the following sections:

- [“Power Dissipation”](#) on page 245
- [“Power Supply Specifications”](#) on page 250
- [“Latencies, Lock Times, and Timing Information”](#) on page 251

### 11.1 Power Dissipation

This section provides power dissipation values for the USB 3.0 femtoPHY.

Typical values have been obtained through simulation at room temperature, using typical process corner, nominal analog and digital power supply levels. Worst-case values have been obtained through simulation at hot temperature, using FF process corner and high power supplies as specified in [Table 6-1](#) on page 190. Power dissipation values for SuperSpeed operation and HS/FS/LS operation are provided separately based on the different operating modes of the USB 3.0 femtoPHY.

The following table provides estimated power consumption values for the USB 3.0 femtoPHY while operating in SuperSpeed operation and while the HS/FS/LS blocks are in their lowest power state (Suspend).

**Table 11-1 USB 3.0 femtoPHY Power Dissipation: SuperSpeed 5-Gbps Operation Only**

| Mode                    | Current from vp<br>(mA) |        | Current from vptx0<br>(mA) |        | Current from vph<br>(mA) |        | Current from DVDD<br>(mA) |       | Current from VDDH0<br>(mA) |       | Current from VDD330<br>(mA) |       | Total Current<br>(mA) |        | Total Power<br>(mW) |         |
|-------------------------|-------------------------|--------|----------------------------|--------|--------------------------|--------|---------------------------|-------|----------------------------|-------|-----------------------------|-------|-----------------------|--------|---------------------|---------|
|                         | Typ                     | WC     | Typ                        | WC     | Typ                      | WC     | Typ                       | WC    | Typ                        | WC    | Typ                         | WC    | Typ                   | WC     | Typ                 | WC      |
| U0                      | 21.520                  | 38.840 | 10.000                     | 13.140 | 21.280                   | 25.110 | 0.036                     | 5.488 | 0.083                      | 0.121 | 0.028                       | 0.048 | 52.960                | 82.740 | 95.850              | 142.300 |
| U1                      | 10.720                  | 26.260 | 1.417                      | 2.332  | 11.490                   | 15.380 | 0.036                     | 5.488 | 0.084                      | 0.121 | 0.028                       | 0.048 | 23.780                | 49.630 | 48.040              | 86.400  |
| U2                      | 9.925                   | 24.420 | 1.416                      | 2.330  | 10.210                   | 13.370 | 0.036                     | 5.488 | 0.083                      | 0.121 | 0.028                       | 0.048 | 21.700                | 45.770 | 43.180              | 77.460  |
| U3                      | 1.457                   | 13.550 | 0.407                      | 0.930  | 1.985                    | 2.928  | 0.036                     | 5.488 | 0.085                      | 0.100 | 0.028                       | 0.056 | 3.999                 | 23.050 | 8.444               | 28.730  |
| Power-down <sup>a</sup> | 0.093                   | 11.880 | 0.007                      | 0.390  | 0.015                    | 0.047  | 0.037                     | 5.469 | 0.084                      | 0.139 | 0.028                       | 0.051 | 0.263                 | 17.980 | 0.528               | 16.430  |

a. HS/FS/LS blocks are in Power-down mode instead of Suspend mode.

The following table provides estimated power consumption values for the USB 3.0 femtoPHY while operating in HS/FS/LS operations and while the SuperSpeed blocks are in their lowest power state (U3).

**Table 11-2 USB 3.0 femtoPHY Power Dissipation: HS/FS/LS Operations Only**

| Mode                                                      | Current from vp<br>(mA) |        | Current from vptx0<br>(mA) |       | Current from vph<br>(mA) |        | Current from DVDD<br>(mA) |       | Current from VDDH0<br>(mA) |        | Current from VDD330<br>(mA) |        | Total Current<br>(mA) |        | Total Power<br>(mW) |         |
|-----------------------------------------------------------|-------------------------|--------|----------------------------|-------|--------------------------|--------|---------------------------|-------|----------------------------|--------|-----------------------------|--------|-----------------------|--------|---------------------|---------|
|                                                           | Typ                     | WC     | Typ                        | WC    | Typ                      | WC     | Typ                       | WC    | Typ                        | WC     | Typ                         | WC     | Typ                   | WC     | Typ                 | WC      |
| HS Transmit<br>Max.<br>transition<br>density <sup>a</sup> | 3.375                   | 16.900 | 0.004                      | 0.370 | 10.420                   | 14.890 | 2.866                     | 9.081 | 12.960                     | 13.930 | 2.597                       | 3.078  | 32.220                | 58.260 | 90.720              | 134.400 |
| HS Transmit<br>Min. transition<br>density <sup>b</sup>    | 3.354                   | 16.910 | 0.004                      | 0.370 | 10.600                   | 14.660 | 2.602                     | 8.586 | 11.560                     | 11.820 | 2.243                       | 2.533  | 30.360                | 54.880 | 85.280              | 124.200 |
| HS Idle <sup>c</sup>                                      | 3.385                   | 16.960 | 0.004                      | 0.370 | 10.490                   | 14.810 | 3.535                     | 9.862 | 1.864                      | 2.174  | 2.086                       | 2.403  | 21.360                | 46.580 | 53.180              | 93.590  |
| FS Transmit<br>Max.<br>transition<br>density <sup>d</sup> | 3.368                   | 16.910 | 0.004                      | 0.370 | 10.570                   | 14.700 | 2.385                     | 8.317 | 1.862                      | 2.169  | 10.960                      | 11.640 | 29.150                | 54.110 | 81.810              | 125.300 |
| LS Transmit<br>Max.<br>transition<br>density <sup>e</sup> | 2.074                   | 14.570 | 0.004                      | 0.370 | 10.310                   | 14.600 | 2.791                     | 8.642 | 1.853                      | 2.205  | 14.670                      | 15.200 | 31.700                | 55.580 | 92.430              | 136.200 |
| Power-down <sup>f</sup>                                   | 0.093                   | 11.880 | 0.007                      | 0.390 | 0.015                    | 0.047  | 0.037                     | 5.469 | 0.084                      | 0.139  | 0.028                       | 0.051  | 0.263                 | 17.980 | 0.528               | 16.430  |
| Suspend <sup>g</sup>                                      | 1.302                   | 14.220 | 0.004                      | 0.371 | 1.316                    | 2.045  | 0.037                     | 5.493 | 0.089                      | 0.160  | 0.028                       | 0.045  | 2.776                 | 22.330 | 5.804               | 25.790  |
| Sleep <sup>h</sup>                                        | 1.414                   | 14.400 | 0.004                      | 0.370 | 1.320                    | 2.040  | 0.038                     | 5.491 | 0.089                      | 0.160  | 0.028                       | 0.045  | 2.893                 | 22.500 | 5.908               | 25.920  |

**Table 11-2 USB 3.0 femtoPHY Power Dissipation: HS/FS/LS Operations Only (Continued)**

| Mode                          | Current from vp<br>(mA) |        | Current from vptx0<br>(mA) |       | Current from vph<br>(mA) |       | Current from DVDD<br>(mA) |       | Current from VDDH0<br>(mA) |       | Current from VDD330<br>(mA) |       | Total Current<br>(mA) |        | Total Power<br>(mW) |        |
|-------------------------------|-------------------------|--------|----------------------------|-------|--------------------------|-------|---------------------------|-------|----------------------------|-------|-----------------------------|-------|-----------------------|--------|---------------------|--------|
|                               | Typ                     | WC     | Typ                        | WC    | Typ                      | WC    | Typ                       | WC    | Typ                        | WC    | Typ                         | WC    | Typ                   | WC     | Typ                 | WC     |
| Battery Charging <sup>i</sup> | 1.414                   | 14.430 | 0.004                      | 0.370 | 1.395                    | 2.151 | 1.538                     | 7.444 | 1.239                      | 1.485 | 1.970                       | 2.344 | 7.561                 | 28.230 | 17.560              | 40.790 |

- Packet transmission by one transceiver operating in Device mode while driving all 0's data (constant JKJK on DP<#>/DM<#>). Loading of 10 pF. Transfers do not include any interpacket delay.
- Packet transmission by one transceiver operating in Device mode while driving all 1's data (alternating 7-bit strings of J, then K on DP<#>/DM<#>). Loading of 10 pF. Transfers do not include any interpacket delay.
- HS Receive mode with no traffic on the line.
- Packet transmission by one transceiver operating in Device mode while driving all 0's data (constant JKJK on DP<#>/DM<#>). Loading of 50 pF. Transfers do not include any interpacket delay.
- Packet transmission by one transceiver operating in Host mode while driving all 0's data (constant JKJK on DP<#>/DM<#>). Loading of 600 pF. Transfers do not include any interpacket delay.
- Power-down when operating in Device mode (while test\_powerdown\_hsp is asserted).
- Suspend when operating in Device mode with no far-side host termination on DP<#>/DM<#> during measurements. Measurements taken when COMMONONN is deasserted.
- Sleep mode when operating in Device mode with no far-side host termination on DP<#>/DM<#> during measurements.
- Battery charging while the high-speed blocks are in Suspend, non-driving mode and the PHY is operating as a portable device connected to a dedicated charger.

The following table provides estimated power consumption values for the USB 3.0 femtoPHY in a hub application where SuperSpeed and HS/FS/LS data are transmitting simultaneously.

**Table 11-3 USB 3.0 femtoPHY Power Dissipation With Voltage Regulators: SuperSpeed Plus HS/FS/LS Operations**

| Mode                                                   | Current from vp<br>(mA) |        | Current from vptx0<br>(mA) |        | Current from vph<br>(mA) |        | Current from DVDD<br>(mA) |       | Current from VDDH0<br>(mA) |        | Current from VDD330<br>(mA) |        | Total Current<br>(mA) |         | Total Power<br>(mW) |         |
|--------------------------------------------------------|-------------------------|--------|----------------------------|--------|--------------------------|--------|---------------------------|-------|----------------------------|--------|-----------------------------|--------|-----------------------|---------|---------------------|---------|
|                                                        | Typ                     | WC     | Typ                        | WC     | Typ                      | WC     | Typ                       | WC    | Typ                        | WC     | Typ                         | WC     | Typ                   | WC      | Typ                 | WC      |
| HS Transmit<br>Max. transition<br>density <sup>a</sup> | 23.890                  | 41.730 | 15.050                     | 19.410 | 24.180                   | 28.290 | 3.128                     | 9.224 | 13.070                     | 13.850 | 12.589                      | 12.971 | 81.910                | 115.500 | 165.100             | 221.100 |
| FS Transmit<br>Max. transition<br>density <sup>b</sup> | 23.720                  | 41.780 | 15.080                     | 19.450 | 23.990                   | 28.490 | 2.539                     | 8.445 | 1.851                      | 2.169  | 11.190                      | 11.750 | 78.360                | 112.100 | 155.300             | 214.500 |
| LS Transmit<br>Max. transition<br>density <sup>c</sup> | 23.720                  | 41.780 | 15.080                     | 19.450 | 23.990                   | 28.490 | 2.791                     | 8.642 | 1.853                      | 2.205  | 14.670                      | 15.200 | 82.090                | 115.800 | 167.000             | 227.300 |

- a. Packet transmission by one transceiver operating in Device mode while driving all 0's data (constant JKJK on DP<#>/DM<#>). Loading of 10 pF. Transfers do not include any interpacket delay.
- b. Packet transmission by one transceiver operating in Device mode while driving all 0's data (constant JKJK on DP<#>/DM<#>). Loading of 50 pF. Transfers do not include any interpacket delay.
- c. Packet transmission by one transceiver operating in Host mode while driving all 0's data (constant JKJK on DP<#>/DM<#>). Loading of 600 pF. Transfers do not include any interpacket delay.

## 11.2 Power Supply Specifications

The following table provides the worst-case maximum current specifications that the USB 3.0 femtoPHY power supplies must be able to handle.

**Table 11-4 Worst-Case Maximum Current**

| vph<#> | vp    | vptx<#> | VDD33<#> | VDDH<#> | DVDD  | Unit |
|--------|-------|---------|----------|---------|-------|------|
| 28.49  | 41.78 | 19.45   | 15.2     | 13.85   | 9.224 | mA   |

The following table provides the fastest ramp-up times allowed for the USB 3.0 femtoPHY power supplies without falsely triggering the ESD circuitry.

**Table 11-5 Maximum Power Supply Slew Rates**

| vph<#> | vp  | vptx<#> | VDD33<#> | VDDH<#> | DVDD | Unit |
|--------|-----|---------|----------|---------|------|------|
| 0.1    | 0.1 | 0.1     | 0.1      | 0.1     | 0.1  | V/μs |

## 11.3 Latencies, Lock Times, and Timing Information

The following table provides timing values for USB 3.0 femtoPHY latencies and transitions.

**Table 11-6 Latencies, Lock Times, and Timing Information**

| Parameter                                              | Description                                                                                                               | PIPE Width                | Value                                                                                                                                                 |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transmit latency                                       | Measured from when the parallel data is sampled at the PIPE interface to when the first word bit is serially transmitted. | 8-bit<br>16-bit<br>32-bit | 28.4 ns<br>24.4 ns<br>32.4 ns                                                                                                                         |
| Receive latency<br>E-Store nominally<br>Half-Full mode | Measured from the first bit of a word being received to when that word is clocked out of the receiver PIPE interface.     | 8-bit<br>16-bit<br>32-bit | 66.4 ns Typical, (min = 50.4 ns, max = 82.4 ns)<br>64.4 ns Typical, (min = 48.4 ns, max = 80.4 ns)<br>72.4 ns Typical, (min = 56.4 ns, max = 88.4 ns) |
| Receive latency<br>E-Store nominally<br>Empty mode     | Measured from the first bit of a word being received to when that word is clocked out of the receiver PIPE interface.     | 8-bit<br>16-bit<br>32-bit | 50.4 ns Typical, (min = 46.4 ns, max = 66.4 ns)<br>48.4 ns Typical, (min = 44.4 ns, max = 64.4 ns)<br>56.4 ns Typical, (min = 52.4 ns, max = 72.4 ns) |
| P1 to P0 transition                                    | Worst-case time from pipeP_powerdown change to P0 until PHY acknowledgment (phy_status goes low)                          |                           | < 40 ns                                                                                                                                               |
| P2 to P0 transition                                    | Worst-case time from pipeP_powerdown change to P0 until PHY acknowledgment (phy_status goes low)                          |                           | 0.93 us (100MHz Ref Clk), 1.46 us (20MHz Ref Clk)                                                                                                     |
| P3 to P0 transition                                    | Worst-case time from pipeP_powerdown change to P0 until PHY acknowledgment (phy_status goes low)                          |                           | 260 us (100MHz Ref Clk), 345 us (19.2MHz Ref Clk)                                                                                                     |
| Reset-to-ready time                                    | Worst-case time from pipeP_reset_n de-assertion to PHY acknowledgment (phy_status goes low)                               |                           | 260 us (100MHz Ref Clk), 345 us (19.2MHz Ref Clk)                                                                                                     |





# 12

## Registers

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This chapter describes the USB 3.0 femtoPHY registers and is organized as follows:

- [“Memory Access Attributes”](#) on page 254
- [“USB3\\_mem\\_map Register Descriptions”](#) on page 284

## 12.1 Memory Access Attributes

The Memory Access attribute is defined as <ReadBehavior>/<WriteBehavior> which are defined in the following table.

**Table 12-1 Possible Read and Write Behaviors**

| Read (or Write) Behavior    | Description                                          |
|-----------------------------|------------------------------------------------------|
| RC                          | A read clears this register field.                   |
| RS                          | A read sets this register field.                     |
| RM                          | A read modifies the contents of this register field. |
| Wo                          | You can only write to this register once field.      |
| W1C                         | A write of 1 clears this register field.             |
| W1S                         | A write of 1 sets this register field.               |
| W1T                         | A write of 1 toggles this register field.            |
| W0C                         | A write of 0 clears this register field.             |
| W0S                         | A write of 0 sets this register field.               |
| W0T                         | A write of 0 toggles this register field.            |
| WC                          | Any write clears this register field.                |
| WS                          | Any write sets this register field.                  |
| WM                          | Any write toggles this register field.               |
| no Read Behavior attribute  | You cannot read this register. It is Write-Only.     |
| no Write Behavior attribute | You cannot write to this register. It is Read-Only.  |

**Table 12-2 Memory Access Examples**

| Memory Access | Description                                                      |
|---------------|------------------------------------------------------------------|
| R             | Read-only register field.                                        |
| W             | Write-only register field.                                       |
| R/W           | Read/write register field.                                       |
| R/W1C         | You can read this register field. Writing 1 clears it.           |
| RC/W1C        | Reading this register field clears it. Writing 1 clears it.      |
| R/Wo          | You can read this register field. You can only write to it once. |

## Special Optional Attributes

Some register fields might use the following optional attributes.

**Table 12-3 Optional Attributes**

| Attribute  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volatile   | As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the core updates the register field contents. |
| Testable   | As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.                                                                                                 |
| Reset Mask | As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.                                                                                                                                                                                                                                                                                                                                               |
| * Varies   | Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.                                                                                                                                                                                                                        |

## 12.2 USB3\_mem\_map Register Descriptions

Lane register addresses are numbered: 1NXX

Where:

- N = [lane] where the value represents the lane number.
- XX = Address

For example, lane-specific registers are addressed as follows:

- Lane0: 0x1000 through 0x1030
- Lane1: 0x1100 through 0x1130
- Lane2: 0x1200 through 0x1230
- LaneX: 0x9000 through 0x9030 (broadcast to all the lanes; for information about broadcasting, see [“Broadcast Addressing”](#))

### Broadcast Addressing

In addition to supporting writes to a single register, the USB 3.0 femtoPHY's register controller also supports broadcast writes. Broadcast writes make it easy (a single register write operation) to write the same value into N instantiations of the same register that exist in N lanes. To perform a broadcast write to all lanes, simply replace the upper byte of the address with 0x90. Note that broadcast reading of registers is not supported.

### USB3\_mem\_map Registers

USB3\_mem\_map Registers. Follow the link for the register to see a detailed description of the register.

All control registers are reset with the assertion of phy\_reset.

**Table 12-4 Registers for Address Block: USB3\_mem\_map**

| Register                                         | Offset | Description                                                            |
|--------------------------------------------------|--------|------------------------------------------------------------------------|
| SUP_DIG_IDCODE_LO on <a href="#">page 263</a>    | 0x0    | <b>Description:</b> Low 16 bits of IDCODE                              |
| SUP_DIG_IDCODE_HI on <a href="#">page 264</a>    | 0x1    | <b>Description:</b> High 16 bits of IDCODE                             |
| SUP_DIG_DEBUG on <a href="#">page 265</a>        | 0x2    | <b>Description:</b> Debug controls                                     |
| SUP_DIG_RTUNE_DEBUG on <a href="#">page 266</a>  | 0x3    | <b>Description:</b> Resistor tuning debug controls                     |
| SUP_DIG_RTUNE_STAT on <a href="#">page 268</a>   | 0x4    | <b>Description:</b> Resistor tuning register status                    |
| SUP_DIG_SS_PHASE on <a href="#">page 269</a>     | 0x5    | <b>Description:</b> Current MPLL phase selector value                  |
| SUP_DIG_SS_FREQ on <a href="#">page 271</a>      | 0x6    | <b>Description:</b> Frequency Control for Spread Spectrum              |
| SUP_DIG_HSP_SS_PHASE on <a href="#">page 272</a> | 0x7    | <b>Description:</b> Current HSP_PMIK SSC phase sel value               |
| SUP_DIG_MPLL_CNT_CTL on <a href="#">page 273</a> | 0x8    | <b>Description:</b> Control for fixed or variable MPLL Count Selection |

**Table 12-4 Registers for Address Block: USB3\_mem\_map (Continued)**

| Register                                              | Offset | Description                                                                                                   |
|-------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------|
| SUP_DIG_CR_PARA_ALU_CTL on <a href="#">page 274</a>   | 0x9    | <b>Description:</b> Control for CR Parallel ALU access                                                        |
| SUP_DIG_CR_PARA_ALU_FLAGS on <a href="#">page 275</a> | 0xa    | <b>Description:</b> Flags from the ALU when CR para access is enabled.                                        |
| SUP_DIG_ATEOVRD on <a href="#">page 276</a>           | 0x10   | <b>Description:</b> Override values for incoming controls from ASIC                                           |
| SUP_DIG_MPLL_OVRD_IN_LO on <a href="#">page 277</a>   | 0x11   | <b>Description:</b> Override values for incoming RTUNE and MPLL controls from ASIC                            |
| SUP_DIG_MPLL_OVRD_IN_HI on <a href="#">page 279</a>   | 0x12   | <b>Description:</b> Override values for incoming VBOOST LVL and MPLL controls from ASIC                       |
| SUP_DIG_SSC_OVRD_IN on <a href="#">page 281</a>       | 0x13   | <b>Description:</b> Override values for incoming SSC controls from ASIC                                       |
| SUP_DIG_BS_OVRD_IN on <a href="#">page 282</a>        | 0x14   | <b>Description:</b> Override values for incoming BS controls from ASIC                                        |
| SUP_DIG_LEVEL_OVRD_IN on <a href="#">page 284</a>     | 0x15   | <b>Description:</b> Override values for acjt_level and los_level                                              |
| SUP_DIG_SUP_OVRD_OUT on <a href="#">page 285</a>      | 0x16   | <b>Description:</b> Override values for support block                                                         |
| SUP_DIG_MPLL_ASIC_IN on <a href="#">page 287</a>      | 0x17   | <b>Description:</b> Current values for incoming MPLL controls from ASIC                                       |
| SUP_DIG_BS_ASIC_IN on <a href="#">page 289</a>        | 0x18   | <b>Description:</b> Current values for incoming boundary scan controls from ASIC                              |
| SUP_DIG_LEVEL_ASIC_IN on <a href="#">page 292</a>     | 0x19   | <b>Description:</b> Current values for incoming los/acjt level, MPLL half rate mode and vreg bypass from ASIC |
| SUP_DIG_SSC_ASIC_IN on <a href="#">page 294</a>       | 0x1a   | <b>Description:</b> Current values for incoming SSC controls from ASIC                                        |
| SUP_DIG_SUP_ASIC_OUT on <a href="#">page 295</a>      | 0x1b   | <b>Description:</b> Current values for outgoing support block controls to ASIC                                |
| SUP_DIG_ATEOVRD_STATUS on <a href="#">page 297</a>    | 0x1c   | <b>Description:</b> USB3_SUP_DIG_ATEOVRD_STATUS                                                               |
| SUP_DIG_HSP_OVRD_IN on <a href="#">page 299</a>       | 0x1d   | <b>Description:</b> Override values for HSP                                                                   |
| SUP_DIG_HSP_ASIC_IN on <a href="#">page 300</a>       | 0x1e   | <b>Description:</b> Current values for incoming HSP & REF XO controls from ASIC                               |
| SUP_DIG_SCOPE_ENABLES on <a href="#">page 301</a>     | 0x20   | <b>Description:</b> Scope Counter configuration                                                               |
| SUP_DIG_SCOPE_SAMPLES on <a href="#">page 302</a>     | 0x21   | <b>Description:</b> Number of samples to count                                                                |

**Table 12-4 Registers for Address Block: USB3\_mem\_map (Continued)**

| Register                                                | Offset | Description                                                  |
|---------------------------------------------------------|--------|--------------------------------------------------------------|
| SUP_DIG_SCOPE_COUNT on <a href="#">page 303</a>         | 0x22   | <b>Description:</b> Results of scope counting                |
| SUP_DIG_SCOPE_CTL on <a href="#">page 304</a>           | 0x23   | <b>Description:</b> Masked scope counting Start and Status   |
| SUP_DIG_SCOPE_MASK_000 on <a href="#">page 305</a>      | 0x24   | <b>Description:</b> Current count value for mask_000         |
| SUP_DIG_SCOPE_MASK_001 on <a href="#">page 306</a>      | 0x25   | <b>Description:</b> Current count value for mask_001         |
| SUP_DIG_SCOPE_MASK_010 on <a href="#">page 307</a>      | 0x26   | <b>Description:</b> Current count value for mask_010         |
| SUP_DIG_SCOPE_MASK_011 on <a href="#">page 308</a>      | 0x27   | <b>Description:</b> Current count value for mask_011         |
| SUP_DIG_SCOPE_MASK_100 on <a href="#">page 309</a>      | 0x28   | <b>Description:</b> Current count value for mask_100         |
| SUP_DIG_SCOPE_MASK_101 on <a href="#">page 310</a>      | 0x29   | <b>Description:</b> Current count value for mask_101         |
| SUP_DIG_SCOPE_MASK_110 on <a href="#">page 311</a>      | 0x2a   | <b>Description:</b> Current count value for mask_110         |
| SUP_DIG_SCOPE_MASK_111 on <a href="#">page 312</a>      | 0x2b   | <b>Description:</b> Current count value for mask_111         |
| SUP_ANA_MPLL_LOOP_CTL on <a href="#">page 313</a>       | 0x30   | <b>Description:</b> MPLL loop control bits                   |
| SUP_ANA_MPLL_ATB_MEAS1 on <a href="#">page 314</a>      | 0x31   | <b>Description:</b> ATB Select bits                          |
| SUP_ANA_MPLL_ATB_MEAS2 on <a href="#">page 316</a>      | 0x32   | <b>Description:</b> ATB Select bits                          |
| SUP_ANA_MPLL_OVRD on <a href="#">page 318</a>           | 0x33   | <b>Description:</b> Local overrides                          |
| SUP_ANA_RTUNE_RTUNE_CTRL on <a href="#">page 320</a>    | 0x34   | <b>Description:</b> RTUNE control bits                       |
| SUP_ANA_ATB_SWITCHYARD_CTRL on <a href="#">page 322</a> | 0x35   | <b>Description:</b> ATB_SWITCHYARD control bits              |
| SUP_ANA_SSC_CLK_CNTRL on <a href="#">page 324</a>       | 0x36   | <b>Description:</b> controls sscf reference clock rate       |
| SUP_ANA_HSP_CLK_DIG_CNTRL0 on <a href="#">page 325</a>  | 0x37   | <b>Description:</b> controls HSP phase clock, and resolution |
| SUP_ANA_HSP_CLK_DIG_CNTRL1 on <a href="#">page 327</a>  | 0x38   | <b>Description:</b> Lower and Upper Phase Increment Values   |

**Table 12-4 Registers for Address Block: USB3\_mem\_map (Continued)**

| Register                                                   | Offset | Description                                                                                                      |
|------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------|
| LANEN_DIG_TX_OVRD_IN_LO on <a href="#">page 328</a>        | 0x1N00 | <b>Description:</b> Override values for incoming TX controls from ASIC                                           |
| LANEN_DIG_TX_OVRD_IN_HI on <a href="#">page 330</a>        | 0x1N01 | <b>Description:</b> Override values for incoming TX controls from ASIC                                           |
| LANEN_DIG_TX_OVRD_DRV_LO on <a href="#">page 332</a>       | 0x1N02 | <b>Description:</b> Override values for incoming TX drive controls from ASIC                                     |
| LANEN_DIG_TX_OVRD_DRV_HI on <a href="#">page 333</a>       | 0x1N03 | <b>Description:</b> Override values for incoming TX drive controls from ASIC                                     |
| LANEN_DIG_TX_OVRD_OUT on <a href="#">page 334</a>          | 0x1N04 | <b>Description:</b> Override values for outgoing TX controls to ASIC                                             |
| LANEN_DIG_RX_OVRD_IN_LO on <a href="#">page 336</a>        | 0x1N05 | <b>Description:</b> Override values for incoming RX controls from ASIC                                           |
| LANEN_DIG_RX_OVRD_IN_HI on <a href="#">page 338</a>        | 0x1N06 | <b>Description:</b> Override values for incoming RX controls from ASIC                                           |
| LANEN_DIG_RX_OVRD_OUT on <a href="#">page 340</a>          | 0x1N07 | <b>Description:</b> Override values for outgoing RX controls to ASIC                                             |
| LANEN_DIG_TX_ASIC_IN on <a href="#">page 342</a>           | 0x1N08 | <b>Description:</b> Current values for incoming TX controls from ASIC                                            |
| LANEN_DIG_TX_ASIC_DRV_LO on <a href="#">page 345</a>       | 0x1N09 | <b>Description:</b> Current values for incoming TX level controls from ASIC                                      |
| LANEN_DIG_TX_ASIC_DRV_HI on <a href="#">page 346</a>       | 0x1N0a | <b>Description:</b> Current values for incoming TX level controls from ASIC                                      |
| LANEN_DIG_TX_ASIC_OUT on <a href="#">page 347</a>          | 0x1N0b | <b>Description:</b> Current values for outgoing TX status controls from PHY                                      |
| LANEN_DIG_RX_ASIC_IN on <a href="#">page 349</a>           | 0x1N0c | <b>Description:</b> Current values for incoming RX controls from ASIC                                            |
| LANEN_DIG_RX_ASIC_OUT on <a href="#">page 352</a>          | 0x1N0d | <b>Description:</b> Current values for outgoing RX status controls from PHY                                      |
| LANEN_DIG_TX_DEBUG on <a href="#">page 353</a>             | 0x1N10 | <b>Description:</b> Transmit debug controls                                                                      |
| LANEN_DIG_TX_CM_WAIT_TIME_OVRD on <a href="#">page 354</a> | 0x1N11 | <b>Description:</b> USB3_LANE0_DIG_TX_CM_WAIT_TIME_OVRD                                                          |
| LANEN_DIG_TX_VMD_FSM_TX_VCM_0 on <a href="#">page 355</a>  | 0x1N12 | <b>Description:</b> Calculated values for N_USE and N_TRISTATE. Also TX ANA configuration shift register output. |
| LANEN_DIG_TX_VMD_FSM_TX_VCM_1 on <a href="#">page 356</a>  | 0x1N13 | <b>Description:</b> Calculated values for N_FIXED and N_TRAILER.                                                 |

**Table 12-4 Registers for Address Block: USB3\_mem\_map (Continued)**

| Register                                                          | Offset | Description                                                                                                                |
|-------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------|
| LANEN_DIG_TX_VMD_FSM_TX_VCM_D EBUG_IN on <a href="#">page 357</a> | 0x1N14 | <b>Description:</b> Access the TX_ANA VMD Control Shift Register                                                           |
| LANEN_DIG_TX_LBERT_CTL on <a href="#">page 359</a>                | 0x1N15 | <b>Description:</b> Pattern Generator controls                                                                             |
| LANEN_DIG_RX_LBERT_CTL on <a href="#">page 360</a>                | 0x1N16 | <b>Description:</b> Pattern Matcher controls                                                                               |
| LANEN_DIG_RX_LBERT_ERR on <a href="#">page 361</a>                | 0x1N17 | <b>Description:</b> Pattern match error counter                                                                            |
| LANEN_DIG_RX_SCOPE_CTL on <a href="#">page 362</a>                | 0x1N18 | <b>Description:</b> Control bits for per-transceiver scope portion                                                         |
| LANEN_DIG_RX_SCOPE_PHASE on <a href="#">page 363</a>              | 0x1N19 | <b>Description:</b> Scope sampling phase.                                                                                  |
| LANEN_DIG_RX_DPLL_FREQ on <a href="#">page 364</a>                | 0x1N1a | <b>Description:</b> Current frequency integrator value.                                                                    |
| LANEN_DIG_RX_CDR_CTL on <a href="#">page 365</a>                  | 0x1N1b | <b>Description:</b> Control bits for receiver in recovered domain                                                          |
| LANEN_DIG_RX_CDR_CDR_FSM_DEB UG on <a href="#">page 367</a>       | 0x1N1c | <b>Description:</b> Current values for CDR and Adaptation FSM.                                                             |
| LANEN_DIG_RX_CDR_LOCK_VEC_OV RD on <a href="#">page 369</a>       | 0x1N1d | <b>Description:</b> Control bits for receiver in recovered domain                                                          |
| LANEN_DIG_RX_CDR_LOCK_VEC on <a href="#">page 371</a>             | 0x1N1e | <b>Description:</b> Current values for the locked vector..                                                                 |
| LANEN_DIG_RX_CDR_ADAP_FSM on <a href="#">page 372</a>             | 0x1N1f | <b>Description:</b> Observation of adaptation FSM register.                                                                |
| LANEN_ANA_RX_ATB0 on <a href="#">page 373</a>                     | 0x1N20 | <b>Description:</b> ATB Control Register                                                                                   |
| LANEN_ANA_RX_ATB1 on <a href="#">page 375</a>                     | 0x1N21 | <b>Description:</b> ATB and Misc Control Register                                                                          |
| LANEN_ANA_RX_ENPWR0 on <a href="#">page 377</a>                   | 0x1N22 | <b>Description:</b> RX State Control Register                                                                              |
| LANEN_ANA_RX_PMI_X_PHASE on <a href="#">page 379</a>              | 0x1N23 | <b>Description:</b> Phase select Value                                                                                     |
| LANEN_ANA_RX_ENPWR1 on <a href="#">page 380</a>                   | 0x1N24 | <b>Description:</b> Additional RX State Control Register                                                                   |
| LANEN_ANA_RX_ENPWR2 on <a href="#">page 382</a>                   | 0x1N25 | <b>Description:</b> Yet another RX State Control Register                                                                  |
| LANEN_ANA_RX_SCOPE_VDCC on <a href="#">page 384</a>               | 0x1N26 | <b>Description:</b> RX Scope controls and clock DCC control voltage probe                                                  |
| LANEN_ANA_TX_TXDRV_CNTRL on <a href="#">page 386</a>              | 0x1N2b | <b>Description:</b> Registers intended for control of TX's drivers states i.e pull_up , pull_dn , vcm_hold etc.Register... |



**Table 12-4 Registers for Address Block: USB3\_mem\_map (Continued)**

| Register                                                   | Offset | Description                                                                                                                   |
|------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------|
| LANEN_ANA_TX_POWER_CTL on <a href="#">page 388</a>         | 0x1N2c | <b>Description:</b> Registers intended for control of TX's sub cells states such as pwrn , divider clk enb , general...       |
| LANEN_ANA_TX_ALT_BLOCK on <a href="#">page 390</a>         | 0x1N2d | <b>Description:</b> Registers intended for control of TX's output states i.e. data transmission mode , LFPS mode , and...     |
| LANEN_ANA_TX_ALT_AND_LOOPBACK on <a href="#">page 392</a>  | 0x1N2e | <b>Description:</b> Registers intended for control of loop back mode , analog test bus enb from various TX's sub cells...     |
| LANEN_ANA_TX_TX_ATB_REG on <a href="#">page 394</a>        | 0x1N2f | <b>Description:</b> Registers intended for analog test bus enabling from various TX's sub cells. Register 7                   |
| LANEN_ANA_TX_TX_BOOST on <a href="#">page 396</a>          | 0x1N30 | <b>Description:</b> Registers intended for control of TX's drivers boost mode , tx_vreg output decap enabling etc.Register... |
| LANEN_DIG_RX_CDR_FSM_INIT_CTL on <a href="#">page 398</a>  | 0x1N40 | <b>Description:</b> Control bits for CDR init phase portion                                                                   |
| LANEN_DIG_RX_CDR_FSM_TRACK_CTL on <a href="#">page 399</a> | 0x1N41 | <b>Description:</b> Control bits for CDR track phase portion                                                                  |
| LANEN_DIG_RX_CDR_FSM_LOCK_CTL on <a href="#">page 400</a>  | 0x1N42 | <b>Description:</b> Control bits for CDR lock phase portion                                                                   |
| PORTN_STAT_RO_0 on <a href="#">page 401</a>                | 0x2000 | <b>Description:</b> Status Register 0                                                                                         |
| PORTN_STAT_RO_1 on <a href="#">page 402</a>                | 0x2001 | <b>Description:</b> Status Register 1                                                                                         |
| PORTN_STAT_RO_2 on <a href="#">page 403</a>                | 0x2002 | <b>Description:</b> Status Register 2                                                                                         |
| PORTN_STAT_RO_3 on <a href="#">page 404</a>                | 0x2003 | <b>Description:</b> Status Register 3                                                                                         |
| PORTN_STAT_RO_4 on <a href="#">page 405</a>                | 0x2004 | <b>Description:</b> Status Register 4                                                                                         |
| PORTN_STAT_RO_5 on <a href="#">page 406</a>                | 0x2005 | <b>Description:</b> Status Register 5                                                                                         |
| PORTN_STAT_RO_6 on <a href="#">page 407</a>                | 0x2006 | <b>Description:</b> Status Register 6                                                                                         |
| PORTN_STAT_RO_7 on <a href="#">page 408</a>                | 0x2007 | <b>Description:</b> Status Register 7                                                                                         |
| PORTN_STAT_RO_8 on <a href="#">page 409</a>                | 0x2008 | <b>Description:</b> Status Register 8                                                                                         |
| PORTN_STAT_RO_9 on <a href="#">page 410</a>                | 0x2009 | <b>Description:</b> Status Register 9                                                                                         |
| PORTN_STAT_RO_10 on <a href="#">page 411</a>               | 0x200a | <b>Description:</b> Status Register 10                                                                                        |
| PORTN_STAT_RO_11 on <a href="#">page 412</a>               | 0x200b | <b>Description:</b> Status Register 11                                                                                        |
| PORTN_STAT_RO_12 on <a href="#">page 413</a>               | 0x200c | <b>Description:</b> Status Register 12                                                                                        |
| PORTN_STAT_RO_13 on <a href="#">page 414</a>               | 0x200d | <b>Description:</b> Status Register 13                                                                                        |
| PORTN_STAT_RO_14 on <a href="#">page 415</a>               | 0x200e | <b>Description:</b> Status Register 14                                                                                        |
| PORTN_STAT_RO_15 on <a href="#">page 416</a>               | 0x200f | <b>Description:</b> Status Register 15                                                                                        |

**Table 12-4 Registers for Address Block: USB3\_mem\_map (Continued)**

| Register                                     | Offset | Description                              |
|----------------------------------------------|--------|------------------------------------------|
| PORTN_OVRD_RW_0 on <a href="#">page 417</a>  | 0x2010 | <b>Description:</b> Override Register 0  |
| PORTN_OVRD_RW_1 on <a href="#">page 418</a>  | 0x2011 | <b>Description:</b> Override Register 1  |
| PORTN_OVRD_RW_2 on <a href="#">page 420</a>  | 0x2012 | <b>Description:</b> Override Register 2  |
| PORTN_OVRD_RW_3 on <a href="#">page 422</a>  | 0x2013 | <b>Description:</b> Override Register 3  |
| PORTN_OVRD_RW_4 on <a href="#">page 424</a>  | 0x2014 | <b>Description:</b> Override Register 4  |
| PORTN_OVRD_RW_5 on <a href="#">page 426</a>  | 0x2015 | <b>Description:</b> Override Register 5  |
| PORTN_OVRD_RW_6 on <a href="#">page 427</a>  | 0x2016 | <b>Description:</b> Override Register 6  |
| PORTN_OVRD_RW_7 on <a href="#">page 428</a>  | 0x2017 | <b>Description:</b> Override Register 7  |
| PORTN_OVRD_RW_8 on <a href="#">page 429</a>  | 0x2018 | <b>Description:</b> Override Register 8  |
| PORTN_OVRD_RW_9 on <a href="#">page 431</a>  | 0x2019 | <b>Description:</b> Override Register 9  |
| PORTN_OVRD_RW_10 on <a href="#">page 433</a> | 0x201a | <b>Description:</b> Override Register 10 |
| PORTN_OVRD_RW_11 on <a href="#">page 435</a> | 0x201b | <b>Description:</b> Override Register 11 |
| PORTN_OVRD_RW_12 on <a href="#">page 437</a> | 0x201c | <b>Description:</b> Override Register 12 |
| PORTN_OVRD_RW_13 on <a href="#">page 439</a> | 0x201d | <b>Description:</b> Override Register 13 |
| PORTN_OVRD_RW_14 on <a href="#">page 440</a> | 0x201e | <b>Description:</b> Override Register 14 |
| PORTN_OVRD_RW_15 on <a href="#">page 442</a> | 0x201f | <b>Description:</b> Override Register 15 |

### 12.2.1 SUP\_DIG\_IDCODE\_LO

- **Description:** Low 16 bits of IDCODE
- **Size:** 16 bits
- **Offset:** 0x0
- **Exists:** Always

**Table 12-5 Fields for Register: SUP\_DIG\_IDCODE\_LO**

| Bits | Name | Memory Access | Description                                                                            |
|------|------|---------------|----------------------------------------------------------------------------------------|
| 15:0 | data | R             | <b>Value After Reset:</b> 0x74cd<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff |

### 12.2.2 SUP\_DIG\_IDCODE\_HI

- **Description:** High 16 bits of IDCODE
- **Size:** 16 bits
- **Offset:** 0x1
- **Exists:** Always

**Table 12-6** Fields for Register: SUP\_DIG\_IDCODE\_HI

| Bits | Name | Memory Access | Description                                                                            |
|------|------|---------------|----------------------------------------------------------------------------------------|
| 15:0 | data | R             | <b>Value After Reset:</b> 0xa22d<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff |

### 12.2.3 SUP\_DIG\_DEBUG

- **Description:** Debug controls
- **Size:** 16 bits
- **Offset:** 0x2
- **Exists:** Always

**Table 12-7 Fields for Register: SUP\_DIG\_DEBUG**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                         |
|------|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:7 | RESERVED_15_7 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                       |
| 6:5  | DTB_SEL       | R/W           | <b>Description:</b> The lane DTB's are OR'd together with the support DTB signals selected with the below encodings 0 - none 1 - reset_ctl DTB output 2 - scope DTB output 3 - rtune DTB output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |
| 4:0  | TX_VREF_SEL   | R/W           | <b>Value After Reset:</b> 0xa<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1f                                                                                                                                                                                                   |

## 12.2.4 SUP\_DIG\_RTUNE\_DEBUG

- **Description:** Resistor tuning debug controls
- **Size:** 16 bits
- **Offset:** 0x3
- **Exists:** Always

**Table 12-8 Fields for Register: SUP\_DIG\_RTUNE\_DEBUG**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                         |
|------|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RESERVED_15_15 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                                                       |
| 14:5 | VALUE          | R/W           | <b>Description:</b> Value to use when triggering SET_VAL field only the 6 LSB's are used when setting rx or tx cal values (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3ff                                                                                                                                                     |
| 4:3  | TYPE           | R/W           | <b>Description:</b> Type of manual tuning or register read/write to execute 0 - ADC, or read/write rt_value 1 - Rx tune, or read/write rx_cal_val (only 6 bits) 2 - Tx tune, or read/write tx_cal_val (only 6 bits) 3 - Resref detect (no affect when triggering SET_VAL field) (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |
| 2    | SET_VAL        | R/W           | <b>Description:</b> Set value Write to a 1 to manually write the register specified by the TYPE field to the value in the VALUE field (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                           |
| 1    | MAN_TUNE       | R/W           | <b>Description:</b> Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know the state of the rtune circuit. (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                             |

**Table 12-8 Fields for Register: SUP\_DIG\_RTUNE\_DEBUG (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                                                                             |
|------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | FLIP_COMP | R/W           | <b>Description:</b> Invert analog comparator output (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.5 SUP\_DIG\_RTUNE\_STAT

- **Description:** Resistor tuning register status
- **Size:** 16 bits
- **Offset:** 0x4
- **Exists:** Always

**Table 12-9 Fields for Register: SUP\_DIG\_RTUNE\_STAT**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                    |
|-------|----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | RESERVED_15_10 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                  |
| 9:0   | STAT           | R             | <b>Description:</b> Current value of the register specified by the TYPE field in the debug control register. Set the TYPE field in the DEBUG register, and then do a read of this register. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |



## 12.2.6 SUP\_DIG\_SS\_PHASE

- **Description:** Current MPLL phase selector value
- **Size:** 16 bits
- **Offset:** 0x5
- **Exists:** Always

**Table 12-10 Fields for Register: SUP\_DIG\_SS\_PHASE**

| Bits  | Name        | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                      |
|-------|-------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | REF_SEL_DIV | R/W           | <b>Description:</b> Reference Divisor to get better resolution and lower PPM errors when synthesizing from a non-integer reference clock. NOTE: Default value is 4. One is added in hardware to give a range of 1 to 8. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x4<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7<br><b>Volatile:</b> true |
| 12    | PH_SEL      | R/W           | <b>Description:</b> Phase Mixer Clock select. 1: is pmix. 0 is VCO. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true                                                                                                                                                     |
| 11    | ZERO_FREQ   | R/W           | <b>Description:</b> Zero frequency register. NOTES: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true                                                                                                                               |
| 10:2  | VAL         | R/W           | <b>Description:</b> Phase value from zero reference (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1ff<br><b>Volatile:</b> true                                                                                                                                                                   |

**Table 12-10 Fields for Register: SUP\_DIG\_SS\_PHASE (Continued)**

| Bits | Name | Memory Access | Description                                                                                                                                                                                                   |
|------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1:0  | DTHR | R/W           | <b>Description:</b> Bits below the useful resolution (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3<br><b>Volatile:</b> true |

## 12.2.7 SUP\_DIG\_SS\_FREQ

- **Description:** Frequency Control for Spread Spectrum
- **Size:** 16 bits
- **Offset:** 0x6
- **Exists:** Always

**Table 12-11 Fields for Register: SUP\_DIG\_SS\_FREQ**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                                           |
|------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RESERVED_15_15 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                         |
| 14   | FREQ_OVRD      | R/W           | <b>Description:</b> Frequency Register Override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true |
| 13:7 | FREQ_PK        | R/W           | <b>Description:</b> Peak Frequency Value (for changing direction) (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x66<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f<br><b>Volatile:</b> true                          |
| 6:0  | FREQ_CNT_INIT  | R/W           | <b>Description:</b> Initial Frequency Counter Value (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x27<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f<br><b>Volatile:</b> true                                        |

## 12.2.8 SUP\_DIG\_HSP\_SS\_PHASE

- **Description:** Current HSP\_PMI<sub>X</sub> SSC phase sel value
- **Size:** 16 bits
- **Offset:** 0x7
- **Exists:** Always

**Table 12-12 Fields for Register: SUP\_DIG\_HSP\_SS\_PHASE**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                       |
|-------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | RESERVED_15_11 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                     |
| 10:4  | VAL            | R/W           | <b>Description:</b> SSC phase value from zero reference (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f<br><b>Volatile:</b> true |
| 3:0   | DTHR           | R/W           | <b>Description:</b> Bits below the useful resolution (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf<br><b>Volatile:</b> true     |

## 12.2.9 SUP\_DIG\_MPLL\_CNT\_CTL

- **Description:** Control for fixed or variable MPLL Count Selection
- **Size:** 16 bits
- **Offset:** 0x8
- **Exists:** Always

**Table 12-13 Fields for Register: SUP\_DIG\_MPLL\_CNT\_CTL**

| Bits | Name          | Memory Access | Description                                                                                                                                                         |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:1 | RESERVED_15_1 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0       |
| 0    | VAR_CNT_EN    | R/W           | <b>Description:</b> Selects variable count when enabled (Default = fixed count)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.10 SUP\_DIG\_CR\_PARA\_ALU\_CTL

- **Description:** Control for CR Parallel ALU access
- **Size:** 16 bits
- **Offset:** 0x9
- **Exists:** Always

**Table 12-14 Fields for Register: SUP\_DIG\_CR\_PARA\_ALU\_CTL**

| Bits | Name                  | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|-----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:5 | RESERVED_15_5         | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4:1  | CR_PARA_ALU_OPCODE    | R/W           | <b>Description:</b> Opcode for the ALU when CR para access is enabled. Valid opcodes are as follows: 0 - Zero the accumulator 1 - Add control register read result to accumulator 2 - Add control register write data to accumulator 3 - Shift accumulator left by 1 bit, shifting in zero at lsb 4 - Shift accumulator right by 1 bit, shifting in zero at msb 5 - Shift accumulator right by 1bit, preserve sign bit 6 - Bit-wise invert accumulator 7 - Negate accumulator 8 - Compute absolute value of accumulator 9 - Add phase select value from READ_RESULT 14 - Trigger a new control register read using address loaded by last CRSEL access. 15 - Trigger a new control register write of accumulator value using address loaded by last CRSEL access.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf |
| 0    | CR_PARA_ALU_ACCESS_EN | R/W           | <b>Description:</b> Enable access to the ALU through the CR parallel bus.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### 12.2.11 SUP\_DIG\_CR\_PARA\_ALU\_FLAGS

- **Description:** Flags from the ALU when CR para access is enabled.
- **Size:** 16 bits
- **Offset:** 0xa
- **Exists:** Always

**Table 12-15 Fields for Register: SUP\_DIG\_CR\_PARA\_ALU\_FLAGS**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3:0  | data          | R             | <b>Description:</b> Flags from the ALU when CR para access is enabled. Valid ALU flags: 0 - Signedness used for next ADD_PHSEL command. (1=signed, 0=unsigned) 1 - Signedness for ADD_PHSEL command has been chosen by a previous ADD_PHSEL command. 2 - Error detected during most recent ALU operation. Currently only used to signal errors from ADD_PHSEL command. 3 - MSB of most recent ALU result (1=negative, 0=positive) (RU - read-only volatile)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf<br><b>Volatile:</b> true |

## 12.2.12 SUP\_DIG\_ATEOVRD

- **Description:** Override values for incoming controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x10
- **Exists:** Always

**Table 12-16 Fields for Register: SUP\_DIG\_ATEOVRD**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                |
|------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:3 | RESERVED_15_3 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                              |
| 2    | ateovrd_en    | R/W           | <b>Description:</b> Override enable for ATE signals. NOTE: This register is only accessible through cregs. Does not retain state during reset or ATPG. (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | ref_usb2_en   | R/W           | <b>Description:</b> Override value for USB HSP PHY ref_clk enable. (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                     |
| 0    | ref_clkdiv2   | R/W           | <b>Description:</b> Override value for SSP ref_clk pre-scaler. (USER2 - read-write and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                         |



### 12.2.13 SUP\_DIG\_MPLL\_OVRD\_IN\_LO

- **Description:** Override values for incoming RTUNE and MPLL controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x11
- **Exists:** Always

**Table 12-17 Fields for Register: SUP\_DIG\_MPLL\_OVRD\_IN\_LO**

| Bits | Name                 | Memory Access | Description                                                                                                                                |
|------|----------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RES_ACK_IN_OVRD      | R/W           | <b>Description:</b> Override enable for res_ack_in<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 14   | RES_ACK_IN           | R/W           | <b>Description:</b> Override value for res_ack_in<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1      |
| 13   | RES_REQ_IN_OVRD      | R/W           | <b>Description:</b> Override enable for res_req_in<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 12   | RES_REQ_IN           | R/W           | <b>Description:</b> Override value for res_req_in<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1      |
| 11   | RTUNE_REQ_OVRD       | R/W           | <b>Description:</b> Override enable for rtune_req<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1      |
| 10   | RTUNE_REQ            | R/W           | <b>Description:</b> Override value for rtune_req<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1       |
| 9    | MPLL_MULTIPLIER_OVRD | R/W           | <b>Description:</b> Override enable for mpll_multipler<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

**Table 12-17 Fields for Register: SUP\_DIG\_MPLL\_OVRD\_IN\_LO (Continued)**

| Bits | Name            | Memory Access | Description                                                                                                                                  |
|------|-----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 8:2  | MPLL_MULTIPLIER | R/W           | <b>Description:</b> Override value for mpll_multiplier<br><b>Value After Reset:</b> 0x13<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f |
| 1    | MPLL_EN_OVRD    | R/W           | <b>Description:</b> Override enable for mpll_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1          |
| 0    | MPLL_EN         | R/W           | <b>Description:</b> Override value for mpll_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1           |

### 12.2.14 SUP\_DIG\_MPLL\_OVRD\_IN\_HI

- **Description:** Override values for incoming VBOOST LVL and MPLL controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x12
- **Exists:** Always

**Table 12-18 Fields for Register: SUP\_DIG\_MPLL\_OVRD\_IN\_HI**

| Bits  | Name                    | Memory Access | Description                                                                                                                                                                                                                                                             |
|-------|-------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | TX_VBOOST_LVL           | R/W           | <b>Description:</b> Override value for tx_vboost_lvl. Override enabled using LEVEL_EN from LEVEL_OVRD_IN register.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7                                                                  |
| 12    | MPLL_REFSSC_CLK_EN_OVRD | R/W           | <b>Description:</b> Override enable for mpll_refssc_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                          |
| 11    | MPLL_REFSSC_CLK_EN      | R/W           | <b>Description:</b> Override value for mpll_refssc_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                           |
| 10    | MPLL_RST                | R/W           | <b>Description:</b> Reset the MPLL state machine Writing the register with this bit set will reset the MPLL power up/down FSM, regardless of the current state of the register bit.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 9     | FSEL_OVRD               | R/W           | <b>Description:</b> Override enable for fsel[2:0]<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                   |
| 8:6   | FSEL                    | R/W           | <b>Description:</b> Override value for fsel [2:0]<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7                                                                                                                                   |

**Table 12-18 Fields for Register: SUP\_DIG\_MPLL\_OVRD\_IN\_HI (Continued)**

| Bits | Name                   | Memory Access | Description                                                                                                                                   |
|------|------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | MPLL_WORD_CLK_EN_OVRD  | R/W           | <b>Description:</b> Override enable for mpll_word_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 4    | MPLL_WORD_CLK_EN       | R/W           | <b>Description:</b> Override value for mpll_word_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |
| 3    | MPLL_DWORD_CLK_EN_OVRD | R/W           | <b>Description:</b> Override enable for mpll_dword_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 2    | MPLL_DWORD_CLK_EN      | R/W           | <b>Description:</b> Override value for mpll_dword_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 1    | MPLL_QWORD_CLK_EN_OVRD | R/W           | <b>Description:</b> Override enable for mpll_qword_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | MPLL_QWORD_CLK_EN      | R/W           | <b>Description:</b> Override value for mpll_qword_clk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |

## 12.2.15 SUP\_DIG\_SSC\_OVRD\_IN

- **Description:** Override values for incoming SSC controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x13
- **Exists:** Always

**Table 12-19 Fields for Register: SUP\_DIG\_SSC\_OVRD\_IN**

| Bits  | Name            | Memory Access | Description                                                                                                                                                   |
|-------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 13    | SSC_OVRD_IN_EN  | R/W           | <b>Description:</b> Override enable for Spread Spectrum generator.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1        |
| 12    | SSC_EN          | R/W           | <b>Description:</b> Override value for SSC enable.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 11:9  | SSC_RANGE       | R/W           | <b>Description:</b> Override value for SSC modulation range.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7              |
| 8:0   | SSC_REF_CLK_SEL | R/W           | <b>Description:</b> Override value for reference clock scaling<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1ff          |

## 12.2.16 SUP\_DIG\_BS\_OVRD\_IN

- **Description:** Override values for incoming BS controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x14
- **Exists:** Always

**Table 12-20 Fields for Register: SUP\_DIG\_BS\_OVRD\_IN**

| Bits  | Name           | Memory Access | Description                                                                                                                                                               |
|-------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:12 | RESERVED_15_12 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0             |
| 11    | EN             | R/W           | <b>Description:</b> Enable override values for all inputs controlled by this register<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 10    | INVERT         | R/W           | <b>Description:</b> Override value for bs_invert<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                      |
| 9     | INIT           | R/W           | <b>Description:</b> Override value for bs_init<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                        |
| 8     | HIGHZ          | R/W           | <b>Description:</b> Override value for bs_highz<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                       |
| 7     | CLAMP          | R/W           | <b>Description:</b> Override value for bs_clamp<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                       |
| 6     | EXTEST_AC      | R/W           | <b>Description:</b> Override value for bs_extest_ac<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |

**Table 12-20 Fields for Register: SUP\_DIG\_BS\_OVRD\_IN (Continued)**

| Bits | Name       | Memory Access | Description                                                                                                                              |
|------|------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | EXTEST     | R/W           | <b>Description:</b> Override value for bs_extest<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 4    | PRELOAD    | R/W           | <b>Description:</b> Override value for bs_preload<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 3    | UPDATE_DR  | R/W           | <b>Description:</b> Override value for bs_update_dr<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 2    | CAPTURE_DR | R/W           | <b>Description:</b> Override value for bs_capture_dr<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | SHIFT_DR   | R/W           | <b>Description:</b> Override value for bs_shift_dr<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |
| 0    | IN         | R/W           | <b>Description:</b> Override value for bs_in<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1         |

## 12.2.17 SUP\_DIG\_LEVEL\_OVRD\_IN

- **Description:** Override values for acjt\_level and los\_level
- **Size:** 16 bits
- **Offset:** 0x15
- **Exists:** Always

**Table 12-21 Fields for Register: SUP\_DIG\_LEVEL\_OVRD\_IN**

| Bits  | Name                | Memory Access | Description                                                                                                                                                                                                                            |
|-------|---------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | LOS_BIAS            | R/W           | <b>Description:</b> Override value for los_bias (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                        |
| 12    | MPLL_HALF_RATE_OVRD | R/W           | <b>Description:</b> Enable override value for mpll_half_rate<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                       |
| 11    | MPLL_HALF_RATE      | R/W           | <b>Description:</b> Override value for mpll_half_rate<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                              |
| 10    | LEVEL_EN            | R/W           | <b>Description:</b> Enable override values for tx_vboost_lvl/los_bias/los_level/acjt_level inputs controlled by this and MPLL_OVRD_IN_HI registers<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 9:5   | ACJT_LEVEL          | R/W           | <b>Description:</b> Override value for acjt_level<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1f                                                                                                 |
| 4:0   | LOS_LEVEL           | R/W           | <b>Description:</b> Override value for los_level<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1f                                                                                                  |



## 12.2.18 SUP\_DIG\_SUP\_OVRD\_OUT

- **Description:** Override values for support block
- **Size:** 16 bits
- **Offset:** 0x16
- **Exists:** Always

**Table 12-22 Fields for Register: SUP\_DIG\_SUP\_OVRD\_OUT**

| Bits  | Name            | Memory Access | Description                                                                                                                                                   |
|-------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | RESERVED_15_10  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 9     | MPLL_STATE_OVRD | R/W           | <b>Description:</b> Override enable for mpll_state output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                 |
| 8     | MPLL_STATE      | R/W           | <b>Description:</b> Override value for mpll_state output<br><b>Value After Reset:</b> 0x1<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                  |
| 7     | BS_OUT_OVRD     | R/W           | <b>Description:</b> Override enable for bs_out output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 6     | BS_OUT          | R/W           | <b>Description:</b> Override value for bs_out output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 5     | RTUNE_ACK_OVRD  | R/W           | <b>Description:</b> Override enable for rtune_ack output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                  |
| 4     | RTUNE_ACK       | R/W           | <b>Description:</b> Override value for rtune_ack output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                   |

**Table 12-22 Fields for Register: SUP\_DIG\_SUP\_OVRD\_OUT (Continued)**

| Bits | Name             | Memory Access | Description                                                                                                                                    |
|------|------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | RES_REQ_OUT_OVRD | R/W           | <b>Description:</b> Override enable for res_req_out output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 2    | RES_REQ_OUT      | R/W           | <b>Description:</b> Override value for res_req_out output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 1    | RES_ACK_OUT_OVRD | R/W           | <b>Description:</b> Override enable for res_ack_out output<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | RES_ACK_OUT      | R/W           | <b>Description:</b> Override value for res_ack_out output<br><b>Value After Reset:</b> 0x1<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |

## 12.2.19 SUP\_DIG\_MPLL\_ASIC\_IN

- **Description:** Current values for incoming MPLL controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x17
- **Exists:** Always

**Table 12-23 Fields for Register: SUP\_DIG\_MPLL\_ASIC\_IN**

| Bits | Name               | Memory Access | Description                                                                                                                                                                                                                   |
|------|--------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RESERVED_15_15     | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                 |
| 14   | MPLL_REFSSC_CLK_EN | R             | <b>Description:</b> Value from ASIC for mpll_refssc_clk_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 13   | MPLL_WORD_CLK_EN   | R             | <b>Description:</b> Value from ASIC for mpll_word_clk_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true   |
| 12   | MPLL_DWORD_CLK_EN  | R             | <b>Description:</b> Value from ASIC for mpll_dword_clk_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 11   | MPLL_QWORD_CLK_EN  | R             | <b>Description:</b> Value from ASIC for mpll_qword_clk_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |

**Table 12-23 Fields for Register: SUP\_DIG\_MPLL\_ASIC\_IN (Continued)**

| Bits | Name            | Memory Access | Description                                                                                                                                                                                                                |
|------|-----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10   | RES_ACK_IN      | R             | <b>Description:</b> Value from ASIC for res_ack_in (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 9    | RES_REQ_IN      | R             | <b>Description:</b> Value from ASIC for res_req_in (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 8    | RTUNE_REQ       | R             | <b>Description:</b> Value from ASIC for rtune_req (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true       |
| 7:1  | MPLL_MULTIPLIER | R             | <b>Description:</b> Value from ASIC for mpll_multiplier (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 0    | MPLL_EN         | R             | <b>Description:</b> Value from ASIC for mpll_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true         |

## 12.2.20 SUP\_DIG\_BS\_ASIC\_IN

- **Description:** Current values for incoming boundary scan controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x18
- **Exists:** Always

**Table 12-24 Fields for Register: SUP\_DIG\_BS\_ASIC\_IN**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                                                                            |
|-------|----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                          |
| 13:11 | TX_VBOOST_LVL  | R             | <b>Description:</b> Value from ASIC for tx_vboost_lvl -- Not as bs_* signal, but we're out of bits. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 10    | INVERT         | R             | <b>Description:</b> Value from ASIC for bs_invert (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                                   |
| 9     | INIT           | R             | <b>Description:</b> Value from ASIC for bs_init (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                                     |
| 8     | HIGHZ          | R             | <b>Description:</b> Value from ASIC for bs_highz (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                                    |

**Table 12-24 Fields for Register: SUP\_DIG\_BS\_ASIC\_IN (Continued)**

| Bits | Name       | Memory Access | Description                                                                                                                                                                                                              |
|------|------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | CLAMP      | R             | <b>Description:</b> Value from ASIC for bs_clamp (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 6    | EXTEST_AC  | R             | <b>Description:</b> Value from ASIC for bs_extest_ac (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 5    | EXTEST     | R             | <b>Description:</b> Value from ASIC for bs_extest (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true     |
| 4    | PRELOAD    | R             | <b>Description:</b> Value from ASIC for bs_preload (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true    |
| 3    | UPDATE_DR  | R             | <b>Description:</b> Value from ASIC for bs_update_dr (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 2    | CAPTURE_DR | R             | <b>Description:</b> Value from ASIC for bs_capture_dr (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

**Table 12-24 Fields for Register: SUP\_DIG\_BS\_ASIC\_IN (Continued)**

| Bits | Name     | Memory Access | Description                                                                                                                                                                                                            |
|------|----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | SHIFT_DR | R             | <b>Description:</b> Value from ASIC for bs_shift_dr (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 0    | IN       | R             | <b>Description:</b> Value from ASIC for bs_in (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true       |

### 12.2.21 SUP\_DIG\_LEVEL\_ASIC\_IN

- **Description:** Current values for incoming los/acjt level, MPLL half rate mode and vreg bypass from ASIC
- **Size:** 16 bits
- **Offset:** 0x19
- **Exists:** Always

**Table 12-25 Fields for Register: SUP\_DIG\_LEVEL\_ASIC\_IN**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                               |
|-------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | RESERVED_15_15 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                             |
| 14:12 | LOS_BIAS       | R             | <b>Description:</b> Value from ASIC for los_bias (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true       |
| 11    | MPLL_HALF_RATE | R             | <b>Description:</b> Value from ASIC for mpll_half_rate (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 10    | VREG_BYPASS    | R             | <b>Description:</b> Value from ASIC for vreg_bypass (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true    |
| 9:5   | ACJT_LEVEL     | R             | <b>Description:</b> Value from ASIC for acjt_level (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true     |



**Table 12-25 Fields for Register: SUP\_DIG\_LEVEL\_ASIC\_IN (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                                                                                                                          |
|------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4:0  | LOS_LEVEL | R             | <b>Description:</b> Value from ASIC for los_level (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

## 12.2.22 SUP\_DIG\_SSC\_ASIC\_IN

- **Description:** Current values for incoming SSC controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1a
- **Exists:** Always

**Table 12-26 Fields for Register: SUP\_DIG\_SSC\_ASIC\_IN**

| Bits  | Name            | Memory Access | Description                                                                                                                                                                                                                |
|-------|-----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | SS_EN           | R             | <b>Description:</b> Value from ASIC for ssc_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true          |
| 14:12 | SSC_RANGE       | R             | <b>Description:</b> Value from ASIC for ssc_range (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true       |
| 11:3  | SSC_REF_CLK_SEL | R             | <b>Description:</b> Value from ASIC for ssc_ref_clk_sel (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 2:0   | FSEL            | R             | <b>Description:</b> Value from ASIC for fsel (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true            |

### 12.2.23 SUP\_DIG\_SUP\_ASIC\_OUT

- **Description:** Current values for outgoing support block controls to ASIC
- **Size:** 16 bits
- **Offset:** 0x1b
- **Exists:** Always

**Table 12-27 Fields for Register: SUP\_DIG\_SUP\_ASIC\_OUT**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                  |
|------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:5 | RESERVED_15_5 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                |
| 4    | MPLL_STATE    | R             | <b>Description:</b> Value from PHY for mpll_state output (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 3    | BS_OUT        | R             | <b>Description:</b> Value from PHY for bs_out output (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 2    | RTUNE_ACK     | R             | <b>Description:</b> Value from PHY for rtune_ack output (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true   |
| 1    | RES_REQ_OUT   | R             | <b>Description:</b> Value from PHY for res_req_out output (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

**Table 12-27 Fields for Register: SUP\_DIG\_SUP\_ASIC\_OUT (Continued)**

| Bits | Name        | Memory Access | Description                                                                                                                                                                                                                  |
|------|-------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | RES_ACK_OUT | R             | <b>Description:</b> Value from PHY for res_ack_out output (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

## 12.2.24 SUP\_DIG\_ATEOVRD\_STATUS

- **Description:** USB3\_SUP\_DIG\_ATEOVRD\_STATUS
- **Size:** 16 bits
- **Offset:** 0x1c
- **Exists:** Always

**Table 12-28 Fields for Register: SUP\_DIG\_ATEOVRD\_STATUS**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                             |
|-------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | RESERVED_15_10 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                           |
| 9:8   | DTB_OUT        | R             | <b>Description:</b> Value from dtb_out bus. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true          |
| 7     | REF_SSP_EN     | R             | <b>Description:</b> Value from ASIC for ref_ssp_en. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 6     | REF_USE_PAD    | R             | <b>Description:</b> Value from ASIC for ref_use_pad. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 5     | PHY_RESET_IN   | R             | <b>Description:</b> Value from ASIC for reset (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true        |

**Table 12-28 Fields for Register: SUP\_DIG\_ATEOVRD\_STATUS (Continued)**

| Bits | Name            | Memory Access | Description                                                                                                                                                                                                                                         |
|------|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | REF_CLKDIV2_IN  | R             | <b>Description:</b> Value from ASIC for ref_clkdiv2 (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                              |
| 3    | REF_USB2_EN_IN  | R             | <b>Description:</b> Value from ASIC for ref_usb2_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                              |
| 2    | ATEOVRD_EN      | R             | <b>Description:</b> When set, values from ATEOVRD register are being sent to PHY (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 1    | REF_CLKDIV2_OUT | R             | <b>Description:</b> Value from ATEOVRD for ref_clkdiv2 (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                           |
| 0    | REF_USB2_EN_OUT | R             | <b>Description:</b> Value from ATEOVRD for ref_usb2_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                           |

## 12.2.25 SUP\_DIG\_HSP\_OVRD\_IN

- **Description:** Override values for HSP
- **Size:** 16 bits
- **Offset:** 0x1d
- **Exists:** Always

**Table 12-29 Fields for Register: SUP\_DIG\_HSP\_OVRD\_IN**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | HSP_PLL_OVRD  | R/W           | <b>Description:</b> Override enable for HSP pll_pu signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                 |
| 2    | HSP_PLL_PU    | R/W           | <b>Description:</b> Override value for HSP pll_pu signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                  |
| 1    | HSP_BIAS_OVRD | R/W           | <b>Description:</b> Override enable for HSP bias_pu signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                |
| 0    | HSP_BIAS_PU   | R/W           | <b>Description:</b> Override value for HSP bias_pu signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                 |

## 12.2.26 SUP\_DIG\_HSP\_ASIC\_IN

- **Description:** Current values for incoming HSP & REF XO controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1e
- **Exists:** Always

**Table 12-30 Fields for Register: SUP\_DIG\_HSP\_ASIC\_IN**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                            |
|------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                          |
| 3    | REF_USE_XO    | R             | <b>Description:</b> Value from ASIC for ref_use_xo (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 2    | REF_XO_EN     | R             | <b>Description:</b> Value from ASIC for ref_xo_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true   |
| 1    | HSP_PLL_PU    | R             | <b>Description:</b> Value from ASIC for hsp_pll_pu (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 0    | HSP_BIAS_PU   | R             | <b>Description:</b> Value from ASIC for hsp_bias_pu (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |



## 12.2.27 SUP\_DIG\_SCOPE\_ENABLES

- **Description:** Scope Counter configuration
- **Size:** 16 bits
- **Offset:** 0x20
- **Exists:** Always

**Table 12-31 Fields for Register: SUP\_DIG\_SCOPE\_ENABLES**

| Bits | Name                 | Memory Access | Description                                                                                                                                                                                                                                                            |
|------|----------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:5 | RESERVED_15_5        | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                          |
| 4:3  | LANE_SELECT          | R/W           | <b>Description:</b> Scope Counter Input select<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                                                                                                                     |
| 2    | MASK_SATURATION_MODE | R/W           | <b>Description:</b> Method of mask saturation. When 1 Saturate when the first mask_counter reaches sample_limit. When 0 Saturate when all mask_counters have reached sample_limit.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | MASK_EN              | R/W           | <b>Description:</b> Enable scope_mask input for tracking count values. Clears registers when deasserted.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                           |
| 0    | XOR_EN               | R/W           | <b>Description:</b> Use scope_xor input for count values.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                          |

### 12.2.28 SUP\_DIG\_SCOPE\_SAMPLES

- **Description:** Number of samples to count
- **Size:** 16 bits
- **Offset:** 0x21
- **Exists:** Always

Table 12-32 Fields for Register: SUP\_DIG\_SCOPE\_SAMPLES

| Bits | Name    | Memory Access | Description                                                                                                                             |
|------|---------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | SAMPLES | R/W           | <b>Description:</b> Number of Sample to count.<br><b>Value After Reset:</b> 0x100<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff |

## 12.2.29 SUP\_DIG\_SCOPE\_COUNT

- **Description:** Results of scope counting
- **Size:** 16 bits
- **Offset:** 0x22
- **Exists:** Always

**Table 12-33 Fields for Register: SUP\_DIG\_SCOPE\_COUNT**

| Bits | Name  | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                            |
|------|-------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | COUNT | R             | <p><b>Description:</b> A write to this register will start the counting process. The value of FFFF indicates counting still in progress. Asserting MASK_EN will also start the counting if in MASK mode. (RU - read-only volatile)</p> <p><b>Value After Reset:</b> 0x0</p> <p><b>Exists:</b> Always</p> <p><b>Reset Mask:</b> 0xffff</p> <p><b>Volatile:</b> true</p> |

### 12.2.30 SUP\_DIG\_SCOPE\_CTL

- **Description:** Masked scope counting Start and Status
- **Size:** 16 bits
- **Offset:** 0x23
- **Exists:** Always

**Table 12-34 Fields for Register: SUP\_DIG\_SCOPE\_CTL**

| Bits | Name            | Memory Access | Description                                                                                                                                                                                                                                                                                                      |
|------|-----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:2 | RESERVED_15_2   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                    |
| 1    | MASK_START      | R/W           | <b>Description:</b> A write to this register will start the counting process The value of FFFF indicates counting still in progress Write will trigger mask counting. (USER1 - read-write volatile)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true |
| 0    | MASK_SATURATION | R/W           | <b>Description:</b> When assert mask registers have saturated. (USER1 - read-write volatile)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true                                                                                                        |

### 12.2.31 SUP\_DIG\_SCOPE\_MASK\_000

- **Description:** Current count value for mask\_000
- **Size:** 16 bits
- **Offset:** 0x24
- **Exists:** Always

**Table 12-35 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_000**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_000 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

### 12.2.32 SUP\_DIG\_SCOPE\_MASK\_001

- **Description:** Current count value for mask\_001
- **Size:** 16 bits
- **Offset:** 0x25
- **Exists:** Always

**Table 12-36 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_001**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_001 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

### 12.2.33 SUP\_DIG\_SCOPE\_MASK\_010

- **Description:** Current count value for mask\_010
- **Size:** 16 bits
- **Offset:** 0x26
- **Exists:** Always

**Table 12-37 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_010**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_010 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

### 12.2.34 SUP\_DIG\_SCOPE\_MASK\_011

- **Description:** Current count value for mask\_011
- **Size:** 16 bits
- **Offset:** 0x27
- **Exists:** Always

**Table 12-38 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_011**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_011 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |



### 12.2.35 SUP\_DIG\_SCOPE\_MASK\_100

- **Description:** Current count value for mask\_100
- **Size:** 16 bits
- **Offset:** 0x28
- **Exists:** Always

**Table 12-39 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_100**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_100 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

### 12.2.36 SUP\_DIG\_SCOPE\_MASK\_101

- **Description:** Current count value for mask\_101
- **Size:** 16 bits
- **Offset:** 0x29
- **Exists:** Always

Table 12-40 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_101

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_101 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

**12.2.37 SUP\_DIG\_SCOPE\_MASK\_110**

- **Description:** Current count value for mask\_110
- **Size:** 16 bits
- **Offset:** 0x2a
- **Exists:** Always

**Table 12-41 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_110**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_110 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

### 12.2.38 SUP\_DIG\_SCOPE\_MASK\_111

- **Description:** Current count value for mask\_111
- **Size:** 16 bits
- **Offset:** 0x2b
- **Exists:** Always

Table 12-42 Fields for Register: SUP\_DIG\_SCOPE\_MASK\_111

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                            |
|------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | MASK_VAL_111 | R/W           | <b>Description:</b> Starting count value of mask register. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xffff<br><b>Volatile:</b> true |

### 12.2.39 SUP\_ANA\_MPLL\_LOOP\_CTL

- **Description:** MPLL loop control bits
- **Size:** 16 bits
- **Offset:** 0x30
- **Exists:** Always

**Table 12-43 Fields for Register: SUP\_ANA\_MPLL\_LOOP\_CTL**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7:4  | PROP_CNTRL    | R/W           | <b>Description:</b> Charge pump proportional current setting<br><b>Value After Reset:</b> 0x8<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf              |
| 3:2  | INT_CNTRL     | R/W           | <b>Description:</b> Charge pump integrating current setting<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3               |
| 1    | GD            | R/W           | <b>Description:</b> Measure MPLL local GD<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                 |
| 0    | VMB           | R/W           | <b>Description:</b> Measure MPLL Master Bias Voltage<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |

## 12.2.40 SUP\_ANA\_MPLL\_ATB\_MEAS1

- **Description:** ATB Select bits
- **Size:** 16 bits
- **Offset:** 0x31
- **Exists:** Always

**Table 12-44 Fields for Register: SUP\_ANA\_MPLL\_ATB\_MEAS1**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | VPSF          | R/W           | <b>Description:</b> puts vpsf on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                  |
| 6    | VREF          | R/W           | <b>Description:</b> puts vref from charge pump on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                 |
| 5    | VCNTRL        | R/W           | <b>Description:</b> puts vcntrl on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                |
| 4    | GD            | R/W           | <b>Description:</b> puts gd on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                    |
| 3    | IVCO          | R/W           | <b>Description:</b> puts VCO current/93 on atb_f_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 2    | VP            | R/W           | <b>Description:</b> puts vpa on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |

**Table 12-44 Fields for Register: SUP\_ANA\_MPLL\_ATB\_MEAS1 (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                                       |
|------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1    | VP_CP     | R/W           | <b>Description:</b> puts vp_cp on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 0    | VREG_MPLL | R/W           | <b>Description:</b> puts vreg_mpll on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.41 SUP\_ANA\_MPLL\_ATB\_MEAS2

- **Description:** ATB Select bits
- **Size:** 16 bits
- **Offset:** 0x32
- **Exists:** Always

**Table 12-45 Fields for Register: SUP\_ANA\_MPLL\_ATB\_MEAS2**

| Bits | Name            | Memory Access | Description                                                                                                                                                             |
|------|-----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0           |
| 7    | IVCO_FILT       | R/W           | <b>Description:</b> puts filtered version of ivco on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 6    | VDCC_M          | R/W           | <b>Description:</b> puts dcc output vdcc_p on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |
| 5    | VDCC_P          | R/W           | <b>Description:</b> puts dcc output vdcc_m on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |
| 4    | ATB_SENSE_SEL   | R/W           | <b>Description:</b> connects internal atb sense bus to external bus<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                 |
| 3    | MEAS_TEMP       | R/W           | <b>Description:</b> tells POR block to measure the temperature<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 2    | FRC_MPMIX_VPMIX | R/W           | <b>Description:</b> Force mpll_pmix_vreg to use atb_s_m instead of vbg as its input<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |



**Table 12-45 Fields for Register: SUP\_ANA\_MPLL\_ATB\_MEAS2 (Continued)**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                     |
|------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | EN_MPMIX_VPMIX | R/W           | <b>Description:</b> Put vreg_pmix on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                |
| 0    | VDCC_IO        | R/W           | <b>Description:</b> when asserted, bit 5, 6 of this register measures i phase clock DCC voltage, otherwise, q phase DCC voltage is measured<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

## 12.2.42 SUP\_ANA\_MPLL\_OVRD

- **Description:** Local overrides
- **Size:** 16 bits
- **Offset:** 0x33
- **Exists:** Always

**Table 12-46 Fields for Register: SUP\_ANA\_MPLL\_OVRD**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | PWRON_LCL     | R/W           | <b>Description:</b> local power_on value<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                  |
| 6    | EN_PWRON_LCL  | R/W           | <b>Description:</b> Enable local control of power_on<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 5    | GS_LCL        | R/W           | <b>Description:</b> local gear_shift value<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                |
| 4    | EN_GS_LCL     | R/W           | <b>Description:</b> Enable local control of gear_shift<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |
| 3    | RST_LCL       | R/W           | <b>Description:</b> local reset value<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                     |
| 2    | EN_RST_LCL    | R/W           | <b>Description:</b> enable local control of reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |

**Table 12-46 Fields for Register: SUP\_ANA\_MPLL\_OVRD (Continued)**

| Bits | Name                | Memory Access | Description                                                                                                                                  |
|------|---------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | PMIX_CLK_SEL_LCL    | R/W           | <b>Description:</b> local pmix_clk_sel value<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1             |
| 0    | EN_PMIX_CLK_SEL_LCL | R/W           | <b>Description:</b> enable local control of pmix_clk_sel<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.43 SUP\_ANA\_RTUNE\_RTUNE\_CTRL

- **Description:** RTUNE control bits
- **Size:** 16 bits
- **Offset:** 0x34
- **Exists:** Always

**Table 12-47 Fields for Register: SUP\_ANA\_RTUNE\_RTUNE\_CTRL**

| Bits | Name            | Memory Access | Description                                                                                                                                                                                                                                                 |
|------|-----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                               |
| 7    | RT_PWRON_FRC_ON | R/W           | <b>Description:</b> When set, forces RTUNE block to be on<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                               |
| 6    | X4_FRC_OFF      | R/W           | <b>Description:</b> When set, do not multiply test current by 4<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                         |
| 5:4  | RT_DAC_MODE     | R/W           | <b>Description:</b> Margin DAC mode control bits(00=powerdown, 01=DAC drives atb_s_p/m directly, 10=DAC drives atb_s_p/m to the RX in margining mode, 11=illegal state)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |
| 3    | RT_DAC_CHOP     | R/W           | <b>Description:</b> Margin DAC chop control bit<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                         |
| 2    | RT_ATB          | R/W           | <b>Description:</b> RTUNE ATB mode control bit (1=RTUNE performs ADC on ATB input, 0=not accessing ATB)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                 |

**Table 12-47 Fields for Register: SUP\_ANA\_RTUNE\_RTUNE\_CTRL (Continued)**

| Bits | Name        | Memory Access | Description                                                                                                                                                     |
|------|-------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | RT_SEL_ATBP | R/W           | <b>Description:</b> RTUNE ATB sense input select bit (1=atb_s_p, 0=atb_s_m)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | RT_SEL_ATBF | R/W           | <b>Description:</b> RTUNE ATB input select bit (1=atb_fm, 0=atb_s_p/m)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1      |

### 12.2.44 SUP\_ANA\_ATB\_SWITCHYARD\_CTRL

- **Description:** ATB\_SWITCHYARD control bits
- **Size:** 16 bits
- **Offset:** 0x35
- **Exists:** Always

**Table 12-48 Fields for Register: SUP\_ANA\_ATB\_SWITCHYARD\_CTRL**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                                |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                              |
| 7    | ASM_GD        | R/W           | <b>Description:</b> Connect gd to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                              |
| 6    | ASP_VP        | R/W           | <b>Description:</b> Connect vp to atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                              |
| 5    | ASP_VBG       | R/W           | <b>Description:</b> The band gap voltage is not muxed on to the ATB, but rather a divided down voltage from a resistor stack is muxed on to the ATB, which is divided down value from a resistor stack<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 4    | ASP_VPHREG    | R/W           | <b>Description:</b> Connect vphreg to atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                          |
| 3    | ATEST_AFM     | R/W           | <b>Description:</b> drive ATEST from atb_f_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                           |

**Table 12-48 Fields for Register: SUP\_ANA\_ATB\_SWITCHYARD\_CTRL (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                                      |
|------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| 2    | ATEST_AFP | R/W           | <b>Description:</b> drive ATEST from atb_f_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | ATEST_ASM | R/W           | <b>Description:</b> Connet ATEST to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 0    | ATEST_ASP | R/W           | <b>Description:</b> Connect ATEST to atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.45 SUP\_ANA\_SSC\_CLK\_CNTRL

- **Description:** controls sscf reference clock rate
- **Size:** 16 bits
- **Offset:** 0x36
- **Exists:** Always

**Table 12-49 Fields for Register: SUP\_ANA\_SSC\_CLK\_CNTRL**

| Bits | Name           | Memory Access | Description                                                                                                                                                   |
|------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | NC             | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 6:0  | SSC_CLK_DIV125 | R/W           | <b>Description:</b> set ssc reference clock to be 20MHz<br><b>Value After Reset:</b> 0x79<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f                 |



## 12.2.46 SUP\_ANA\_HSP\_CLK\_DIG\_CNTRL0

- **Description:** controls HSP phase clock, and resolution
- **Size:** 16 bits
- **Offset:** 0x37
- **Exists:** Always

**Table 12-50 Fields for Register: SUP\_ANA\_HSP\_CLK\_DIG\_CNTRL0**

| Bits | Name             | Memory Access | Description                                                                                                                                                                                                                  |
|------|------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8    | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                |
| 7    | NUM_OVERRIDE     | R/W           | <b>Description:</b> Set the value to 1 to enable using values from creg bit 6 downto 0<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                   |
| 6:4  | GATE_PMIK_BITS   | R/W           | <b>Description:</b> Set gate_pmix_bits to gate the corresponding MSB of the 7-bit PMIX phase select bits<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7                                 |
| 3    | HALF_CLK_EN      | R/W           | <b>Description:</b> Set to 1 for 1.25GHz updates to HSP PMIX. Set to 0 for 2.5GHz updates to PMIX.<br><b>Value After Reset:</b> 0x1<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                       |
| 2    | UCLK_EN_N        | R/W           | <b>Description:</b> Set to 1 to disable update clock. Set to 0 to enable update clock. When clock is enabled, phase updates are allowed.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | EN_HSPPMIX_VPMIX | R/W           | <b>Description:</b> Set to 1 to connect the regulated HSP PMIX supply to the atb_s_p bus<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                 |

**Table 12-50 Fields for Register: SUP\_ANA\_HSP\_CLK\_DIG\_CNTRL0 (Continued)**

| Bits | Name                 | Memory Access | Description                                                                                                                                                                            |
|------|----------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | EN_HSPPMIX_FRC_VPMIX | R/W           | <b>Description:</b> Set to 1 to force override of HSP PMIX regulator voltage to voltage on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

## 12.2.47 SUP\_ANA\_HSP\_CLK\_DIG\_CNTRL1

- **Description:** Lower and Upper Phase Increment Values
- **Size:** 16 bits
- **Offset:** 0x38
- **Exists:** Always

**Table 12-51 Fields for Register: SUP\_ANA\_HSP\_CLK\_DIG\_CNTRL1**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                 |
| 7:4  | UPPER_INT     | R/W           | <b>Description:</b> These bits are only used when num_override is set to 1. Upper integer for phase increments of 2.5 to 2.4 GHz conversion. This is the phase increment that is applied 6 times in every 25, for 1.25 GHz increments, and 3 times in every 25 for 2.5GHz<br><b>Value After Reset:</b> 0xb<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf |
| 3:0  | LOWER_INT     | R/W           | <b>Description:</b> These bits are only used when num_override is set to 1. Lower integer for phase increments of 2.5 to 2.4 GHz conversion. This phase increment is applied 19 times in every 25, for 1.25 GHz increments, and 22 times in every 25 for 2.5GHz.<br><b>Value After Reset:</b> 0xa<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf          |

## 12.2.48 LANEN\_DIG\_TX\_OVRD\_IN\_LO

- **Description:** Override values for incoming TX controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N00
- **Exists:** Always

**Table 12-52 Fields for Register: LANEN\_DIG\_TX\_OVRD\_IN\_LO**

| Bits | Name                  | Memory Access | Description                                                                                                                                      |
|------|-----------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | LANE_IF_WIDTH_OVRD    | R/W           | <b>Description:</b> Override enable for lane interface width<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 14   | LANE_IF_WIDTH         | R/W           | <b>Description:</b> Override value for lane interface width<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 13   | TX_DETECT_RX_REQ_OVRD | R/W           | <b>Description:</b> Override enable for tx_detect_rx_req<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 12   | TX_DETECT_RX_REQ      | R/W           | <b>Description:</b> Override value for tx_detect_rx_req<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1      |
| 11   | TX_BEACON_EN_OVRD     | R/W           | <b>Description:</b> Override enable for tx_beacon_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1         |
| 10   | TX_BEACON_EN          | R/W           | <b>Description:</b> Override value for tx_beacon_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1          |
| 9    | TX_CM_EN_OVRD         | R/W           | <b>Description:</b> Override enable for tx_cm_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1             |

**Table 12-52 Fields for Register: LANEN\_DIG\_TX\_OVRD\_IN\_LO (Continued)**

| Bits | Name              | Memory Access | Description                                                                                                                            |
|------|-------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 8    | TX_CM_EN          | R/W           | <b>Description:</b> Override value for tx_cm_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 7    | TX_EN_OVRD        | R/W           | <b>Description:</b> Override enable for tx_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1      |
| 6    | TX_EN             | R/W           | <b>Description:</b> Override value for tx_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1       |
| 5    | TX_DATA_EN_OVRD   | R/W           | <b>Description:</b> Override enable for tx_data_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 4    | TX_DATA_EN        | R/W           | <b>Description:</b> Override value for tx_data_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 3    | TX_INVERT_OVRD    | R/W           | <b>Description:</b> Override enable for tx_invert<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 2    | TX_INVERT         | R/W           | <b>Description:</b> Override value for tx_invert<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |
| 1    | TX_LOOPBK_EN_OVRD | R/W           | <b>Description:</b> Override enable for loopbk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 0    | LOOPBK_EN         | R/W           | <b>Description:</b> Override value for loopbk_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |

## 12.2.49 LANEN\_DIG\_TX\_OVRD\_IN\_HI

- **Description:** Override values for incoming TX controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N01
- **Exists:** Always

**Table 12-53 Fields for Register: LANEN\_DIG\_TX\_OVRD\_IN\_HI**

| Bits  | Name               | Memory Access | Description                                                                                                                                                   |
|-------|--------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | RESERVED_15_10     | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 9     | TX_VBOOST_EN_OVRD  | R/W           | <b>Description:</b> Override enable for tx_vboost_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 8     | TX_VBOOST_EN       | R/W           | <b>Description:</b> Override value for tx_vboost_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |
| 7     | TX_RESET_OVRD      | R/W           | <b>Description:</b> Override enable for tx_reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 6     | TX_RESET           | R/W           | <b>Description:</b> Override value for tx_reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 5     | TX_NYQUIST_DATA    | R/W           | <b>Description:</b> Override incoming data to nyquist<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 4     | TX_CLK_OUT_EN_OVRD | R/W           | <b>Description:</b> Override enable for tx_clk_out_en.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |

**Table 12-53 Fields for Register: LANEN\_DIG\_TX\_OVRD\_IN\_HI (Continued)**

| Bits | Name          | Memory Access | Description                                                                                                                              |
|------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | TX_CLK_OUT_EN | R/W           | <b>Description:</b> Override incoming tx_clk_out_en.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 2    | TX_RATE_OVRD  | R/W           | <b>Description:</b> Override enable for tx_rate.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 1:0  | TX_RATE       | R/W           | <b>Description:</b> Override incoming tx lane rate.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3  |

### 12.2.50 LANEN\_DIG\_TX\_OVRD\_DRV\_LO

- **Description:** Override values for incoming TX drive controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N02
- **Exists:** Always

**Table 12-54 Fields for Register: LANEN\_DIG\_TX\_OVRD\_DRV\_LO**

| Bits | Name           | Memory Access | Description                                                                                                                                                               |
|------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RESERVED_15_15 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0             |
| 14   | EN             | R/W           | <b>Description:</b> Enable override values for all inputs controlled by this register<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 13:7 | PREEMPH        | R/W           | <b>Description:</b> Override value for transmit preemphasis.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f                         |
| 6:0  | AMPLITUDE      | R/W           | <b>Description:</b> Override value for transmit amplitude.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7f                           |



### 12.2.51 LANEN\_DIG\_TX\_OVRD\_DRV\_HI

- **Description:** Override values for incoming TX drive controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N03
- **Exists:** Always

**Table 12-55 Fields for Register: LANEN\_DIG\_TX\_OVRD\_DRV\_HI**

| Bits | Name          | Memory Access | Description                                                                                                                                                               |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:6 | RESERVED_15_6 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0             |
| 5    | EN            | R/W           | <b>Description:</b> Enable override values for all inputs controlled by this register<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 4:0  | TERM_OFFSET   | R/W           | <b>Description:</b> Override value for tx_term_offset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1f                                |

## 12.2.52 LANEN\_DIG\_TX\_OVRD\_OUT

- **Description:** Override values for outgoing TX controls to ASIC
- **Size:** 16 bits
- **Offset:** 0x1N04
- **Exists:** Always

**Table 12-56 Fields for Register: LANEN\_DIG\_TX\_OVRD\_OUT**

| Bits | Name                  | Memory Access | Description                                                                                                                                                   |
|------|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8         | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | TX_STATE_OVRD         | R/W           | <b>Description:</b> Override enable for tx_state<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 6    | TX_STATE              | R/W           | <b>Description:</b> Override value for tx_state<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 5    | TX_CM_STATE_OVRD      | R/W           | <b>Description:</b> Override enable for tx_cm_state<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |
| 4    | TX_CM_STATE           | R/W           | <b>Description:</b> Override value for tx_cm_state<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 3    | TX_DETECT_RX_ACK_OVRD | R/W           | <b>Description:</b> Override enable for tx_detect_rx_ack<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                  |
| 2    | TX_DETECT_RX_ACK      | R/W           | <b>Description:</b> Override value for tx_detect_rx_ack<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                   |

**Table 12-56 Fields for Register: LANEN\_DIG\_TX\_OVRD\_OUT (Continued)**

| Bits | Name               | Memory Access | Description                                                                                                                                  |
|------|--------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | DETECT_RX_RES_OVRD | R/W           | <b>Description:</b> Override enable for tx_detect_rx_res<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | DETECT_RX_RES      | R/W           | <b>Description:</b> Override value for tx_detect_rx_res<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |

### 12.2.53 LANEN\_DIG\_RX\_OVRD\_IN\_LO

- **Description:** Override values for incoming RX controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N05
- **Exists:** Always

**Table 12-57 Fields for Register: LANEN\_DIG\_RX\_OVRD\_IN\_LO**

| Bits  | Name              | Memory Access | Description                                                                                                                                                   |
|-------|-------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14    | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 13    | RX_LOS_EN_OVRD    | R/W           | <b>Description:</b> Override enable for rx_los_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |
| 12    | RX_LOS_EN         | R/W           | <b>Description:</b> Override value for rx_los_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 11    | RX_TERM_EN_OVRD   | R/W           | <b>Description:</b> Override enable for rx_term_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 10    | RX_TERM_EN        | R/W           | <b>Description:</b> Override value for rx_term_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |
| 9     | RX_BIT_SHIFT_OVRD | R/W           | <b>Description:</b> Override enable for rx_bit_shift<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 8     | RX_BIT_SHIFT      | R/W           | <b>Description:</b> Override value for rx_bit_shift<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |

**Table 12-57 Fields for Register: LANEN\_DIG\_RX\_OVRD\_IN\_LO (Continued)**

| Bits | Name             | Memory Access | Description                                                                                                                             |
|------|------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 7    | RX_ALIGN_EN_OVRD | R/W           | <b>Description:</b> Override enable for rx_align_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 6    | RX_ALIGN_EN      | R/W           | <b>Description:</b> Override value for rx_align_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 5    | RX_DATA_EN_OVRD  | R/W           | <b>Description:</b> Override enable for rx_data_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1  |
| 4    | RX_DATA_EN       | R/W           | <b>Description:</b> Override value for rx_data_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |
| 3    | RX_PLL_EN_OVRD   | R/W           | <b>Description:</b> Override enable for rx_pll_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |
| 2    | RX_PLL_EN        | R/W           | <b>Description:</b> Override value for rx_pll_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 1    | RX_INVERT_OVRD   | R/W           | <b>Description:</b> Override enable for rx_invert<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1   |
| 0    | RX_INVERT        | R/W           | <b>Description:</b> Override value for rx_invert<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |

### 12.2.54 LANEN\_DIG\_RX\_OVRD\_IN\_HI

- **Description:** Override values for incoming RX controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N06
- **Exists:** Always

**Table 12-58 Fields for Register: LANEN\_DIG\_RX\_OVRD\_IN\_HI**

| Bits  | Name           | Memory Access | Description                                                                                                                                                   |
|-------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 13    | RX_RESET_OVRD  | R/W           | <b>Description:</b> Override enable for rx_reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 12    | RX_RESET       | R/W           | <b>Description:</b> Override value for rx_reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 11    | RX_EQ_OVRD     | R/W           | <b>Description:</b> Override enable for rx_eq<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                             |
| 10:8  | RX_EQ          | R/W           | <b>Description:</b> Override value for rx_eq<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7                              |
| 7     | RX_EQ_EN_OVRD  | R/W           | <b>Description:</b> Override enable for rx_eq_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 6     | RX_EQ_EN       | R/W           | <b>Description:</b> Override value for rx_eq_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |

**Table 12-58 Fields for Register: LANEN\_DIG\_RX\_OVRD\_IN\_HI (Continued)**

| Bits | Name               | Memory Access | Description                                                                                                                               |
|------|--------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | RX_LOS_FILTER_OVRD | R/W           | <b>Description:</b> Override enable for rx_los_filter<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 4:3  | RX_LOS_FILTER      | R/W           | <b>Description:</b> Override value for rx_los_filter<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3  |
| 2    | RX_RATE_OVRD       | R/W           | <b>Description:</b> Override enable for rx_rate<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1       |
| 1:0  | RX_RATE            | R/W           | <b>Description:</b> Override value for rx_rate<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3        |

## 12.2.55 LANEN\_DIG\_RX\_OVRD\_OUT

- **Description:** Override values for outgoing RX controls to ASIC
- **Size:** 16 bits
- **Offset:** 0x1N07
- **Exists:** Always

**Table 12-59 Fields for Register: LANEN\_DIG\_RX\_OVRD\_OUT**

| Bits | Name           | Memory Access | Description                                                                                                                                                   |
|------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:7 | RESERVED_15_7  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 6    | ZERO_DATA      | R/W           | <b>Description:</b> Override data output to all zeros<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 5    | LOS_OVRD       | R/W           | <b>Description:</b> Override value for rx_los<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                             |
| 4    | LOS            | R/W           | <b>Description:</b> Override value for rx_los<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                             |
| 3    | PLL_STATE_OVRD | R/W           | <b>Description:</b> Override enable for rx_pll_state<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 2    | PLL_STATE      | R/W           | <b>Description:</b> Override value for rx_pll_state<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |
| 1    | VALID_OVRD     | R/W           | <b>Description:</b> Override enable for rx_valid<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |



**Table 12-59 Fields for Register: LANEN\_DIG\_RX\_OVRD\_OUT (Continued)**

| Bits | Name  | Memory Access | Description                                                                                                                         |
|------|-------|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 0    | VALID | R/W           | <b>Description:</b> Override value for rx_valid<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

## 12.2.56 LANEN\_DIG\_TX\_ASIC\_IN

- **Description:** Current values for incoming TX controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N08
- **Exists:** Always

**Table 12-60 Fields for Register: LANEN\_DIG\_TX\_ASIC\_IN**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                              |
|-------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                            |
| 13    | LANE_IF_WIDTH  | R             | <b>Description:</b> Value from ASIC for lane_if_width (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 12    | DATA_EN_S      | R             | <b>Description:</b> Value from ASIC for tx_data_en_s (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 11    | TX_VBOOST_EN   | R             | <b>Description:</b> Value from ASIC for tx_vboost_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 10    | TX_CLK_OUT_EN  | R             | <b>Description:</b> Value from ASIC for tx_clk_out_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

**Table 12-60 Fields for Register: LANEN\_DIG\_TX\_ASIC\_IN (Continued)**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                 |
|------|---------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9    | DETECT_RX_REQ | R             | <b>Description:</b> Value from ASIC for tx_detect_rx_req (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 8    | BEACON_EN     | R             | <b>Description:</b> Value from ASIC for tx_beacon_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true     |
| 7    | CM_EN         | R             | <b>Description:</b> Value from ASIC for tx_cm_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true         |
| 6    | TX_EN         | R             | <b>Description:</b> Value from ASIC for tx_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true            |
| 5    | DATA_EN       | R             | <b>Description:</b> Value from ASIC for tx_data_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true       |
| 4    | TX_RESET      | R             | <b>Description:</b> Value from ASIC for tx_reset (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true         |

**Table 12-60 Fields for Register: LANEN\_DIG\_TX\_ASIC\_IN (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                                                                                                                          |
|------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | INVERT    | R             | <b>Description:</b> Value from ASIC for tx_invert (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 2    | LOOPBK_EN | R             | <b>Description:</b> Value from ASIC for loopbk_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 1:0  | TX_RATE   | R             | <b>Description:</b> Value from ASIC for tx_rate (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true   |

## 12.2.57 LANEN\_DIG\_TX\_ASIC\_DRV\_LO

- **Description:** Current values for incoming TX level controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N09
- **Exists:** Always

**Table 12-61 Fields for Register: LANEN\_DIG\_TX\_ASIC\_DRV\_LO**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                                              |
|-------|----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                            |
| 13:7  | PREEMPH        | R             | <b>Description:</b> Value from ASIC for tx_preemph, MSBit is not used (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 6:0   | AMPLITUDE      | R             | <b>Description:</b> Value from ASIC for tx_amplitude (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                  |

### 12.2.58 LANEN\_DIG\_TX\_ASIC\_DRV\_HI

- **Description:** Current values for incoming TX level controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N0a
- **Exists:** Always

**Table 12-62 Fields for Register: LANEN\_DIG\_TX\_ASIC\_DRV\_HI**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                               |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:5 | RESERVED_15_5 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                             |
| 4:0  | TERM_OFFSET   | R             | <b>Description:</b> Value from ASIC for tx_term_offset (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

## 12.2.59 LANEN\_DIG\_TX\_ASIC\_OUT

- **Description:** Current values for outgoing TX status controls from PHY
- **Size:** 16 bits
- **Offset:** 0x1N0b
- **Exists:** Always

**Table 12-63 Fields for Register: LANEN\_DIG\_TX\_ASIC\_OUT**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                |
|------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:5 | RESERVED_15_5 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                              |
| 4    | STATE         | R             | <b>Description:</b> Value from PHY for tx_state (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true         |
| 3    | CM_STATE      | R             | <b>Description:</b> Value from PHY for tx_cm_state (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 2    | DETECT_RX_ACK | R             | <b>Description:</b> Value from PHY for tx_detect_rx_ack (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 1    | RESERVED      | R             | <b>Description:</b> Always returns 0 (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                    |

**Table 12-63 Fields for Register: LANEN\_DIG\_TX\_ASIC\_OUT (Continued)**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                |
|------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | DETECT_RX_RES | R             | <b>Description:</b> Value from PHY for tx_detect_rx_res (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |



## 12.2.60 LANEN\_DIG\_RX\_ASIC\_IN

- **Description:** Current values for incoming RX controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N0c
- **Exists:** Always

**Table 12-64 Fields for Register: LANEN\_DIG\_RX\_ASIC\_IN**

| Bits  | Name       | Memory Access | Description                                                                                                                                                                                                              |
|-------|------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | RX_EQ_EN   | R             | <b>Description:</b> Value from ASIC for rx_eq_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 14:12 | RX_EQ      | R             | <b>Description:</b> Value from ASIC for rx_eq (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true         |
| 11:10 | LOS_FILTER | R             | <b>Description:</b> Value from ASIC for rx_los_filter (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 9     | LOS_EN     | R             | <b>Description:</b> Value from ASIC for rx_los_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true     |
| 8     | TERM_EN    | R             | <b>Description:</b> Value from ASIC for rx_term_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true    |

**Table 12-64 Fields for Register: LANEN\_DIG\_RX\_ASIC\_IN (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                                                                                                                             |
|------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | CLK_SHIFT | R             | <b>Description:</b> Value from ASIC for rx_bit_shift (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 6    | ALIGN_EN  | R             | <b>Description:</b> Value from ASIC for rx_align_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true  |
| 5    | DATA_EN   | R             | <b>Description:</b> Value from ASIC for rx_data_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true   |
| 4    | PLL_EN    | R             | <b>Description:</b> Value from ASIC for rx_pll_en (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true    |
| 3    | RX_RESET  | R             | <b>Description:</b> Value from ASIC for rx_reset (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true     |
| 2    | INVERT    | R             | <b>Description:</b> Value from ASIC for rx_invert (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true    |

**Table 12-64 Fields for Register: LANEN\_DIG\_RX\_ASIC\_IN (Continued)**

| Bits | Name    | Memory Access | Description                                                                                                                                                                                                        |
|------|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1:0  | RX_RATE | R             | <b>Description:</b> Value from ASIC for rx_rate (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

### 12.2.61 LANEN\_DIG\_RX\_ASIC\_OUT

- **Description:** Current values for outgoing RX status controls from PHY
- **Size:** 16 bits
- **Offset:** 0x1N0d
- **Exists:** Always

**Table 12-65 Fields for Register: LANEN\_DIG\_RX\_ASIC\_OUT**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                            |
|------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:3 | RESERVED_15_3 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                          |
| 2    | LOS           | R             | <b>Description:</b> Value from PHY for rx_los (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true       |
| 1    | PLL_STATE     | R             | <b>Description:</b> Value from PHY for rx_pll_state (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 0    | VALID         | R             | <b>Description:</b> Value from PHY for rx_valid (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true     |

## 12.2.62 LANEN\_DIG\_TX\_DEBUG

- **Description:** Transmit debug controls
- **Size:** 16 bits
- **Offset:** 0x1N10
- **Exists:** Always

**Table 12-66 Fields for Register: LANEN\_DIG\_TX\_DEBUG**

| Bits  | Name             | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                             |
|-------|------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | RESERVED_15_13   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                           |
| 12    | KEEP_TX_REFGEN   | R/W           | <b>Description:</b> Keep TX voltage regulator ON until TX clocks are OFF<br><b>Value After Reset:</b> 0x1<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                                            |
| 11:4  | RXDET_MEAS_TIME  | R/W           | <b>Description:</b> Time to wait for rxdet measurement (* 4/2/1 refclk cycles based on multiplier)<br><b>Value After Reset:</b> 0x10<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xff                                                                                                                                                                                |
| 3     | DETECT_RX_ALWAYS | R/W           | <b>Description:</b> Always signal 1 for rx_detect ignoring analog<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                                                   |
| 2:0   | DTB_SEL          | R/W           | <b>Description:</b> Selects data to drive on DTB 0 - disabled 1 - mag_update/sub signals 2 - mag_high/add signals 3 - clock aligner early/shift signals 4 - analog/asic clocks 5 - tx_state counter/lbert strobe 6 - nocl_ref_rst/dig_rst 7 - tx_ana_rxdetp_high/tx_ana_rxdetm_high<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7 |

### 12.2.63 LANEN\_DIG\_TX\_CM\_WAIT\_TIME\_OVRD

- **Description:** USB3\_LANE0\_DIG\_TX\_CM\_WAIT\_TIME\_OVRD
- **Size:** 16 bits
- **Offset:** 0x1N11
- **Exists:** Always

**Table 12-67 Fields for Register: LANEN\_DIG\_TX\_CM\_WAIT\_TIME\_OVRD**

| Bits  | Name            | Memory Access | Description                                                                                                                                                                                                                        |
|-------|-----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | RESERVED_15_11  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                      |
| 10    | CM_TIME_OVRD_EN | R/W           | <b>Description:</b> Override enable for common mode charging wait time.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                        |
| 9:0   | CM_TIME_OVRD    | R/W           | <b>Description:</b> Time to wait, in ref clock cycles, for LFPS common mode charging (This value, x32, is used for VCMHOLD charging method)<br><b>Value After Reset:</b> 0x9f<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3ff |

## 12.2.64 LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_0

- **Description:** Calculated values for N\_USE and N\_TRISTATE. Also TX ANA configuration shift register output.
- **Size:** 16 bits
- **Offset:** 0x1N12
- **Exists:** Always

**Table 12-68 Fields for Register: LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_0**

| Bits | Name       | Memory Access | Description                                                                                                                                                                                                                                     |
|------|------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | SHIFT_OUT  | R             | <b>Description:</b> Current value from TX_ANAs configuration shift register. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 14   | DONE       | R             | <b>Description:</b> Configuration is done (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                    |
| 13:7 | N_USE      | R             | <b>Description:</b> Value from VMD for legs to use. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                          |
| 6:0  | N_TRISTATE | R             | <b>Description:</b> Value from VMD for number of tristate legs. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true              |

### 12.2.65 LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_1

- **Description:** Calculated values for N\_FIXED and N\_TRAILER.
- **Size:** 16 bits
- **Offset:** 0x1N13
- **Exists:** Always

**Table 12-69 Fields for Register: LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_1**

| Bits | Name       | Memory Access | Description                                                                                                                                                                                                                            |
|------|------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | FIXED_DONE | R             | <b>Description:</b> N_FIXED Multiplication has completed. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true           |
| 14   | TRA_DONE   | R             | <b>Description:</b> N_TRAILER Multiplication has completed. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true         |
| 13:7 | N_FIXED    | R             | <b>Description:</b> Value from VMD for number of fixed driver legs. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 6:0  | N_TRAILER  | R             | <b>Description:</b> Value from VMD for number of trailer legs. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |



**12.2.66 LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_DEBUG\_IN**

- **Description:** Access the TX\_ANA VMD Control Shift Register
- **Size:** 16 bits
- **Offset:** 0x1N14
- **Exists:** Always

**Table 12-70 Fields for Register: LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_DEBUG\_IN**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                                      |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                    |
| 3    | CONFIG_OVRD   | R/W           | <b>Description:</b> Override the Voltage Mode Driver Configuration FSM and access the shift chain directly. (volatile and 2 reads needed to read value) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 2    | CONFIG_LOAD   | R/W           | <b>Description:</b> Override value for the Voltage Mode Driver Configuration FSM's config load. (volatile and 2 reads needed to read value) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true             |
| 1    | CONFIG_CLK    | R/W           | <b>Description:</b> Override value for the Voltage Mode Driver Configuration FSM's config clk. (volatile and 2 reads needed to read value) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true              |

**Table 12-70 Fields for Register: LANEN\_DIG\_TX\_VMD\_FSM\_TX\_VCM\_DEBUG\_IN (Continued)**

| Bits | Name        | Memory Access | Description                                                                                                                                                                                                                                                                          |
|------|-------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | CONFIG_DATA | R/W           | <b>Description:</b> Override value for the Voltage Mode Driver Configuration FSM's config data. (volatile and 2 reads needed to read value) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |

## 12.2.67 LANEN\_DIG\_TX\_LBERT\_CTL

- **Description:** Pattern Generator controls
- **Size:** 16 bits
- **Offset:** 0x1N15
- **Exists:** Always

**Table 12-71 Fields for Register: LANEN\_DIG\_TX\_LBERT\_CTL**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RESERVED_15_15 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                                                                                                                                                                                       |
| 14:5 | PAT0           | R/W           | <b>Description:</b> 10 bit pattern for modes using PAT0 (5-7). Ignored for other MODEs.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3ff                                                                                                                                                                                                                                                                                                                                                       |
| 4    | TRIGGER_ERR    | R/W           | <b>Description:</b> Single shot inversion the LSB of the current symbol. Any write of a 1 to this bit will insert an error. Maintain mode and pattern from setup.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                                                                                                               |
| 3:0  | MODE           | R/W           | <b>Description:</b> Pattern to generate When changing modes, you must change to disabled first 0 - disabled 1 - lfsr31. $X^{31} + X^{28} + 1$ 2 - lfsr23. $X^{23} + X^{18} + 1$ 3 - lfsr15. $X^{15} + X^{14} + 1$ 4 - lfsr7. $X^7 + X^6 + 1$ 5 - fixed word (PAT0) 6 - DC balanced word (PAT0, ~PAT0) 7 - Fixed pattern: (000, PAT0, 3ff, ~PAT0) 8 - USB3.0 TSEQ (rx equalizer training ordered sets) Transmits Even, then Odd. 9-15 - RESERVED<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf |

## 12.2.68 LANEN\_DIG\_RX\_LBERT\_CTL

- **Description:** Pattern Matcher controls
- **Size:** 16 bits
- **Offset:** 0x1N16
- **Exists:** Always

**Table 12-72 Fields for Register: LANEN\_DIG\_RX\_LBERT\_CTL**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:5 | RESERVED_15_5 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                                                                                             |
| 4    | SYNC          | R/W           | <b>Description:</b> Synchronize pattern matcher with incoming data A write of a one to this bit will reset the error counter and start a synchronization of the PM. There is no need to write this back to zero to run normally.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                      |
| 3:0  | MODE          | R/W           | <b>Description:</b> Pattern to match When changing modes, you must change to disabled first 0 - disabled 1 - lfsr31: $X^{31} + X^{28} + 1$ 2 - lfsr23: $X^{23} + X^{18} + 1$ 3 - lfsr15: $X^{15} + X^{14} + 1$ 4 - lfsr7 : $X^7 + X^6 + 1$ 5 - d[n] = d[n-10] 6 - d[n] = !d[n-10] 7 - d[n] = !d[n-20] 8 - USB 3.0 TSEQ (NOT SUPPORTED IN PM. PG Only!<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf |

## 12.2.69 LANEN\_DIG\_RX\_LBERT\_ERR

- **Description:** Pattern match error counter
- **Size:** 16 bits
- **Offset:** 0x1N17
- **Exists:** Always

**Table 12-73 Fields for Register: LANEN\_DIG\_RX\_LBERT\_ERR**

| Bits | Name  | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|-------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | OV14  | R/W           | <b>Description:</b> A read of this register, or a sync of the Pattern Matcher resets the error count. If all bits of this register are 1, the error counter has saturated. If the status If active, multiply COUNT by 128. If OV14=1 and COUNT=2 <sup>15</sup> -1, signals overflow of counter (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true |
| 14:0 | COUNT | R/W           | <b>Description:</b> Current error count If OV14 field is active, then multiply count by 128 to get the actual count. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7fff<br><b>Volatile:</b> true                                                                                                                                                                        |

### 12.2.70 LANEN\_DIG\_RX\_SCOPE\_CTL

- **Description:** Control bits for per-transceiver scope portion
- **Size:** 16 bits
- **Offset:** 0x1N18
- **Exists:** Always

**Table 12-74 Fields for Register: LANEN\_DIG\_RX\_SCOPE\_CTL**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | RESERVED_15_14 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                                                                                                                                                                                           |
| 13:12 | RX_VALID_CTL   | R/W           | <b>Description:</b> When to Start Sampling control based on rx_valid. 0 = Ignore rx_valid. 1 = Heed rx_valid. 2 = Wait for rising_edge on rx_valid.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                                                                                                                                                                                                                                                                 |
| 11:3  | DELAY          | R/W           | <b>Description:</b> number of cycles to skip between samples<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1ff                                                                                                                                                                                                                                                                                                                                                                      |
| 2:0   | MODE           | R/W           | <b>Description:</b> sampling mode of counters NOTE: WORD is 20 bits. 0 = off 1 = sample data every WORD*(1+DELAY) bits from the digital recovered path. 2 = sample data every WORD*(1+DELAY) + 1 bits from the digital recovered path. 3 = sample data every WORD*(1+DELAY) + 2 bits from the digital recovered path. 4 = sample data every WORD*(1+DELAY) bits from the analog probe. Asserts XOR and MASK increments accordingly.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7 |

## 12.2.71 LANEN\_DIG\_RX\_SCOPE\_PHASE

- **Description:** Scope sampling phase.
- **Size:** 16 bits
- **Offset:** 0x1N19
- **Exists:** Always

**Table 12-75 Fields for Register: LANEN\_DIG\_RX\_SCOPE\_PHASE**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                  |
|-------|----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | RESERVED_15_15 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                |
| 14:10 | BASE           | R/W           | <b>Description:</b> which bit to sample when MODE = 1 or 4<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1f                              |
| 9:8   | SCOPE_DELAY    | R/W           | <b>Description:</b> How many clocks to delay the analog scope_data.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                      |
| 7     | SCOPE_SEL      | R/W           | <b>Description:</b> Select Sampling Mode. 0 - Before AFE Sampling 1 - After AFE Sampling<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 6     | UPDATE         | R/W           | <b>Description:</b> Update Sampling phase. Write a 1.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                    |
| 5:0   | SAMPLE_PHASE   | R/W           | <b>Description:</b> Sampling Phase<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3f                                                      |

## 12.2.72 LANEN\_DIG\_RX\_DPLL\_FREQ

- **Description:** Current frequency integrator value.
- **Size:** 16 bits
- **Offset:** 0x1N1a
- **Exists:** Always

**Table 12-76 Fields for Register: LANEN\_DIG\_RX\_DPLL\_FREQ**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                     |
|-------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | RESERVED_15_13 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                   |
| 12:1  | VAL            | R/W           | <b>Description:</b> Freq is $15.26 \times \text{VAL}$ ppm from the reference<br>When mpll_slow is set, the ppm is half the eqn above (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xfff<br><b>Volatile:</b> true |
| 0     | DTHR           | R/W           | <b>Description:</b> Bits below the useful resolution (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true                                                                                   |



### 12.2.73 LANEN\_DIG\_RX\_CDR\_CTL

- **Description:** Control bits for receiver in recovered domain
- **Size:** 16 bits
- **Offset:** 0x1N1b
- **Exists:** Always

**Table 12-77 Fields for Register: LANEN\_DIG\_RX\_CDR\_CTL**

| Bits  | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:12 | DTB_SEL        | R/W           | <b>Description:</b> select to drive various signals onto the dtb 0 - disabled 1 - pll_ana_rst,pll_count from rx_pwr_ctl 2 - com_good_high/low from aligner 3 - com_bad_high/low from aligner 4 - shift_in_prog,ana_odd_data from aligner 5 - low bits of XAUI align FSM state 6 - error_high/low from lbert_pm 7 - ana_los,los_filter from los block 8 - coast_dppll,eios_det from los block 9 - cdr_en and cdr_timeout from cdr_ctl A - cdr_en_adap and cdr_rx_valid from cdr_ctl B - cdr_en_eq and locked_vector_en from cdr_ctl C - dig_word_rst,rx_dig_en D - dig_dword_rst,dig_dword_clk<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf |
| 11    | ALWAYS_REALIGN | R/W           | <b>Description:</b> realign on any misaligned comma<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 10    | FAST_START     | R/W           | <b>Description:</b> decrease startup steps by 50%<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 9:8   | FRUG_VALUE     | R/W           | <b>Description:</b> override value for FRUG<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7:6   | PHUG_VALUE     | R/W           | <b>Description:</b> override value for PHUG<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**Table 12-77 Fields for Register: LANEN\_DIG\_RX\_CDR\_CTL (Continued)**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                                |
|------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | OVRD_DPLL_GAIN | R/W           | <b>Description:</b> Override PHUG and FRUG values<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                      |
| 4    | PHDET_POL      | R/W           | <b>Description:</b> Reverse polarity of phase error<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                    |
| 3:2  | PHDET_EDGE     | R/W           | <b>Description:</b> Edges to use for phase detection. 11 - Use both edges 10 - Use rising edges only 01 - Use falling edges only 00 - Ignore all edges<br><b>Value After Reset:</b> 0x3<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |
| 1:0  | PHDET_EN       | R/W           | <b>Description:</b> Enable phase detector. top bit is odd slicers, bottom is even<br><b>Value After Reset:</b> 0x3<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                                                      |

## 12.2.74 LANEN\_DIG\_RX\_CDR\_CDR\_FSM\_DEBUG

- **Description:** Current values for CDR and Adaptation FSM.
- **Size:** 16 bits
- **Offset:** 0x1N1c
- **Exists:** Always

**Table 12-78 Fields for Register: LANEN\_DIG\_RX\_CDR\_CDR\_FSM\_DEBUG**

| Bits  | Name          | Memory Access | Description                                                                                                                                                                                                                        |
|-------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | adap_rx_eq    | R             | <b>Description:</b> Equalization setting from adaptation FSM. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true   |
| 12:10 | rx_eq_ctr     | R             | <b>Description:</b> Initial centre point from equalization FSM. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 9:7   | rx_ana_eq     | R             | <b>Description:</b> Equalization setting to Analog. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true             |
| 6     | adap_rx_valid | R             | <b>Description:</b> Adaptation has completed and locked. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true        |
| 5     | cdr_en_adap   | R             | <b>Description:</b> Adaptation loop is enabling the CDR. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true        |

**Table 12-78 Fields for Register: LANEN\_DIG\_RX\_CDR\_CDR\_FSM\_DEBUG (Continued)**

| Bits | Name         | Memory Access | Description                                                                                                                                                                                                                                |
|------|--------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | cdr_en_eq    | R             | <b>Description:</b> Equalization loop is enabling the CDR. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true              |
| 3    | aligned      | R             | <b>Description:</b> Datapath is bit-aligned. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                            |
| 2    | cdr_rx_valid | R             | <b>Description:</b> CDR has locked to incoming data stream. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true             |
| 1    | cdr_timeout  | R             | <b>Description:</b> CDR has not locked to datastream and has timed-out. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 0    | cdr_en       | R             | <b>Description:</b> CDR has been enabled. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                               |

## 12.2.75 LANEN\_DIG\_RX\_CDR\_LOCK\_VEC\_OVRD

- **Description:** Control bits for receiver in recovered domain
- **Size:** 16 bits
- **Offset:** 0x1N1d
- **Exists:** Always

**Table 12-79 Fields for Register: LANEN\_DIG\_RX\_CDR\_LOCK\_VEC\_OVRD**

| Bits  | Name             | Memory Access | Description                                                                                                                                                                                                                                                                                                 |
|-------|------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | adap_ctr_level   | R/W           | <b>Description:</b> Ammount of earlies that increment the adaptation counter (times 32, Default is 512). (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x10<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1f<br><b>Volatile:</b> true                                         |
| 10    | adap_polarity    | R/W           | <b>Description:</b> If asserted invert default adaptation adjustment for equalization. IF early decrease equalization. Normal mode is to decrease. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true |
| 9     | lock_vector_ovrd | R/W           | <b>Description:</b> Override enable for the rx_eq outputs. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true                                                                                         |
| 8     | lock_vector_en   | R/W           | <b>Description:</b> Override value for the locked_vector output completion. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1<br><b>Volatile:</b> true                                                                        |

**Table 12-79 Fields for Register: LANEN\_DIG\_RX\_CDR\_LOCK\_VEC\_OVRD (Continued)**

| Bits | Name        | Memory Access | Description                                                                                                                                                                                                         |
|------|-------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | lock_vector | R/W           | <b>Description:</b> Override value for the locked_vector. (volatile and 2 reads needed to read value)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xff<br><b>Volatile:</b> true |

## 12.2.76 LANEN\_DIG\_RX\_CDR\_LOCK\_VEC

- **Description:** Current values for the locked vector..
- **Size:** 16 bits
- **Offset:** 0x1N1e
- **Exists:** Always

**Table 12-80 Fields for Register: LANEN\_DIG\_RX\_CDR\_LOCK\_VEC**

| Bits  | Name                | Memory Access | Description                                                                                                                                                                                                                             |
|-------|---------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:12 | RESERVED_15_12      | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                           |
| 11:9  | eq_rx_eq            | R             | <b>Description:</b> Equalization setting from the Equalization Loop. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |
| 8     | eq_locked_vector_en | R             | <b>Description:</b> Equalization locked vector has been filled. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true      |
| 7:0   | eq_locked_vector    | R             | <b>Description:</b> Results of equalization loop. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                    |

## 12.2.77 LANEN\_DIG\_RX\_CDR\_ADAP\_FSM

- **Description:** Observation of adaptation FSM register.
- **Size:** 16 bits
- **Offset:** 0x1N1f
- **Exists:** Always

**Table 12-81 Fields for Register: LANEN\_DIG\_RX\_CDR\_ADAP\_FSM**

| Bits  | Name       | Memory Access | Description                                                                                                                                                                                                                                                                                                                       |
|-------|------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | mstr_ctr   | R             | <b>Description:</b> Master count register. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                                                                                                     |
| 10:7  | loop_ctr   | R             | <b>Description:</b> Loop count register. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                                                                                                       |
| 6:3   | adap_ctr   | R             | <b>Description:</b> Adaptation count register. (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true                                                                                                                 |
| 2:0   | adap_state | R             | <b>Description:</b> Adaptation State. NOTE: ADAP_RESET = 3'b000 ADAP_LOCK = 3'b001 ADAP_SUFF = 3'b010 ADAP_LOOP = 3'b011 ADAP_MSTR = 3'b100 ADAP_DONE = 3'b101 (RU - read-only volatile) (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0<br><b>Volatile:</b> true |



## 12.2.78 LANEN\_ANA\_RX\_ATB0

- **Description:** ATB Control Register
- **Size:** 16 bits
- **Offset:** 0x1N20
- **Exists:** Always

**Table 12-82 Fields for Register: LANEN\_ANA\_RX\_ATB0**

| Bits | Name          | Memory Access | Description                                                                                                                                                                         |
|------|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                       |
| 7    | EN_ATB        | R/W           | <b>Description:</b> Enable ATB sensing and forcing on internal RX nodes<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |
| 6    | EN_MARG       | R/W           | <b>Description:</b> Enable margining mode in receiver; it puts EQ common mode output on atb_f_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 5    | EN_ATB_RM_F   | R/W           | <b>Description:</b> Enable atb_f_p on negative-side termination resistor<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 4    | EN_ATB_RM_S   | R/W           | <b>Description:</b> Enable atb_s_m on negative-side termination resistor<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 3    | EN_ATB_RP_F   | R/W           | <b>Description:</b> Enable atb_f_p on positive-side termination resistor<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |

**Table 12-82 Fields for Register: LANEN\_ANA\_RX\_ATB0 (Continued)**

| Bits | Name        | Memory Access | Description                                                                                                                                                            |
|------|-------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | EN_ATB_RP_S | R/W           | <b>Description:</b> Enable atb_s_p on positive-side termination resistor<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1           |
| 1    | EN_ATB_VOS  | R/W           | <b>Description:</b> Put rxafe filtered outputs vo_p on atb_s_p and vo_m on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | EN_ATB_VOFF | R/W           | <b>Description:</b> Put rxafe voff_p on atb_s_p and voff_m on atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1              |

**12.2.79 LANEN\_ANA\_RX\_ATB1**

- **Description:** ATB and Misc Control Register
- **Size:** 16 bits
- **Offset:** 0x1N21
- **Exists:** Always

**Table 12-83 Fields for Register: LANEN\_ANA\_RX\_ATB1**

| Bits | Name                   | Memory Access | Description                                                                                                                                                                                                                                   |
|------|------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8          | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                 |
| 7:6  | rx_dcc_iq_offset_h_4_3 | R/W           | <b>Description:</b> Code which helps to manually tune reference voltage (r-string dac) for IQ opamp and correct input offset (rx_dcc_iq_offset_h_4_3 bit)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |
| 5    | EN_ATB_VLOS            | R/W           | <b>Description:</b> Enable sensing of LOS reference voltage on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                    |
| 4    | EN_ATB_VRF             | R/W           | <b>Description:</b> Enable sensing of vref_rx (RX Bias B.G. Voltage) on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                           |
| 3    | MEAS_GD                | R/W           | <b>Description:</b> Enable sensing of local gd in RX; ties gd to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                  |
| 2    | MEAS_VP                | R/W           | <b>Description:</b> Enable sensing of local vp in RX; ties vp to atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                  |

**Table 12-83 Fields for Register: LANEN\_ANA\_RX\_ATB1 (Continued)**

| Bits | Name                   | Memory Access | Description                                                                                                                                                                                                                                   |
|------|------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1:0  | rx_dcc_iq_offset_h_2_1 | R/W           | <b>Description:</b> Code which helps to manually tune reference voltage (r-string dac) for IQ opamp and correct input offset (rx_dcc_iq_offset_h_2_1 bit)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |

## 12.2.80 LANEN\_ANA\_RX\_ENPWR0

- **Description:** RX State Control Register
- **Size:** 16 bits
- **Offset:** 0x1N22
- **Exists:** Always

**Table 12-84 Fields for Register: LANEN\_ANA\_RX\_ENPWR0**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                            |
|------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                          |
| 7    | CTL_RXPWRON   | R/W           | <b>Description:</b> MUX select for AFE en to select between CTL_RXPWRON=0 rx_afe_en (External signal) or CTL_RXPWRON=1 LCL_RXPWRON<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 6    | LCL_RXPWRON   | R/W           | <b>Description:</b> Enable/Disable RX AFE .This bit is only selected when CTL_RXPWRON = 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                          |
| 5    | CTL_EN_LOS    | R/W           | <b>Description:</b> MUX select for LOS en to select between CTL_EN_LOS=0 rx_los_en (External signal) or CTL_EN_LOS=1 LCL_EN_LOS<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 4    | LCL_EN_LOS    | R/W           | <b>Description:</b> Enable/Disable LOS.This bit is only selected when CTL_EN_LOS = 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                               |

**Table 12-84 Fields for Register: LANEN\_ANA\_RX\_ENPWR0 (Continued)**

| Bits | Name     | Memory Access | Description                                                                                                                                                                                                           |
|------|----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | CTL_RXCK | R/W           | <b>Description:</b> MUX select for RX Clock en to select between CTL_RXCK=0 rx_clock_en (External signal) or CTL_RXCK=1 LCL_RXCK.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 2    | LCL_RXCK | R/W           | <b>Description:</b> Enable/Disable en_rx_clock (RX clock enable). This bit is only selected when CTL_RXCK = 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 1    | CTL_ACJT | R/W           | <b>Description:</b> MUX select for ACJTAG to select between CTL_ACJTG=0 rx_acjtg_en (External signal) or CTL_ACJTG=1 LCL_ACJT.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 0    | LCL_ACJT | R/W           | <b>Description:</b> Enable/Disable acjtag block, This bit is only selected when CTL_ACJTG = 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                     |

### 12.2.81 LANEN\_ANA\_RX\_PMI\_X\_PHASE

- **Description:** Phase select Value
- **Size:** 16 bits
- **Offset:** 0x1N23
- **Exists:** Always

**Table 12-85 Fields for Register: LANEN\_ANA\_RX\_PMI\_X\_PHASE**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                              |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                            |
| 7:0  | PHASE         | R/W           | <b>Description:</b> Write to bits 8 to 1 of the phase select register in the phase mixer through the CDR (USER2 volatile - read-write volatile and custom)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xff<br><b>Volatile:</b> true |

## 12.2.82 LANEN\_ANA\_RX\_ENPWR1

- **Description:** Additional RX State Control Register
- **Size:** 16 bits
- **Offset:** 0x1N24
- **Exists:** Always

**Table 12-86 Fields for Register: LANEN\_ANA\_RX\_ENPWR1**

| Bits | Name              | Memory Access | Description                                                                                                                                                                                                                                                   |
|------|-------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8     | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                 |
| 7    | CTL_PHASE_REG_RST | R/W           | <b>Description:</b> MUX select for Phase Register Reset to select between CTL_PHASE_REG_RST=0 rx_phase_reg_rst (External signal) or CTL_PHASE_REG_RST=1 LCL_PHASE_REG_RST<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 6    | LCL_PHASE_REG_RST | R/W           | <b>Description:</b> Reset phase register. This bit is only selected when CTL_PHASE_REG_RST= 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                             |
| 5    | CTL_BST           | R/W           | <b>Description:</b> MUX select for AFE Boost to select between CTL_BST=0 rx_aeq[2:0] (External signal) or CTL_BST=1 LCL_BST<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                               |
| 4:2  | LCL_BST           | R/W           | <b>Description:</b> RX Boost (equalization) Value<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7                                                                                                                         |



**Table 12-86 Fields for Register: LANEN\_ANA\_RX\_ENPWR1 (Continued)**

| Bits | Name       | Memory Access | Description                                                                                                                                                                                                                  |
|------|------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | CTL_RXTERM | R/W           | <b>Description:</b> MUX select for RX Term enable to select between CTL_RXTERM=0 rx_term_en (External signal) or CTL_RXTERM=1 LCL_RXTERM<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | LCL_RXTERM | R/W           | <b>Description:</b> Enable/disable RX termination when CTL_RXTERM = 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                    |

### 12.2.83 LANEN\_ANA\_RX\_ENPWR2

- **Description:** Yet another RX State Control Register
- **Size:** 16 bits
- **Offset:** 0x1N25
- **Exists:** Always

**Table 12-87 Fields for Register: LANEN\_ANA\_RX\_ENPWR2**

| Bits | Name                  | Memory Access | Description                                                                                                                                                                                                                                 |
|------|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8         | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                               |
| 7    | EN_RXPMIX_TST         | R/W           | <b>Description:</b> Enable XOR gate to test linearity of RX phase mixer using atb_s_p and atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                       |
| 6    | EN_RXPMIX_VPMIX       | R/W           | <b>Description:</b> Put vreg_pmix on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                            |
| 5    | EN_RXPMIX_VRX         | R/W           | <b>Description:</b> Put vreg_rx on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                              |
| 4    | rx_dcc_iq_offset_h_0  | R/W           | <b>Description:</b> Code which helps to manually tune reference voltage (r-string dac) for IQ opamp and correct input offset (rx_dcc_iq_offset_h_0 bit)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 3    | en_rxpmix_frc_vpmix_h | R/W           | <b>Description:</b> Connecting vbg_in to vbg (in case of 0) or to atb_s_m_int (in case of 1)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                            |

**Table 12-87 Fields for Register: LANEN\_ANA\_RX\_ENPWR2 (Continued)**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                |
|------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | RX_SCOPE_ATB_2 | R/W           | <b>Description:</b> Put XOR of RX SCOPE PMIX input and output on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |
| 1    | RX_SCOPE_ATB_1 | R/W           | <b>Description:</b> Put RX SCOPE regulated VP on atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                               |
| 0    | RX_SCOPE_ATB_0 | R/W           | <b>Description:</b> Have RX SCOPE reference for regulator use atb_f_p as a reference instead of VP/VBG<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.84 LANEN\_ANA\_RX\_SCOPE\_VDCC

- **Description:** RX Scope controls and clock DCC control voltage probe
- **Size:** 16 bits
- **Offset:** 0x1N26
- **Exists:** Always

**Table 12-88 Fields for Register: LANEN\_ANA\_RX\_SCOPE\_VDCC**

| Bits | Name            | Memory Access | Description                                                                                                                                                                                                                                                 |
|------|-----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                               |
| 7:6  | VDCC            | R/W           | <b>Description:</b> Probe DCC control voltage<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                                                                                                           |
| 5    | NC5             | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                            |
| 4    | RX_SCOPE_SLEW   | R/W           | <b>Description:</b> Set high for low RX clock frquencies (625MHz) for RX Scope to work correctly<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                        |
| 3    | RX_SCOPE_FDIV20 | R/W           | <b>Description:</b> Divide scope output clock by 20 instead of 10<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                       |
| 2    | rx_los_lfps_en  | R/W           | <b>Description:</b> In case of rx_los_lfps_en_cntrl = 1/0 reg_rx_los_lfps_en/rx_ana_los_lfps_en signal directly applied to rx_los circuit through lfps_mux multiplexer.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

**Table 12-88 Fields for Register: LANEN\_ANA\_RX\_SCOPE\_VDCC (Continued)**

| Bits | Name                  | Memory Access | Description                                                                                                                                                                                                                   |
|------|-----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | rx_dcc_iq_offset_en_h | R/W           | <b>Description:</b> Select ref voltage for IQ opamp from testmode part (in case of 1) or from ordinary resistive dividers (in case of 0).<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | RX_LOS_LFPS_EN        | R/W           | <b>Description:</b> Enables the RX LOS LFPS mode, use when you want to detect incoming LFPS signals, USB3.0<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |

### 12.2.85 LANEN\_ANA\_TX\_TXDRV\_CNTRL

- **Description:** Registers intended for control of TX's drivers states i.e pull\_up , pull\_dn , vcm\_hold etc.Register 3
- **Size:** 16 bits
- **Offset:** 0x1N2b
- **Exists:** Always

**Table 12-89 Fields for Register: LANEN\_ANA\_TX\_TXDRV\_CNTRL**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                                                          |
|------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                        |
| 7    | PULL_DN_REG   | R/W           | <b>Description:</b> register bit that causes the calibrated TX bits to pull down in a common mode fashion. If pull_dn_reg and tx_pull_up are both high, then pull_dn_reg wins<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 6    | PULL_UP_REG   | R/W           | <b>Description:</b> Register override for tx_pull_up. Used when ovr_d_pull_up is HIGH, causes calibrated TX bits to pull up in a common mode fashion, unless pull_dn_reg is HIGH<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 5    | OVRD_PULL_UP  | R/W           | <b>Description:</b> Selects local value of pull_up_reg instead of tx_pull_up<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                     |
| 4    | VCM_HOLD_REG  | R/W           | <b>Description:</b> Register override for tx_vcm_hold. Used when ovr_d_vcm_hold is HIGH, controls the TX common mode hold circuitry<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |

**Table 12-89 Fields for Register: LANEN\_ANA\_TX\_TXDRV\_CNTRL (Continued)**

| Bits | Name                | Memory Access | Description                                                                                                                                                                                                                                     |
|------|---------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | OVRD_VCM_HOLD       | R/W           | <b>Description:</b> Selects local value of vcm_hold_reg instead of tx_vcm_hold to control state of TX common mode hold circuitry<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                            |
| 2    | REFGEN_PWRDN_OVRD_N | R/W           | <b>Description:</b> Set LOW for TX regulator not to turn on when RX clock is enabled. Default is for TX regulator to turn on when RX or TX clock is enabled<br><b>Value After Reset:</b> 0x1<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | NOCONN_1            | R/W           | <b>Description:</b> Not used but reserved to use for future in case of new functionality<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                    |
| 0    | NOCONN_0            | R/W           | <b>Description:</b> Not used but reserved to use for future in case of new functionality<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                    |

## 12.2.86 LANEN\_ANA\_TX\_POWER\_CTL

- **Description:** Registers intended for control of TX's sub cells states such as pwn , divider clk enb , general TX's clk enb etc, tx\_serializer enable etc.Register 4
- **Size:** 16 bits
- **Offset:** 0x1N2c
- **Exists:** Always

**Table 12-90 Fields for Register: LANEN\_ANA\_TX\_POWER\_CTL**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                              |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                            |
| 7    | OVRD_EN       | R/W           | <b>Description:</b> Enables local overrides for all signals in this register except for TX_DIV_CLK_EN which doesn't need this signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 6    | SERIAL_EN_REG | R/W           | <b>Description:</b> Forces state of tx_serial_en when OVRD_EN is 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                   |
| 5    | CLK_EN_REG    | R/W           | <b>Description:</b> Forces state of tx_clk_en when OVRD_EN is 1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                      |
| 4    | DATA_EN_REG   | R/W           | <b>Description:</b> Forces state of tx_data_en when OVRD_EN is HIGH<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                  |



**Table 12-90 Fields for Register: LANEN\_ANA\_TX\_POWER\_CTL (Continued)**

| Bits | Name               | Memory Access | Description                                                                                                                                                                                                                                                                                                           |
|------|--------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | REFGEN_EN_REG      | R/W           | <b>Description:</b> Register override value for tx_refgen_en. Controls the TX regulator, biasing circuitry and the rxdetect comparators when OVRD_EN is HIGH. This function may be overridden by the state of refgen_pwdn_reg<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |
| 2    | TX_DIV_CLK_EN      | R/W           | <b>Description:</b> Enables the div clock that goes from the TX to the undersampler, more appropriately called tx_sampler_clk_en, this clock comes after the optional divide by 2 or 4, tx_clk_en must be HIGH to get a clock out<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 1    | REFGEN_PDN_REG     | R/W           | <b>Description:</b> Register override for refgen pwn. Used when OVRD_EN is HIGH. Controls the TX regulator, biasing circuitry and the rxdetect comparators when OVRD_EN is HIGH<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                   |
| 0    | LFPS_high_priority | R/W           | <b>Description:</b> When this is set high LFPS priority becomes higher than tx_data<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                               |

## 12.2.87 LANEN\_ANA\_TX\_ALT\_BLOCK

- **Description:** Registers intended for control of TX's output states i.e. data transmission mode , LFPS mode , and enb TX for alt bus mode etc.Register 5
- **Size:** 16 bits
- **Offset:** 0x1N2d
- **Exists:** Always

**Table 12-91 Fields for Register: LANEN\_ANA\_TX\_ALT\_BLOCK**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                        |
|------|----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                                                                                                                                      |
| 7    | EN_ALT_BUS     | R/W           | <b>Description:</b> Enables the TX for alt bus mode, powers up the TX regulator, biasing circuitry and the rxdetect comparators. Needed if manually running the alt bus features<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                               |
| 6:5  | DRV_SOURCE_REG | R/W           | <b>Description:</b> Forces state of tx_data_source when OVRD_ALT_BUS is HIGH. 00 = serializer data or alt oscillator vp/vph/vphreg/vptx if selected, 01 = jtag data differential, 10 = LFPS oscillator differential, 11 = jtag data common mode drive for test<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3 |
| 4    | JTAG_DATA_REG  | R/W           | <b>Description:</b> Forces state of jtag_data when OVRD_ALT_BUS is HIGH<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                        |

**Table 12-91 Fields for Register: LANEN\_ANA\_TX\_ALT\_BLOCK (Continued)**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                                                                                                                                                                                       |
|------|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3:1  | TX_RINGO_CNTRL | R/W           | <b>Description:</b> Enables and connects the various test oscillators to the transmit pins, must set drv_source_reg bus correctly. 0XX = All oscillators are disabled, 100 = HV oscillator enabled, 101 = SVT oscillator enabled, 110 = LVT oscillator enabled, 111 = ULVT oscillator enabled<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7 |
| 0    | OVRD_ALT_BUS   | R/W           | <b>Description:</b> Enables local overrides for alt-bus control signals: DRV_SOURCE_REG and JTAG_DATA_REG<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                                                                                                     |

### 12.2.88 LANEN\_ANA\_TX\_ALT\_AND\_LOOPBACK

- **Description:** Registers intended for control of loop back mode , analog test bus enb from various TX's sub cells such as tx\_vreg , tx\_dcc , enb top decpas ot tx etc.Register 6
- **Size:** 16 bits
- **Offset:** 0x1N2e
- **Exists:** Always

**Table 12-92 Fields for Register: LANEN\_ANA\_TX\_ALT\_AND\_LOOPBACK**

| Bits | Name          | Memory Access | Description                                                                                                                                                                |
|------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0              |
| 7    | OVRD_TX_LB    | R/W           | <b>Description:</b> Enables the override of the tx_lb_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |
| 6    | TX_LB_EN_REG  | R/W           | <b>Description:</b> Forces state of tx_lb_en when OVRD_TX_LB is enabled<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                |
| 5    | ATB_VPTX      | R/W           | <b>Description:</b> Connects vptx to atb_s_p and gd to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |
| 4    | ATB_VREG_TX   | R/W           | <b>Description:</b> Connects TX regulator input reference to atb_s_p and gd to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 3    | ATB_VDCCP     | R/W           | <b>Description:</b> Connects vdcc_p to atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                         |

**Table 12-92 Fields for Register: LANEN\_ANA\_TX\_ALT\_AND\_LOOPBACK (Continued)**

| Bits | Name            | Memory Access | Description                                                                                                                                            |
|------|-----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | ATB_VDCCM       | R/W           | <b>Description:</b> Connects vdcc_m to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 1    | NOCONN_01       | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                       |
| 0    | select_pmix_clk | R/W           | <b>Description:</b> select pmix_clk for TX clock for ATE test mode<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.89 LANEN\_ANA\_TX\_TX\_ATB\_REG

- **Description:** Registers intended for analog test bus enabling from various TX's sub cells. Register 7
- **Size:** 16 bits
- **Offset:** 0x1N2f
- **Exists:** Always

**Table 12-93 Fields for Register: LANEN\_ANA\_TX\_TX\_ATB\_REG**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                                                          |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                                                        |
| 7    | ATB_PBIAS     | R/W           | <b>Description:</b> Connect real pmos_bias voltage for TX PMOS driver pull up path to atb_s_p and local ground at the pmos_bias block to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 6    | ATB_VCM_REP   | R/W           | <b>Description:</b> Connect common mode replica voltage in pmos_bias block to atb_s_p and local ground to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                |
| 5    | ATB_RXDETREF  | R/W           | <b>Description:</b> Connect RX detect block reference voltage to atb_s_p and local ground to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                             |
| 4    | ATB_TXFP      | R/W           | <b>Description:</b> Connect tx_p to atb_f_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                      |
| 3    | ATB_TXFM      | R/W           | <b>Description:</b> Connect tx_m to atb_f_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                                      |

**Table 12-93 Fields for Register: LANEN\_ANA\_TX\_TX\_ATB\_REG (Continued)**

| Bits | Name     | Memory Access | Description                                                                                                                                                                              |
|------|----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | ATB_TXSP | R/W           | <b>Description:</b> Connect tx_p to atb_s_p<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                          |
| 1    | ATB_TXSM | R/W           | <b>Description:</b> Connect tx_m to atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                          |
| 0    | ATB_VCM  | R/W           | <b>Description:</b> Connect tx_p/tx_m common mode voltage onto atb_s_p and local ground onto atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.90 LANEN\_ANA\_TX\_TX\_BOOST

- **Description:** Registers intended for control of TX's drivers boost mode , tx\_vreg output decap enabling etc.Register 8
- **Size:** 16 bits
- **Offset:** 0x1N30
- **Exists:** Always

**Table 12-94 Fields for Register: LANEN\_ANA\_TX\_TX\_BOOST**

| Bits | Name           | Memory Access | Description                                                                                                                                                       |
|------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0     |
| 7    | OVRD_VBOOST_EN | R/W           | <b>Description:</b> Enables the overrides on VBOOST_EN_REG<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |
| 6    | VBOOST_EN_REG  | R/W           | <b>Description:</b> Forces state of tx_vboost_en, when OVRD_VBOOST_EN is HIGH<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 5    | VCON_EN_N      | R/W           | <b>Description:</b> Forces state of vcon_en_n<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                 |
| 4    | ATB_VBOOST     | R/W           | <b>Description:</b> Connects vboost to atb_s_p and local ground to Atb_s_m<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1    |
| 3    | ATB_VCMBOOST   | R/W           | <b>Description:</b> Connects vcmboost to atb_s_p and local ground to atb_s_m.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |



**Table 12-94 Fields for Register: LANEN\_ANA\_TX\_TX\_BOOST (Continued)**

| Bits | Name                | Memory Access | Description                                                                                                                                                                                                                       |
|------|---------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | ATB_F_VBOOST_REF    | R/W           | <b>Description:</b> Enables forcing of tx_vboost_vref through atb_f_p from external<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                           |
| 1    | OVRD_VREG_DECAP     | R/W           | <b>Description:</b> Enables override of vreg_decap_en_n signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                                                |
| 0    | VREG_DECAP_EN_N_REG | R/W           | <b>Description:</b> Forces state of vreg_decap_en_n control signal. Controls whether extra decoupling is switched in for TX regulator output.<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

### 12.2.91 LANEN\_DIG\_RX\_CDR\_FSM\_INIT\_CTL

- **Description:** Control bits for CDR init phase portion
- **Size:** 16 bits
- **Offset:** 0x1N40
- **Exists:** Always

**Table 12-95 Fields for Register: LANEN\_DIG\_RX\_CDR\_FSM\_INIT\_CTL**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                               |
|------|---------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:9 | RESERVED_15_9 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                             |
| 8    | INIT_CNT_OVRD | R/W           | <b>Description:</b> Active high override enable bit to select the override INIT_CNT register 0 = Disabled 1 = Enabled<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 7:0  | INIT_CNT      | R/W           | <b>Description:</b> Override value for CDR Init Phase<br><b>Value After Reset:</b> 0x10<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xff                                                               |

**12.2.92 LANEN\_DIG\_RX\_CDR\_FSM\_TRACK\_CTL**

- **Description:** Control bits for CDR track phase portion
- **Size:** 16 bits
- **Offset:** 0x1N41
- **Exists:** Always

**Table 12-96 Fields for Register: LANEN\_DIG\_RX\_CDR\_FSM\_TRACK\_CTL**

| Bits | Name           | Memory Access | Description                                                                                                                                                                                                |
|------|----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:9 | RESERVED_15_9  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                              |
| 8    | TRACK_CNT_OVRD | R/W           | <b>Description:</b> Active high override enable bit to select the override TRACK_CNT register 0 = Disabled 1 = Enabled<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 7:0  | TRACK_CNT      | R/W           | <b>Description:</b> Override value for CDR Track Phase<br><b>Value After Reset:</b> 0x18<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xff                                                               |

### 12.2.93 LANEN\_DIG\_RX\_CDR\_FSM\_LOCK\_CTL

- **Description:** Control bits for CDR lock phase portion
- **Size:** 16 bits
- **Offset:** 0x1N42
- **Exists:** Always

**Table 12-97 Fields for Register: LANEN\_DIG\_RX\_CDR\_FSM\_LOCK\_CTL**

| Bits | Name          | Memory Access | Description                                                                                                                                                                                               |
|------|---------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:9 | RESERVED_15_9 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                                             |
| 8    | LOCK_CNT_OVRD | R/W           | <b>Description:</b> Active high override enable bit to select the override LOCK_CNT register 0 = Disabled 1 = Enabled<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 7:0  | LOCK_CNT      | R/W           | <b>Description:</b> Override value for CDR Lock Phase<br><b>Value After Reset:</b> 0x40<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xff                                                               |

## 12.2.94 PORTN\_STAT\_RO\_0

- **Description:** Status Register 0
- **Size:** 16 bits
- **Offset:** 0x2000
- **Exists:** Always

**Table 12-98 Fields for Register: PORTN\_STAT\_RO\_0**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | PLL_Lock      | R             | <b>Description:</b> analog PLL Lock Signal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                |
| 2    | RX_Squelch    | R             | <b>Description:</b> analog RX Squelch Signal (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0  |
| 1    | RESERVED_1    | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 0    | RESERVED_0    | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |

## 12.2.95 PORTN\_STAT\_RO\_1

- **Description:** Status Register 1
- **Size:** 16 bits
- **Offset:** 0x2001
- **Exists:** Always

**Table 12-99 Fields for Register: PORTN\_STAT\_RO\_1**

| Bits | Name            | Memory Access | Description                                                                                                                                                   |
|------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | vbus_valid_pu   | R             | <b>Description:</b> DRVVBUS Input status<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                  |
| 2    | otg_pu          | R             | <b>Description:</b> OTGDISABLE Input status<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |
| 1    | tx_vref_vreg_pu | R             | <b>Description:</b> TX hs_driver Comparator Power Up<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 0    | sqr_x_pu        | R             | <b>Description:</b> RX squelch Comparator Power Up<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |

## 12.2.96 PORTN\_STAT\_RO\_2

- **Description:** Status Register 2
- **Size:** 16 bits
- **Offset:** 0x2002
- **Exists:** Always

**Table 12-100 Fields for Register: PORTN\_STAT\_RO\_2**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | comp_pu       | R             | <b>Description:</b> resistor tune Comparator Power Up<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 2    | bias_pu       | R             | <b>Description:</b> MPLL Power Up<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                         |
| 1    | pll_pu        | R             | <b>Description:</b> MPLL Reset, active low<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                |
| 0    | xo_pu         | R             | <b>Description:</b> XO Power Up, require for External Crystal<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1             |

## 12.2.97 PORTN\_STAT\_RO\_3

- **Description:** Status Register 3
- **Size:** 16 bits
- **Offset:** 0x2003
- **Exists:** Always

**Table 12-101 Fields for Register: PORTN\_STAT\_RO\_3**

| Bits | Name             | Memory Access | Description                                                                                                                                                   |
|------|------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4    | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | shrtser_shortdet | R             | <b>Description:</b> Register status for Serial Interface USB short<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1        |
| 2    | TxValid          | R             | <b>Description:</b> UTMI+ Input TXVALID (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0       |
| 1    | TxValidH         | R             | <b>Description:</b> UTMI+ Input TXVALIDH (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0      |
| 0    | TxReady          | R             | <b>Description:</b> UTMI+ Input TXREADY<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |



**12.2.98 PORTN\_STAT\_RO\_4**

- **Description:** Status Register 4
- **Size:** 16 bits
- **Offset:** 0x2004
- **Exists:** Always

**Table 12-102 Fields for Register: PORTN\_STAT\_RO\_4**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | RxActive      | R             | <b>Description:</b> UTMI+ Output RXACTIVE<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                 |
| 2    | RxValid       | R             | <b>Description:</b> UTMI+ Output RXVALID<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                  |
| 1    | RxValidH      | R             | <b>Description:</b> UTMI+ Output RXVALIDH<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                 |
| 0    | RxError       | R             | <b>Description:</b> UTMI+ Output RXERROR<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                  |

## 12.2.99 PORTN\_STAT\_RO\_5

- **Description:** Status Register 5
- **Size:** 16 bits
- **Offset:** 0x2005
- **Exists:** Always

Table 12-103 Fields for Register: PORTN\_STAT\_RO\_5

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | DataOut       | R             | <b>Description:</b> UTMI+ Output DATAOUT<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf                                  |

**12.2.100 PORTN\_STAT\_RO\_6**

- **Description:** Status Register 6
- **Size:** 16 bits
- **Offset:** 0x2006
- **Exists:** Always

**Table 12-104 Fields for Register: PORTN\_STAT\_RO\_6**

| Bits | Name            | Memory Access | Description                                                                                                                                                   |
|------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | tx_sliver_en_ct | R             | <b>Description:</b> Tx Parallel Resistance Circuit<br><b>Value After Reset:</b> 0x8<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf                        |

## 12.2.101 PORTN\_STAT\_RO\_7

- **Description:** Status Register 7
- **Size:** 16 bits
- **Offset:** 0x2007
- **Exists:** Always

**Table 12-105 Fields for Register: PORTN\_STAT\_RO\_7**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | DataIn        | R             | <b>Description:</b> UTMI+ input DATAIN (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0        |

**12.2.102 PORTN\_STAT\_RO\_8**

- **Description:** Status Register 8
- **Size:** 16 bits
- **Offset:** 0x2008
- **Exists:** Always

**Table 12-106 Fields for Register: PORTN\_STAT\_RO\_8**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | DMinus        | R             | <b>Description:</b> Single-Ended Rx D- Output (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 2    | DPlus         | R             | <b>Description:</b> Single-Ended Rx D+ Output (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 1    | HSData        | R             | <b>Description:</b> HS Differential Rx Data (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0   |
| 0    | FSData        | R             | <b>Description:</b> FS Differential Rx Data (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0   |

### 12.2.103 PORTN\_STAT\_RO\_9

- **Description:** Status Register 9
- **Size:** 16 bits
- **Offset:** 0x2009
- **Exists:** Always

Table 12-107 Fields for Register: PORTN\_STAT\_RO\_9

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | RESERVED_3    | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 2    | tx_short      | R             | <b>Description:</b> HS Tx Short Detect<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                    |
| 1    | bistDone      | R             | <b>Description:</b> BIST Done<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                             |
| 0    | bistError     | R             | <b>Description:</b> BIST Error<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                            |

**12.2.104 PORTN\_STAT\_RO\_10**

- **Description:** Status Register 10
- **Size:** 16 bits
- **Offset:** 0x200a
- **Exists:** Always

**Table 12-108 Fields for Register: PORTN\_STAT\_RO\_10**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | RESERVED_3_0  | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf                                              |

## 12.2.105 PORTN\_STAT\_RO\_11

- **Description:** Status Register 11
- **Size:** 16 bits
- **Offset:** 0x200b
- **Exists:** Always

Table 12-109 Fields for Register: PORTN\_STAT\_RO\_11

| Bits | Name               | Memory Access | Description                                                                                                                                                      |
|------|--------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4      | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0    |
| 3    | RESERVED_3         | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                 |
| 2    | comp_dis_latch_enb | R             | <b>Description:</b> Disconnect Comparator Latch Enable<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |
| 1    | comp_dis           | R             | <b>Description:</b> Disconnect Comparator Output (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 0    | RESERVED_0         | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                 |



**12.2.106 PORTN\_STAT\_RO\_12**

- **Description:** Status Register 12
- **Size:** 16 bits
- **Offset:** 0x200c
- **Exists:** Always

**Table 12-110 Fields for Register: PORTN\_STAT\_RO\_12**

| Bits | Name           | Memory Access | Description                                                                                                                                                   |
|------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4  | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3    | underOverDebug | R             | <b>Description:</b> Indicates ElasticBuf Under/Over<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |
| 2    | receive_start  | R             | <b>Description:</b> Indicates UTMIP start receiving packet<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                |
| 1    | RESERVED_1     | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 0    | RESERVED_0     | R             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |

### 12.2.107 PORTN\_STAT\_RO\_13

- **Description:** Status Register 13
- **Size:** 16 bits
- **Offset:** 0x200d
- **Exists:** Always

**Table 12-111 Fields for Register: PORTN\_STAT\_RO\_13**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | tap_val       | R             | <b>Description:</b> TAP Value<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf                                             |

**12.2.108 PORTN\_STAT\_RO\_14**

- **Description:** Status Register 14
- **Size:** 16 bits
- **Offset:** 0x200e
- **Exists:** Always

**Table 12-112 Fields for Register: PORTN\_STAT\_RO\_14**

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | RESERVED_3_0  | R             | <b>Description:</b> Reserved (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                  |

## 12.2.109 PORTN\_STAT\_RO\_15

- **Description:** Status Register 15
- **Size:** 16 bits
- **Offset:** 0x200f
- **Exists:** Always

Table 12-113 Fields for Register: PORTN\_STAT\_RO\_15

| Bits | Name          | Memory Access | Description                                                                                                                                                   |
|------|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | RESERVED_15_4 | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 3:0  | RESERVED_3_0  | R             | <b>Description:</b> Reserved (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                  |

**12.2.110 PORTN\_OVRD\_RW\_0**

- **Description:** Override Register 0
- **Size:** 16 bits
- **Offset:** 0x2010
- **Exists:** Always

**Table 12-114 Fields for Register: PORTN\_OVRD\_RW\_0**

| Bits | Name                | Memory Access | Description                                                                                                                                                   |
|------|---------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8       | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7:4  | overrd_addr_OVRD_EN | W             | <b>Description:</b> Override Enable for overrd_addr [7:4]<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf                 |
| 3:0  | overrd_addr         | R/W           | <b>Description:</b> Analog Test Control Override Address [3:0]<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf            |

### 12.2.111 PORTN\_OVRD\_RW\_1

- **Description:** Override Register 1
- **Size:** 16 bits
- **Offset:** 0x2011
- **Exists:** Always

**Table 12-115 Fields for Register: PORTN\_OVRD\_RW\_1**

| Bits | Name                    | Memory Access | Description                                                                                                                                                   |
|------|-------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8           | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | vbus_valid_pu_OVRD_EN   | W             | <b>Description:</b> Override Enable for vbus_valid_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 6    | otg_pu_OVRD_EN          | W             | <b>Description:</b> Override Enable for otg_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                            |
| 5    | tx_vref_vreg_pu_OVRD_EN | W             | <b>Description:</b> Override Enable for tx_vref_vreg_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                   |
| 4    | sqr_x_pu_OVRD_EN        | W             | <b>Description:</b> Override Enable for sqr_x_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 3    | vbus_valid_pu           | R/W           | <b>Description:</b> DRVVBUS Input (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0             |
| 2    | otg_pu                  | R/W           | <b>Description:</b> OTGDISABLE Input (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0          |

**Table 12-115 Fields for Register: PORTN\_OVRD\_RW\_1 (Continued)**

| Bits | Name            | Memory Access | Description                                                                                                                                                          |
|------|-----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | tx_vref_vreg_pu | R/W           | <b>Description:</b> TX hs_driver Comparator Power Up (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 0    | sqr_x_pu        | R/W           | <b>Description:</b> RX squelch Comparator Power Up (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0   |

### 12.2.112 PORTN\_OVRD\_RW\_2

- **Description:** Override Register 2
- **Size:** 16 bits
- **Offset:** 0x2012
- **Exists:** Always

**Table 12-116 Fields for Register: PORTN\_OVRD\_RW\_2**

| Bits | Name            | Memory Access | Description                                                                                                                                                           |
|------|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0         |
| 7    | comp_pu_OVRD_EN | W             | <b>Description:</b> Override Enable for comp_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |
| 6    | bias_pu_OVRD_EN | W             | <b>Description:</b> Override Enable for bias_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |
| 5    | pll_pu_OVRD_EN  | W             | <b>Description:</b> Override Enable for pll_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                    |
| 4    | xo_pu_OVRD_EN   | W             | <b>Description:</b> Override Enable for xo_pu<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                     |
| 3    | comp_pu         | R/W           | <b>Description:</b> resistor tune Comparator Power Up (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 2    | bias_pu         | R/W           | <b>Description:</b> MPLL Power Up (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                     |



**Table 12-116 Fields for Register: PORTN\_OVRD\_RW\_2 (Continued)**

| Bits | Name   | Memory Access | Description                                                                                                                                                |
|------|--------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | pll_pu | R/W           | <b>Description:</b> MPLL Reset, active low (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 0    | xo_pu  | R/W           | <b>Description:</b> XO Power Up (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0            |

### 12.2.113 PORTN\_OVRD\_RW\_3

- **Description:** Override Register 3
- **Size:** 16 bits
- **Offset:** 0x2013
- **Exists:** Always

**Table 12-117 Fields for Register: PORTN\_OVRD\_RW\_3**

| Bits | Name                  | Memory Access | Description                                                                                                                                                   |
|------|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8         | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | RESERVED_7            | W             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 6    | XO_always_on_OVRD_EN  | W             | <b>Description:</b> Override Enable for XO_always_on<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 5    | serclk_rx_enb_OVRD_EN | W             | <b>Description:</b> Override Enable for serclk_rx_enb<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 4    | serclk_tx_enb_OVRD_EN | W             | <b>Description:</b> Override Enable for serclk_tx_enb<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 3    | RESERVED_3            | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 2    | XO_always_on          | R/W           | <b>Description:</b> Set for XO Power Up<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |

**Table 12-117 Fields for Register: PORTN\_OVRD\_RW\_3 (Continued)**

| Bits | Name          | Memory Access | Description                                                                                                              |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------|
| 1    | serclk_rx_enb | R/W           | <b>Description:</b> Enable RX Serclk<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | serclk_tx_enb | R/W           | <b>Description:</b> Enable TX Serclk<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

## 12.2.114 PORTN\_OVRD\_RW\_4

- **Description:** Override Register 4
- **Size:** 16 bits
- **Offset:** 0x2014
- **Exists:** Always

Table 12-118 Fields for Register: PORTN\_OVRD\_RW\_4

| Bits | Name                | Memory Access | Description                                                                                                                                                   |
|------|---------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8       | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | LoopbackEnb_OVRD_EN | W             | <b>Description:</b> Override Enable for LoopbackEnb<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                       |
| 6    | RESERVED_6          | W             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 5    | VDPU_EN2_OVRD_EN    | W             | <b>Description:</b> Override Enable for VDPU_EN2<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 4    | VDPU_EN1_OVRD_EN    | W             | <b>Description:</b> Override Enable for VDPU_EN1<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 3    | LoopbackEnb         | R/W           | <b>Description:</b> Enable Lookback (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0           |
| 2    | RESERVED_2          | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |

**Table 12-118 Fields for Register: PORTN\_OVRD\_RW\_4 (Continued)**

| Bits | Name     | Memory Access | Description                                                                                                                                               |
|------|----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | VDPU_EN2 | R/W           | <b>Description:</b> D+ Pull-Up Override 2 (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 0    | VDPU_EN1 | R/W           | <b>Description:</b> D+ Pull-Up Override 1 (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |

### 12.2.115 PORTN\_OVRD\_RW\_5

- **Description:** Override Register 5
- **Size:** 16 bits
- **Offset:** 0x2015
- **Exists:** Always

**Table 12-119 Fields for Register: PORTN\_OVRD\_RW\_5**

| Bits | Name                    | Memory Access | Description                                                                                                                                                          |
|------|-------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8           | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0        |
| 7:6  | xo_clk_div_OVRD_EN      | W             | <b>Description:</b> Override Enable for xo_clk_div<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                               |
| 5    | mstr_overrd_bit_OVRD_EN | W             | <b>Description:</b> Override Enable for mstr_overrd_bit<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 4    | overrd_addr_4_OVRD_EN   | W             | <b>Description:</b> Override Enable for overrd_addr_4<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                            |
| 3:2  | xo_clk_div              | R/W           | <b>Description:</b> setting reference clock division (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 1    | mstr_overrd_bit         | R/W           | <b>Description:</b> Analog Test Control Master Override<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 0    | overrd_addr_4           | R/W           | <b>Description:</b> Analog Test Control Override Address [4]<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |

**12.2.116 PORTN\_OVRD\_RW\_6**

- **Description:** Override Register 6
- **Size:** 16 bits
- **Offset:** 0x2016
- **Exists:** Always

**Table 12-120 Fields for Register: PORTN\_OVRD\_RW\_6**

| Bits | Name                 | Memory Access | Description                                                                                                                                                   |
|------|----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8        | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7:4  | tx_sliver_en_OVRD_EN | W             | <b>Description:</b> Override Enable for tx_sliver_en [7:4]<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf                |
| 3:0  | tx_sliver_en         | R/W           | <b>Description:</b> Signal for Tx Parallel Resistance Circuit [3:0]<br><b>Value After Reset:</b> 0x8<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0xf       |

## 12.2.117 PORTN\_OVRD\_RW\_7

- **Description:** Override Register 7
- **Size:** 16 bits
- **Offset:** 0x2017
- **Exists:** Always

**Table 12-121 Fields for Register: PORTN\_OVRD\_RW\_7**

| Bits | Name                    | Memory Access | Description                                                                                                                                                         |
|------|-------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8           | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0       |
| 7    | tx_ls_en_OVRD_EN        | W             | <b>Description:</b> Override Enable for tx_ls_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                |
| 6:4  | pwr_seq_clk_div_OVRD_EN | W             | <b>Description:</b> Override Enable for pwr_seq_clk_div<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x7                         |
| 3    | tx_ls_en                | R/W           | <b>Description:</b> TX LS Enable<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                |
| 2:0  | pwr_seq_clk_div         | R/W           | <b>Description:</b> setting power_sequence division (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |



**12.2.118 PORTN\_OVRD\_RW\_8**

- **Description:** Override Register 8
- **Size:** 16 bits
- **Offset:** 0x2018
- **Exists:** Always

**Table 12-122 Fields for Register: PORTN\_OVRD\_RW\_8**

| Bits | Name            | Memory Access | Description                                                                                                                                                   |
|------|-----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8   | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | bistEn_OVRD_EN  | W             | <b>Description:</b> Override Enable for bistEn<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                            |
| 6    | hs_bist_OVRD_EN | W             | <b>Description:</b> Override Enable for hs_bist<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 5    | fs_bist_OVRD_EN | W             | <b>Description:</b> Override Enable for fs_bist<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 4    | ls_bist_OVRD_EN | W             | <b>Description:</b> Override Enable for ls_bist<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 3    | bistEn          | R/W           | <b>Description:</b> BIST Enable<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                           |
| 2    | hs_bist         | R/W           | <b>Description:</b> HS BIST<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                               |

**Table 12-122 Fields for Register: PORTN\_OVRD\_RW\_8 (Continued)**

| Bits | Name    | Memory Access | Description                                                                                                     |
|------|---------|---------------|-----------------------------------------------------------------------------------------------------------------|
| 1    | fs_bist | R/W           | <b>Description:</b> FS BIST<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | ls_bist | R/W           | <b>Description:</b> LS BIST<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

**12.2.119 PORTN\_OVRD\_RW\_9**

- **Description:** Override Register 9
- **Size:** 16 bits
- **Offset:** 0x2019
- **Exists:** Always

**Table 12-123 Fields for Register: PORTN\_OVRD\_RW\_9**

| Bits | Name                   | Memory Access | Description                                                                                                                                                   |
|------|------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8          | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | inst_bist_err_OVRD_EN  | W             | <b>Description:</b> Override Enable for inst_bist_err`<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |
| 6    | pll_lock_sts_OVRD_EN   | W             | <b>Description:</b> Override Enable for pll_lock_sts<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 5    | dm_pulldown_en_OVRD_EN | W             | <b>Description:</b> Override Enable for dm_pulldown_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |
| 4    | dp_pulldown_en_OVRD_EN | W             | <b>Description:</b> Override Enable for dp_pulldown_en<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |
| 3    | inst_bist_err          | R/W           | <b>Description:</b> Insert BIST Error<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                     |
| 2    | pll_lock_sts           | R/W           | <b>Description:</b> analog MPLL Lock Status<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                               |

**Table 12-123 Fields for Register: PORTN\_OVRD\_RW\_9 (Continued)**

| Bits | Name           | Memory Access | Description                                                                                                                                            |
|------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | dm_pulldown_en | R/W           | <b>Description:</b> D- Pulldown Enable (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 0    | dp_pulldown_en | R/W           | <b>Description:</b> D+ Pulldown Enable (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |

**12.2.120 PORTN\_OVRD\_RW\_10**

- **Description:** Override Register 10
- **Size:** 16 bits
- **Offset:** 0x201a
- **Exists:** Always

**Table 12-124 Fields for Register: PORTN\_OVRD\_RW\_10**

| Bits | Name                  | Memory Access | Description                                                                                                                                                   |
|------|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8         | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | RESERVED_7            | W             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 6    | fs_assert_se0_OVRD_EN | W             | <b>Description:</b> Override Enable for fs_assert_se0<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 5    | fs_output_enb_OVRD_EN | W             | <b>Description:</b> Override Enable for fs_output_enb<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 4    | fs_dataout_OVRD_EN    | W             | <b>Description:</b> Override Enable for fs_dataout<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 3    | RESERVED_3            | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                              |
| 2    | fs_assert_se0         | R/W           | <b>Description:</b> FS TX SE0 (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                 |

**Table 12-124 Fields for Register: PORTN\_OVRD\_RW\_10 (Continued)**

| Bits | Name          | Memory Access | Description                                                                                                                                      |
|------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | fs_output_enb | R/W           | <b>Description:</b> FS TX Enable (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 0    | fs_dataout    | R/W           | <b>Description:</b> FS TX Data (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0   |

**12.2.121 PORTN\_OVRD\_RW\_11**

- **Description:** Override Register 11
- **Size:** 16 bits
- **Offset:** 0x201b
- **Exists:** Always

**Table 12-125 Fields for Register: PORTN\_OVRD\_RW\_11**

| Bits | Name                    | Memory Access | Description                                                                                                                                                            |
|------|-------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8           | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0          |
| 7    | res_req_in_OVRD_EN      | W             | <b>Description:</b> Override Enable for res_req_in<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                 |
| 6    | incr_tune_dis_OVRD_EN   | W             | <b>Description:</b> Override Enable for incr_tune_dis<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                              |
| 5    | comp_sel_analog_OVRD_EN | W             | <b>Description:</b> Override Enable for comp_sel_analog<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                            |
| 4    | RESERVED_4              | W             | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                                       |
| 3    | res_req_in              | R/W           | <b>Description:</b> resistor tune external requirement (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 2    | incr_tune_dis           | R/W           | <b>Description:</b> Disable Incremental Tune<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                       |

**Table 12-125 Fields for Register: PORTN\_OVRD\_RW\_11 (Continued)**

| Bits | Name            | Memory Access | Description                                                                                                                                   |
|------|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | comp_sel_analog | R/W           | <b>Description:</b> Disconnect Block Comparator Test Mode<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | RESERVED_0      | R/W           | <b>Description:</b> Reserved<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                              |



**12.2.122 PORTN\_OVRD\_RW\_12**

- **Description:** Override Register 12
- **Size:** 16 bits
- **Offset:** 0x201c
- **Exists:** Always

**Table 12-126 Fields for Register: PORTN\_OVRD\_RW\_12**

| Bits | Name                | Memory Access | Description                                                                                                                                                                |
|------|---------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8       | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0              |
| 7    | res_ack_out_OVRD_EN | W             | <b>Description:</b> Override Enable for res_ack_out<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                    |
| 6:5  | pll_clk_div_OVRD_EN | W             | <b>Description:</b> Override Enable for pll_clk_div<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                                    |
| 4    | pll_reset_OVRD_EN   | W             | <b>Description:</b> Override Enable for pll_reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                      |
| 3    | res_ack_out         | R/W           | <b>Description:</b> Ack resistor tune external requirement (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 2:1  | pll_clk_div         | R/W           | <b>Description:</b> setting MPLL division (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0                  |

**Table 12-126 Fields for Register: PORTN\_OVRD\_RW\_12 (Continued)**

| Bits | Name      | Memory Access | Description                                                                                                             |
|------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------|
| 0    | pll_reset | R/W           | <b>Description:</b> MPLL gear_shift<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |

**12.2.123 PORTN\_OVRD\_RW\_13**

- **Description:** Override Register 13
- **Size:** 16 bits
- **Offset:** 0x201d
- **Exists:** Always

**Table 12-127 Fields for Register: PORTN\_OVRD\_RW\_13**

| Bits | Name                 | Memory Access | Description                                                                                                                                                   |
|------|----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8        | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | bist_mode_OVRD_EN    | W             | <b>Description:</b> Override Enable for bist_mode<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |
| 6    | test_pkt_rpt_OVRD_EN | W             | <b>Description:</b> Override Enable for test_pkt_rpt<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 5:4  | xo_clk_sel_OVRD_EN   | W             | <b>Description:</b> Override Enable for xo_clk_sel<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x3                        |
| 3    | bist_mode            | R/W           | <b>Description:</b> BIST Mode<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                             |
| 2    | test_pkt_rpt         | R/W           | <b>Description:</b> BIST Test Packet Mode<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                 |
| 1:0  | xo_clk_sel           | R/W           | <b>Description:</b> Reference Clock Selection (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |

## 12.2.124 PORTN\_OVRD\_RW\_14

- **Description:** Override Register 14
- **Size:** 16 bits
- **Offset:** 0x201e
- **Exists:** Always

**Table 12-128 Fields for Register: PORTN\_OVRD\_RW\_14**

| Bits | Name                  | Memory Access | Description                                                                                                                                                   |
|------|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8         | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | tap_sel_ovrd_OVRD_EN  | W             | <b>Description:</b> Override Enable for tap_sel_ovrd<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                      |
| 6    | tap_shift_enb_OVRD_EN | W             | <b>Description:</b> Override Enable for tap_shift_enb<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 5    | tap_shift_dir_OVRD_EN | W             | <b>Description:</b> Override Enable for tap_shift_dir<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                     |
| 4    | Squelch_OVRD_EN       | W             | <b>Description:</b> Override Enable for Squelch<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                           |
| 3    | tap_sel_ovrd          | R/W           | <b>Description:</b> TAP Select Override<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                   |
| 2    | tap_shift_enb         | R/W           | <b>Description:</b> TAP Shift Enable<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                                      |

**Table 12-128 Fields for Register: PORTN\_OVRD\_RW\_14 (Continued)**

| Bits | Name          | Memory Access | Description                                                                                                                                    |
|------|---------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | tap_shift_dir | R/W           | <b>Description:</b> TAP Shift Direction<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                    |
| 0    | Squelch       | R/W           | <b>Description:</b> RX Squelch (actual reset value is 0xX)<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |

## 12.2.125 PORTN\_OVRD\_RW\_15

- **Description:** Override Register 15
- **Size:** 16 bits
- **Offset:** 0x201f
- **Exists:** Always

**Table 12-129 Fields for Register: PORTN\_OVRD\_RW\_15**

| Bits | Name                    | Memory Access | Description                                                                                                                                                   |
|------|-------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | RESERVED_15_8           | R             | <b>Description:</b> Reserved for Future use and actual reset value is 0xX<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x0 |
| 7    | ClkUsable_OVRD_EN       | W             | <b>Description:</b> Override Enable for ClkUsable<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                         |
| 6    | scale_down_mode_OVRD_EN | W             | <b>Description:</b> Override Enable for scale_down_mode<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                   |
| 5    | pll_lock_OVRD_EN        | W             | <b>Description:</b> Override Enable for pll_lock<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 4    | reset_tm_OVRD_EN        | W             | <b>Description:</b> Override Enable for reset_tm<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                          |
| 3    | ClkUsable               | R/W           | <b>Description:</b> Signal for gating PLLDiv Clock<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1                        |
| 2    | scale_down_mode         | R/W           | <b>Description:</b> Scale-Down Enable for Disconnect Timers<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1               |

**Table 12-129 Fields for Register: PORTN\_OVRD\_RW\_15 (Continued)**

| Bits | Name     | Memory Access | Description                                                                                                                      |
|------|----------|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1    | pll_lock | R/W           | <b>Description:</b> PLL Lock Enable for MPLL<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1 |
| 0    | reset_tm | R/W           | <b>Description:</b> Test Interface Reset<br><b>Value After Reset:</b> 0x0<br><b>Exists:</b> Always<br><b>Reset Mask:</b> 0x1     |





# A

## JTAG Port

This appendix includes the following sections:

- [“Configuring the JTAG Port”](#) on page 445
- [“Using 1149.6 AC JTAG Boundary Scan”](#) on page 451
- [“DFTMAX Boundary Scan Flow for USB 3.0 femtoPHY”](#) on page 462

### A.1 Configuring the JTAG Port

This section provides information about configuring the JTAG port.

#### A.1.1 JTAG Port

The USB 3.0 femtoPHY includes a JTAG TAP controller with an 18-bit CRSEL register that can be used to access the internal control registers. While access to these registers is not required for normal PHY operation, this interface is used to access the BIST and diagnostic features of the PHY during ATE testing in production.

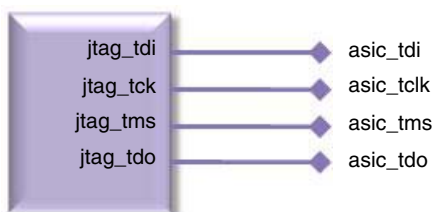
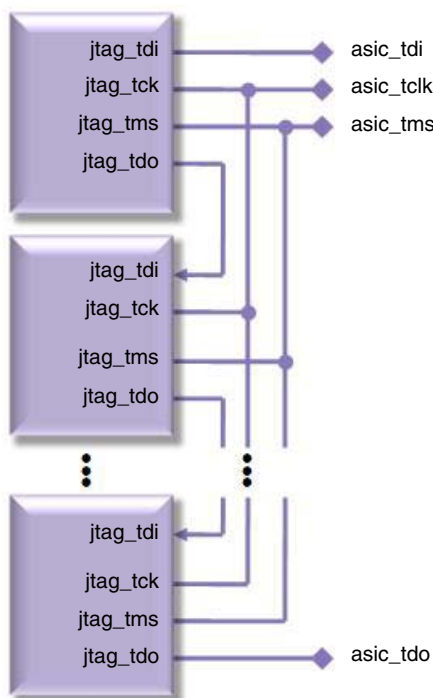
For details about the testing methodology and the control registers used to implement analog testing of the PHY, see [“Production Test and Debug Implementation”](#) on page 215 and [“Registers”](#) on page 253, respectively.

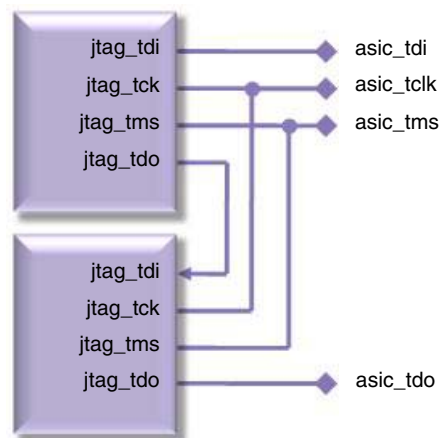
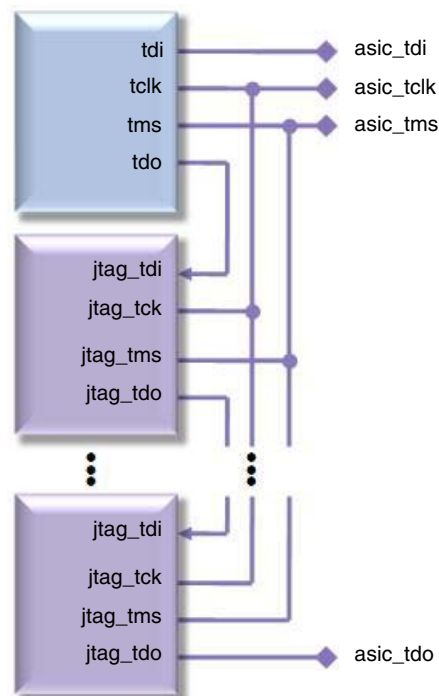
The USB 3.0 femtoPHY IP includes software that connects to the PHY through a USB 2-to-JTAG dongle and provides access to the `jtag_tdi`, `jtag_tdo`, `jtag_tms`, and `jtag_tck` signals. [Figure A-1](#) through [Figure A-5](#) on page 448 show several ways to provide access to these signals from off-chip for both single and multiple PHY instances on the chip.

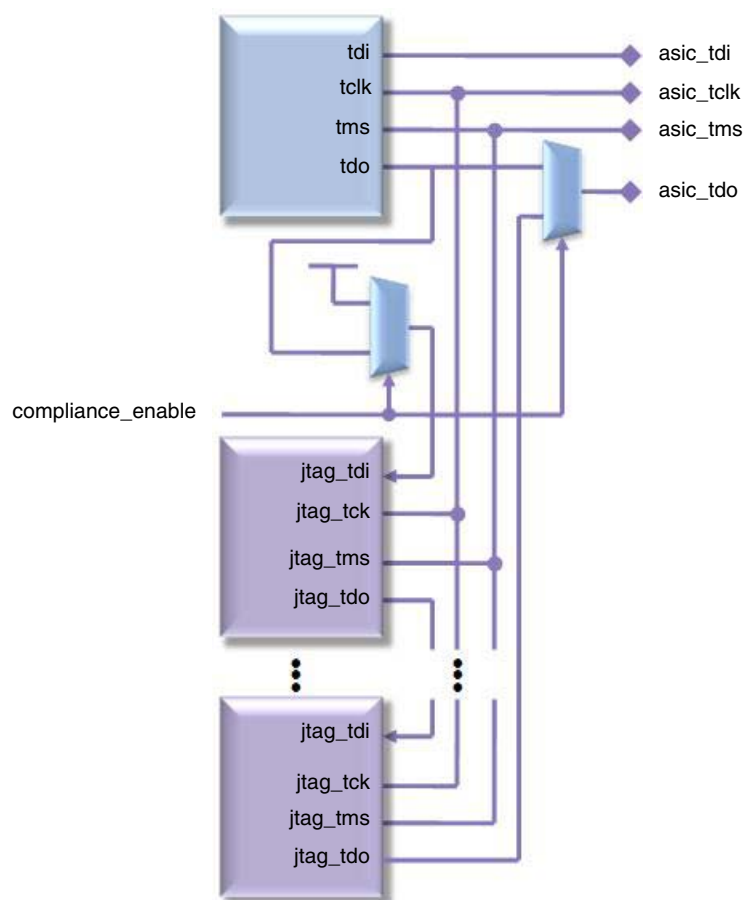
[Figure A-5](#) shows connectivity where a JTAG TAP controller on the ASIC implements boundary scan (that requires one TAP controller per chip in Boundary Scan mode) and a compliance enable pin (or register output bit) hides the PHYs that would otherwise be in the JTAG chain. For ATE and diagnostic access, the compliance pin is simply set low, and the PHYs appear in the JTAG chain. This implementation is compliant with IEEE standard 1149.1, which requires a single TAP controller during boundary scan testing.

**Note**

jtag\_tdo\_en is an output enable signal for the TDO output driver. In the event that the USB 3.0 femtoPHY IP is the last block in the JTAG chain and this block drives the TDO output driver, jtag\_tdo\_en can be used to enable the tri-state driver. Otherwise, jtag\_tdo\_en can be left floating.

**Figure A-1 Dedicated JTAG Pins: Single PHY****Figure A-2 Dedicated Pads With  $N$  PHYs**

**Figure A-3 Dedicated JTAG Pins: Dual PHYs****Figure A-4 Multiple Tap Controllers (No Boundary Scan)**

**Figure A-5 Multiple Tap Controllers With Single Boundary Scan “Compliance Enable” Pin**

## A.1.2 Accessing the Control Registers Through JTAG

Direct access to the control registers through JTAG without using the USB 2-to-JTAG dongle can be made through the five-wire JTAG port by shifting a command and either register address or data into the CRSEL register in the TAP controller in the IP. The CMD field determines what type of access is performed. The following table provides the encodings defined for the CMD field in the CRSEL register. All actions are initiated during the UPDATE-DR phase of the JTAG state machine when the CRSEL register is accessed.

**Table A-1 CRSEL Register Command**

| CMD   | Encoding | Description                                                                                                                                                                                                                          |
|-------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADDR  | 2'b00    | Captures ADDRESS field for use in a subsequent WRITE command. Shifts current value from most recently accessed register (via READ or WRITE) out on TDO.                                                                              |
| WRITE | 2'b01    | Writes the value in the DATA field to the register selected by the address captured during the most recent ADDR or READ command. Shifts current value from most recently accessed register (via READ or WRITE) out on TDO.           |
| DATA  | 2'b10    | Shifts current value from most recently accessed register (via READ or WRITE) out on TDO. DATA value shifted in is not used.                                                                                                         |
| READ  | 2'b11    | Performs a read operation to the register selected by the ADDRESS field to make its current value available for subsequent CRSEL accesses. Shifts current value from most recently accessed register (via READ or WRITE) out on TDO. |

To perform a control register read, simply shift a READ command into the CRSEL register along with the target register address in the ADDRESS field. Any subsequent CRSEL access causes the data returned by the read to be shifted out of TDO. This process enables a limited form of pipelining READ commands, because the second access can be shifting in the command and address for the subsequent read while shifting out the data from the previous read.

To perform a control register write, the address of the register to be written must first be selected by performing either an ADDR or READ command and specifying the register address in the ADDRESS field. A second access using the WRITE command and specifying the data to be written in the DATA field causes the write to finally be performed in the target register.

### A.1.3 JTAG Run-Test-Idle (RTI) Cycles

When performing JTAG write and read operations, RTI cycles are used between instructions to ensure the PHY is either ready to return data (in the case of a read) or has sufficient time to capture the incoming data (in the case of a write).

The PHY contains a datapath pipeline that each JTAG READ and WRITE command must go through. This pipeline is clocked on the internal ref\_clk, and as a result, the time taken to propagate through the pipeline is proportional to both the jtag\_tck and the ref\_clk.

#### Example A-1 JTAG READ and WRITE Command Sequences

JTAG READ:

1. Shift in READ command with read address.
2. Wait for nRTI cycles.

For information about calculating the nRTI, see [“Calculating nRTI”](#) on page 450.

3. Shift in next command and shift out read data.

JTAG WRITE:

1. Shift in ADDRESS command with write address.
2. Shift in WRITE command with write data.
3. Wait for nRTI cycles.

For information about calculating the nRTI, see [“Calculating nRTI”](#) on page 450.

4. Shift in next command.

#### A.1.3.1 Calculating nRTI

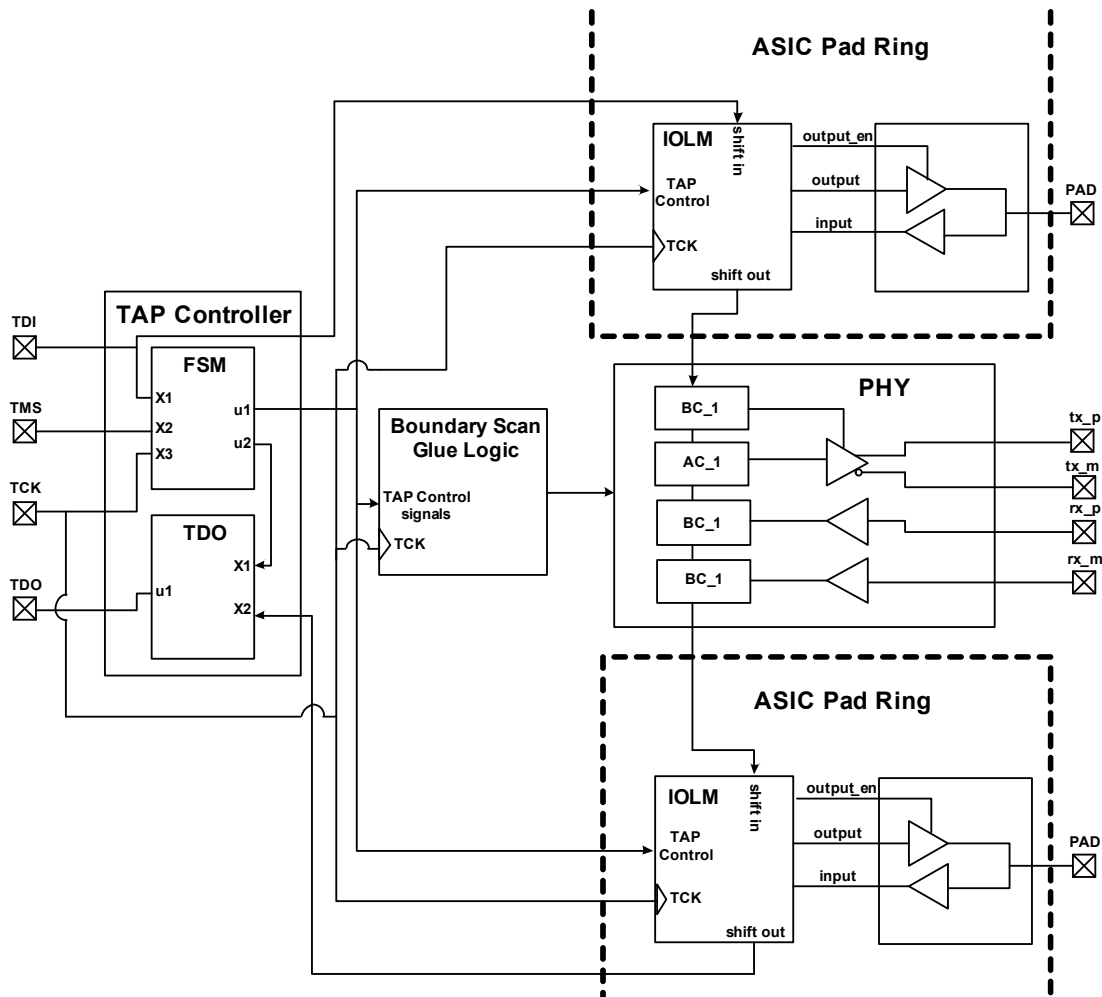
The number of RTI cycles depends on the ratio of the jtag\_tck and ref\_clk. Specifically, when the ratio is at or near 1:1 (jtag\_tck clock to ref\_clk), the minimum number of RTI cycles is 16.5. This value can be scaled to accommodate any jtag\_tck-to-reference clock ratio as follows:

$$\text{Min RTI cycles} = \text{ceil} (14 \times T_{\text{ref}} / T_{\text{tck}} + 2.5)$$

## A.2 Using 1149.6 AC JTAG Boundary Scan

The SuperSpeed serial link differential pairs for the USB 3.0 femtoPHY include AC coupling capacitors that are soldered on the board in series with the high-speed signals. To test for proper soldering of these external components on the board, the PHY supports IEEE standard 1149.6 AC JTAG Boundary Scan. As shown in the following figure, the USB 3.0 femtoPHY boundary scan includes its own I/O logic module (IOLM). This cell includes the boundary shift register and primary logic to support I/O testing of the differential transmitter and receiver pins.

**Figure A-6 Boundary Scan ASIC Hierarchy**



During integration and testing, consider the following:

- This implementation is flop-based and therefore, requires a boundary scan clock.
- For tool-based boundary scan chain creation, see [“DFTMAX Boundary Scan Flow for USB 3.0 femtoPHY”](#) on page 462.
- You can find Design Compiler information for automated BSDL chain construction in the BSD Compiler User Guide or the Test Automation Collection on SolvNet. In particular, pay attention to the “Inserting Boundary- Scan Components” chapter.

**Note**

The maximum frequency of boundary scan clock is 50 MHz.

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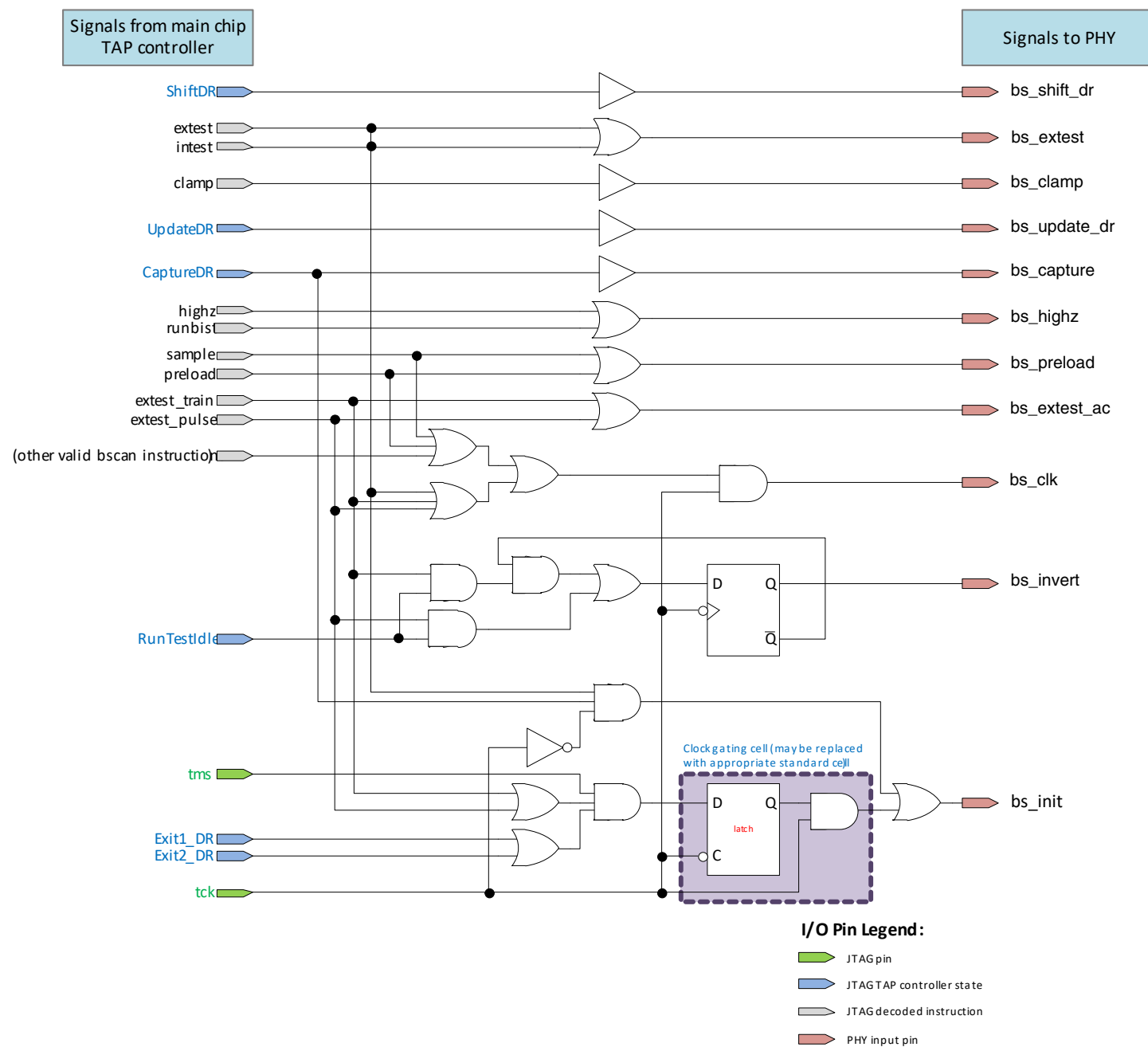
The main TAP controller on the chip controls the boundary scan inputs to the PHY, and these inputs are independent of the USB 3.0 femtoPHY’s JTAG port. [Figure A-7](#) shows the chip-level glue logic required to connect the chip TAP controller to the boundary scan pins of the IP.

**Note**

These JTAG pins belong to the main chip boundary scan TAP controller, not to the USB 3.0 femtoPHY’s JTAG port.

---



**Figure A-7 Required Boundary Scan Glue Logic**

The glue logic uses the following TAP controller states and decoded JTAG instructions from the chip's TAP controller.

**Table A-2     JTAG TAP Controller States and JTAG Instructions**

| JTAG TAP Controller States                                              | JTAG Decoded Instructions                                                                         | JTAG Input Pins   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------|
| ShiftDR<br>CaptureDR<br>UpdateDR<br>RunTestIdle<br>EXIT1_DR<br>EXIT2_DR | sample<br>preload<br>extest<br>etest_train<br>extest_pulse<br>clamp<br>highz<br>intest<br>runbist | tms<br>tdi<br>tck |

With this glue logic controlling the boundary scan inputs to the USB 3.0 femtoPHY, the boundary scan chain itself is connected to bs\_in and bs\_out. The actual scan chain order is bs\_in -> Tx\_transmit\_enable -> tx\_p\_m data -> Rx\_p -> Rx\_m -> bs\_out. tx\_p\_m data drives the shifted-in value on \_p and the inverse on \_m. [Figure A-8](#) shows a functional diagram of the AC JTAG chain.

Figure 10 is a detailed block diagram of the RX and TX logic. It shows the internal structure of the RX and TX blocks, including registers, multiplexers, and logic gates. The RX block has inputs RXM and RXP, and outputs bs\_out and bs\_shift\_dr. The TX block has inputs TXP and TXM, and outputs bs\_highz and bs\_in. The diagram includes a legend for register\_enable and bs\_update\_dr signals, and a note about \*\_dr signals.

Legend:

- register\_enable = bs\_preload | bs\_extest | bs\_extest\_ac
- bs\_update\_dr = bs\_extest | bs\_extest\_ac | bs\_clamp | bs\_highz

Note: \*\_dr signals are not clocks. These signals are connected to the enable pin of the flip-flop.

Figure A-9 shows a functional diagram of the comparator associated with the receive pins shown in Figure A-8. During testing, the AC JTAG level sets the cell's hysteresis trip point for comparing the initial value to the initial RX value in the boundary scan cell. During AC testing, the Hysteresis block's output is then used to desensitize the cell after the pulse is triggered. For DC testing, to ensure that a continuous value above or below the set threshold is available for sampling, the initial value is looped back to form a memory element.

### Figure A-9 RX Cell

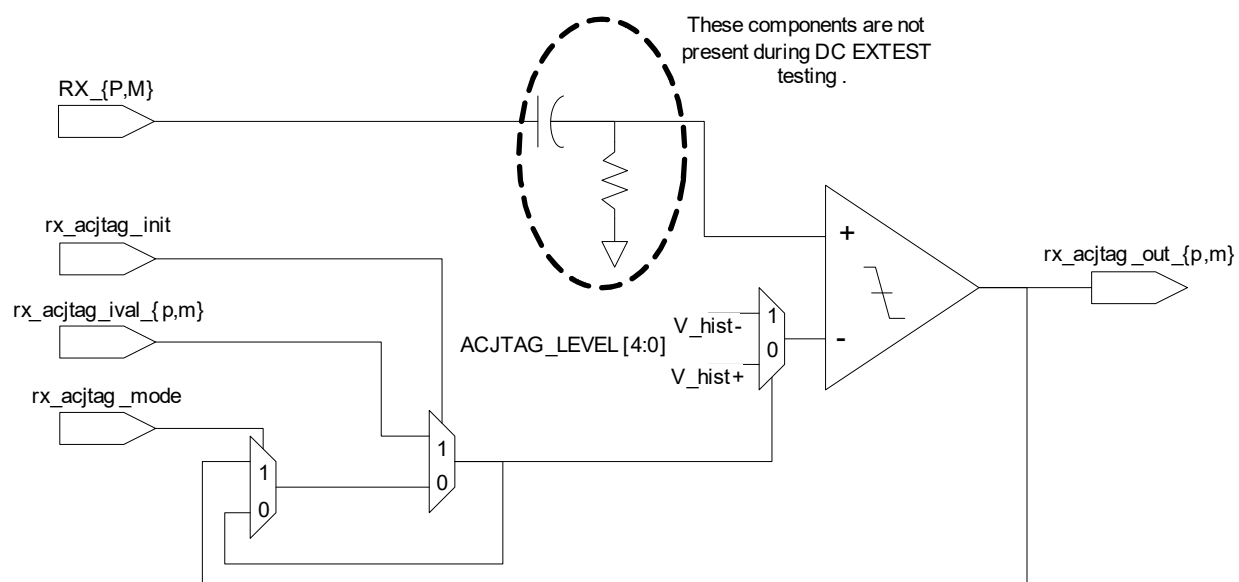
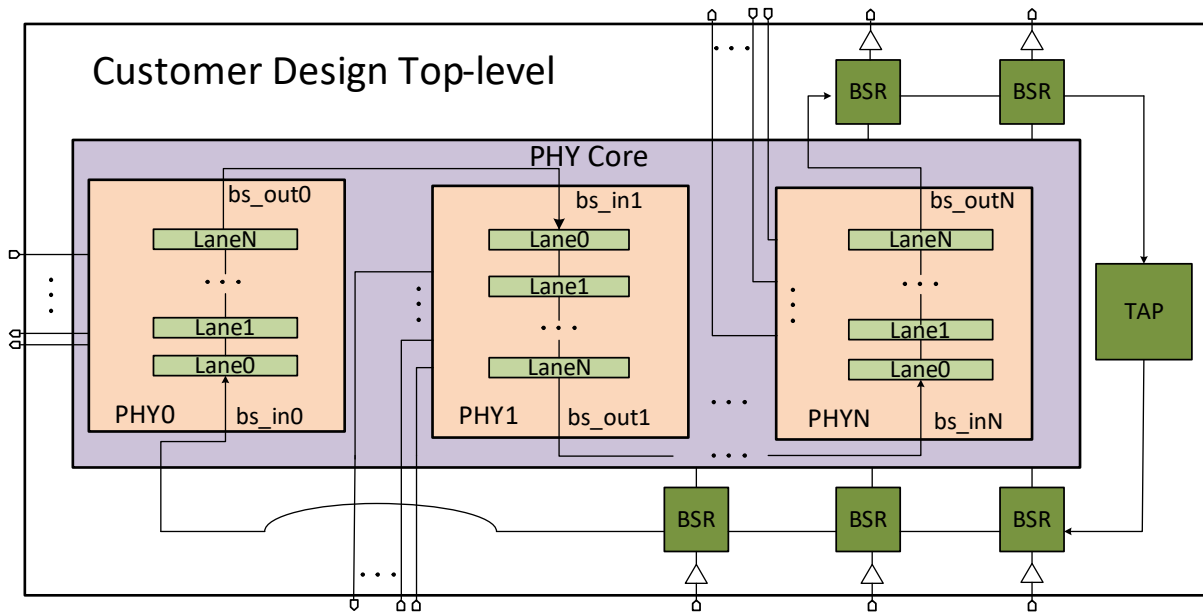
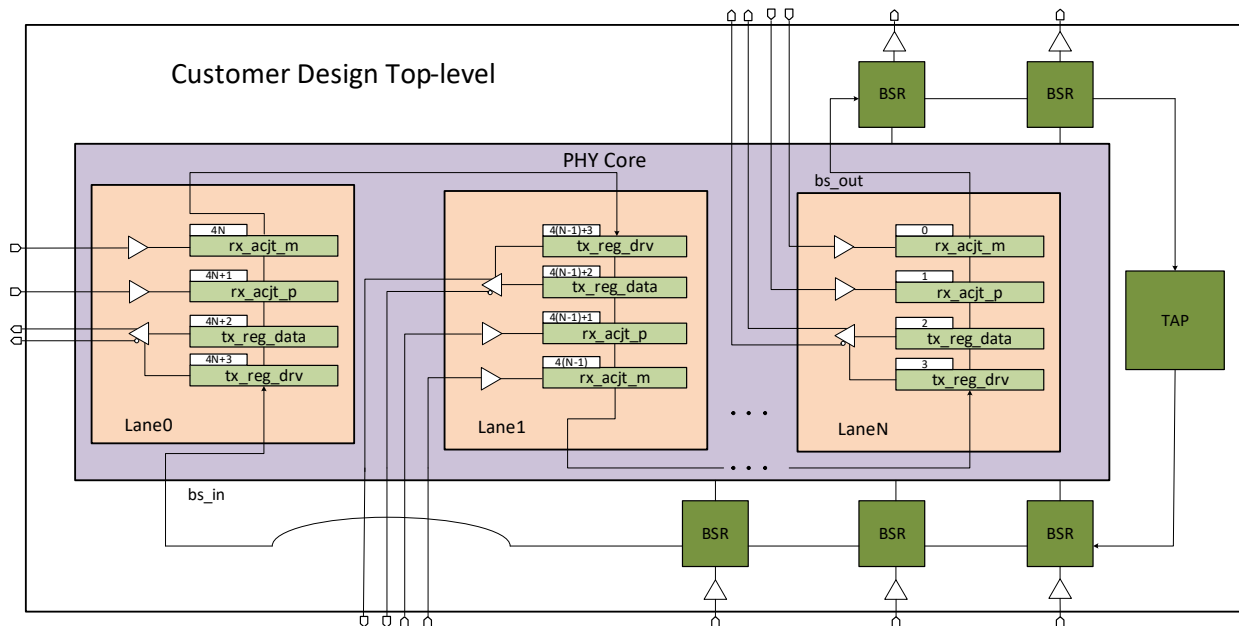


Figure A-10 and Figure A-11 show connectivity of a single PHY/multiple PHYs with multiple lanes. In Figure A-10, boundary scan registers (BSR) of multiple PHYs are connected to a daisy chain. In Figure A-11, *bs\_in* is connected to Lane0 and *bs\_out* is connected to LaneN; therefore, the BSR values of LaneN are shifted out first, and the values of Lane0 are shifted out last. Note that in the following figures, the dark green colored BSR's represent top-level basic boundary scan registers within the SoC.

**Figure A-10 BSR Connections for Multiple PHYs**



**Figure A-11 BSR Connections for a Single PHY With Multiple Lanes**



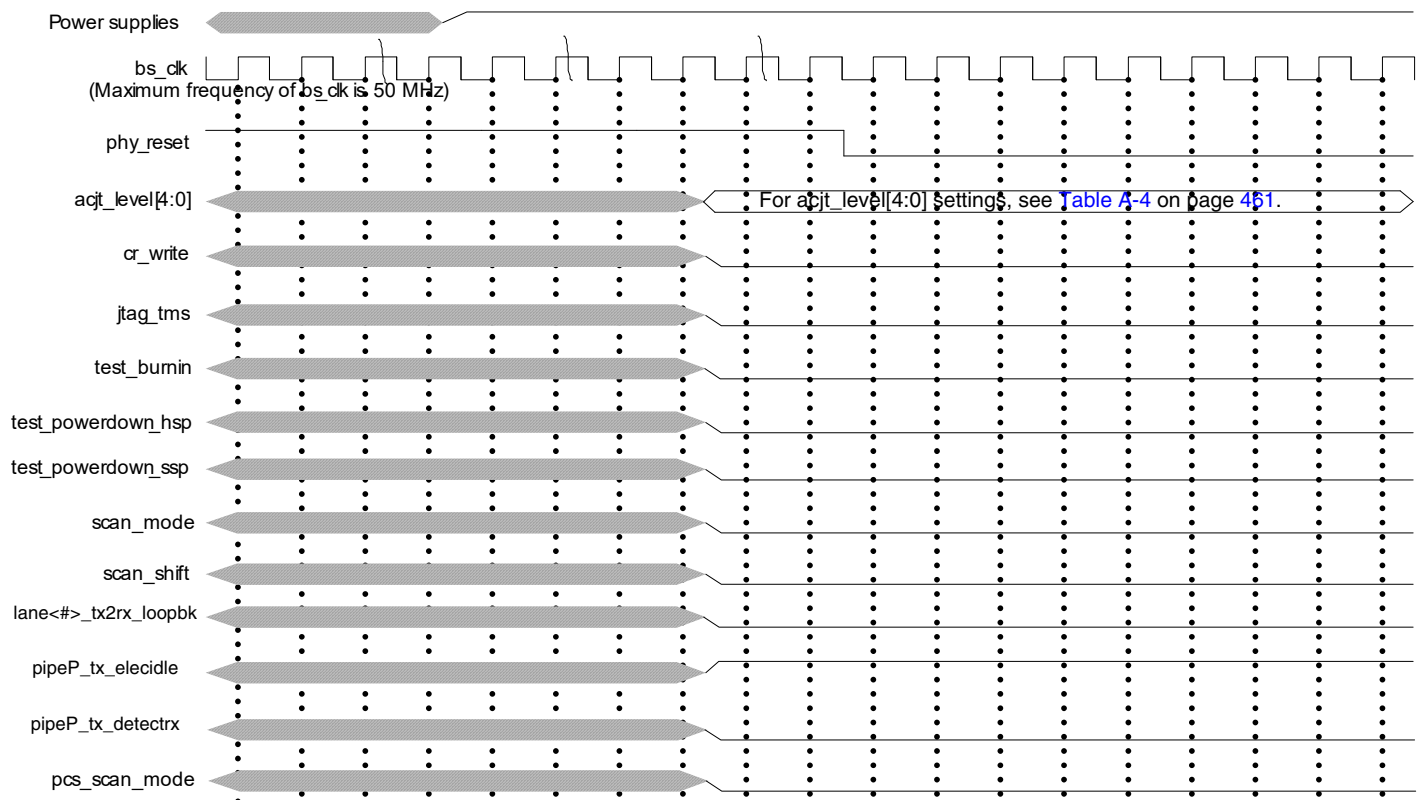
In addition to adding the glue logic to control the boundary scan input signals, the signals shown in the following table must also be driven to their indicated values.

**Table A-3 PHY Compliance Pin Setting**

| Signal Name          | Value                   |
|----------------------|-------------------------|
| cr_write             | 0                       |
| phy_reset            | Apply high pulse (PoR). |
| jtag_tms             | 0                       |
| scan_mode            | 0                       |
| scan_shift           | 0                       |
| test_burnin          | 0                       |
| test_powerdown_ssp   | 0                       |
| test_powerdown_hsp   | 0                       |
| pipeP_tx_elecidle    | 1                       |
| pipeP_tx_detectrx    | 0                       |
| lane<#>_tx2rx_loopbk | 0                       |
| pcs_scan_mode        | 0                       |
| ref_pad_clk_(p,m)    | Parked                  |
| ref_alt_clk_(p,m)    | Parked                  |

The following figure shows the initialization state of the PHY for boundary scan.

**Figure A-12 Boundary Scan Initialization State**



It is strongly recommended that you use a BSD Compiler flow to insert the glue logic and TAP and to generate the BSDL file automatically. If this flow is not used, for information about implementing the TAP, see the timing diagrams in “DFTMAX Boundary Scan Flow for USB 3.0 femtoPHY” on page 462. The following code snippet represents the AC JTAG chain that can be inserted into the BSDL file.

```
-- Example BSDL File snippets for the Customer ASIC referencing
-- the high speed serial IO.
```

```
entity customer_ASIC is
```

```
-- This section specifies the differential IO port groupings.
```

```
attribute PORT_GROUPING of Customer_ASIC: entity is
  "Differential_Voltage ( " &
  "(rx<#>_p,rx<#>_m) , " & "(tx<#>_p,tx<#>_m) )";
```

```
attribute BOUNDARY_REGISTER of Customer_ASIC: entity is
```

```
--
-- num cell port function safe [ccell disval rslt]
```

--

```
"33 (BC_1, *, control, 0), " &
"32 (AC_1, tx0_p, output3, X, 33, 0, Z), " &
"31 (BC_1, rx0_p, observe_only, X), " &
"30 (BC_1, rx0_m, observe_only, X) ";
```

```
AIO_COMPONENT_CONFORMANCE of usb3_phy: entity is "STD_1149_6_2003";
```

```
attribute AIO_Pin_Behavior of Customer_ASIC: entity is
"tx0_p ;"& "rx0_p[1] : HP_time=60.0e-9, LP_time=10e-9;"
```

```
end Customer_ASIC;
```

During BSDL testing, observe the PHY transmitter states for simulation, as listed in [Table 5-8](#) on page 181, and consider the following behavior:

- The CLAMP instruction results in a differential transmitter output.
- The HIGHZ instruction results in <0,0> being present on the differential transmitter outputs. When active for AC coupling, a keeper circuit establishes and maintains common mode.
  - You might need to modify the BSDL that is used for test bench generation.
  - For simulation, you can modify the transmitter BSDL definition so that it has a default state of 0 instead of Z.
  - You can re-encode the test bench wrapper so that it has the transmitter output of Z when both outputs are 0.

For example:

```
assign {tx_p_tb,tx_m_tb} = (tx0_p == 1'b0 && tx0_m ==
1'b0) ? 2'bzz : {tx0_p, tx0_m};
```

- The CONTROL bit for TX\_P is the BC\_1 cell in the chain.

The AC JTAG receiver has an adjustable receiver sensitivity that must be pin-programmed based on the input signal. For valid settings, see [Table A-4](#).



Determine the correct setting by doing the following:

1. Identify the lowest differential output from the TX driving the AC JTAG receiver (this is usually on another chip or part of the test setup).
2. In [Table A-4](#), determine the row that has high and low values that encompass the transmit level. For example, for a PIPE driver, the lowest differential TX output is 800 mV, which corresponds to an `acjt_level[4:0]` value of 00110.

**Table A-4** `acjt_level[4:0]` Settings for Signal Levels to Detect AC JTAG Pulse

| Differential Input Signal Swing (mV <sub>pp</sub> ) |       | <code>acjt_level[4:0]</code> Setting to Center 50% and 90% Trip Levels |
|-----------------------------------------------------|-------|------------------------------------------------------------------------|
| Above                                               | Below |                                                                        |
| 235                                                 | 375   | 00001                                                                  |
| 288                                                 | 463   | 00010                                                                  |
| 344                                                 | 554   | 00011                                                                  |
| 401                                                 | 649   | 00100                                                                  |
| 460                                                 | 747   | 00101                                                                  |
| 521                                                 | 849   | 00110                                                                  |
| 432                                                 | 740   | 10001                                                                  |
| 525                                                 | > 967 | 10010                                                                  |

If boundary scan is not implemented, simply tie off the following signals:

- `acjt_level[4:0] = 5'b0`
- `bs_capture_dr = 1'b0`
- `bs_clamp = 1'b0`
- `bs_clk = 1'b0`
- `bs_extest = 1'b0`
- `bs_extest_ac = 1'b0`
- `bs_highz = 1'b0`
- `bs_in = 1'b0`
- `bs_init = 1'b0`
- `bs_invert = 1'b0`
- `bs_preload = 1'b0`
- `bs_shift_dr = 1'b0`
- `bs_update_dr = 1'b0`

## A.3 DFTMAX Boundary Scan Flow for USB 3.0 femtoPHY

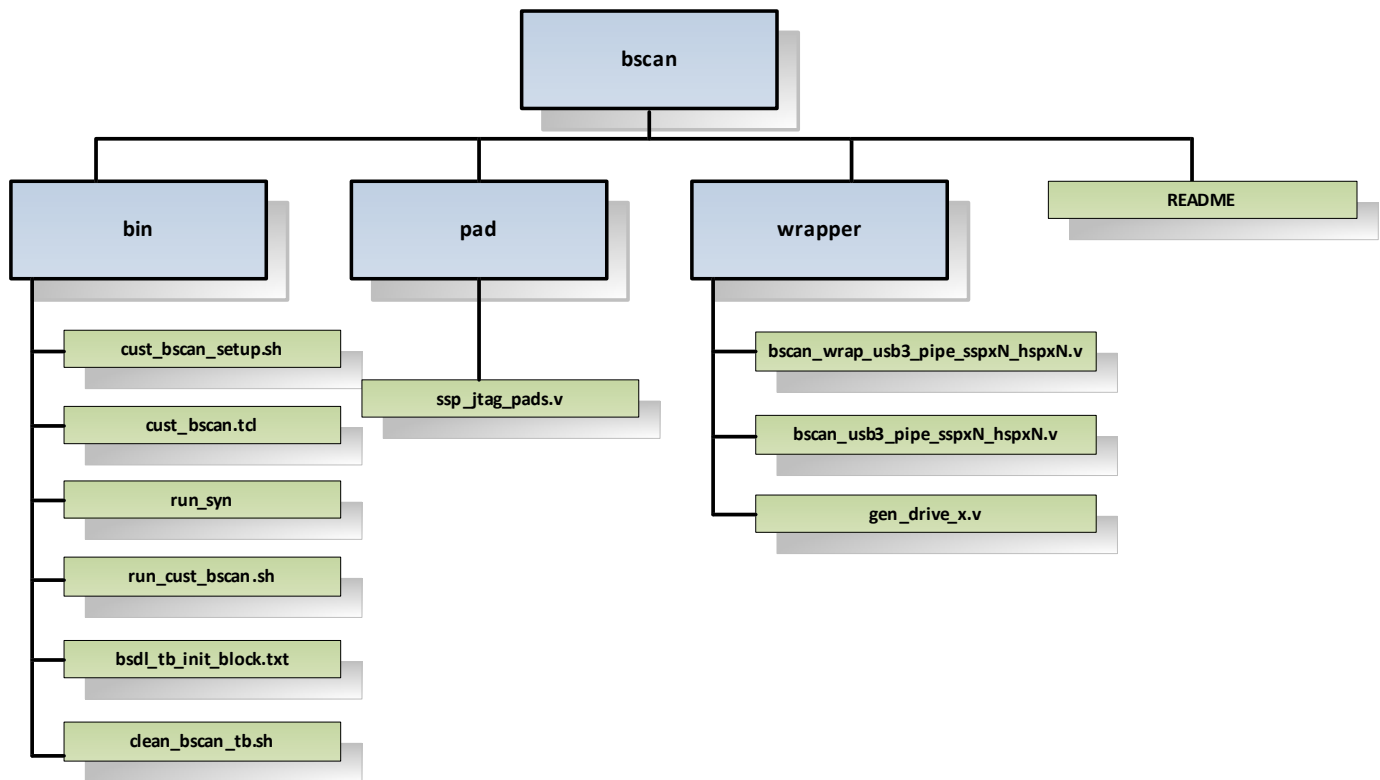
This section (and its accompanying support files) documents the flow for integrating the USB 3.0 femtoPHY IP design into the chip-level boundary scan register (BSR) created by the DFTMAX boundary scan insertion feature.

The USB 3.0 femtoPHY IP contains a BSR segment that uses AC-coupled signals, as supported by IEEE Std 1149.6-2003 (JTAG). This section provides an automated flow to integrate the USB 3.0 femtoPHY's BSR segment to the chip-level IEEE 1149.6 design as well as an automated method to generate the chip-level BSDL file and BSD patterns in Verilog, STIL, or WGL format.

### A.3.1 Directory Structure

This section provides some example files. The following figure shows the directory structure for these files. Green boxes represent files, and blue boxes represent directories.

Figure A-13 Directory Structure of Example Files



Files to note are:

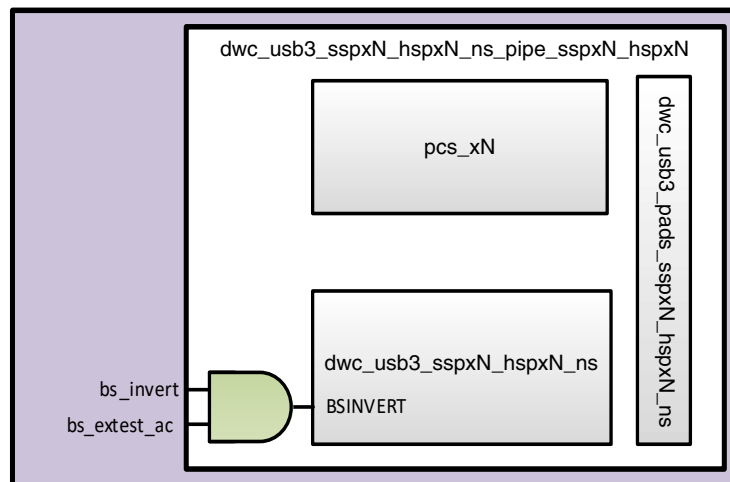
- **cust\_bscan\_setup.sh**: Master boundary scan script to run the BSD insertion flow
- **cust\_bscan.tcl**: Design Compiler script
- **run\_syn**: Runs the Design compiler script.
- **run\_cust\_bscan.sh**: Runs the JTAG test bench simulation script.
- **bsdl\_tb\_init\_block.txt**, **clean\_bscan\_tb.sh**: Post-processes the test bench generated by DFTMAX.
- **ssp\_jtag\_pads.v**: JTAG pad designs; also includes some GTECH cells used in this example.
- **bscan\_wrap\_dwc\_usb3\_sspaN\_hspaN\_ns\_pipe\_sspaN\_hspaN.v**: Chip-level wrapper for boundary scan
- **bscan\_dwc\_usb3\_sspaN\_hspaN\_ns\_pipe\_sspaN\_hspaN.v**: USB 3.0 femtoPHY top-level file

The `bscan_dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN.v` file is the USB 3.0 femtoPHY IP core top-level module. This file contains the PHY hard macro, PHY pads, and PIPE, as shown in the following figure. In this design, a single AND gate is instantiated, which masks the `bs_invert` signal from the boundary scan logic to the PHY with the `bs_extest_ac` pin for all boundary scan instructions.



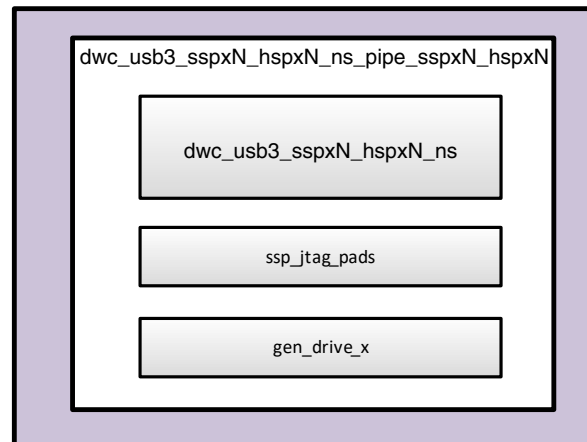
**Note** You must implement or synthesize this AND gate in your target technology library for your design.

**Figure A-14 USB 3.0 femtoPHY IP Top-Level Module**



The `bscan_dwc_usb3_sspaN_hspaN_ns_pipe_sspaN_hspaN.v` file is the wrapper for the USB 3.0 femtoPHY core file. This file instantiates the IP instance and pads for all of the ASIC-side pins of the PHY, as shown in the following figure. For portability of this script to all customer environments, some GTECH cells are also used within the I/O pad cells.

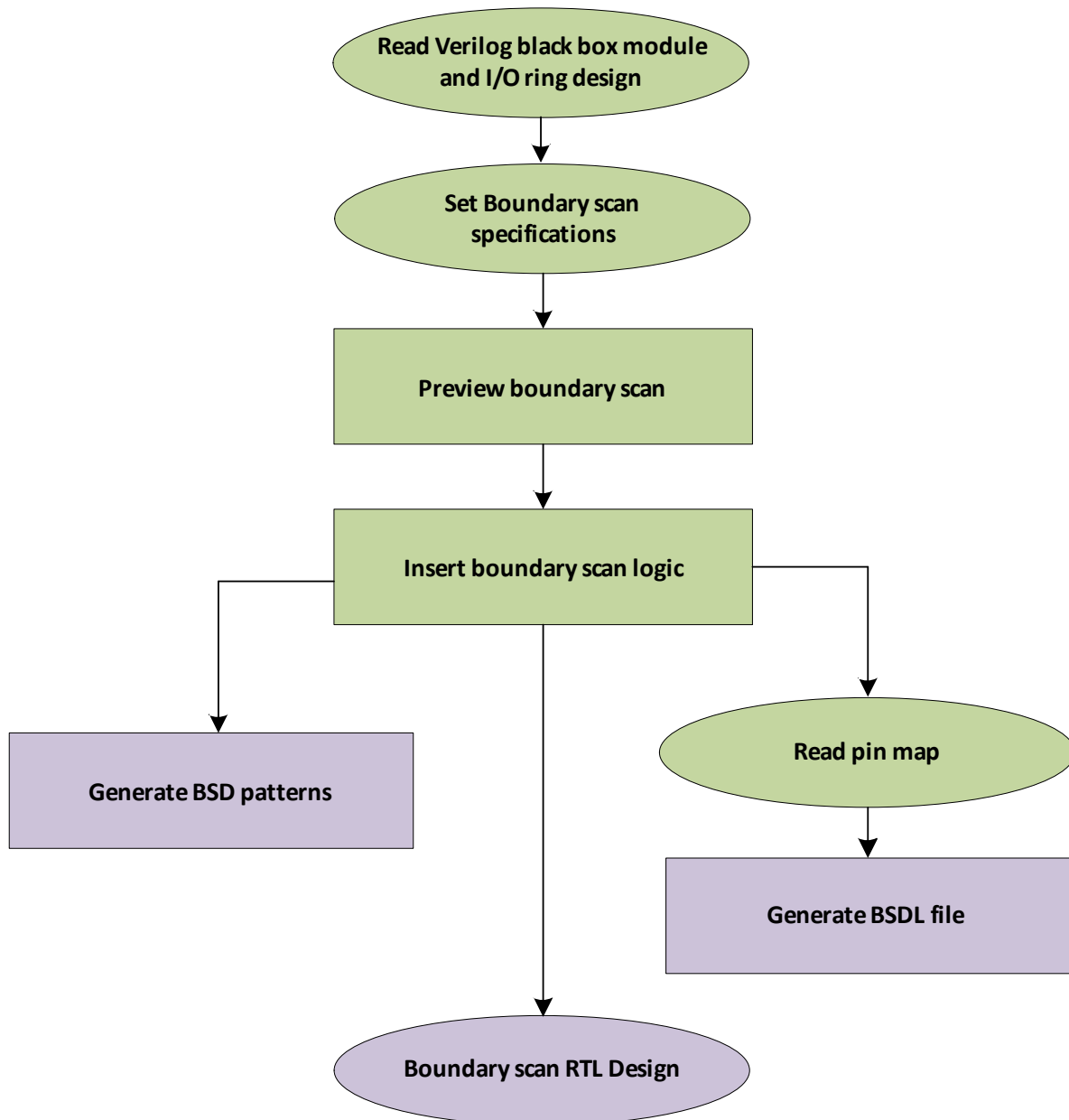
**Figure A-15 Top-Level Wrapper Module for Boundary Scan Design**



### A.3.2 Getting Started

The DFTMAX tool supports two boundary scan insertion flows: an RTL-level insertion flow and a gate-level insertion flow. This section focuses on the RTL-level insertion flow, as illustrated in the following figure.

**Figure A-16 RTL-Level Boundary Scan Insertion Flow**



In the RTL-level boundary scan insertion flow, the tool inserts the TAP controller, the BSR cells, and the AC-coupled signal JTAG glue logic. The tool requires the top-level Verilog module of the IP as input, as well as the I/O ring along with the port definitions. The pad IP in this flow can be a library model or a black-box soft macro module.

Boundary scan compliance for IEEE Std 1149.6 designs is determined by ensuring boundary scan vectors pass VCS simulation; the `check_bsd` command is not supported for these designs. After inserting BSD, the correctness of design can be validated via simulation.

The steps in the RTL-level boundary scan insertion flow are:

1. Synthesis: Integrate the PHY BSR segment with the top-level BSR chain.
2. Pattern Generation: Generate the boundary scan Verilog test bench.
3. Post-processing: Modify the boundary scan Verilog test bench to add the `phy_reset` cycle.
4. Simulation: Run VCS simulation using the modified BSD Verilog test bench.

### A.3.2.1 Synthesis

The `cust_bscan.tcl` file is the TCL constraint file for Design Compiler.

The `cust_bscan_setup.sh` file is the run script for this example. The BSDC tool is called by this script, which also specifies the output logs, greps for error messages from the logs, and post-processes the Verilog test bench.

To begin the RTL-level boundary scan insertion flow, the PHY and PADs WC corner `.lib` files must be provided as options for the following run script.

```
% cd ./bscan/bin
% ./cust_bscan_setup.sh -synth -phy_wc_lib <PHY WC corner lib> -pads_wc_lib <PADS WC corner lib>
```

The shell script invokes Design Compiler with the `cust_bscan.tcl` script to insert the JTAG logic in your chip-level design, generate the boundary-scan-inserted netlist, the BSDL file, the STIL patterns, and STIL verification test bench files.

In the script, the `define_dft_design` command specifies the access interface signals and boundary scan cell order of the BSR segment for integration with the chip-level BSR chain.

```
define_dft_design -type PAD -design_name ${scan_wrap_design} \
  -interface { \
    ac_mode bs_extest_ac h \
    ac_init_clk bs_init h \
    ac_test bs_invert h \
    shift_dr bs_shift_dr h \
    capture_clk bs_clk h \
    si bs_in h \
    so bs_out h \
    low cr_write h \
    low jtag_tms h \
    low test_burnin h \
    low test_powerdown_ssp h \
    low test_powerdown_hsp h \
    high pipe0_tx_elecidle h \
    low scan_mode h \
  }
low pcs_scan_mode h \
```

```

low scan_shift h \
low lane0_tx2rx_loopbk h \
low pipe0_tx_detectrx h \
} \
-params { $lib_cell$ string false $pad_type$ string hybrid \
    $differential$ boolean true \
    $diff_port_pairs$ list string "rx<#>_p rx<#>_m" "tx0_p
    tx<#>_m" $end_list$ \
    $bsr_segment$ list string \
    "0 BC_1 - tx0_p 0 -" \
    "1 AC_1 - tx0_p 0 0" \
    "2 BC_1 rx0_p - X -" \
    "3 BC_1 rx0_m - X -" $end_list$ }

```

DFTMAX boundary scan insertion supports the generation of RTL files for the TAP controller and BSR logic. When doing so, before writing the final boundary scan RTL files, remove any non-JTAG blocks as follows.

```

remove_design $scan_wrap_design
write_bsd_rtl -format verilog -all -out ./${proc}/${design}_bsd.v

```

The BSDL file generated by the tool uses the following syntax to describe the differential signaling for all TX and RX pins.

```

attribute PORT_GROUPING of top: entity is
"Differential_Voltage ( (rx0_p,rx0_m), (tx0_p,tx0_m))";

```

The BSDL file uses the following syntax to describe each boundary scan cell in the BSR segment.

```
"33  (BC_1, *, control, 0), " &
"32  (AC_1, tx0_p, output3, X, 33, 0, Z), " &
"31  (BC_1, rx0_p, observe_only, X), " &
"30  (BC_1, rx0_m, observe_only, X), " &
```

The BC\_1 boundary scan cell is attached to an output-enable control signal for the driver tx0\_(p,m) differential pairs. The AC\_1 boundary scan cell holds the data bit that is transmitted on the differential TX output. The tx0\_m signal's value is implied here, because tx0\_p and tx0\_m have been defined as differential signals. The two BC\_4 boundary scan cells are used to set the expected bit for the RX and read back the captured result. The IEEE STD 1149.6-2003 and AC timing attributes for the IP are defined in this BSDL syntax.

-- Advanced I/O Description

```
attribute AIO_COMPONENT_CONFORMANCE of bscan_wrap_usb3_pipe_ssp1_hsp1
: entity is "STD_1149_6_2003";

attribute AIO_Pin_Behavior of bscan_wrap_usb3_pipe_ssp1_hsp1: entity
is "rx0_p : LP_time=10.000000e-9 HP_time=60.000000e-9 ;" &
"tx0_p  ";
```

The AC timing attributes are associated to AC pins as previously listed; the high-pass filter time and low-pass filter time are defined as follows:

```
LP_time=10e-9;    HP_time=60e-9;
```

The BC\_1 boundary scan cell is attached to an output-enable control signal for the driver tx0\_(p,m) differential pairs. The AC\_1 boundary scan cell holds the data bit that is transmitted on the differential TX output. The tx0\_m signal's value is implied here, because tx0\_p and tx0\_m have been defined as differential signals. The two BC\_4 boundary scan cells are used to set the expected bit for the RX and read back the captured result. The IEEE STD 1149.6-2003 and AC timing attributes for the IP are defined in this BSDL syntax.

-- Advanced I/O Description

```
attribute AIO_COMPONENT_CONFORMANCE of bscan_wrap_usb3_pipe_ssp1_hsp1
: entity is "STD_1149_6_2003";

attribute AIO_Pin_Behavior of bscan_wrap_usb3_pipe_ssp1_hsp1: entity
is "rx0_p : LP_time=10.000000e-9 HP_time=60.000000e-9 ;" &
"tx0_p  ";
```

The AC timing attributes are associated to AC pins as previously listed; the high-pass filter time and low-pass filter time are defined as follows:

```
LP_time=10e-9;    HP_time=60e-9;
```



### A.3.2.2 Post-Processing

After performing boundary scan insertion and generating the test bench files, the following post-processing steps are required.

1. To instruct the Verilog test bench generated by boundary scan insertion to use a transmitter output value of Z when both outputs are 0, modify the test bench as follows:

```
assign {tx_p_tb,tx_m_tb} = (tx_p == 1'b0 && tx_m == 1'b0) ?
2'bzz : {tx_p, tx_m};
```

2. To toggle the phy\_reset pin and initialize the PHY, modify the BSDL-generated test bench.

For example, in the following test bench excerpt, the connection of phy\_reset is modified to connect to phy\_reset\_ovrd.

```
bscan_wrap_usb3_pipe_sspN_hspN T1 ( .phy_reset(phy_reset_ovrd), ... ...)
```

To toggle this signal, simply insert the initial block into the test bench.

```
initial begin
phy_reset_ovrd = 0;
#1;
phy_reset_ovrd = 1;
#310;
phy_reset_ovrd = 0;
end
```

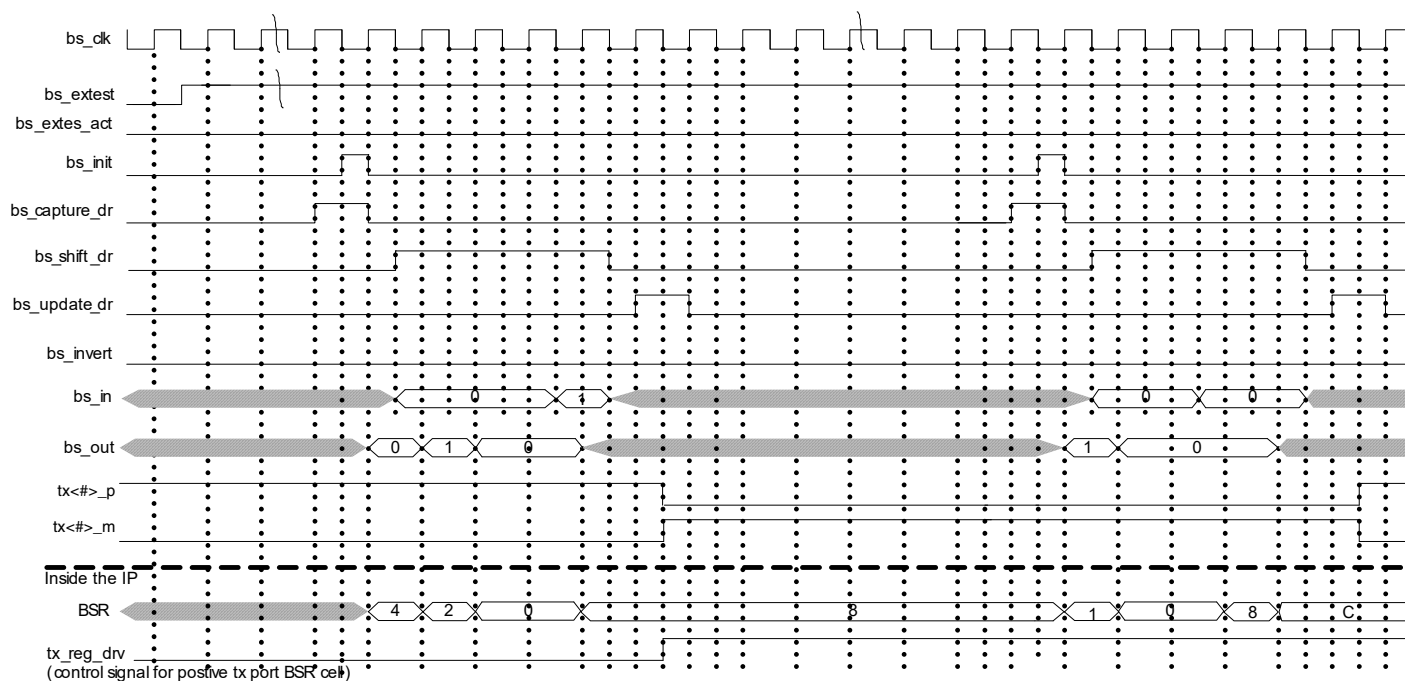
3. Insert waveform dumping system tasks.

This step is integrated with the synthesis step; Detailed information can be found in the clean\_bscan\_tb.sh file.



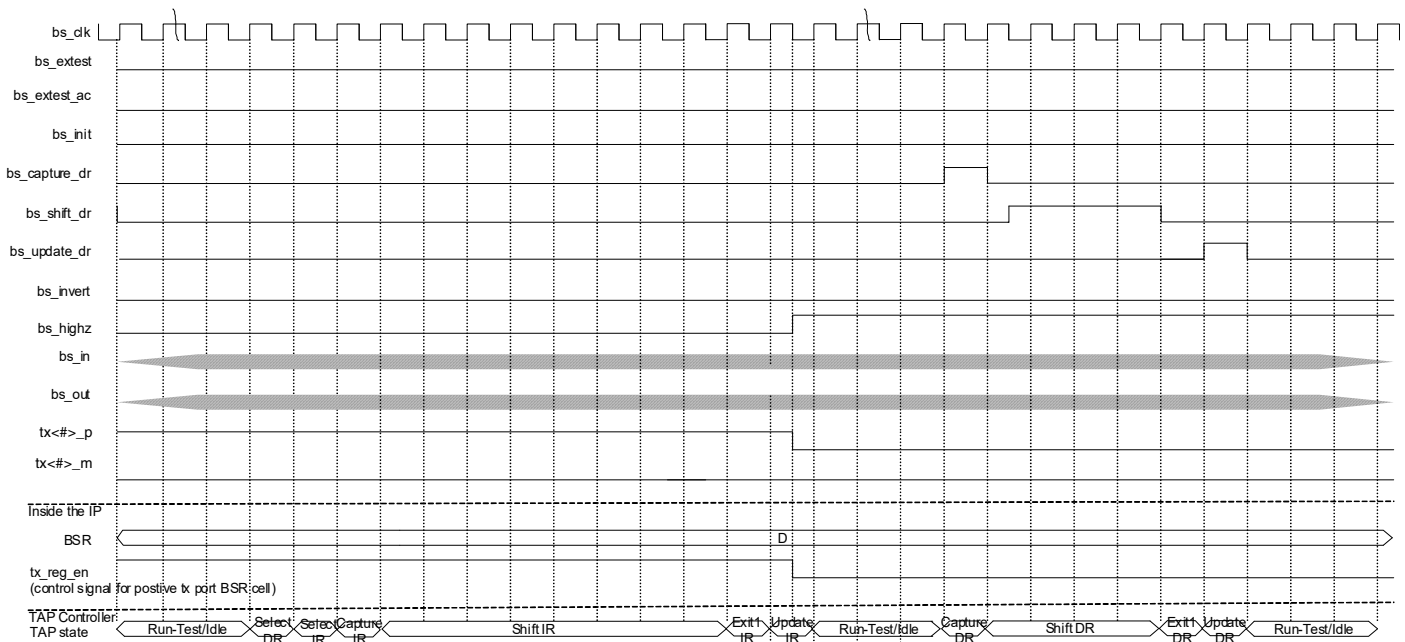
The EXTEST instruction is used in DC JTAG testing to check interconnections by forcing a test pattern at output pins and capturing test results at the input pins. When Capture-DR is active, the status of rx\_(p,m) is loaded into the BSR. After phy\_bs\_init is asserted, the hysteresis trip point in the RX cell is initialized, and the initial value is looped back for DC testing. The phy\_bs\_invert signal is masked in this process.

**Figure A-18 Timing for EXTEST Instruction**



The HIGHZ instruction places the 1-bit Bypass register between the TDI and TDO pins, which enables the boundary scan test data to pass synchronously through selected devices to adjacent devices during normal device operation, while tristating all the I/O pins. During execution of the HIGHZ instruction, the BSR is not shifted, `phy_bs_in` is not sampled, and `phy_bs_out` is not changed. The PHY has a keeper circuit due to the differential nature of transmitter. This keeper circuit is not required to be disabled in the HIGHZ state, so the PHY asserts common mode. All TX outputs “00” are the modeling of the common mode. In real silicon, the common mode ( $v_{ptx} / 2$ ) can be observed.

**Figure A-19 Timing for HIGHZ Instruction**

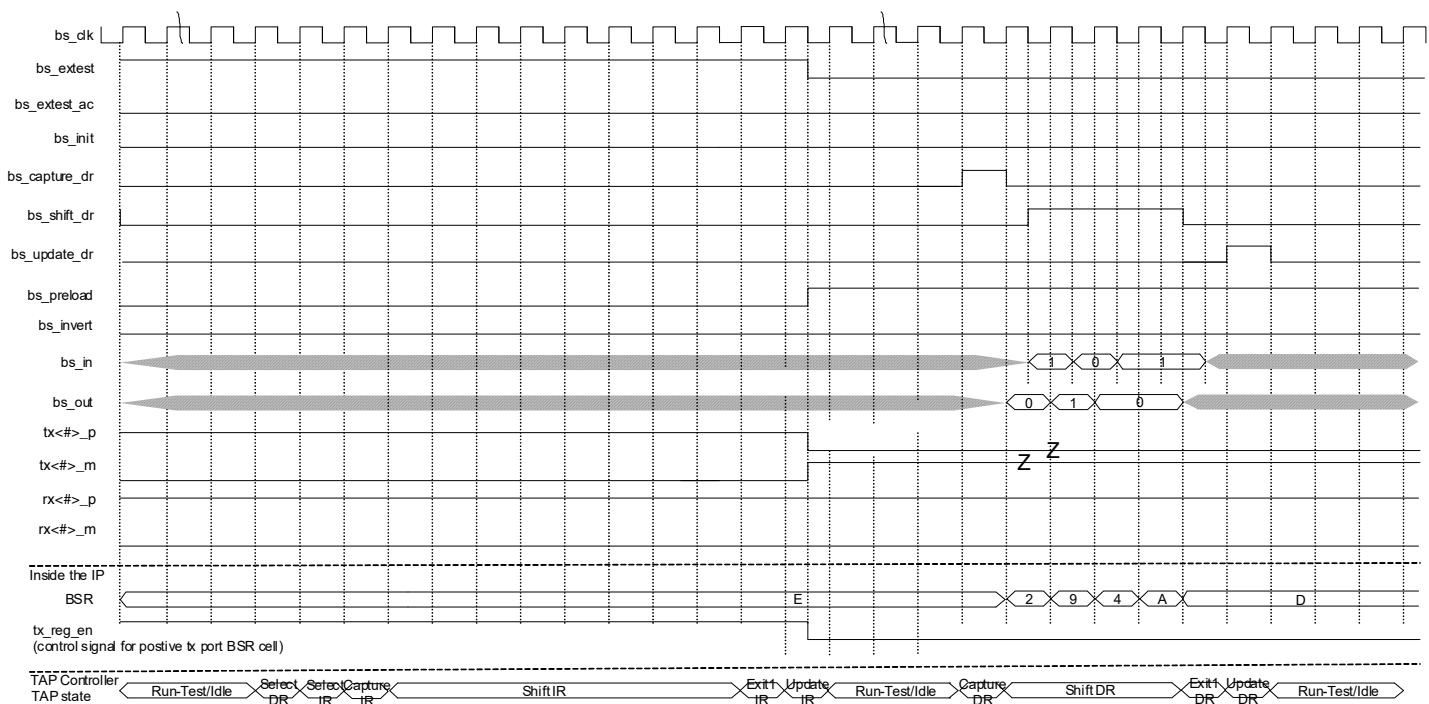


The SAMPLE instruction enables you to take a snapshot of the states of the component's input and output signals without interfering with normal operation of the assembled board.

The PRELOAD instruction enables scanning of the boundary scan register without causing interference to normal operation of the on-chip system logic. (The tx\_<#>(p,m) values are changed to "Z" in the following figure, but their values can also be 0/1 based on the input settings.)

The SAMPLE and PRELOAD instructions can be merged into one instruction as supported by previous versions of the IEEE Std 1149.1 standard. When in the CaptureDR state, the input and output snapshot can be captured using the SAMPLE instruction. When in the ShiftDR state, the boundary scan input data can be shifted using the PRELOAD instruction through TDI. Meanwhile, the sampled input/output snapshot can be shifted out through TDO. In the PHY Boundary Scan interface, the phy\_bs\_preload signal is used for the SAMPLE/PRELOAD instruction.

**Figure A-20 Timing for SAMPLE/PRELOAD Instruction**



### A.3.3 Reset Implementation

Before any AC-coupled JTAG testing can be performed, the PHY must have a `phy_reset` event. Due to the mixed-signal nature of the PHY, this is an absolute requirement. This reset pulse must have a minimum duration as noted in [Table 5-1](#) on page 167. To remain compliant with IEEE Std 1149.1, it is recommended that users implement power-on reset circuitry to reset the PHY.

**Note**

Note that using the TAP controller's test-logic-reset state (because the TAP might be placed in reset during functional mode) or using the optional TRST\* pin (which is prohibited in IEEE 1149.1 section 4.6.1 rule C) is discouraged unless total compliance with the 1149.1 standard is not necessary.

### A.3.4 Simulation Limitations

Note the following simulation limitations:

- Version F-2011.09-SP4 or later of DFTMAX must be used.
- When simulating the STIL patterns using the given test bench, you must add a `phy_reset` event to the stimulus.
- When simulating the STIL patterns using the given test bench, it reports errors for the HIGHZ instruction and in some cases where the test bench is expecting a Z state on the TX driver. According to IEEE STD 1149.1-2001, section 8.15.1, note 1, if present, pull-up, pull-down, and keeper circuits are not required to be disabled in the HIGHZ state. The TX common mode hold circuit falls into this category. The TX differential driver has four states: 1, 0, Z, and common mode. The PHY model encoding for common mode is `tx_p/tx_m = 1'b0/1'b0`. Currently, the Synopsys BSD Compiler does not have a method to program its test bench checker with this non-differential and non-Z state encoding.

# B

## Scan Implementation

---

This appendix includes the following sections:

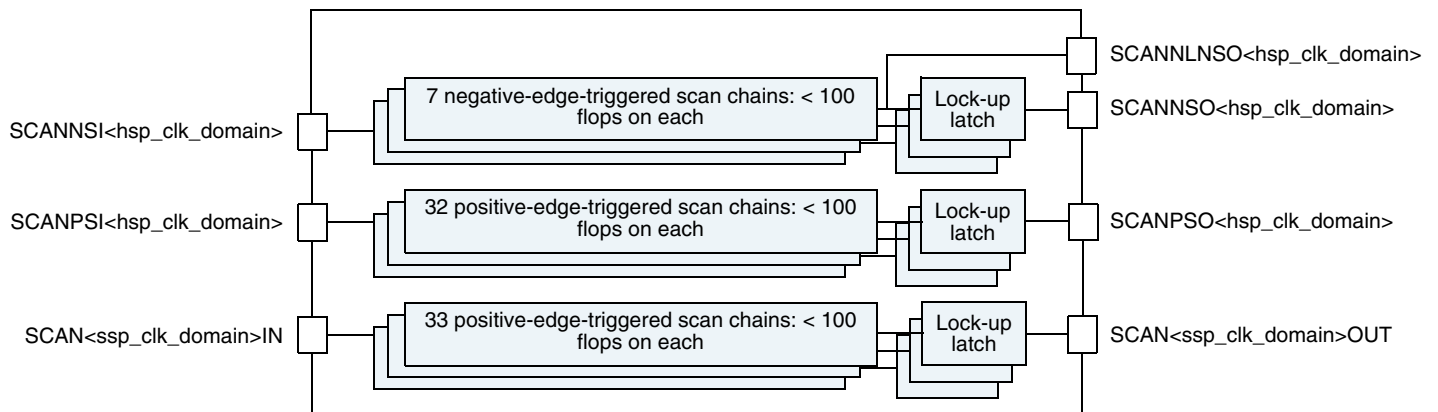
- [“PHY Scan Implementation”](#) on page 476
- [“PCS Scan Implementation”](#) on page 482

## B.1 PHY Scan Implementation

The PHY scan implementation has one scan enable, one test enable, and 41 clocks to support at-speed scan testing (ASST). [Table 8-2](#) on page 219 outlines the clocks and their associated at-speed clock frequencies. The macro incorporates 65 positive-edge-triggered scan chains and 7 negative-edge-triggered scan chains. [Table 8-3](#) on page 221 outlines the scan chains. The maximum scan shift and stuck-at fault testing frequency is 100 MHz.

The following figure shows the PHY scan implementation.

**Figure B-1** PHY Scan Chain Implementation



For negative-edge-triggered scan chains (for clock domains that contain negative-edge-triggered scan chains, see [Table 8-3](#) on page 221), all flip-flops are triggered by the negative edge of the associated scan clock. The input to these chains are the associated `SCANNSI<hsp_clk_domain>` pins, and the outputs are the associated `SCANNNSO<hsp_clk_domain>` and `SCANNLNSO<hsp_clk_domain>` pins. `SCANNNSO<hsp_clk_domain>` are the outputs of lock-up latches at the end of the negative-edge-triggered scan chain. If a negative-edge-triggered scan chain is stitched to the input of another negative-edge-triggered scan chain external to the PHY, use the `SCANNNSO<hsp_clk_domain>` outputs and leave the `SCANNLNSO<hsp_clk_domain>` outputs floating. The `SCANNLNSO<hsp_clk_domain>` outputs are tied to the output of the last flip-flop in the scan chain and bypass the lock-up latch. If a negative-edge-triggered scan chain is stitched to the input of a positive-edge-triggered scan chain external to the PHY, use the `SCANNLNSO<hsp_clk_domain>` outputs and leave the `SCANNNSO<hsp_clk_domain>` outputs floating.

The inputs to the positive-edge-triggered scan chains are the `SCANPSI<hsp_clk_domain>`, as well as the `SCAN<ssp_clk_domain>IN` inputs (for clock domains that contain positive-edge-triggered scan chains, see [Table 8-3](#) on page 221). Each positive-edge-triggered scan chain is terminated by a lock-up latch. The outputs of these chains are `SCANPSO<hsp_clk_domain>` and `SCAN<ssp_clk_domain>OUT`.

To ensure high scan-test coverage, the scan implementation requires access to all USB 3.0 femtoPHY primary digital inputs and outputs for control and observation. This accessibility must be established by either deterministically driven logic interconnection or direct pin mappings to external pins. If some or all primary inputs and outputs are not controllable or observable, the scan test works, but test coverage decreases significantly.

Internal digital inputs and outputs connected to USB 3.0 femtoPHY analog components are multiplexed to be either testable or not testable during the scan test.



## B.1.1 Generating Vectors

The following steps describe the ATPG flow using Synopsys's TetraMAX ATPG tool. A sample TetraMAX script embodying this flow is included in the scan/tmax/tmax\_atpg.scr file.

1. In TetraMAX, read in the design netlist (scan/vectors/<top\_level\_design\_name>\_scan.v).
2. A number of USB 3.0 femtoPHY pins, such as the power and ground pins, clock selects, analog test signals, and parameter override bits, do not have to be included in every vector. Therefore, tie these pins to logic 0, logic 1, logic X, or logic Z as appropriate.

The bidirectional pins, DP and DM, are considered to be tied off to logic X in Input mode and are masked off in Output mode. TX0P and TX0M are also masked off. Therefore, these pins are not used for vector comparisons.

3. Build the scan model (using the TetraMAX build\_model command).
4. Define the clocks. Also define SCANRST and PHYRESET as scan reset signals.

Table 8-2 on page 219 outlines all scan clocks and their associated frequencies. Note that when defining clocks at the PHY interface, all signal names are all-caps with no spaces (for example, scan\_mpll\_word\_clk becomes SCANMPLLWORDCLK).

5. Define the scan chains.
  - a. Define SCANSHIFT as the scan enable function.
  - b. Define each negative-edge-triggered scan chain for each clock domain.  
 Example: For the SCANUTMICK domain, define SCANNSIUTMICK0 and SCANNSOUTMICK0 as the scan-in and scan-out functions of the negative-edge-triggered scan chain. Table 8-3 on page 221 lists the scan chains and associated bus widths.
  - c. Define each positive-edge-triggered scan chain for each clock domain.  
 Example: For the SCANUTMICK domain, define SCANPSIUTMICK0 and SCANPSOUTMICK0 as the scan-in and scan-out functions of the positive-edge-triggered scan chain. Table 8-3 on page 221 lists the scan chains and associated bus widths.
  - d. Specify the primary input constraints on SCANMODE (logic 1), TESTBURNIN, PDDQ, and TESTBYPASS (logic 0).
6. Run the DRC command in TetraMAX.
7. Add all faults.
8. Configure the ATPG command to enable full\_sequential ATPG.
9. Run the ATPG command in TetraMAX.
10. Verify the vectors with TetraMAX's run simulation command.
11. Write out vectors in different formats.

## B.1.2 Design for Testability Model

To simplify scan integration, Synopsys provides a Design for Testability (DFT) model of the USB 3.0 femtoPHY that includes information sufficient for typical scan-integration flows, such as Hierarchical Scan Synthesis (HSS), and so on.

The USB 3.0 femtoPHY's DFT model is located in the `/atpg/ctl/` directory and is provided in plain-text format (\*.ctl).

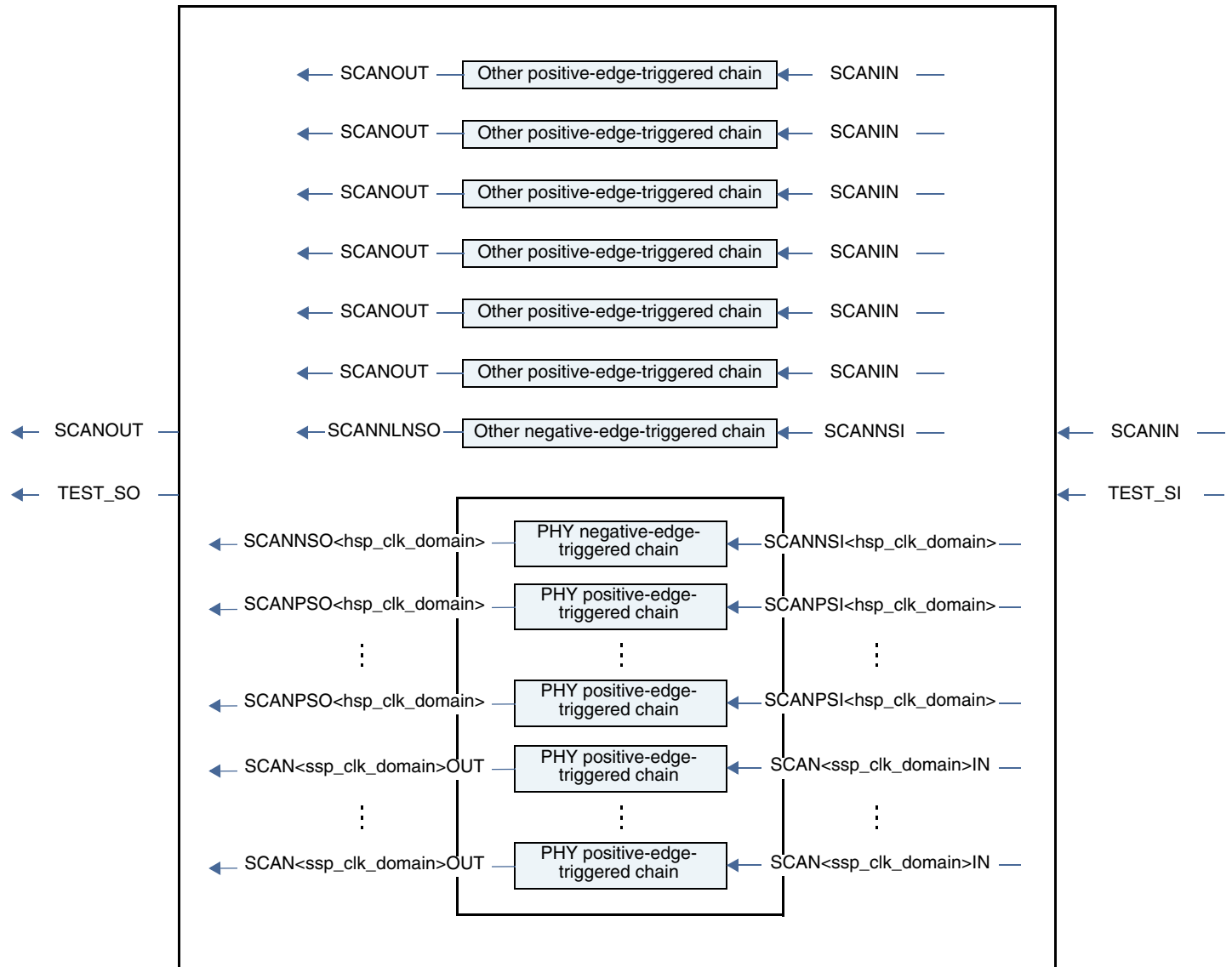
In addition, the test-protocol file of the USB 3.0 femtoPHY is provided in Standard Test Interface Language (STIL) format (`atpg/ctl/*.spf`).

For information about using DFT models, refer to the *DFT Compiler User Guide*, “Scan” section.

## Example DFT Compiler Script for Hierarchical Scan Synthesis

This example uses the USB 3.0 femtoPHY's DFT model and DFT models of other sub-designs in an ASIC to illustrate how the scan chains are stitched between these blocks. The following figure shows a conceptual block diagram of the ASIC. The example is for reference purpose only and the details provided might vary depending on your rest of ASIC and process used.

**Figure B-2 Scan Chains in an ASIC With a USB 3.0 femtoPHY**



As illustrated, the ASIC has two groups of scan chains: the USB 3.0 femtoPHY scan chains, and TEST\_SI through TEST\_SO. The USB 3.0 femtoPHY scan chains are previously described, and the ASIC has several instances of a second design with positive-edge-triggered scan chains only.

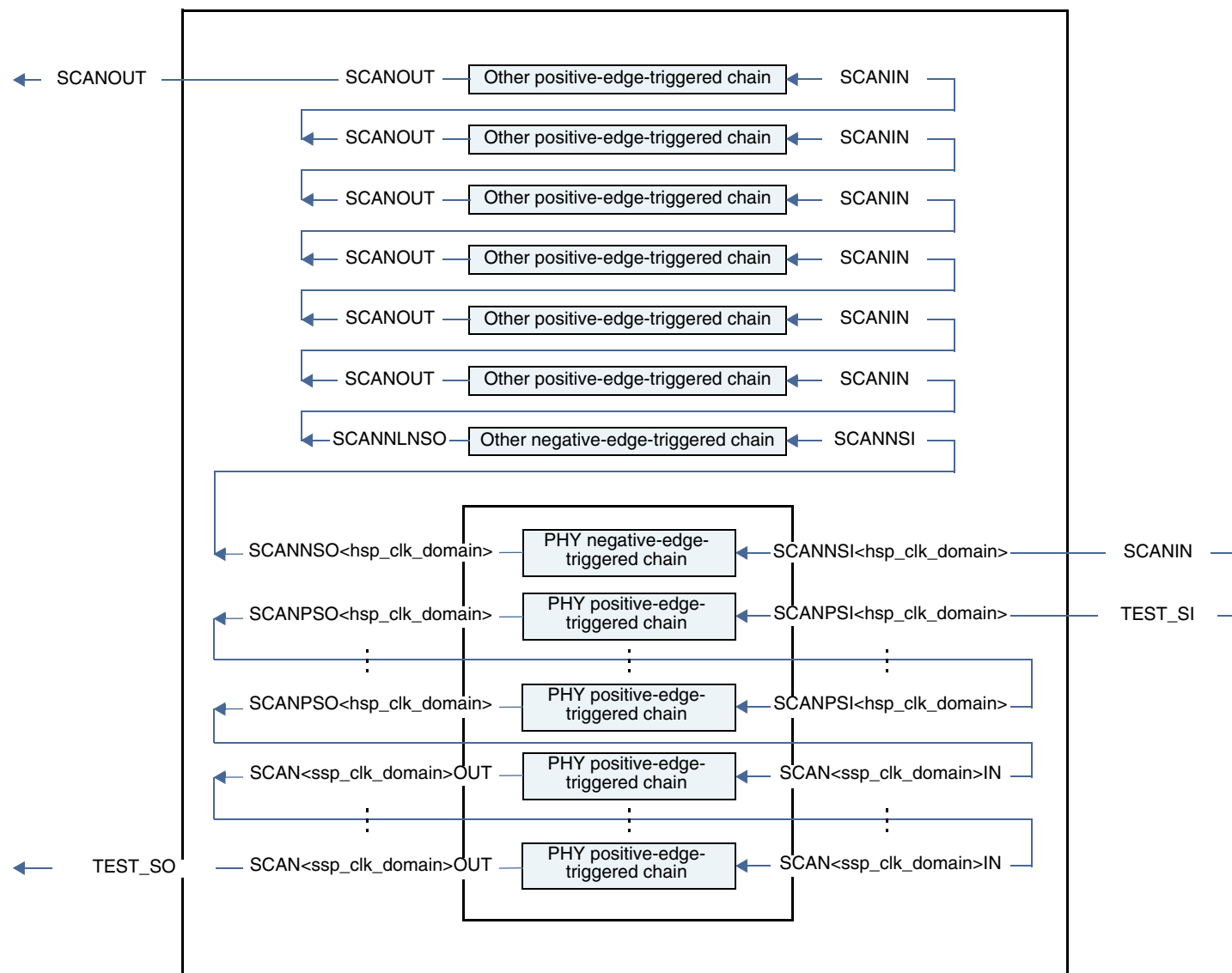
The Hierarchical Scan Synthesis (HSS) process involves using DFT models for all sub-designs in the ASIC to stitch together the final scan chains for the ASIC.

The following sample DFT Compiler script performs this specific task. Note that for simplicity, only a single ASST scan chain is shown.

```
set link_library "<path to library>/std_cell_bc.db dw_foundation.sldb *"
set target_library "<path to library>/std_cell_wc.db"
define_design_lib work -path synth/WORK;
set test_enable_dft_drc true
set test_use_test_models true
set test_xg_use_models true read_verilog verilog/top_soc.v
read_test_model -format ctl -design <counter_pos> <path_to_ctl_model>/counter_pos.ctl
read_test_model -format ctl -design <counter_neg> <path_to_ctl_model>/counter_neg.ctl
read_test_model -format ctl -design <usb3_phy_top>
<path_to_ctl_model>/<usb3_phy_top>_dft.ctl
use_test_model -true counter_pos
use_test_model -true counter_neg
use_test_model -true <usb3_phy_top>
current_design top_soc
link
set_dft_signal -view spec -port SCANMPLLWORDIN -type ScanDataIn
set_dft_signal -view spec -port TEST_SI -type ScanDataIn
set_dft_signal -view spec -port SCANMPLLWORDOUT -type ScanDataOut
set_dft_signal -view spec -port TEST_SO -type ScanDataOut
set_dft_signal -view spec -port SCANSHIFT -type ScanEnable -active_state 1
set_dft_signal -view spec -port SCANMODE -type TestMode -active_state 1
set_dft_signal -view spec -port SCANMODE -type Constant -active_state 1
set_dft_signal -view existing_dft -port SCANMPLLWORDCLK -type ScanClock -timing {8 13}
set_dft_signal -view existing_dft -port SCANRST -type Reset -active_state 1
set_dft_signal -view existing_dft -port PHYRESET -type Reset -active_state 1
set_scan_path chain1 -view spec -scan_data_in SCANMPLLWORDIN -scan_data_out
SCANMPLLWORDOUT
set_scan_path chain2 -view spec -scan_data_in TEST_SI -scan_data_out TEST_SO
set_scan_configuration -chain_count 2
set_scan_configuration -clock_mixing mix_clocks_not_edges
create_test_protocol
dft_drc preview_dft -show all
insert_dft
write_test_protocol -o hss_output/top_soc.spf
report_scan_path -view existing_dft -chain chain1 -test_mode Internal_scan write -f
verilog -o hss_output/top_soc_scan.v
write_test_model -f ctl -o hss_output/top_soc_scan.ctl
quit
```

At the end of this process, the final scan-stitched output of the ASIC can be schematically depicted, as shown in the following figure.

**Figure B-3 Final Scan-Stitched Output of ASIC**



## B.2 PCS Scan Implementation

Scan control signals have been designed into the PCS RTL to enable end users to implement ATPG stuck-at fault scan testing for the logic when performing physical implementation of the soft macro portions of the ASIC.

These PCS signals are as follows.

- **pcs\_scan\_mode**: PCS enable for test mode. When this signal is asserted, the clock sources are mapped to their equivalent scan pins.
- **pcs\_scan\_shift**: Enables scan shift logic paths within the PCS for testability. During test insertion at synthesis, map to the equivalent shift control.
- **pcs\_scan\_rst**: Scan-specific reset. Provides control of internal reset nets and bypasses phy\_reset and pipeP\_reset\_n.
- **pcs\_scan\_clk**: Scan clock connected to the pcs\_clk domain
- **pcs\_scan\_pclk**: Scan clock connected to the pipe\_pclk domain
- **pcs\_scan\_refclk**: Scan clock connected to the refclk domain
- **pcs\_scan\_rx<#>\_clk**: Scan clock connected to the RX clock domain
- **pcs\_scan\_tx<#>\_clk**: Scan clock connected to the TX clock domain

You can treat the entire PIPE as a single scan domain by performing the following steps.



### Note

The following details are provided for reference; the final method of DFT insertion is left to the users' discretion.

1. Tie pcs\_scan\_mode to the SCANMODE input of the PHY.
2. Tie pcs\_scan\_shift to the SCANSHIFT input of the PHY.
3. Tie pcs\_scan\_rst to the SCANRST signal input of the PHY.
4. Tie pcs\_scan\_clk, pcs\_scan\_pclk, pcs\_scan\_refclk, pcs\_scan\_rx<#>\_clk, and pcs\_scan\_tx<#>\_clk to the SCANCLK input of the PHY.
5. Target DFT insertion for the PCS to a maximum scan clock frequency of 100 MHz.
6. You can add PCS scan input and scan output pins at the PIPE level.

# C

## ATE Test Bench

This appendix describes the ATE Verilog test bench and includes the following sections:

- [“ATE Verilog Test Bench Overview”](#) on page 483
- [“ATE Testing Details”](#) on page 506
- [“Bench Testing Details”](#) on page 536
- [“Installing Python in Windows”](#) on page 541
- [“Flowchart for ate\\_initialize ATE Subtask”](#) on page 543
- [“RJ45 BreakOut Board Pinout”](#) on page 544

### C.1 ATE Verilog Test Bench Overview

Developing a production test for an ASIC has become more complicated. ASICs contain more functions than previously, and many of the functions are purchased as IP from outside sources. Test engineers must understand these IP functions to test them effectively. To compound the issue, test engineers usually have little or no direct contact with the people who developed these functions. Digital scan tools have been developed to solve part of this problem. Users can use these scan tools to generate test vectors to test the digital portion of ASICs without knowing details about the function of the circuits being tested. These scan tools can compute how effectively the vectors test the digital portion of the IP. Often, these scan patterns are delivered with the IP to simplify the matter.

The challenge is testing analog functions that digital scan does not test effectively. No industry tools are available to assist with this task. Test engineers have to develop and implement a test plan that effectively and efficiently tests the functions. Test engineers must use their knowledge, experience, and information they can gather to develop the test plan. This task can be daunting if many different pieces of IP are on the ASIC with which the test engineers are not familiar. After the test plan is developed, the test must be implemented for the specific tester using the appropriate tester analog resources available. The whole analog test development process can be difficult and labor intensive.

Synopsys has solved the analog test issue for the USB 3.0 femtoPHY by adding test features inside the IP and developing software that turns the analog testing portion of the IP into a pseudo digital scan. For analog testing, vectors are scanned in and scanned out of a JTAG port, similar to a digital scan. No complicated test code or analog test resources are required. Analog testing is a simple pass/fail operation.

## C.1.1 Methodology

The USB 3.0 femtoPHY IP was designed with ATE testing in mind. Features were added inside the IP to aid the testing process. Tests were developed early on in development to ensure sufficient test coverage of blocks inside the IP, and external requirements were kept to a minimum. All that is needed on a load board is a simple AC-coupled serial loopback for the USB 3.0 femtoPHY, a reference clock, and a 200-Ω resistor.

Typically, each test is configured through the JTAG port by writing to internal IP registers. A measurement is then made and compared against limits, and a pass/fail value is returned through the JTAG port. The limits are embedded inside the JTAG vector to account for variations between load boards or ASIC implementations. In the Synopsys solution, no vector capture is required. Every test returns a simple pass/fail result.

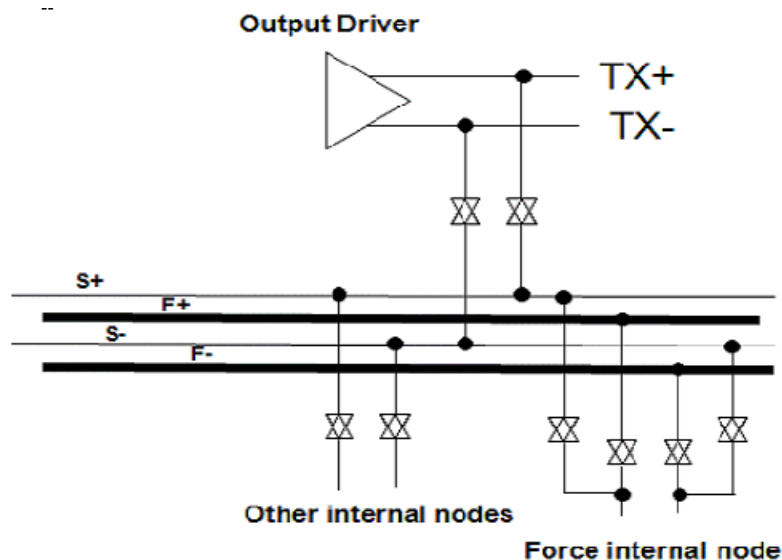
## C.1.2 Built-In Features

Test features were designed using some of the PHY's existing circuitry, enabling the design to minimize the impact to die size. These features were also designed to have no effect on power when the PHY is in its normal mode of operation. This section describes the test features included in the IP.

### C.1.2.1 Analog Test Bus

In digital scan, registers are chained together and accessed through a single port. The USB 3.0 femtoPHY includes a four-wire (two force and two sense lines) analog test bus (ATB). Analog signals can be placed on the bus via pass gates. After these signals are placed on the bus, they can be routed to other portions of the IP or to an on-board ADC for measurement. The following figure shows a conceptual diagram of the ATB. With Synopsys's methodology, the entire analog portion of the PHY can be tested in less than 50 ms.

**Figure C-1 Analog Test Bus**





### **C.1.2.2 DAC/ADC**

An on-board 10-bit ADC/DAC is included in the IP. To minimize the effect on area, this circuit reuses some features of the transmitter's and receiver's termination calibration circuitry. The DAC can be applied to the test bus to force internal nodes. This feature is used during the EYEMASK test. The DAC enables measurement of internal nodes.

### **C.1.2.3 Arithmetic Logic Unit**

The Arithmetic Logic Unit (ALU) is a feature included in USB 3.0 femtoPHYs. This feature enables a simple math function to be made on registers inside the IP for the addition, subtraction, averaging, and comparison of registers. The ALU can also place a result in any register in the PHY. The ALU is mainly used to compare results against limits and return pass/fail results as well as to add offsets to registers in the PHY.

### **C.1.2.4 LBERT Pattern Generator**

The USB 3.0 femtoPHY provides a pattern generator that enables several different patterns to be sent out of the transmitter during testing.

### **C.1.2.5 LBERT Pattern Matcher**

The USB 3.0 femtoPHY provides a pattern matcher that is used to verify the transmit-to-receive path of the IP.

### **C.1.2.6 Receiver Eye Voltage and Phase Control**

The USB 3.0 femtoPHY includes registers to control various voltage and phase functions. The receiver sampling point can be moved both in voltage and phase to margin the TX/RX channel.

### C.1.3 CREG Access

The CREG can be accessed through the JTAG interface and the CR Parallel port.

#### C.1.3.1 CREG Access Through the JTAG Interface

Control register access is performed by shifting a command and either register address or data into the CRSEL register. The CMD field determines what type of access is performed. The following table describes the encodings defined for the CRSEL.CMD field. All actions are initiated during the UPDATE-DR phase of the JTAG state machine while the CRSEL register is being accessed.

**Table C-1 CRSEL Register Commands**

| CMD   | Encoding | Description                                                                                                                                                                                                                         |
|-------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADDR  | 2'b00    | Capture ADDRESS field for use in a subsequent WRITE command. Shifts current value from most recently accessed register (via READ or WRITE) out on TDO.                                                                              |
| WRITE | 2'b01    | Write the value in the DATA field to the register selected by the address captured during the most recent ADDR or READ command. Shifts current value from most recently accessed register (via READ or WRITE) out on TDO.           |
| DATA  | 2'b10    | Shifts current value from most recently accessed register (via READ or WRITE) out on TDO. DATA value shifted in is not used.                                                                                                        |
| READ  | 2'b11    | Perform a read operation to the register selected by the ADDRESS field to make its current value available for subsequent CRSEL accesses. Shifts current value from most recently accessed register (via READ or WRITE) out on TDO. |

To perform a control register read, simply shift a READ command into the CRSEL register along with the register address in the ADDRESS field. Any subsequent CRSEL access causes the data returned by the READ command to be shifted out of TDO. This action enables a limited form of pipelining READ commands, because the second access can be shifting in the command and address for the subsequent READ command while shifting out the data from the previous READ command.

To perform a control register write, the ADDRESS of the register to be written must first be selected by performing either a ADDR or READ command and specifying the register address in the ADDRESS field. A second access using the WRITE command and specifying the data to be written in the DATA field causes the write to finally be performed in the target register.

### C.1.3.2 JTAG Run-Test-Idle (RTI) Cycles

When performing JTAG write and read operations, RTI cycles are used between instructions to ensure the PHY is either ready to return data (in the case of a read) or has sufficient time to capture the incoming data (in the case of a write).

The PHY contains a data-path pipeline that each JTAG READ and WRITE command must go through. This pipeline is clocked on the internal ref\_clk, and as a result, the time taken to propagate through the pipeline is proportional to both the jtag\_tck and ref\_clk. The following are example JTAG READ and WRITE command sequences.

#### JTAG READ

1. Shift in READ command with read address.
2. Wait for nRTI cycles.
3. Shift in next command and shift out read data.

#### JTAG WRITE

1. Shift in ADDRESS command with write address.
2. Shift in WRITE command with write data.
3. Wait for nRTI cycles.
4. Shift in next command.

#### Calculating nRTI

The number of RTI cycles depends on the clock ratio of jtag\_tck to ref\_clk. Specifically, when the ratio is at or near 1:1 (jtag\_tck clock to ref\_clk clock), the minimum number of RTI cycles is 16.5. This 16.5 value can be scaled to accommodate any jtag\_tck-to-reference-clock ratio using the following equation:

$$\text{Min RTI cycles} = \text{ceil} (14 \times T_{\text{ref}} / T_{\text{tck}} + 2.5)$$

When the ref\_clkdiv2 signal is set to 1, the reference clock frequency is divided by 2. To calculate the RTI cycles, double the Tref parameter value.

### C.1.3.3 CREG Access Through the CR Parallel Port

The CR port is a simple, 16-bit data/address parallel port provided for on-chip access to control registers inside the PHY. While access to these registers is not required for normal PHY operation, this interface is included for users who want to access some of the PHY's diagnostic features during normal operation or to override some of the PHY's control signals. This interface is completely asynchronous, using a handshake between `phy_cr_cap_addr`, `phy_cr_cap_data`, `phy_cr_read`, and `phy_cr_write` input commands with `phy_cr_ack` acknowledgments and `phy_cr_data_out[15:0]` outputs from the PHY. CR port access is divided into address, read, and write transactions.

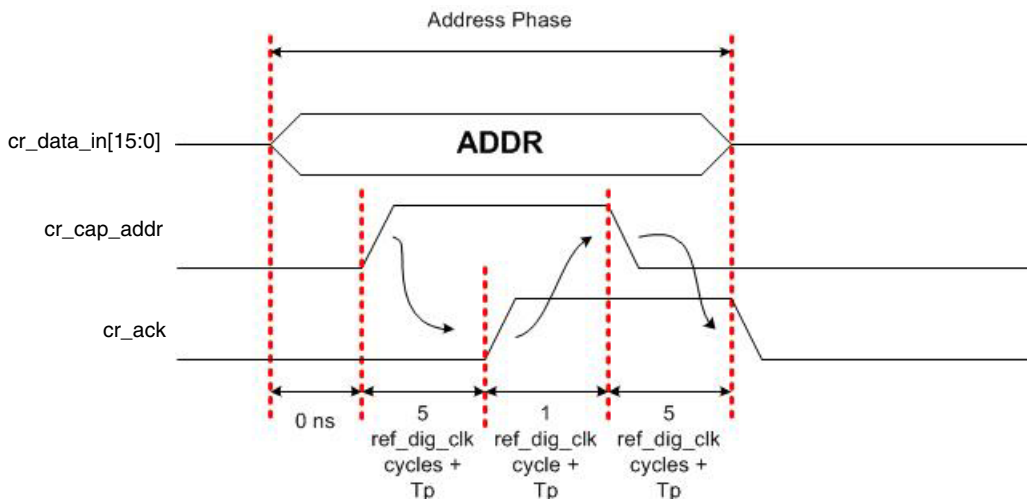
For details about accessing the CR parallel interface, see [“Control Register Access”](#) on page 159.

To access the CR port, minimum timing requirements exist; an example to illustrate the requirements follows.

#### Example: Minimum Timing Requirements When Reference Clock Frequency is 100 MHz

For the current ATE vector that the test bench generates, to ensure that the minimum timing requirement is met, the `cr_ack` signal is detected in the test bench. If `cr_ack` is not used in the user-defined vector, the minimum timing requirement is as follows.

**Figure C-2 Address Phase: 100-MHz Reference Clock Frequency, `cr_ack` Not Used**



Minimum timing requirement for Address phase:

- 1 ref\_dig\_clk cycle = 10 ns (based on 100-MHz reference clock)
- $T_p = 1$  ns

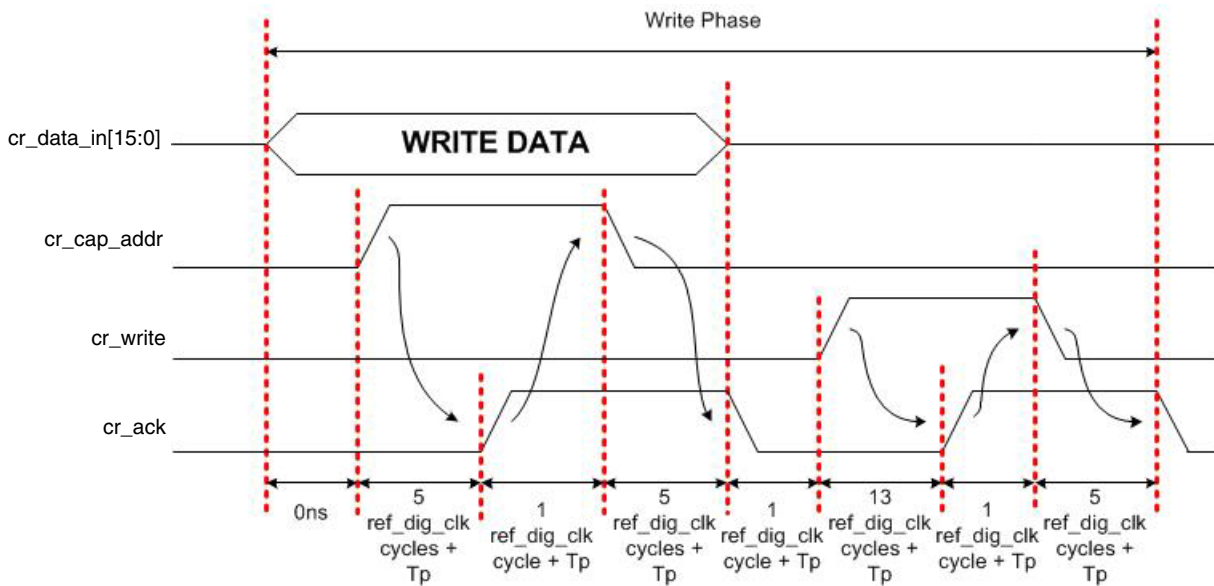
Assumptions:

- cr\_cap\_addr is asserted at the same time as cr\_data\_in[15:0] = 0 ns setup.
- cr\_cap\_addr is de-asserted one ref\_dig\_clk cycle after the assertion of cr\_ack.

$$\begin{aligned}\text{Minimum time} &= 0 \text{ ns} + (5 \text{ ref\_dig\_clk cycles} + T_p) + (1 \text{ ref\_dig\_clk cycle} + T_p) + (5 \text{ ref\_dig\_clk cycles} + T_p) \\ &= 11 \text{ ref\_dig\_clk cycles} + (3 \times T_p) \\ &= 11 \times 10 \text{ ns} + (3 \times 1 \text{ ns}) \\ &= 113 \text{ ns}\end{aligned}$$

Summary:

- cr\_data\_in must be asserted for a minimum of 11 ref\_dig\_clk cycles from time 0.
- cr\_cap\_addr must be asserted for 6 ref\_dig\_clk cycles from time 0.
- Address phase requires a minimum of 11 ref\_dig\_clk cycles from time 0.

**Figure C-3 Write Phase: 100-MHz Reference Clock Frequency, cr\_ack Not Used**

Minimum timing requirement for Write phase:

- One ref\_dig\_clk cycle = 10 ns (based on 100-MHz reference clock)
- $T_p = 1$  ns

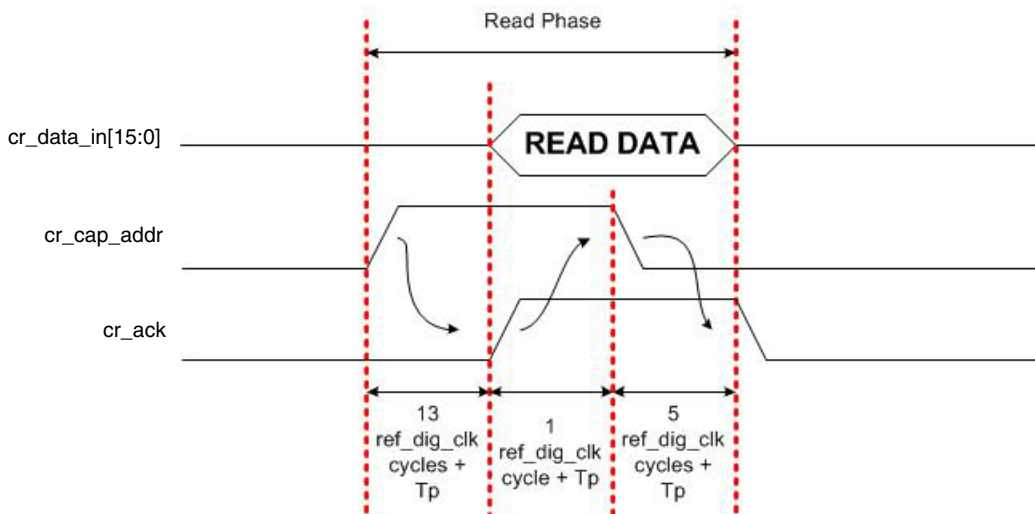
Assumptions:

- cr\_cap\_data is asserted at the same time as cr\_data\_in[15:0].
- cr\_cap\_data is de-asserted one ref\_dig\_clk cycle after the assertion of cr\_ack.
- cr\_write is de-asserted one ref\_dig\_clk cycle after the assertion of cr\_ack.

$$\begin{aligned}
 \text{Minimum time} &= 0 \text{ ns} + (5 \text{ ref\_dig\_clks} + T_p) + (1 \text{ ref\_dig\_clk} + T_p) + (5 \text{ ref\_dig\_clks} + T_p) + \\
 &\quad (1 \text{ ref\_dig\_clk} + T_p) + (13 \text{ ref\_dig\_clks} + T_p) + (1 \text{ ref\_dig\_clk} + T_p) + (5 \text{ ref\_dig\_clks} + T_p) \\
 &= 31 \text{ ref\_dig\_clk cycles} + (7 \times T_p) \\
 &= (31 \times 10 \text{ ns}) + (7 \times 1 \text{ ns}) \\
 &= 317 \text{ ns}
 \end{aligned}$$

Summary:

- cr\_cap\_data must be asserted for a minimum 6 ref\_dig\_clk cycles from time 0.
- cr\_data\_in must have valid write data for a minimum of 11 ref\_dig\_clk cycles from time 0.
- cr\_write must be asserted after a minimum of 12 ref\_dig\_clk cycles and for a minimum of 14 ref\_dig\_clk cycles from time 0.
- Write phase requires a minimum of 31 ref\_dig\_clk cycles from time 0.

**Figure C-4 Read Phase: 100-MHz Reference Clock Frequency, cr\_ack Not Used**

Minimum timing requirement for Read phase:

- One `ref_dig_clk` cycle = 10 ns (based on 100-MHz reference clock)
- $T_p = 1$  ns

$$\begin{aligned}
 \text{Minimum time} &= (13 \text{ ref\_dig\_clks} + T_p) + (1 \text{ ref\_dig\_clk} + T_p) + (5 \text{ ref\_dig\_clks} + T_p) \\
 &= 19 \text{ ref\_dig\_clk cycles} + (3 \times T_p) \\
 &= (19 \times 10 \text{ ns}) + 3 \times 1 \text{ ns} \\
 &= 193 \text{ ns}
 \end{aligned}$$

Summary:

- `cr_read` must be asserted for a minimum of 14 `ref_dig_clk` cycles from time 0.
- `cr_data_out` is valid after 13 `ref_dig_clk` cycles from time 0.
- Read phase requires a minimum of 19 `ref_dig_clk` cycles from time 0.

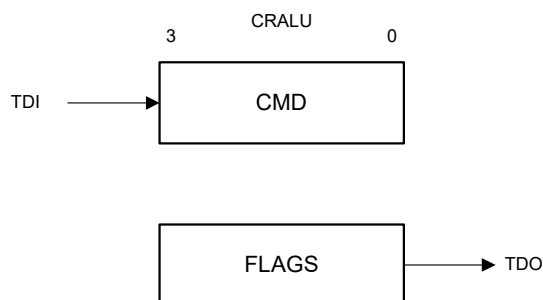
## C.1.4 Arithmetic Logic Unit

This section describes the Arithmetic Logic Unit (ALU).

### C.1.4.1 ALU Access Through the JTAG Interface

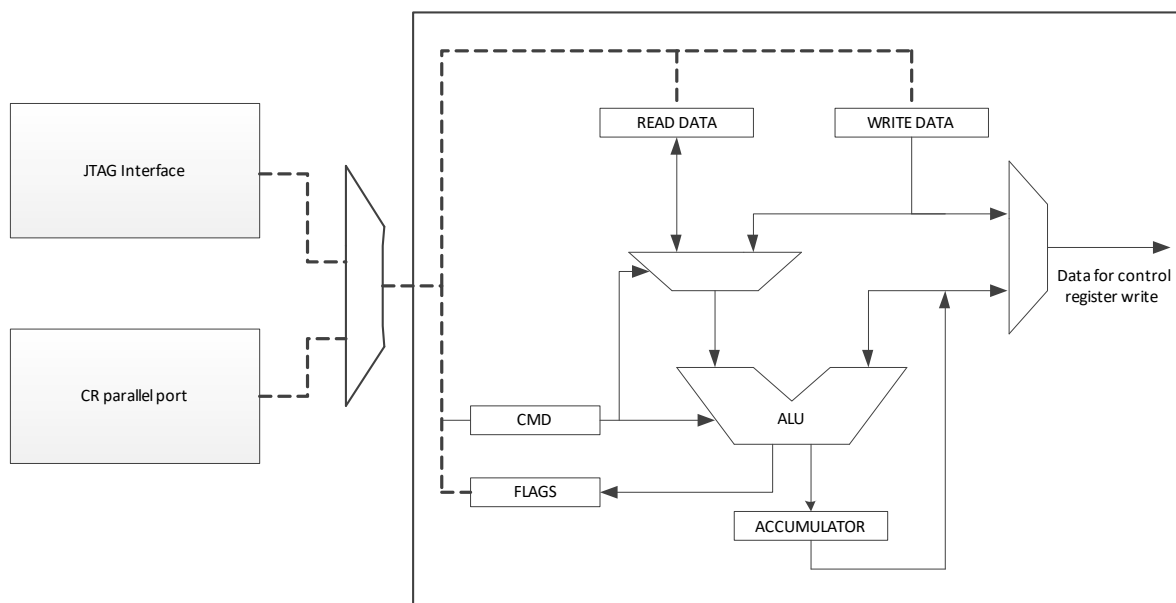
This register is a 4-bit command register that enables values from internal control registers as well as user-supplied values from the CRSEL DATA field to be manipulated using simple arithmetic operations. The following figure shows the format of the CRALU register.

Figure C-5 CRALU Register



An ALU with an associated 16-bit accumulator is controlled by commands written to this register, and the results can be stored back in the PHY's control registers. The following figure shows the data and control flow between the ALU, the CRSEL register, and the CRALU register.

Figure C-6 ALU Functional Diagram





The following table describes the commands that the ALU supports via the CRALU register.

**Table C-2 CRALU Register Commands**

| Command     | Encoding | Description                                                                                                                      |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------|
| ZERO        | 4'b0000  | Zero the accumulator.<br>$ACC := 0$                                                                                              |
| ADD_RD_DATA | 4'b0001  | Add control register read result to accumulator.<br>$ACC := ACC + CRSEL.READ\_RESULT$                                            |
| ADD_WR_DATA | 4'b0010  | Add control register write data to accumulator.<br>$ACC := ACC + CRSEL.DATA$                                                     |
| SHL1        | 4'b0011  | Shift accumulator left by 1 bit, shifting in zero at LSB.<br>$ACC := ACC << 1$                                                   |
| SHR1        | 4'b0100  | Shift accumulator right by 1 bit, shifting in zero at MSB.<br>$ACC := ACC >> 1$                                                  |
| ASHR1       | 4'b0101  | Shift accumulator right by 1 bit, preserve sign bit.<br>$ACC[14:0] := ACC[15:1]$                                                 |
| INV         | 4'b0110  | Bit-wise invert accumulator.<br>$ACC := \sim ACC$                                                                                |
| NEG         | 4'b0111  | Negate accumulator.<br>$ACC := -ACC$                                                                                             |
| ABS         | 4'b1000  | Compute absolute value of accumulator.<br>$ACC :=  ACC $                                                                         |
| ADD_PHSEL   | 4'b1001  | Add phase select value from READ_RESULT. For more information regarding this command, see <a href="#">Table C-3</a> .            |
| READ        | 4'b1110  | Trigger a new control register read using address loaded by last CRSEL access.<br>$CRSEL.READ\_RESULT := READ(CRSEL.ADDR)$       |
| WRITE       | 4'b1111  | Trigger a new control register write of accumulator value using address loaded by last CRSEL access.<br>$REG(CRSEL.ADDR) := ACC$ |

The following table describes the flags defined in the 4-bit CRALU.FLAGS field. This field is captured and shifted out on TDO during each access to the CRALU register.

**Table C-3 ALU Flags (CRALU.FLAGS)**

| Command  | Encoding | Description                                                                                                   |
|----------|----------|---------------------------------------------------------------------------------------------------------------|
| SIGN     | 3        | MSB of most recent ALU result (1 = negative; 0 = positive)                                                    |
| ERROR    | 2        | Error detected during most recent ALU operation. Currently used only to signal errors from ADD_PHSEL command. |
| PH_VALID | 1        | Signedness for ADD_PHSEL command has been chosen by a previous ADD_PHSEL command.                             |
| PH_SIGN  | 0        | Signedness used for next ADD_PHSEL command (1 = signed; 0 = unsigned).                                        |

### C.1.4.2 CREG Access Through CR Parallel Port

You can also perform ALU operations through the CR parallel port. Two registers, ALU\_CTL and ALU\_FLAGS, are used to access the ALU through the CR parallel port.

**Table C-4 SUP.CR\_PARA\_ALU\_CTL**

| Register/Address             | Description                                                                                  | Register Bit Details                                                      |
|------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| SUP.CR_PARA_ALU_CTL<br>16'h9 | <b>OPCODE</b><br><b>Function:</b> Opcode for the ALU when CR parallel access is enabled.     | <b>Bit range:</b> 4:1<br><b>Value at reset:</b> 4'h0<br><b>Access:</b> RW |
|                              | <b>ACCESS_EN</b><br><b>Function:</b> Enables access to the ALU through the CR parallel port. | <b>Bit range:</b> 0<br><b>Value at reset:</b> 1'h0<br><b>Access:</b> RW   |

**Table C-5 SUP.CR\_PARA\_ALU\_FLAGS**

| Register/Address                | Description                                                                                 | Register Bit Details                                                      |
|---------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| SUP.CR_PARA_ALU_FLAGS<br>16'h10 | <b>ALU_FLAGS</b><br><b>Function:</b> Flags from the ALU when CR parallel access is enabled. | <b>Bit range:</b> 3:0<br><b>Value at reset:</b> 4'h0<br><b>Access:</b> RU |

To set up the ALU to operate in CR parallel access mode:

1. Write 1'b1 to the CR\_PARA\_ALU\_ACCESS\_EN field.
2. Write an opcode to the CR\_PARA\_ALU\_OPCODE register.
3. Read flags from the CR\_PARA\_ALU\_FLAGS register.

It is important to note that the using the ALU in CR parallel access 'borrows' some signals and registers that were previously used exclusive by the JTAG interface. As a result, when operating the ALU in CR parallel mode, you must ensure that JTAG operations are not performed, because this can result in random register reads or writes.

Also note that clock/reset signals for the ALU are multiplexed based on the value of CR\_PARA\_ALU\_ACCESS\_EN. Therefore, it is highly recommended that you do not toggle the jtag\_tck signal while writing the value of CR\_PARA\_ALU\_ACCESS\_EN.

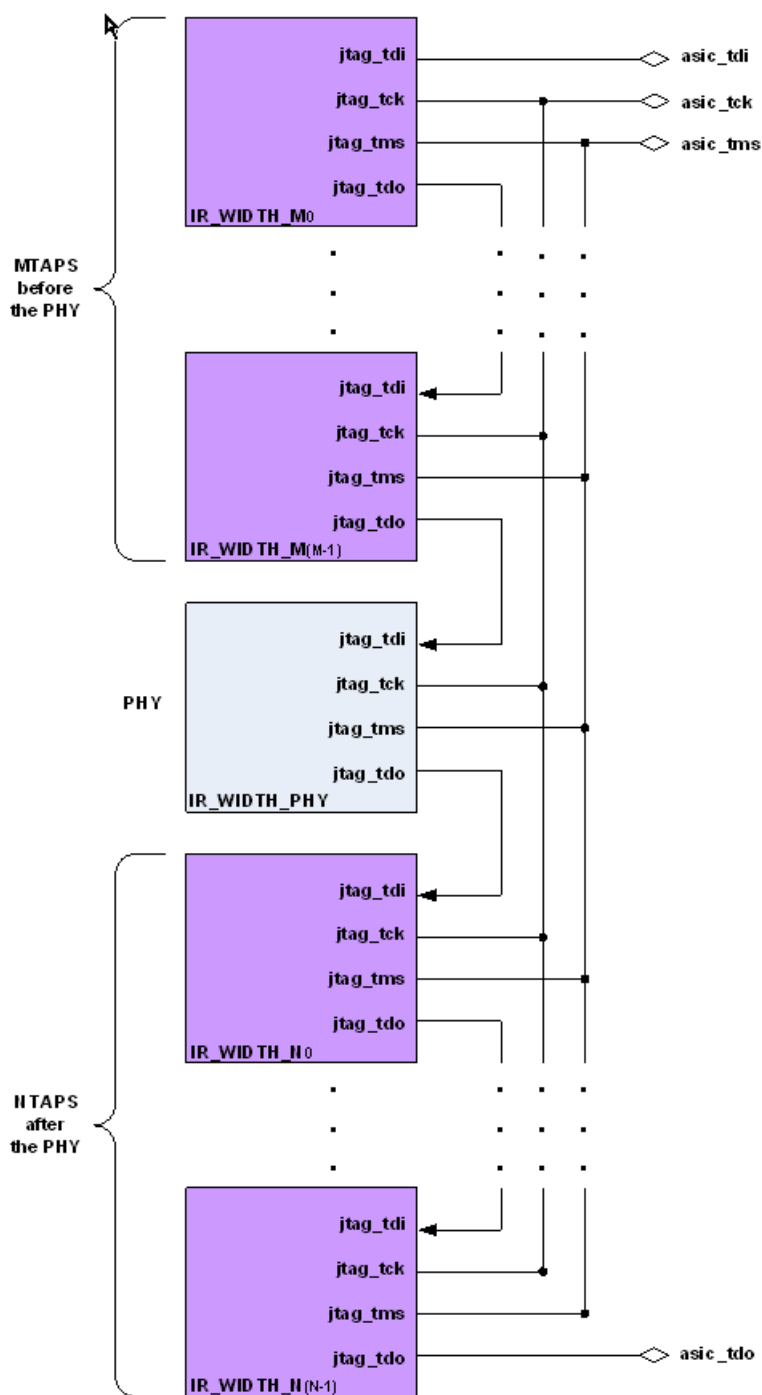
The ACCESS\_TYPE define in each ATE test controls the ATE test running through the JTAG or CR parallel port.

- ACCESS\_TYPE=1: For JTAG access; the default definition for simulation.
- ACCESS\_TYPE=0: For CR parallel access. If running the test through the CR parallel port, +define+ACCESS\_TYPE=0 must be defined for CR parallel ALU access.

## C.1.5 JTAG Configurability

The USB 3.0 femtoPHY can be configured to support multi-TAP configurations and different IR widths in each TAP controller, as shown in the following figure.

**Figure C-7 Multi-TAP Configuration**



To configure the USB 3.0 femtoPHY, the JTAG chain information must be entered in the `tb_tap_config.v` file (located in `./macro/Latest/testbench/testbench/`).

### C.1.5.1 Example Configuration

This section provides an example configuration and the steps needed to configure the USB 3.0 femtoPHY.

In the example, the following configuration is required:

- MTAPS = 1
- IR\_WIDTH\_M0 = 8
- NTAPS = 0
- IR\_WIDTH\_N0 = 0

The required steps are as follows.

1. Modify the content of the `tb_tap_config.v` file as follows.

```
`define MTAPS 1 //Number of tap controllers before the PHY
`define IR_WIDTH_M0 8 //Width of IR for tap controller 0 before the PHY

`define NTAPS 0 //Number of tap controllers after the PHY
`define IR_WIDTH_N0 0 //Width of IR for tap controller 0 after the PHY
`define NONZEROTC_M //Must be defined if MTAPS >= 1
`define MTAPS_IRW_SUM 8 //Sum of width of IRs of all the tap controllers before the PHY
//`define NONZEROTC_N //Must be defined if NTAPS >= 1 [Commented out since NTAPS = 0]
`define NTAPS_IRW_SUM 0 //Sum of width of IRs of all the tap controllers after the PHY
```

2. Run simulation with the ATE Verilog test bench.



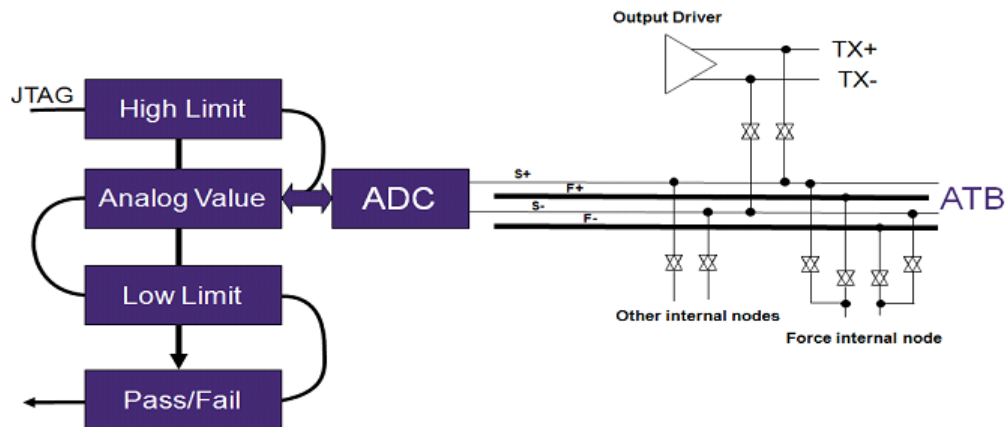
#### Note

The TAP controllers before and after the PHY are placed in BYPASS mode.

## C.1.6 Example Test

The TX\_DC\_LEVELS test is a good example to illustrate features inside the USB 3.0 femtoPHY IP in use. In this test, the pattern generator is set up to send an “all 1’s” pattern. The transmit positive value is then placed on the test bus and measured with the ADC. This measurement is averaged and stored in the ALU. The transmit negative value is measured and then subtracted from the positive value. The resultant value is compared against limits placed in the ALU during the setup phase. A pass/fail result is then returned to the tester to complete the test. The following figure shows an example of the TX\_DC\_LEVELS test setup.

Figure C-8 Example of TX\_DC\_LEVELS Test



## C.1.7 Vector File Definitions

To generate a vector file from the USB 3.0 femtoPHY, define the VEC\_TRACE macro and undefine the ANI\_SHORT\_RESET and ANI\_SIM\_MODE macros at run time.

You can use the following code snippet:

```
\ifdef VEC_TRACE
\undef ANI_SHORT_RESET
\undef ANI_SIM_MODE
\endif
```

Where:

- **define VEC\_TRACE:** Generates JTAG signals dump format pattern.
- **undefine ANI\_SHORT\_RESET:** Simulates the same as silicon (that is, PLL calibration and so on are not skipped).
- **undefine ANI\_SIM\_MODE:** The wait time matches the wait time used for silicon. If defined, this macro uses a short wait time for simulation purpose.

The following example creates a vector file for the Register test.

```
./run_ate_pipe ../testbench/ate_register_test/ate_register_test.v +define+VEC_TRACE
```

Note that the 5-Gbps baud rate is always used.

The vector file name is `ate.vec`, and the file contains bitstreams in the format, `tms_tdi_tdo`. Each line in the vector file corresponds to a positive edge of the JTAG clock (`tck`). The USB 3.0 femtoPHY internally masks and unmask the `tdo` output.

## C.1.8 Verilog Test Bench Configurability

This section provides information about configuring the ATE Verilog test bench.

### C.1.8.1 JTAG Clock Frequency

The default JTAG clock frequency is 16.666 MHz. As shown in the following code snippet, this frequency is specified in the Verilog file, `./macro/Latest/testbench/testbench/tb_jtag_driver.v`, using the `tck_phase` parameter.

```
parameter  JTAG_TCK_PHASE = 30;
//-----
real tck_phase;
initial begin
    tck_pin = 0;
    tck_phase = JTAG_TCK_PHASE;
    forever begin
        if(installed) begin
            #(tck_phase) tck_pin = ~tck_pin;
        end
        else begin
            // need something to not have 0 delay loop
            #(tck_phase);
        end
    end
end
end
```

Jtag\_clock frequency =  $1 / (2 * \text{tck\_phase})$ . For instance, to set the JTAG clock frequency to 10 MHz, set `tck_phase` to 50.



#### Note

The ratio between the reference clock frequency ( $F_{ref}$ ) and JTAG clock frequency ( $F_{tck}$ ) can affect JTAG read operations. This effect is due to a clock domain crossing between the JTAG clock domain and reference clock domain.

For proper JTAG read operations, the following is required.

- For a ratio of  $F_{ref} / F_{tck} = 1$ , a minimum of 12.5 run-test/idle (RTI) cycles is needed.
- For other ratios, a minimum of  $\lceil 11 * F_{tck} / F_{ref} + 1.5 \rceil$  RTI cycles is needed.

No restriction exists with respect to the relationship between the JTAG clock frequency and the reference clock frequency.



### C.1.8.2 Configuration Parameters

To configure the relevant parameters, you can modify the Verilog test bench. These parameters are specified in the Verilog file, `./macro/Latest/testbench/testbench/tb_ate_top.v`, as shown in the following table.

**Table C-6 Configuration Parameters**

| Parameter in <code>tb_ate_top.v</code> | Supported Range                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Description                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ate_ssc_range = 0;</code>        | <ul style="list-style-type: none"> <li>■ 000: -4,980 ppm</li> <li>■ 001: -4,492 ppm</li> <li>■ 010: -4,003 ppm</li> <li>■ 011: -2,000 ppm</li> <li>■ 100: 4,980 ppm</li> <li>■ 101: 4,492 ppm</li> <li>■ 110: 4,003 ppm</li> <li>■ 111: 2,000 ppm</li> </ul>                                                                                                                                                                                                                                                                                        | <p>Selects the range of spread spectrum modulation.</p> <p>When <code>ssc_en</code> is asserted and the PHY is spreading the high-speed transmit clocks, applies a fixed offset to the phase accumulator.</p> |
| <code>ate_loss_lvl = 9;</code>         | This bus must be set to 5'b01001.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sets the sensitivity level for the Loss-of-Signal detector.                                                                                                                                                   |
| <code>ate_acjt_level = 13;</code>      | For <code>acjt_level[4:0]</code> settings, see <a href="#">Table A-4</a> on page 461.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Selects the sensitivity level for the 1149.6 receiver for AC JTAG.                                                                                                                                            |
| <code>ate_lbert_mode = 4;</code>       | <ul style="list-style-type: none"> <li>■ 1: lfsr31. <math>X^{31} + X^{28} + 1</math></li> <li>■ 2: lfsr23. <math>X^{23} + X^{18} + 1</math></li> <li>■ 3: lfsr15. <math>X^{15} + X^{14} + 1</math></li> <li>■ 4: lfsr7. <math>X^7 + X^6 + 1</math></li> <li>■ 5: fixed word (PAT0)</li> <li>■ 6: DC balanced word (PAT0, <math>\sim</math>PAT0)</li> <li>■ 7: Fixed pattern: (000, PAT0, 3ff, <math>\sim</math>PAT0)</li> <li>■ 8: USB3.0 TSEQ (RX equalizer training ordered sets). Transmits even, then odd.</li> <li>■ 9–15: Reserved</li> </ul> | LBERT pattern to be generated when the mode is changed.                                                                                                                                                       |
| <code>ate_lbert_pattern = 302;</code>  | 10-bit pattern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Pattern for LBERT (integer representing binary pattern of 10 bits, 0–1,023) used for LBERT mode 5, 6, and 7.                                                                                                  |
| <code>ate_hsp_ext_ref = 0;</code>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enables USB reference clock if required.                                                                                                                                                                      |
| <code>ate_tx_lvl = 7'h7f;</code>       | Integer value between 0 and 127                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Transmit level</p> <p>Output amplitude is set by <math>v_{ptx} * (1 + tx\_lvl) / 128</math>.</p>                                                                                                           |
| <code>ate_tx_boost = 23;</code>        | Integer value between 0 and 127                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Transmit boost level</p> <p><math>tx\_eq\_ratio = (128 - 2 * tx\_boost) / 128</math></p> <p><math>tx\_eq\_dB = -20 * \log_{10}(tx\_eq\_ratio)</math></p>                                                   |
| <code>ate_rx_eq = 4;</code>            | Integer value between 0 and 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RX equalization setting                                                                                                                                                                                       |

**Table C-6 Configuration Parameters (Continued)**

| Parameter in tb_ate_top.v | Supported Range                                                                                                                            | Description                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| ate_flip_one = 0;         | <ul style="list-style-type: none"> <li>0: PHY does not perform polarity inversion.</li> <li>1: PHY performs polarity inversion.</li> </ul> | Receive Polarity Inversion |
| ate_loopback = 1;         | <ul style="list-style-type: none"> <li>0: External loopback</li> <li>1: Internal loopback</li> </ul>                                       |                            |
| ate_vph_lvl = 3.3         | 3.3 V (+10%, –7%)                                                                                                                          |                            |

**Note**

Some parameters in the preceding table are set to a generic value; you must update them to match the project-specific settings. For example, set `ate_vph_lvl` to match the vph level you use during ATE testing.

### C.1.8.3 Configuring the Test Limits

Several tests in the Verilog test bench use test limits to define the pass or fail criteria. These test limits are centralized and specified in the Verilog file, `./macro/Latest/testbench/testbench/tb_ate_top.v`.



#### Note

These default test limits are set for normal operating conditions. Optimize the limits through characterization of the specific test environment. In addition, some tests require that you set the supply levels (for example, the `ate_vp_cm` parameter) to match the levels being used during ATE testing.

The following table describes the configuration parameters for various tests.

**Table C-7 Test Configuration Parameters**

| Parameter                                             | Description                                                                                |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Settings for <code>ate_async_test</code></b>       |                                                                                            |
| <code>ate_ppm_offset_a = 2500;</code>                 | Not supported.                                                                             |
| <code>ate_init_ref_clk_sel_a = 0;</code>              | Not supported.                                                                             |
| <b>Settings for <code>ate_eyemask_test</code></b>     |                                                                                            |
| <code>ate_ppm_offset_em = 100;</code>                 | Not supported.                                                                             |
| <code>ate_noffset_em = -0.0775;</code>                | Negative voltage offset of eye amplitude                                                   |
| <code>ate_poffset_em = 0.0775;</code>                 | Positive voltage offset of eye amplitude                                                   |
| <code>ate_left0_margin_em = 2;</code>                 | Margins of pmix phase values that shift left in the Eye Mask test.                         |
| <code>ate_right0_margin_em = 2;</code>                | Margins of pmix phase values that shift right in the Eye Mask test                         |
| <code>ate_leftp_margin_em = 2;</code>                 | Margins of pmix phase values that shift left in the Eye Mask Positive Voltage Corner test  |
| <code>ate_rightp_margin_em = 2;</code>                | Margins of pmix phase values that shift right in the Eye Mask Positive Voltage Corner test |
| <code>ate_leftn_margin_em = 2;</code>                 | Margins of pmix phase values that shift left in the Eye Mask Negative Voltage Corner test  |
| <code>ate_rightn_margin_em = 2;</code>                | Margins of pmix phase values that shift right in the Eye Mask Negative Voltage Corner test |
| <b>Settings for <code>ate_common_mode_test</code></b> |                                                                                            |
| <code>ate_vp_cm = 1.0;</code>                         | Supply voltage value for common mode test                                                  |
| <b>Settings for <code>ate_ifps_los_test</code></b>    |                                                                                            |
| <code>ate_ifpsh_II = 7'b111_1000; //120</code>        | Set for high tx_amplitude.                                                                 |
| <code>ate_ifpsl_II = 7'b000_0111; //7</code>          | Set for low tx_amplitude.                                                                  |

**Table C-7 Test Configuration Parameters (Continued)**

| Parameter                                 | Description                                                                                                                          |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ate_txlvl_ll = 7'b111_1111; //127         | Set for regular tx_amplitude.                                                                                                        |
| <b>Settings for ate_tx_dc_levels_test</b> |                                                                                                                                      |
| ate_tx_preemph_tdl = 7'd22;               | Set for TX boost setting case1.                                                                                                      |
| ate_tx_preemph2_tdl = 7'd29;              | Set for TX boost setting case2.                                                                                                      |
| ate_txlvl_limit_val_lo_tdl = 0.80;        | Set for TX DC level low.                                                                                                             |
| ate_txlvl_limit_val_hi_tdl = 1.20;        | Set for TX DC level high.                                                                                                            |
| ate_txlvl_limit_val_lo_boost_tdl = 0.50;  | txlvl_lowerlim=txlvl_limit_val_lo*1024/vph, the voltage upper limit for the ATE TX DC Level test when preemph is ate_tx_preemph_tdl  |
| ate_txlvl_limit_val_hi_boost_tdl = 0.90;  | txlvl_upperlim=txlvl_limit_val_hi*1024/vph, the voltage upper limit for the ATE TX DC Level test when preemph is ate_tx_preemph_tdl  |
| ate_txlvl_limit_val_lo_boost2_tdl = 0.40; | txlvl_lowerlim=txlvl_limit_val_lo*1024/vph, the voltage upper limit for the ATE TX DC Level test when preemph is ate_tx_preemph_tdl2 |
| ate_txlvl_limit_val_hi_boost2_tdl = 0.80; | txlvl_upperlim=txlvl_limit_val_hi*1024/vph, the voltage upper limit for the ATE TX DC Level test when preemph is ate_tx_preemph_tdl2 |
| ate_adc_exp_val_hi_tdl = 1.1;             | The voltage codes converted from the ADC; high when preemp is 0                                                                      |
| ate_adc_exp_val_lo_tdl = 0.1;             | The voltage codes converted from the ADC; low when preemp is 0                                                                       |
| ate_adc_exp_val_hi_boost_tdl = 0.85;      | The voltage codes converted from the ADC; high when preemp is ate_tx_preemph_tdl                                                     |
| ate_adc_exp_val_lo_boost_tdl = 0.15;      | The voltage codes converted from the ADC; low when preemp is ate_tx_preemph_tdl                                                      |
| ate_adc_exp_val_hi_boost2_tdl = 0.8;      | The voltage codes converted from the ADC; high when preemp is ate_tx_preemph_tdl2                                                    |
| ate_adc_exp_val_lo_boost2_tdl = 0.2;      | The voltage codes converted from the ADC; low when preemp is ate_tx_preemph_tdl2                                                     |

#### C.1.8.4 Modifying the BERT Test

The BERT test is an external loopback test that exercises the TX and RX paths in the PHY. The LFSR7 pattern is the default pattern used in the BERT test.

A different pattern can be created by modifying the `ate_lbert_mode` and `ate_lbert_pattern` parameters, as described in “[Configuration Parameters](#)” on page 501. For more information about the different pattern modes that the USB 3.0 femtoPHY supports, see [Table C-6](#) on page 501.

#### C.1.8.5 Modifying the LOS Test

The LOS test provided in the Verilog test bench is written as a ‘go-no-go’ test. This test is run in two parts. In the first part, the transmitted amplitude is set high, and the LOS bit is verified to be 1'b0 (that is, there is no loss of signal). In the second part, the transmitted amplitude is set low, and the LOS bit is verified to be 1'b1.

This test can be modified to get the exact amplitude for the LOS threshold using the following pseudocode.

1. Set amplitude to `lfps_val = 1`.
2. While LOS bit != 0  
`lfps_val = lfps_val + 1`

The corresponding analog amplitude is calculated as:

$$\text{Tx amplitude (V)} = (\text{corresponding launch amplitude based on tx\_vboost\_lvl}[2:0]) \times \left\lceil \frac{\text{lfps\_val} + 1}{128} \right\rceil$$

where `vptx` is the `vptx` level at which the test is run.

## C.2 ATE Testing Details

This section describes each ATE test in detail.

### C.2.1 Tester Requirements

The USB 3.0 femtoPHY's test methodology uses minimal tester resources. The tester's hardware requirements include two power supplies, four digital pins, and a differential reference clock. In most applications, two digital tester pins can be used as the reference clock source. The reference clock must be present before the PHY is released from reset.



#### Note

If at any time during the test sequence the reference clock is stopped/removed, the PHY must be reinitialized.

The following table lists the minimum tester requirements.

**Table C-8 Minimum Tester Requirements**

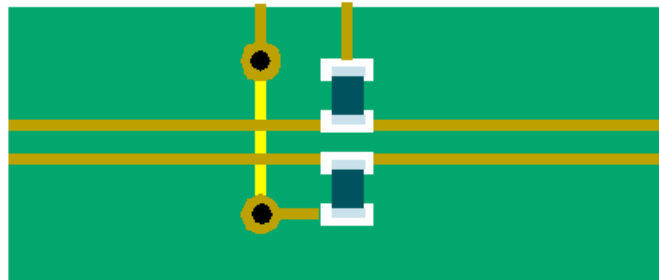
| Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Details                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Two power supplies: one at high voltage, the other at core voltage                                                                                                                    |
| Digital force/compare (JTAG TDI, TMS, TCK, and TDO)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Four pins typically at 10 MHz (The maximum JTAG clock frequency is 50 MHz.)                                                                                                           |
| Differential reference clock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 19.2–200 MHz, 500 mV peak-peak<br><b>Note:</b> If you are using <code>ref_alt_clk</code> or an oscillator on the load board, this resource does not need to be sourced by the tester. |
| <b>Note:</b> <ul style="list-style-type: none"> <li>■ The PHY uses two power supplies and a four-pin JTAG interface running at 10 MHz.</li> <li>■ If the ASIC has a JTAG port, the JTAG pins from the ASIC can be shared.</li> <li>■ Slow-speed digital pins can be used, but test times increase.</li> <li>■ One of the digital pins must have the ability to compare vectors against a known pattern.</li> <li>■ The reference clock requires a source.</li> <li>■ If the ATE cannot generate a high-quality reference clock, an external oscillator can be used.</li> </ul> |                                                                                                                                                                                       |

## C.2.2 Load Board Requirements

Load board requirements are minimal. The USB 3.0 femtoPHY requires an external 200- $\Omega$  resistor on the resref pin. To take advantage of the built-in test capabilities, an external AC-coupled loopback between the USB 3.0 femtoPHY's transmit and receive pins is required. If continuity/leakage measurements are required on the High-Speed (HS) function's and/or SS function's serial interfaces, place an optional relay on the signal path. To minimize any discontinuities in the signal path, take care when placing the relay. If only a continuity check is required, you can place a high-value resistor on the SS function's loopback path.

The following figure shows the recommended layout.

**Figure C-9 Continuity Resistor**



The horizontal lines are TX(p,m). The vertical lines go to a probe point or connector to enable an ATE tester to perform a continuity test. The typical value for a high-value resistor is above 10 k $\Omega$ .

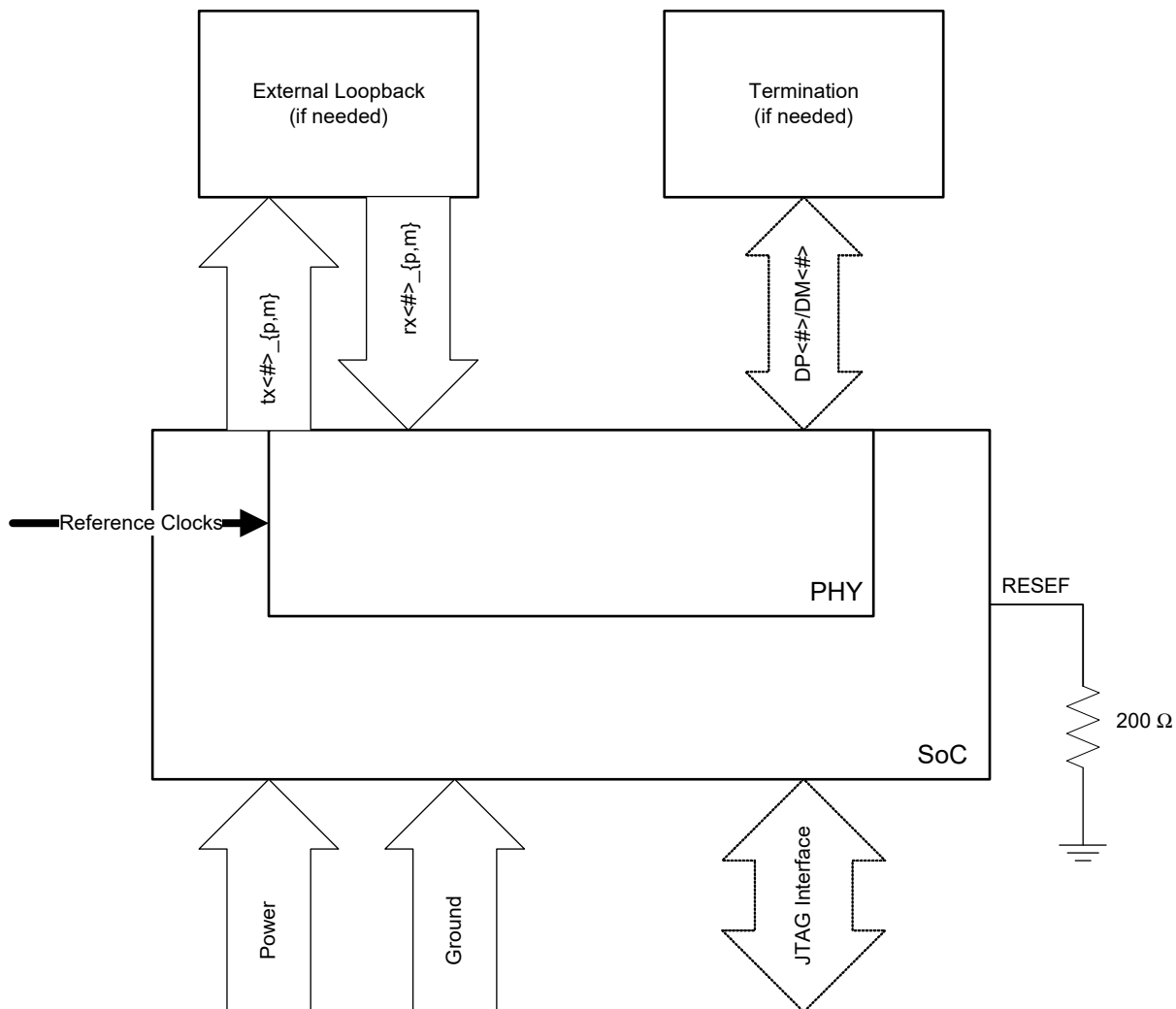
### C.2.3 Test Mode Pin Requirements

The USB 3.0 femtoPHY deliverables provides a pin list (<protocol>\_<configuration>\_ate\_pinlist.txt), that lists the external requirements of the IP's top-level pins to place the IP in test mode.

The top-level pins that are connected to a physical pad are noted along with the required setting for control pins. Note the JTAG pins (tck, tdi, tcm, and tdo) are control pins that must be accessible to users. Access can be provided either by direct connection or through a typical JTAG chain.

The following figure shows the external board requirements to place the USB 3.0 femtoPHY in test mode.

**Figure C-10 External Board Requirements**





The following table lists the termination requirements for various tests.

**Table C-9 Termination Requirements**

| Test              | Termination Required          |
|-------------------|-------------------------------|
| REGISTER          | None                          |
| JTAG_NO_CLOCK     | None                          |
| JTAG_CLOCK        | None                          |
| BERT              | External loopback on SS ports |
| EYEMASK           | External loopback on SS ports |
| ASYNC             | External loopback on SS ports |
| LFPS_LOS          | External loopback on SS ports |
| TX_DC_LEVELS      | None                          |
| INTERNAL_LOOPBACK | None                          |
| COMMON_MODE       | None                          |
| TX_RX_DETECT      | External loopback on SS ports |
| RX_SCOPE_TEST     | External loopback on SS ports |
| POWER_STATES_TEST | External loopback on SS ports |
| BIST              | 130 $\Omega$ on DP<#>/DM<#>   |
| CDR               | 130 $\Omega$ on DP<#>/DM<#>   |
| TX_REF            | None                          |

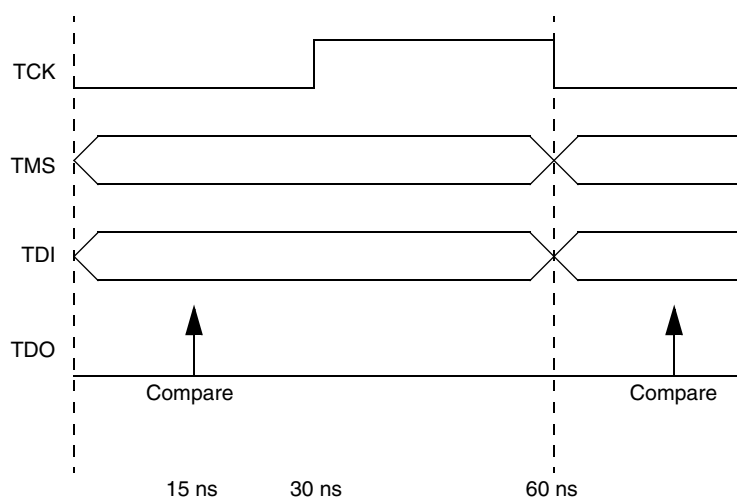
## C.2.4 ATE Tests

This section describes the timing requirements and the ATE test suite supported by the USB 3.0 femtoPHY.

### C.2.4.1 Timing

Timing for sampling and comparing TDO occurs before the rising edge of the TCK. The following figure shows the timing with a 16.6-MHz clock as TCK. Users must ensure that the TDO comparison with the expected TDO (as provided in the ate.vec file) occurs a quarter period after the falling edge of jtag\_tck, which ensures that TDO is stable at that point.

**Figure C-11 JTAG Timing Diagram**



#### Note

The comparison of TDO occurs before the clock edge of TCK.

### C.2.4.2 ATE Test Suite

The following table lists the ATE tests and their test times (assuming a 16.6-MHz JTAG clock frequency). Tests that require limits to be set are also indicated.



#### Note

Unless indicated otherwise, use your discretion to determine whether to include each test in your test plan, based on your specific test plan requirements.

**Table C-10 ATE Tests and Test Times**

| Test              | Approximate Test Time (ms) <sup>a</sup> | Limits Required (Y/N) | Can be Simulated in Verilog (Y/N) | Verilog Task                 |
|-------------------|-----------------------------------------|-----------------------|-----------------------------------|------------------------------|
| REGISTER          | 1.1                                     | N                     | Y                                 | ate_register_test.v          |
| JTAG_NO_CLOCK     | 0.01                                    | N                     | Y                                 | ate_jtag_no_clock_test.v     |
| JTAG_CLOCK        | 0.01                                    | N                     | Y                                 | ate_jtag_clock_test.v        |
| BERT              | 6.03                                    | N                     | Y                                 | ate_bert_test.v              |
| EYEMASK           | 11.15                                   | Y                     | N                                 | ate_eyemask_test.v           |
| ASYNC             | 4.58                                    | Y                     | N                                 | ate_async_test.v             |
| LFPS_LOS          | 0.08                                    | Y                     | N                                 | ate_lfps_los_test.v          |
| TX_DC_LEVELS      | 5.13                                    | Y                     | N                                 | ate_tx_dc_levels_test.v      |
| INTERNAL_LOOPBACK | 2.56                                    | N                     | Y                                 | ate_internal_loopback_test.v |
| COMMON_MODE       | 0.51                                    | Y                     | N                                 | ate_common_mode_test.v       |
| TX_RX_DETECT      | 0.05                                    | N                     | N                                 | ate_tx_rx_detect_test.v      |
| RX_SCOPE_TEST     | 8,420                                   | N                     | N                                 | ate_rx_scope_test.v          |
| POWER_STATES_TEST | 0.7                                     | N                     | N                                 | ate_power_states_test.v      |
| BIST              | 2.75                                    | N                     | Y                                 | ate_bist_test.v              |
| CDR               | 3.68                                    | Y                     | N                                 | ate_cdr_test.v               |
| TX_REF            | 0.27                                    | Y                     | N                                 | ate_tx_ref_test.v            |

a. Test times can vary based on your specific application.



#### Note

The *ate\_initialize* subtask that resets and initializes the PHY for ATE testing is located in the Verilog *tb\_ate\_tasks.v* file.

The following subsections describe these tests. In addition to the test descriptions, the generated vector files contain detailed comments and step-by-step instructions for each register read and write operation.

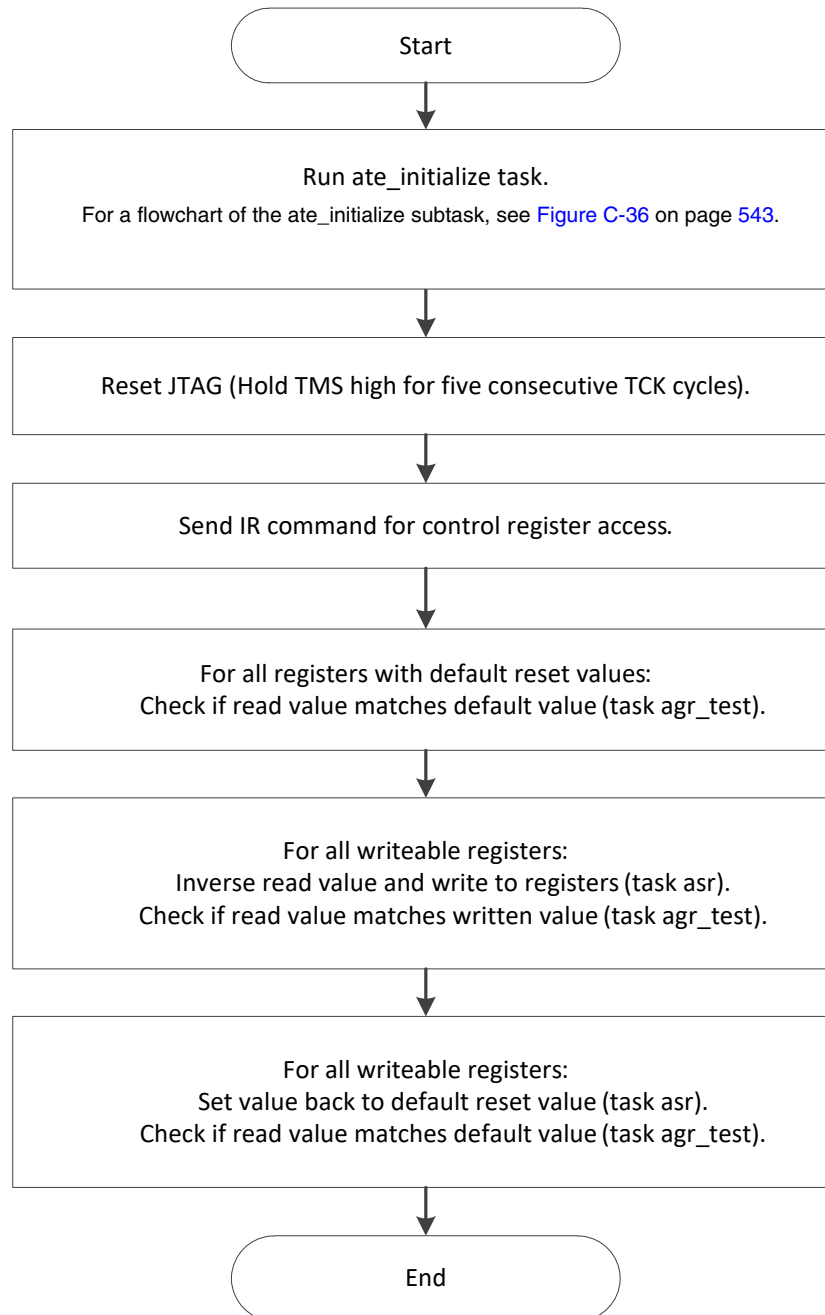
### C.2.4.2.1 REGISTER

This test reads the default value of all readable registers in the PHY, then writes the inverse pattern and verifies the result. Finally, the test writes the default value back to the registers and verifies the result.

**External Loopback Required:** N

**Wafer Sort:** Y

**Figure C-12** ate\_register\_test



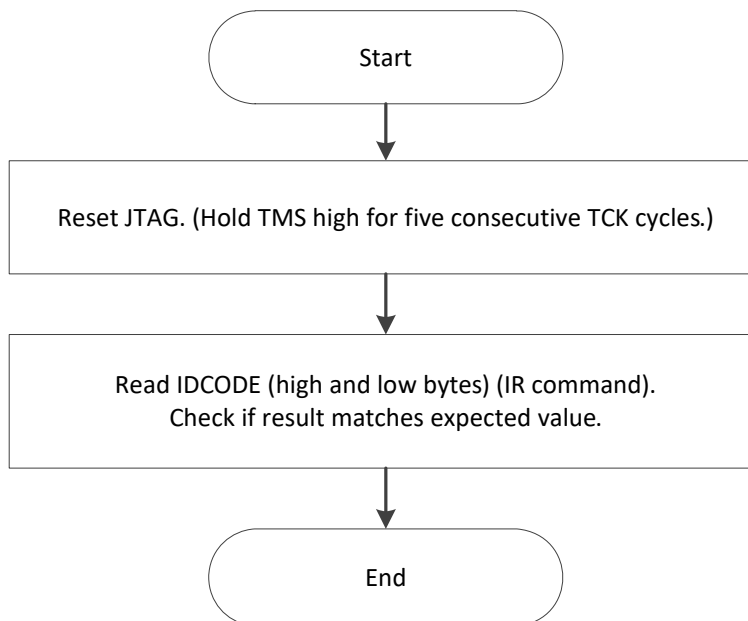
### C.2.4.2.2 JTAG\_NO\_CLOCK

**External Loopback Required:** N

**Wafer Sort:** Y

This test verifies basic functions of the JTAG state machine by sending an IR IDCODE command and reading the IDCODE. This test requires only that the power to the PHY is present. This test does not provide much test coverage, but it verifies that the PHY can be accessed through the JTAG port. This is a good 'first test' to rule out connection issues from the tester to the PHY.

**Figure C-13** ate\_jtag\_no\_clock\_test



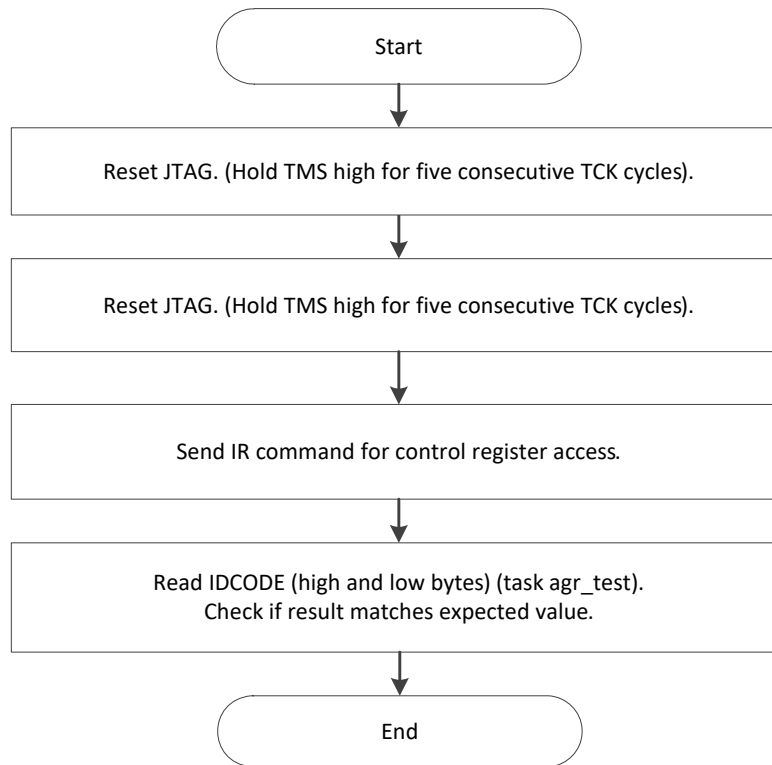
### C.2.4.2.3 JTAG\_CLOCK

**External Loopback Required:** N

**Wafer Sort:** Y

This test is similar to the JTAG\_NO\_CLOCK test, but JTAG\_CLOCK reads the JTAG ID value from registers in the USB 3.0 femtoPHY. To read registers in the USB 3.0 femtoPHY, a reference clock must be present, and the USB 3.0 femtoPHY must be enabled. After running JTAG\_NO\_CLOCK, JTAG\_CLOCK is a good second test to run to verify that both the reference clock is present and the USB 3.0 femtoPHY is enabled.

**Figure C-14** ate\_jtag\_clock\_test



### C.2.4.2.4 BERT

**External Loopback Required:** Y

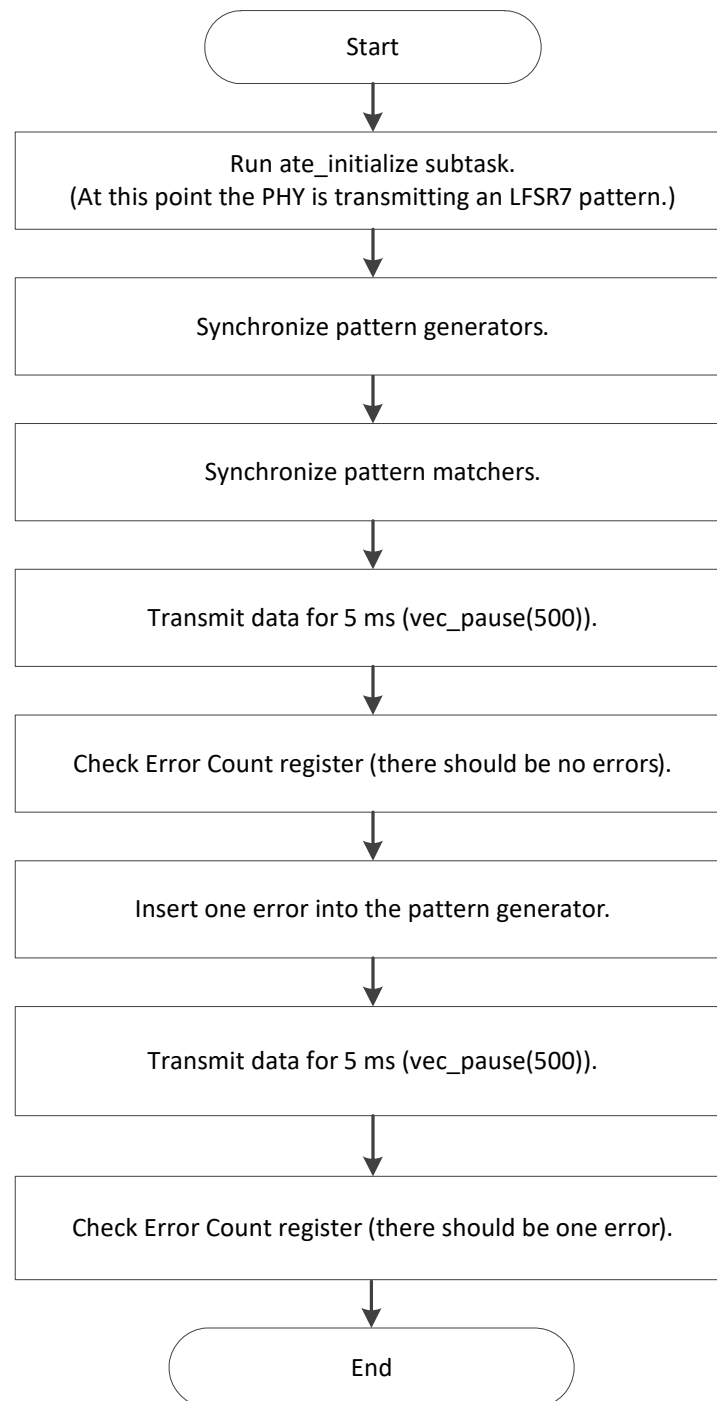
**Wafer Sort:** N

It is highly recommended that you include this test, because it exercises the entire datapath.

The BERT test checks the following in the SS function:

- Signal path
- Pattern matchers
- Pattern generators
- Error counters to be used in subsequent tests

This test uses an external loopback. A known LFSR7 pattern (default) is transmitted and matched at the receiver to verify that data can be passed from the transmitter to the receiver.

**Figure C-15** ate\_bert\_test

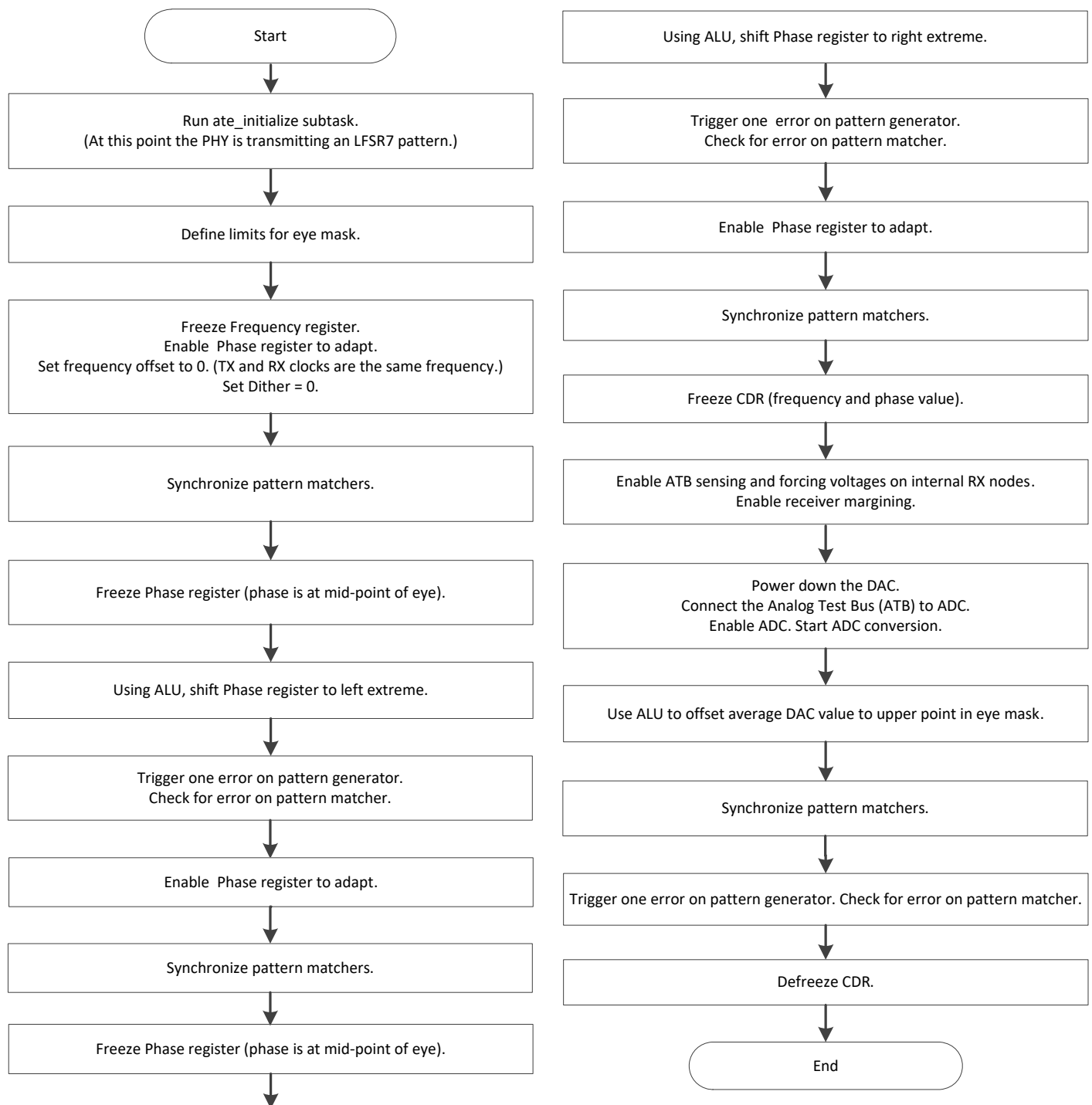
### C.2.4.2.5 EYEMASK

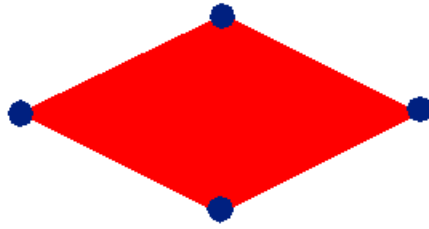
**External Loopback Required:** Y

**Wafer Sort:** N

This test relies on the load board to have the channel loopbacked to itself. The external loopback trace should be a representative trace that the SS function encounters in normal operation. The EYEMASK test enables users to define a diamond-shaped mask for template testing on the USB 3.0 femtoPHY. The EYEMASK test is accomplished in the USB 3.0 femtoPHY by moving the data strobe point in both voltage and phase at the receiver. Similar to the BERT test, the pattern generator creates a known pattern, which the pattern matcher matches. However, instead of sampling in the center of the eye, the strobe points used are the offset defined by the mask. The EYEMASK test enables users to test beyond a normal loopback test, in which the bit error rate is not guaranteed.



**Figure C-16** ate\_eyemask\_test

**Figure C-17 Eye Mask Template Configurations**

The left point in the eye diagram is related to the left margin phase value. The right point is related to the right margin phase value.

The upper point in the eye diagram is related to the upper offset average DAC value. The lower point is related to the lower offset average DAC value.

### C.2.4.2.6 ASYNC

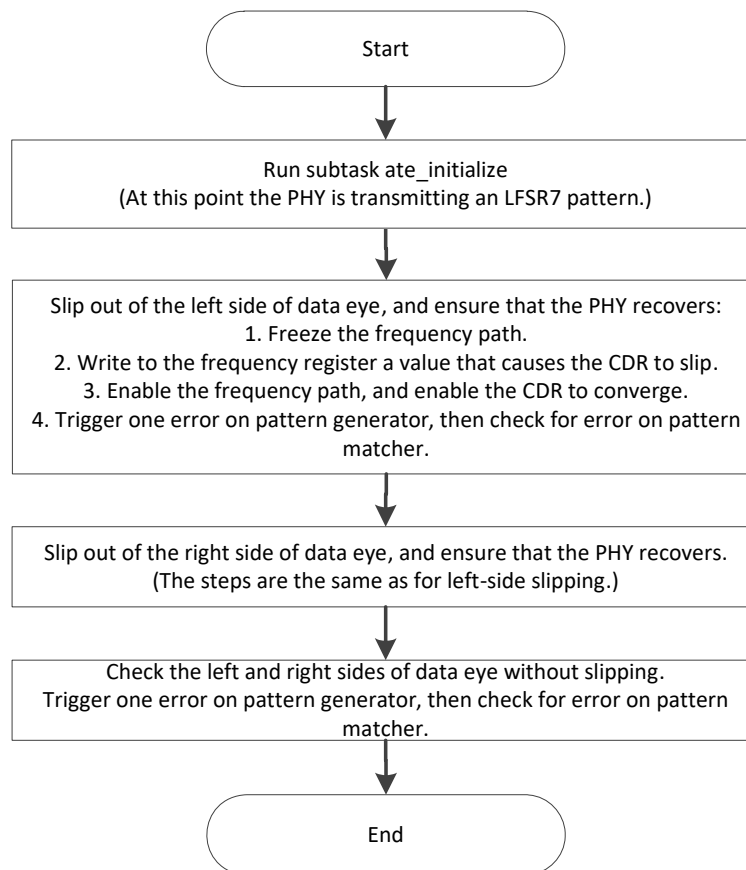
**External Loopback Required:** Y

**Wafer Sort:** N

This test checks the USB 3.0 femtoPHY receiver. In most transceiver ATE testing, all tests are done in a synchronous environment. In the past, external equipment was required to perform any type of asynchronous testing. This requirement usually involved having some type of relay matrix to MUX the external signal to all the channels. This setup takes time to implement and requires a device be tested on a tester with such equipment.

The USB 3.0 femtoPHY IP is designed to enable an alternate transmit clock to be generated with a user-defined ppm offset from the single reference clock source. This feature enables the transmitter to transit a signal that is asynchronous to the reference clock of the receiver's CDR, providing more robust testing on the CDR of the receiver. Similar to the BERT test, the pattern generator and pattern matcher are used.

**Figure C-18** ate\_async\_test



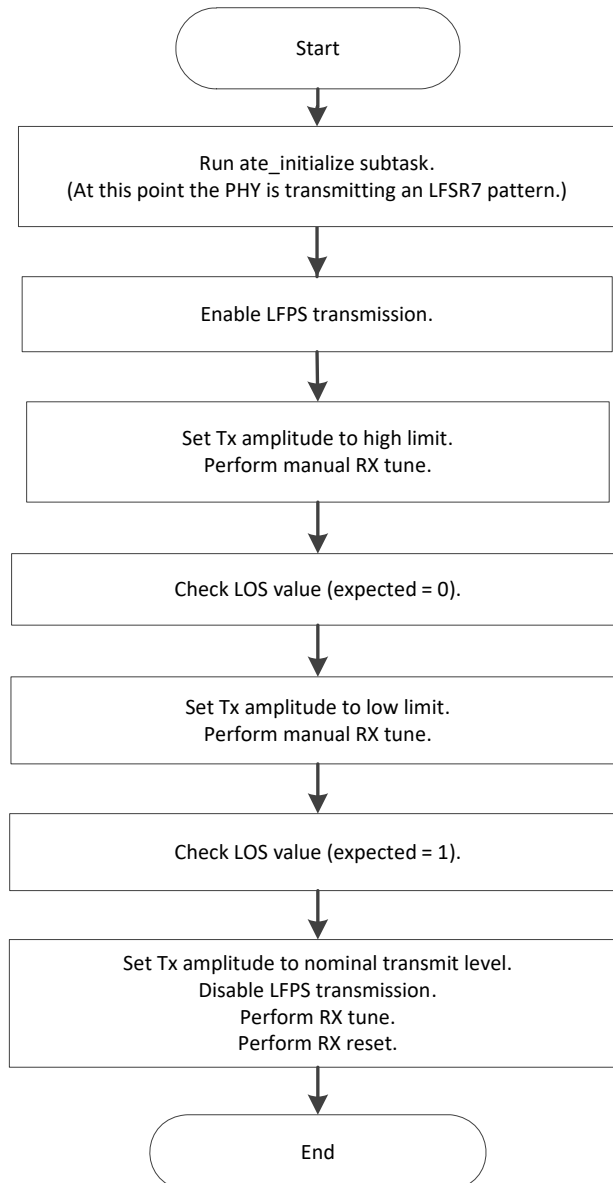
### C.2.4.2.7 LFPS\_LOS

**External Loopback Required:** Y

**Wafer Sort:** N

This test sends Low Frequency Periodic Signaling (LFPS) out the transmitter and measures the Loss-of-Signal (LOS) level of the SuperSpeed function's receiver. Users can set the upper and lower comparison points. When transmitting the upper voltage limit, the LOS should de-assert. When transmitting the lower limit, the LOS should remain asserted.

**Figure C-19** ate\_lfps\_los\_test

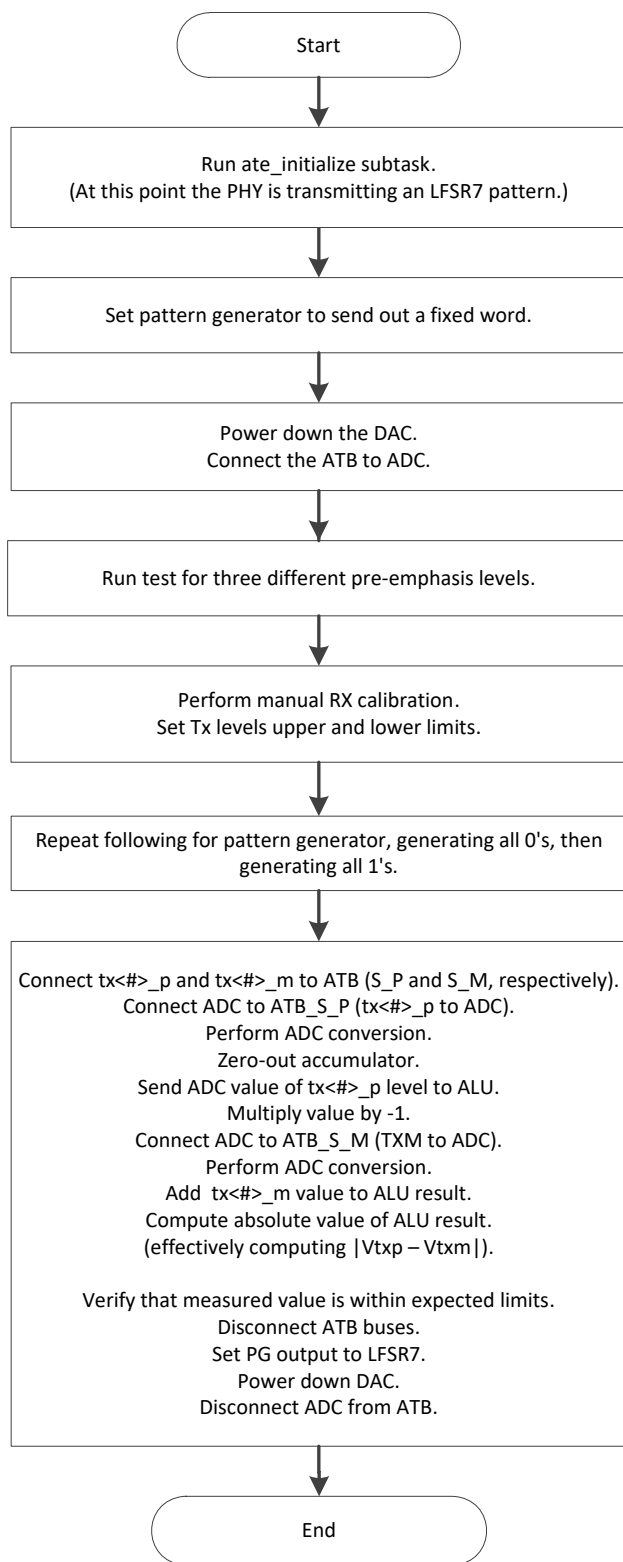


#### C.2.4.2.8 TX\_DC\_LEVELS

**External Loopback Required:** N

**Wafer Sort:** Y

The differential DC voltage of the USB 3.0 femtoPHY transmitter is measured without boost, the default boost level, and a user-defined boost value. This test uses the internally available ADC to measure the voltages and compares them to user-defined limits using the ALU.

**Figure C-20** ate\_tx\_dc\_levels\_test

#### **C.2.4.2.9 INTERNAL\_LOOPBACK**

**External Loopback Required:** N

**Wafer Sort:** Y

This test verifies the function of the USB 3.0 femtoPHY's TX-to-RX digital serial loopback. This test is identical to the BERT/External Loopback test, except that the RX AFE is not exercised. An internal loopback path is exercised, bypassing the TX drivers and the RX AFE. If BERT is being run, INTERNAL\_LOOPBACK does not have to be run at board level; instead, this test can be run before packaging at wafer sort.

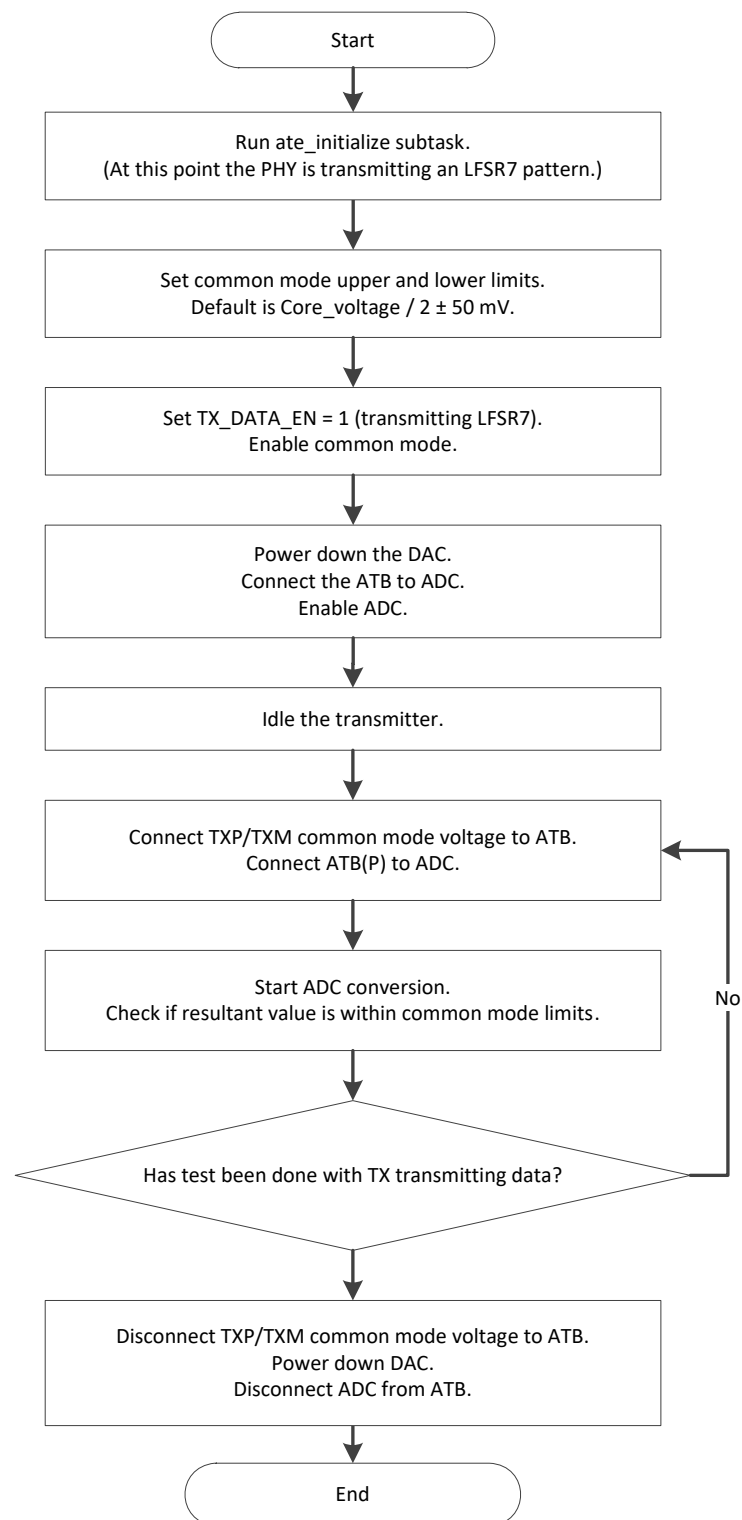
#### **C.2.4.2.10 COMMON\_MODE**

**External Loopback Required:** N

**Wafer Sort:** Y

**Test Plan Inclusion:** Optional

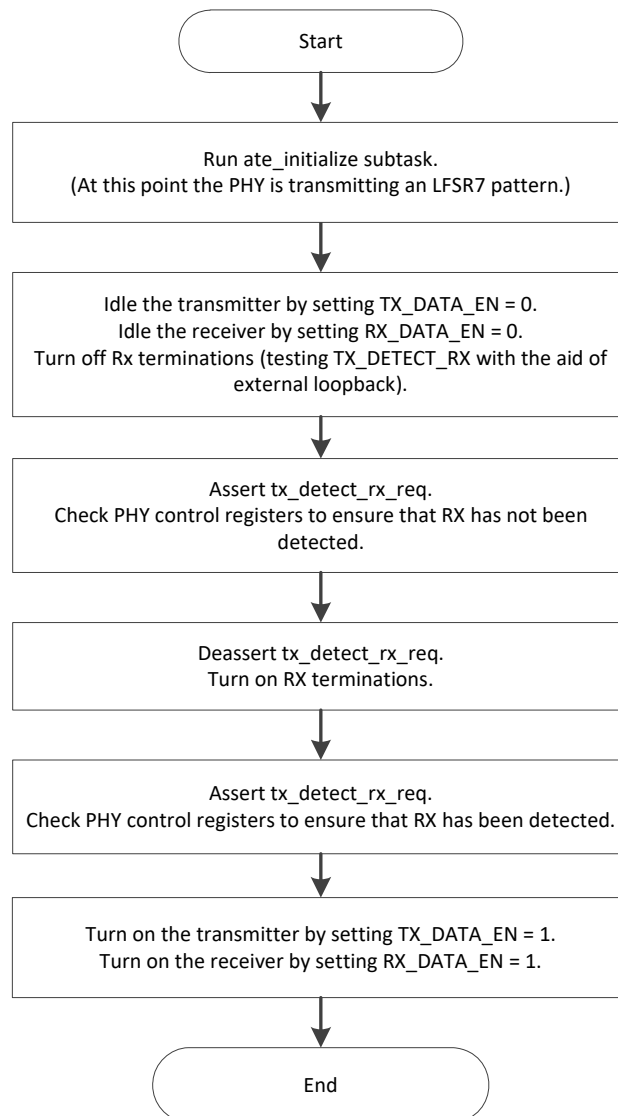
This test measures the DC transmit common-mode voltage during transmit and idle states of the USB 3.0 femtoPHY.

**Figure C-21** ate\_common\_mode\_test



**C.2.4.2.11 TX\_RX\_DETECT****External Loopback Required:** Y**Wafer Sort:** N

TX/RX detection is tested by turning off the USB 3.0 femtoPHY receiver terminations and verifying that the transmitter does not detect a receiver. The receiver terminations are then enabled, and the transmitter is verified to have detected a receiver. External loopback is not required for this test; however, high-impedance DC from the transmitter is required. Therefore, the external loopback with the AC-coupling capacitors provides the high-impedance DC, because the inline capacitors are high-impedance and DC. Also note that the USB 3.0 femtoPHY transmitter has sampling capability that is used for this test.

**Figure C-22 ate\_tx\_rx\_detect\_test**

#### C.2.4.2.12 RX\_SCOPE\_TEST

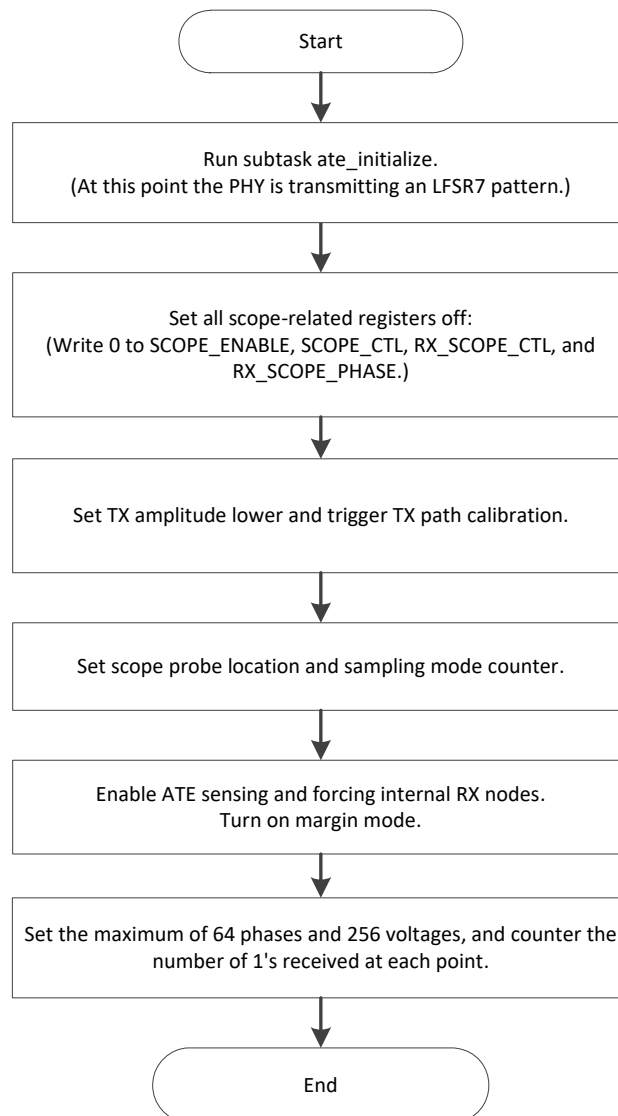
**External Loopback Required:** Y

**Wafer Sort:** N

The PHY has a built-in, on-chip scope that can be used to generate a receive eye diagram. For detailed information, see [“On-Chip Scope”](#) on page 537.

In the `ate_rx_scope_test` directory, the files are as follows.

- **ate\_rx\_scope\_test.py**: Python file that generates the ATE vector  
Because RX\_SCOPE\_TEST requires a lot of time to generate the vector by Verilog simulation, the Python file is used to generate the ATE vector here.
- **eye.py**: Python file that generates the eye diagram
- **override.txt**: User parameter control for MPLL multiplier, SSC `ref_clk_sel`, TX amplitude, and JTAG clock period

**Figure C-23** ate\_rx\_scope\_test

### C.2.4.2.13 POWER\_STATES\_TEST

**External Loopback Required:** Y

**Wafer Sort:** N

This test measures the power consumption for the various power states. To measure power consumption of the expected power states, you must define the corresponding definition for this test.

The following definitions control the initial power state or destination power state.

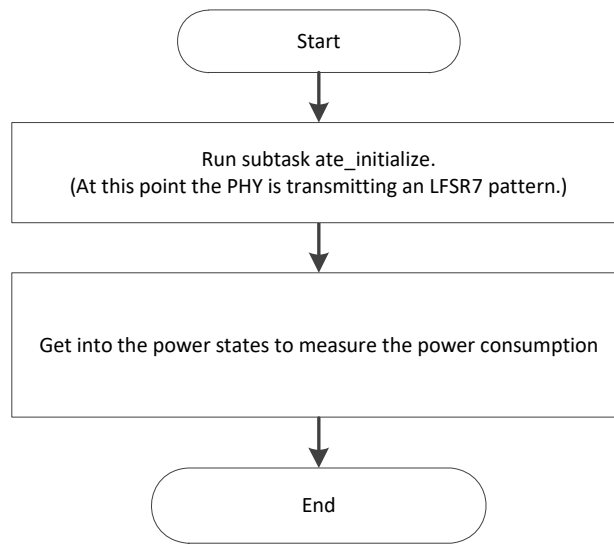
- **PSTATE\_INIT:** Initial power state from which the PHY is powered up  
The default PSTATE\_INIT setting is the lowest power state.
- **PSTATE\_DEST:** The destination power state for the PHY  
The default PSTATE\_DEST setting is the full, active power state.
- **PSTATE\_HSP\_INIT:** Initial power state from which the PHY's HS portion is powered up  
The default PSTATE\_HSP\_INIT setting is Suspend mode.
- **PSTATE\_HSP\_DEST:** The destination power state for the PHY's HS portion  
The default PSTATE\_HSP\_DEST setting is Idle mode.

**Table C-11 Power State Definitions**

| Power State    | Definition         |
|----------------|--------------------|
| P0 (U0)        | PSTATE_P0          |
| P1 (U1)        | PSTATE_P1          |
| P2 (U2)        | PSTATE_P2          |
| P3 (U3)        | PSTATE_P3          |
| L0_IDLE (Idle) | PSTATE_HSP_L0_IDLE |
| L0_TX (Data)   | PSTATE_HSP_L0_TX   |
| L1 (Sleep)     | PSTATE_HSP_L1      |
| L2 (Suspend)   | PSTATE_HSP_L2      |

For example, to test the P0 power state powered up from the P2 state, plus the HS L0\_TX power state powered up from the L1 state, the run command is:

```
./run_ate_pipe ../testbench/ate_power_states_test/ate_power_states_test.v
+define+VEC_TRACE +define+PSTATE_INIT=PSTATE_P2 +define+PSTATE_DEST=PSTATE_P0
+define+PSTATE_HSP_INIT=PSTATE_HSP_L1 +define+PSTATE_HSP_DEST=PSTATE_HSP_L0_TX
```

**Figure C-24** ate\_power\_states\_test**C.2.4.2.14 BIST**

**External Loopback Required:** N (130-Ω termination required)

**Wafer Sort:** N

This test is highly recommended, because it exercises the HS/FS and LS portions in the HS function of the USB 3.0 femtoPHY.

This test uses the BIST feature of the PHY's HS function. This test runs BIST in HS, FS, and LS modes.

**C.2.4.2.15 CDR**

**External Loopback Required:** N (130-Ω termination required)

**Wafer Sort:** N

This test measures the phase opening of the HS function's transmit eye at the receiver. The transmitter is internally connected to the receiver. To determine the eye opening, the phase is stepped left and right from its nominal value while BIST is running.

**C.2.4.2.16 TX\_REF**

**External Loopback Required:** N

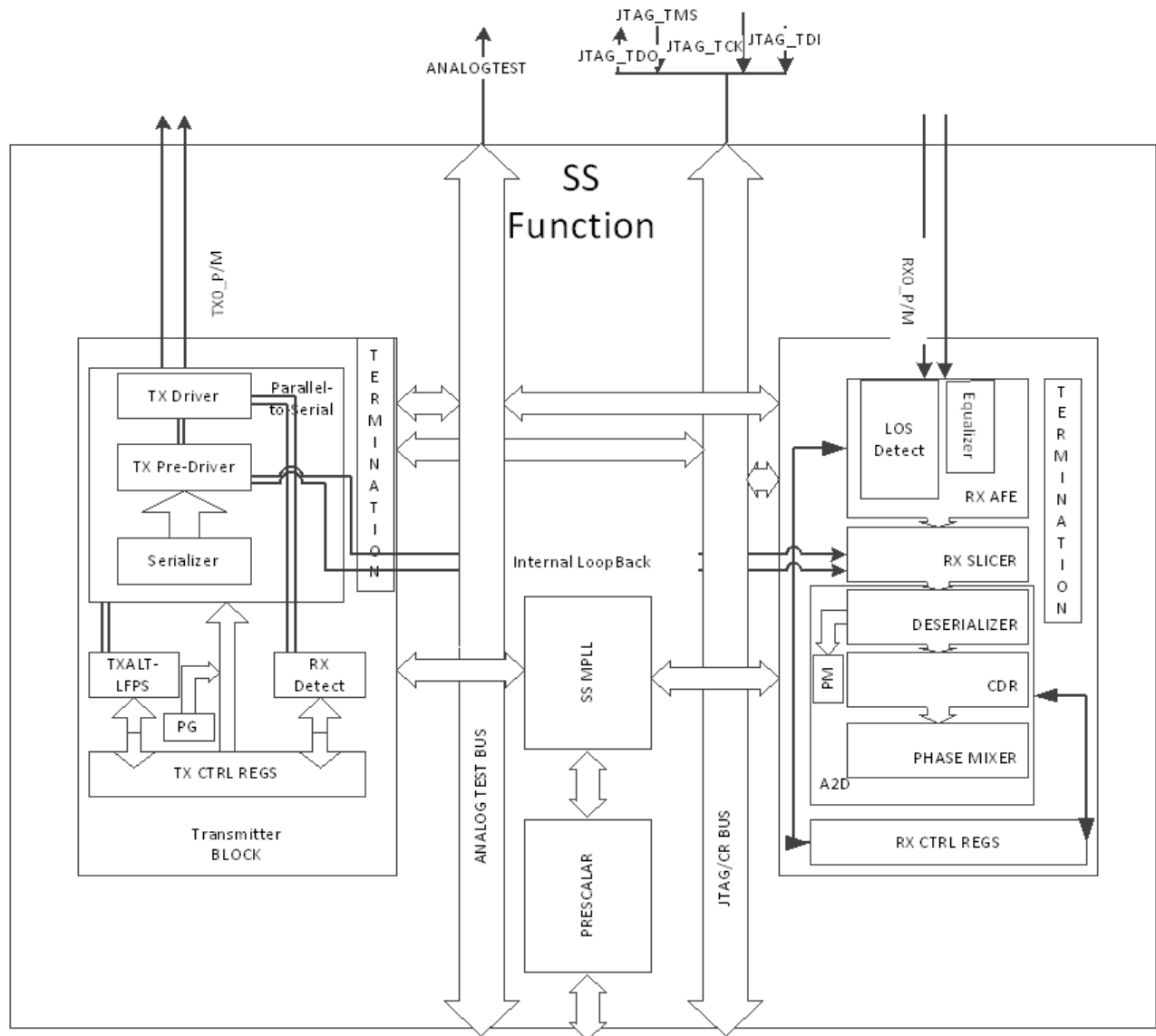
**Wafer Sort:** N

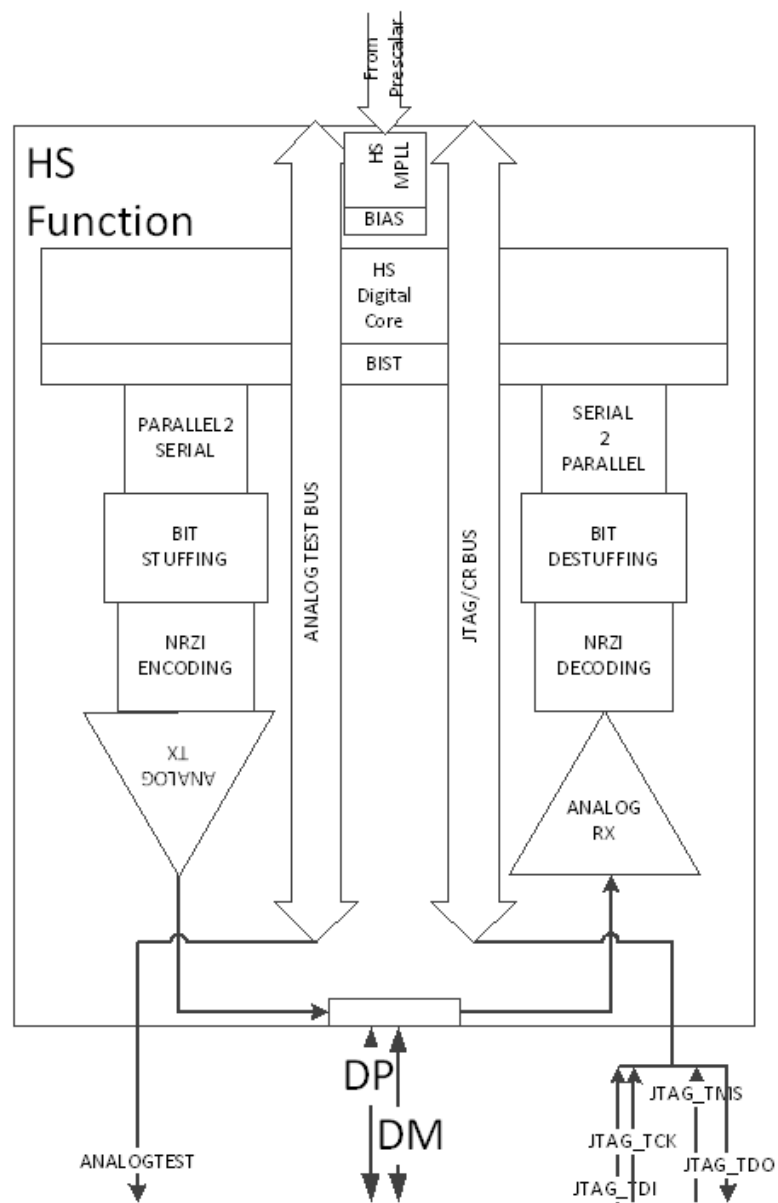
Using the on-board ADC, this test measures the transmitter voltage reference of the HS function.

## C.2.5 ATE Test Coverage

The following two figures show a block diagram of the USB 3.0 femtoPHY macro.

**Figure C-25 USB 3.0 femtoPHY SS Function Block Diagram**



**Figure C-26 USB 3.0 femtoPHY HS Function Block Diagram**

The following table lists the individual tests and corresponding blocks that are exercised in each test.

**Table C-12 ATE Test Coverage**

| Block                   | ATE Tests |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
|-------------------------|-----------|---------------|------------|------|---------|-------|----------|--------------|-------------------|-------------|--------------|------|-----|--------|----------|--------------|
|                         | REGISTER  | JTAG_NO_CLOCK | JTAG_CLOCK | BERT | EYEMASK | ASYNC | LFPS_LOS | TX_DC_LEVELS | INTERNAL_LOOPBACK | COMMON_MODE | TX_RX_DETECT | BIST | CDR | TX_REF | RX_SCOPE | POWER_STATES |
| SS Function Receiver    |           |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| Equalizer               |           |               |            | X    |         |       |          |              |                   |             |              |      |     |        |          |              |
| CDR                     |           |               |            | X    | X       | X     |          |              | X                 |             |              |      |     |        | X        | X            |
| Termination             |           |               |            | X    | X       | X     |          |              |                   |             | X            |      |     |        | X        | X            |
| Mixer                   |           |               |            | X    | X       | X     |          |              | X                 |             |              |      |     |        | X        | X            |
| Slicer                  |           |               |            | X    | X       | X     |          |              | X                 |             |              |      |     |        | X        | X            |
| Deserializer            |           |               |            | X    | X       | X     |          |              | X                 |             |              |      |     |        | X        | X            |
| SS Function Transmitter |           |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| Driver                  |           |               |            | X    | X       | X     | X        | X            |                   | X           | X            |      |     |        | X        | X            |
| LFPS                    |           |               |            |      |         |       | X        |              |                   |             |              |      |     |        |          |              |
| RX Detection            |           |               |            |      |         |       |          |              |                   |             | X            |      |     |        |          |              |
| Serializer              |           |               |            | X    | X       | X     |          |              | X                 |             |              |      |     |        | X        | X            |
| Termination             |           |               |            | X    | X       | X     | X        |              |                   |             | X            |      |     |        | X        | X            |
| Loopback                |           |               |            |      |         |       |          |              | X                 |             |              |      |     |        | X        | X            |
| SS Function Support     |           |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| Power-On Reset          | X         |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| Resistor Tuning         | X         |               |            | X    | X       | X     | X        |              |                   |             | X            |      |     |        | X        | X            |
| Common Blocks           |           |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| Prescaler               |           |               |            | X    | X       | X     |          |              | X                 |             | X            |      |     |        | X        | X            |
| MPLL                    |           |               |            | X    | X       | X     |          |              | X                 |             | X            |      |     |        | X        | X            |
| Test Bus                |           |               |            |      | X       | X     |          | X            |                   |             |              |      |     |        | X        |              |



**Table C-12 ATE Test Coverage (Continued)**

| Block                   | ATE Tests |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
|-------------------------|-----------|---------------|------------|------|---------|-------|----------|--------------|-------------------|-------------|--------------|------|-----|--------|----------|--------------|
|                         | REGISTER  | JTAG_NO_CLOCK | JTAG_CLOCK | BERT | EYEMASK | ASYNC | LFPS_LOS | TX_DC_LEVELS | INTERNAL_LOOPBACK | COMMON_MODE | TX_RX_DETECT | BIST | CDR | TX_REF | RX_SCOPE | POWER_STATES |
| JTAG                    | X         | X             | X          | X    | X       | X     | X        | X            | X                 | X           | X            | X    | X   | X      | X        | X            |
| HS Function Transceiver |           |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| Transmitter             |           |               |            |      |         |       |          |              |                   |             |              | X    | X   |        |          |              |
| Squelch/Receiver        |           |               |            |      |         |       |          |              |                   |             |              | X    | X   |        |          |              |
| Disconnect/Tuning       |           |               |            |      |         |       |          |              |                   |             |              |      |     | X      |          |              |
| Digital Core            |           |               |            |      |         |       |          |              |                   |             |              | X    | X   |        |          |              |
| HS Function Support     |           |               |            |      |         |       |          |              |                   |             |              |      |     |        |          |              |
| PLL                     |           |               |            |      |         |       |          |              |                   |             |              | X    | X   |        |          |              |
| BIAS                    |           |               |            |      |         |       |          |              |                   |             |              | X    | X   | X      |          |              |
| Oscillator              |           |               |            |      |         |       |          |              |                   |             |              | X    | X   |        |          |              |

### C.2.5.1 Test Insertions

Some tests rely on the presence of external loopback from the SS transmitter to the SS receiver with the appropriate AC-coupling capacitors in order to function. During wafer sort, the use of probes capable of operating at multi-Gbps speeds can be costly. The following table lists each test and indicates whether the test can be run with/without external loopback on the SuperSpeed lines.



#### Note

For vector generation purposes and for ATE tests that can be run without external loopback, enable the ANI\_WAFER\_SORT define macro to enable the internal loopback path during the ate\_initialize\_test subtask. For details on the specific tests, see “ATE Tests” on page 510.

**Table C-13 Tests That Can Be Run at Wafer Sort Without External Loopback**

| Test              | PHY Function Tested | Can Be Run Without External Loopback (Y/N) |
|-------------------|---------------------|--------------------------------------------|
| REGISTER          | SS                  | Y                                          |
| JTAG_NO_CLOCK     | SS/HS               | Y                                          |
| JTAG_CLOCK        | SS                  | Y                                          |
| BERT              | SS                  | N                                          |
| EYEMASK           | SS                  | N                                          |
| ASYNC             | SS                  | N                                          |
| LFPS_LOS          | SS                  | N                                          |
| TX_DC_LEVELS      | SS                  | Y                                          |
| INTERNAL_LOOPBACK | SS                  | Y                                          |
| COMMON_MODE       | SS                  | Y                                          |
| TX_RX_DETECT      | SS                  | N                                          |
| RX_SCOPE_TEST     | SS                  | N                                          |
| POWER_STATES_TEST | SS                  | N                                          |
| BIST              | HS                  | Y                                          |
| CDR               | HS                  | Y                                          |
| TX_REF            | HS                  | Y                                          |

## C.2.6 ATE Test Limits

The following table describes the recommended test limits for the ATE tests where applicable.



### Note

These test limits are recommendations only—these limits can be optimized. For more information about how to change the test limits, see [“Configuring the Test Limits”](#) on page 503.

**Table C-14 ATE Test Limits**

| Test              | Low Limit | High Limit |
|-------------------|-----------|------------|
| REGISTER          | N/A       | N/A        |
| JTAG_NO_CLOCK     | N/A       | N/A        |
| JTAG_CLOCK        | N/A       | N/A        |
| BERT              | N/A       | N/A        |
| EYEMASK           | N/A       | N/A        |
| ASYNC             | N/A       | N/A        |
| LFPS_LOS          | 0.1 V     | 0.9 V      |
| TX_DC_LEVELS      | 0.8 V     | 1.2 V      |
| INTERNAL_LOOPBACK | N/A       | N/A        |
| COMMON_MODE       | 0.05 V    | 0.05 V     |
| TX_RX_DETECT      | N/A       | N/A        |
| RX_SCOPE_TEST     | N/A       | N/A        |
| POWER_STATES_TEST | N/A       | N/A        |
| BIST              | N/A       | N/A        |
| CDR               | N/A       | N/A        |
| TX_REF            | 0.725 V   | 0.875 V    |

## C.3 Bench Testing Details

This section describes how to use the ASCII vectors from the ATE Verilog Test Bench for bench testing. Familiarity with the ATE Verilog Test Bench, its JTAG chain configurability and its vector generation capability are prerequisites for this section.

### C.3.1 Hardware

This section describes the hardware requirements and setup for using the ASCII vectors from the USB 3.0 femtoPHY.

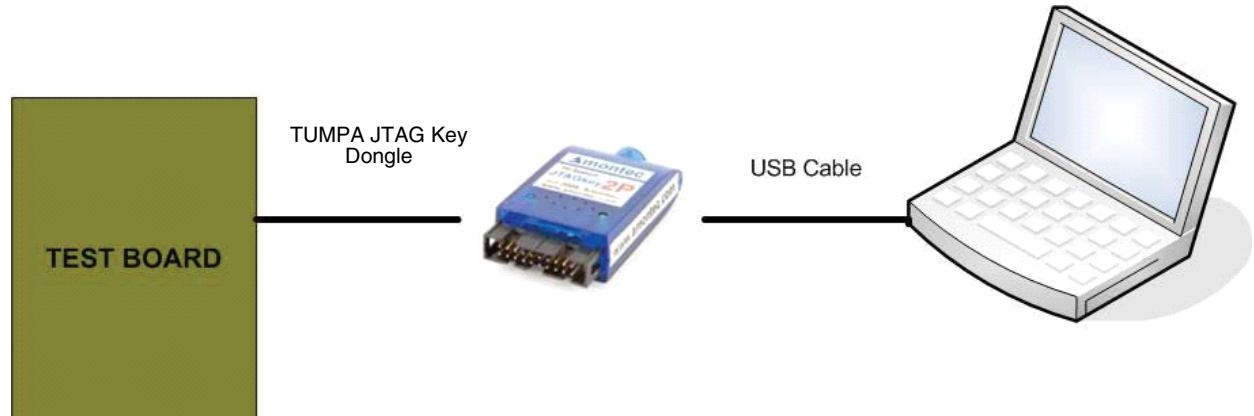
#### C.3.1.1 Hardware Required

The hardware required to run the ASCII vectors is as follows:

- PC running Windows OS
- Test board: This can be either the Synopsys PHY daughter card or a custom board with access to the PHY's JTAG interface.
- USB cable
- USB-to-JTAG dongle: Synopsys recommends using the TIAO USB Multi-Purpose Adapter (TUMPA).

The following figure shows the dongle connection between test board and PC.

**Figure C-27 Dongle Connection**



## C.3.2 Software

For information about the TUMPA, refer to *Using the TIAO USB Multi-Purpose Adapter to Drive JTAG on Synopsys IP Application Note*, which is accessible on your USB 3.0 femtoPHY component page at [DesignWare.com](http://DesignWare.com).

## C.3.3 On-Chip Scope

This section describes how to use the USB 3.0 femtoPHY to generate a receive eye diagram.

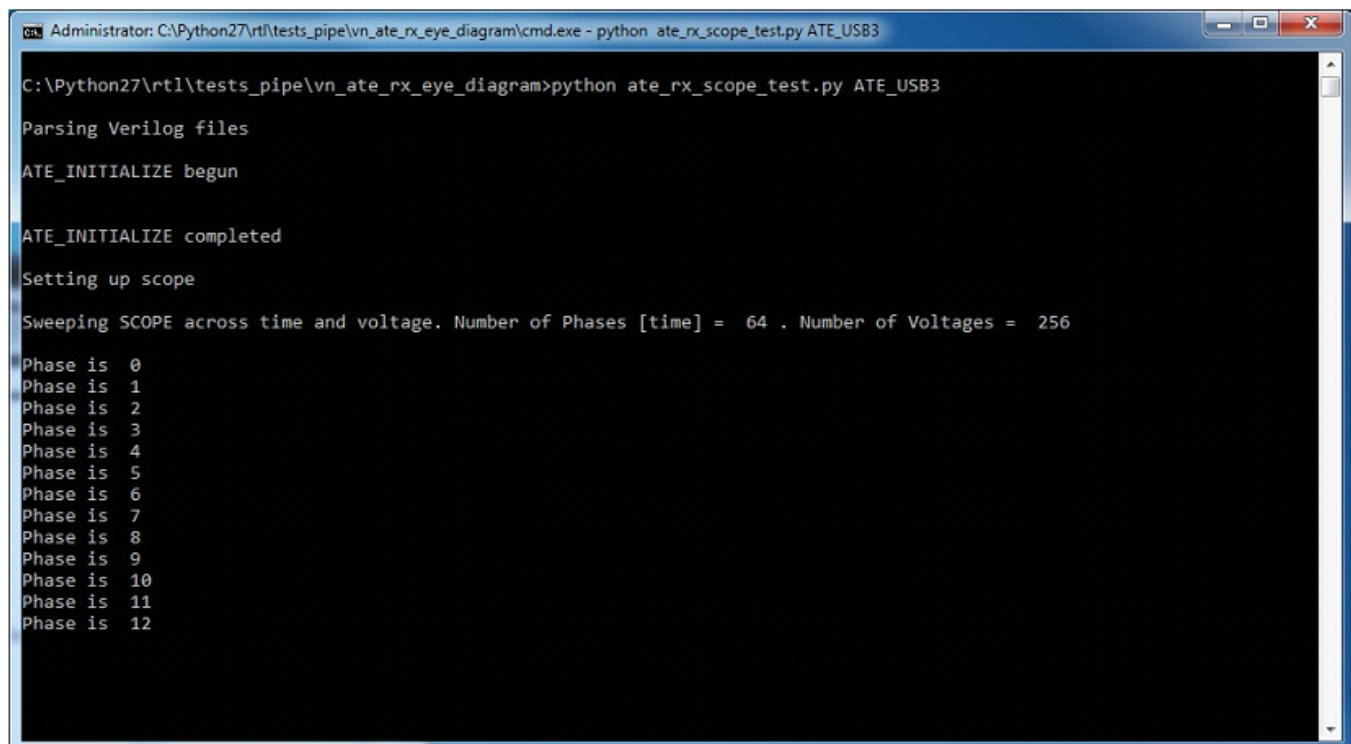
### C.3.3.1 Vector Generation

The PHY has a built-in, on-chip scope that can be used to generate a receive eye diagram. From the `ate_rx_scope_test/` directory, run the `ate_rx_scope_test.py` Python script.

This test varies the phase of the CDR and performs voltage margining at each phase. A maximum of 64 phases and 256 voltages are used, and at each point the scope counter is used to count the number of 1's received. Users must enter parameters, such as reference clock period, MPLL multiplier, TX amplitude, and JTAG\_TCK period, in the `override.txt` file that is provided. Users can choose the resolution (coarse, medium, or fine) and whether to plot the eye before or after the AFE.

The following two figures show a run of the `ate_rx_scope_test.py` script.

**Figure C-28 Example Run of `ate_rx_scope_test.py`**



```
Administrator: C:\Python27\rt1\tests_pipe\vn_ate_rx_eye_diagram\cmd.exe - python ate_rx_scope_test.py ATE_USB3

C:\Python27\rt1\tests_pipe\vn_ate_rx_eye_diagram>python ate_rx_scope_test.py ATE_USB3

Parsing Verilog files
ATE_INITIALIZE begun
ATE_INITIALIZE completed
Setting up scope
Sweeping SCOPE across time and voltage. Number of Phases [time] = 64 . Number of Voltages = 256

Phase is 0
Phase is 1
Phase is 2
Phase is 3
Phase is 4
Phase is 5
Phase is 6
Phase is 7
Phase is 8
Phase is 9
Phase is 10
Phase is 11
Phase is 12
```

**Figure C-29 Example Run of ate\_rx\_scope\_test.py, Continued**

```

Administrator: C:\Python27\rt\tests_pipe\vn_ate_rx_eye_diagram\cmd.exe
Phase is 40
Phase is 41
Phase is 42
Phase is 43
Phase is 44
Phase is 45
Phase is 46
Phase is 47
Phase is 48
Phase is 49
Phase is 50
Phase is 51
Phase is 52
Phase is 53
Phase is 54
Phase is 55
Phase is 56
Phase is 57
Phase is 58
Phase is 59
Phase is 60
Phase is 61
Phase is 62
Phase is 63

Making sure all scope related registers is OFF

Vectors are available in ate.vec

C:\Python27\rt\tests_pipe\vn_ate_rx_eye_diagram>

```

The following figure shows the user parameter control setting in override.txt. The Python script generates a vector file, ate.vec, that can be used by the TUMPA console.

**Figure C-30 Controlling Parameters in override.txt**

```

LTF (Large Text File) Viewer 5.2u - [C:\Python27\rt\tests_pipe\vn_ate_rx_eye_diagram\override.txt]
//PERIOD_REF_CLK in ns
50
//MPLL MULTIPLIER
125
//SSC_REF_CLK_SEL -> Please refer to frequency tables in databook
0
//TX amplitude [valid values 0 to 127]
90
//Eye Diagram @ Before AFE = 0 or AFTER AFE = 1
0
//PERIOD_JTAG_TCK in ns
1000
//Resolution 0 -> coarse[32x64] 1-> medium[64x128 2-> fine[64x256]
2

```

### C.3.3.2 Running the Scope Vector

For detailed information about running the ATE scope test, refer to *Using the TIAO USB Multi-Purpose Adapter to Drive JTAG on Synopsys IP Application Note*, “Quick Start Ate Scope Reference” section, which is accessible on your USB 3.0 femtoPHY component page at [DesignWare.com](http://DesignWare.com).

### C.3.3.3 Eye Diagram Generation

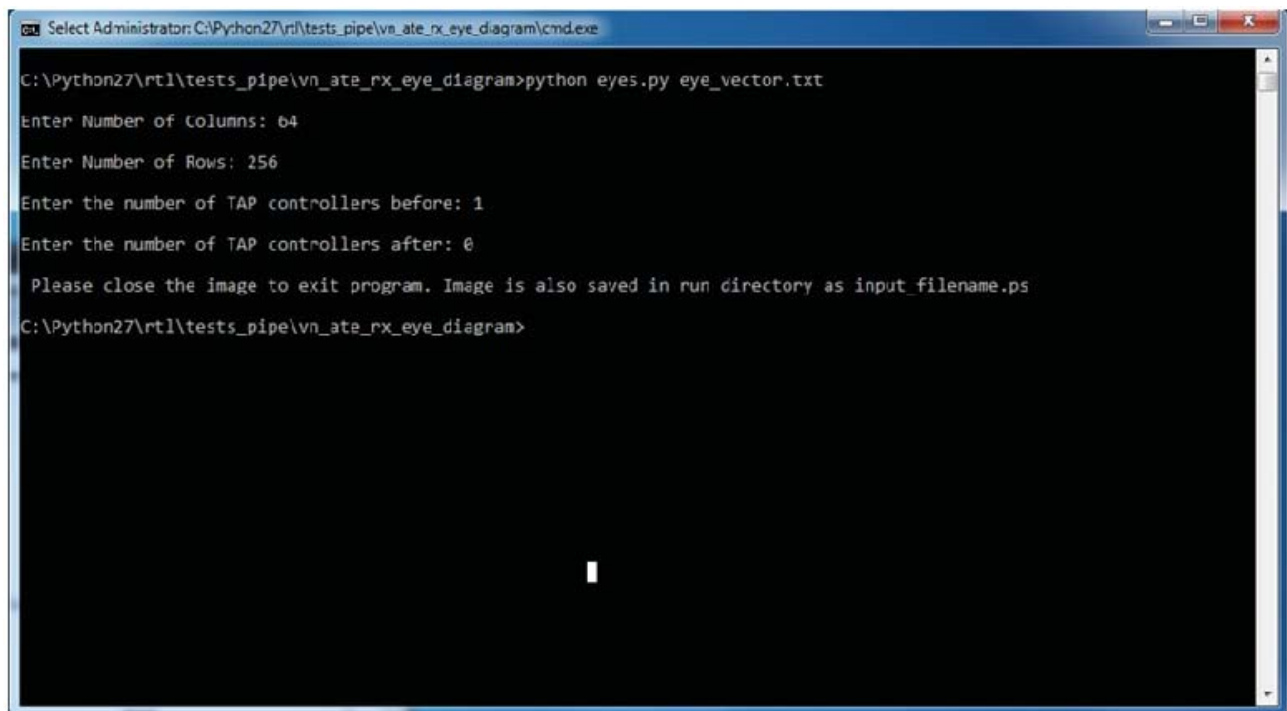
Using the following procedure, run the Python script to post-process the raw TDO output (from the Amontec dongle) and to plot an eye diagram.

1. From the command line, run the Python script using the command:  
`python eyes.py Name_Of_TDO_output_File`
2. As prompted by the script, specify the number of columns, rows, number of TAP controllers before the PHY, and number of TAP controllers after the PHY.

The number of rows and columns is determined by the eye diagram resolution specified in the `override.txt` file.

The following figure shows usage of the post-processing Python script.

**Figure C-31 Usage of Post-Processing Python Script**



```
Select Administrator: C:\Python27\rt1\tests_pipe\vn_ate_rx_eye_diagram\cmd.exe

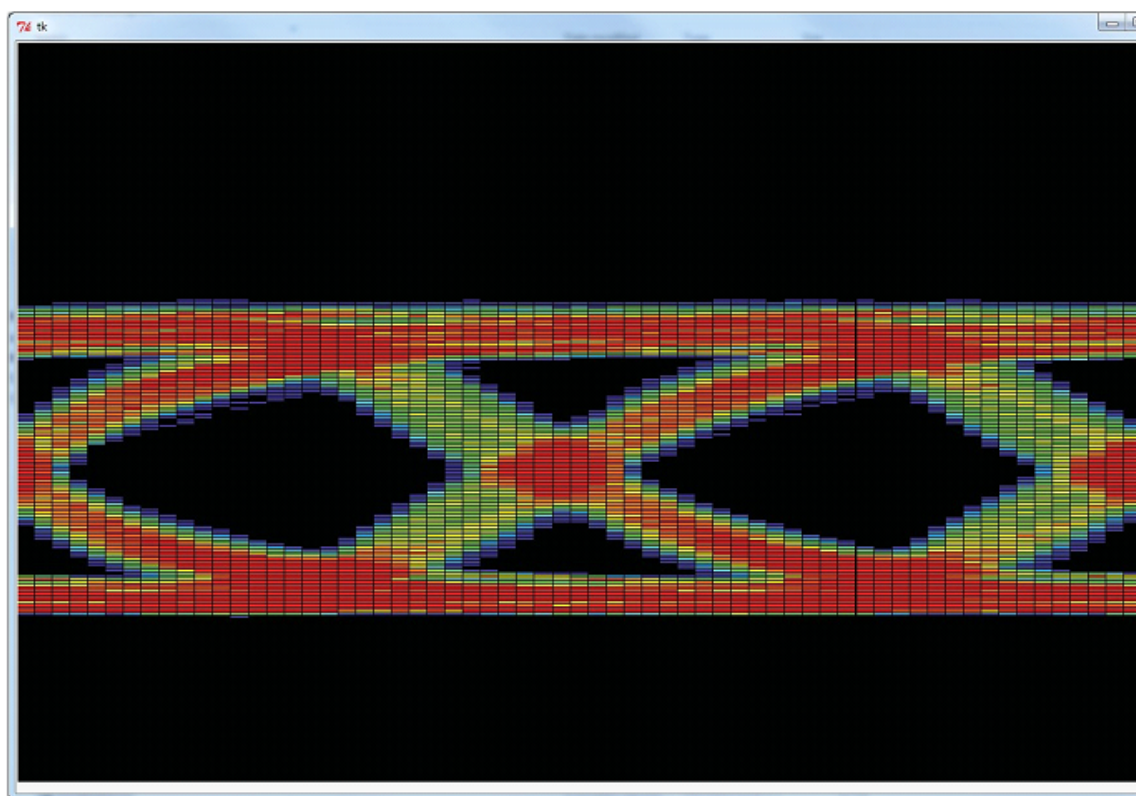
C:\Python27\rt1\tests_pipe\vn_ate_rx_eye_diagram>python eyes.py eye_vector.txt
Enter Number of Columns: 64
Enter Number of Rows: 256
Enter the number of TAP controllers before: 1
Enter the number of TAP controllers after: 0
Please close the image to exit program. Image is also saved in run directory as input_filename.ps
C:\Python27\rt1\tests_pipe\vn_ate_rx_eye_diagram>
```

The eye diagram is generated and saved in the `run/` directory as `input_filename.ps`. This postscript file can be converted to PDF format and viewed with any PDF reader.



The following figure shows the generated eye diagram.

**Figure C-32 Generated Eye Diagram**





## C.4 Installing Python in Windows

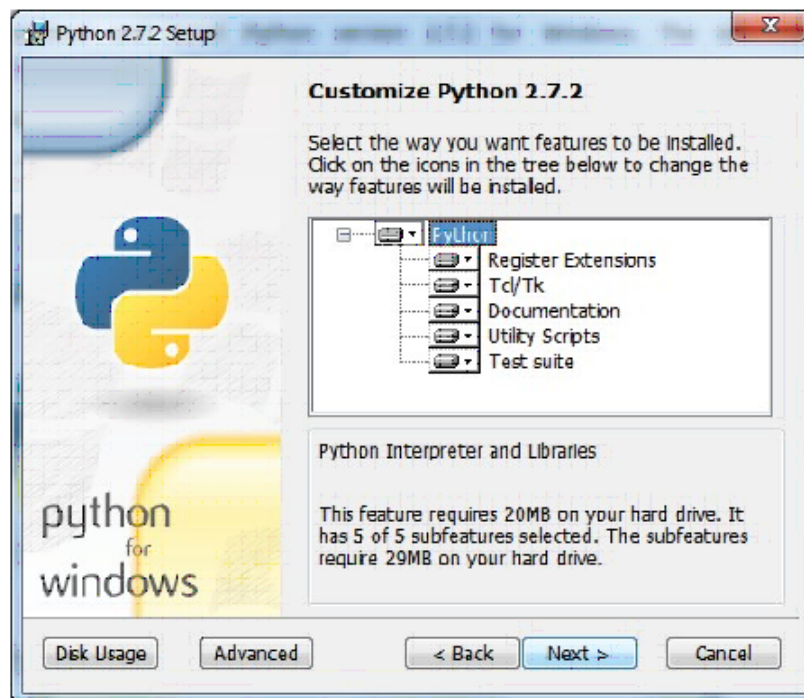
You can download and install Python version 2.7.2 for Windows from:

<http://python.org/download/releases/2.7.2/>

Figure C-33 through Figure C-35 on page 542 show the steps for installing Python.

**Figure C-33** Installing Python, Step 1

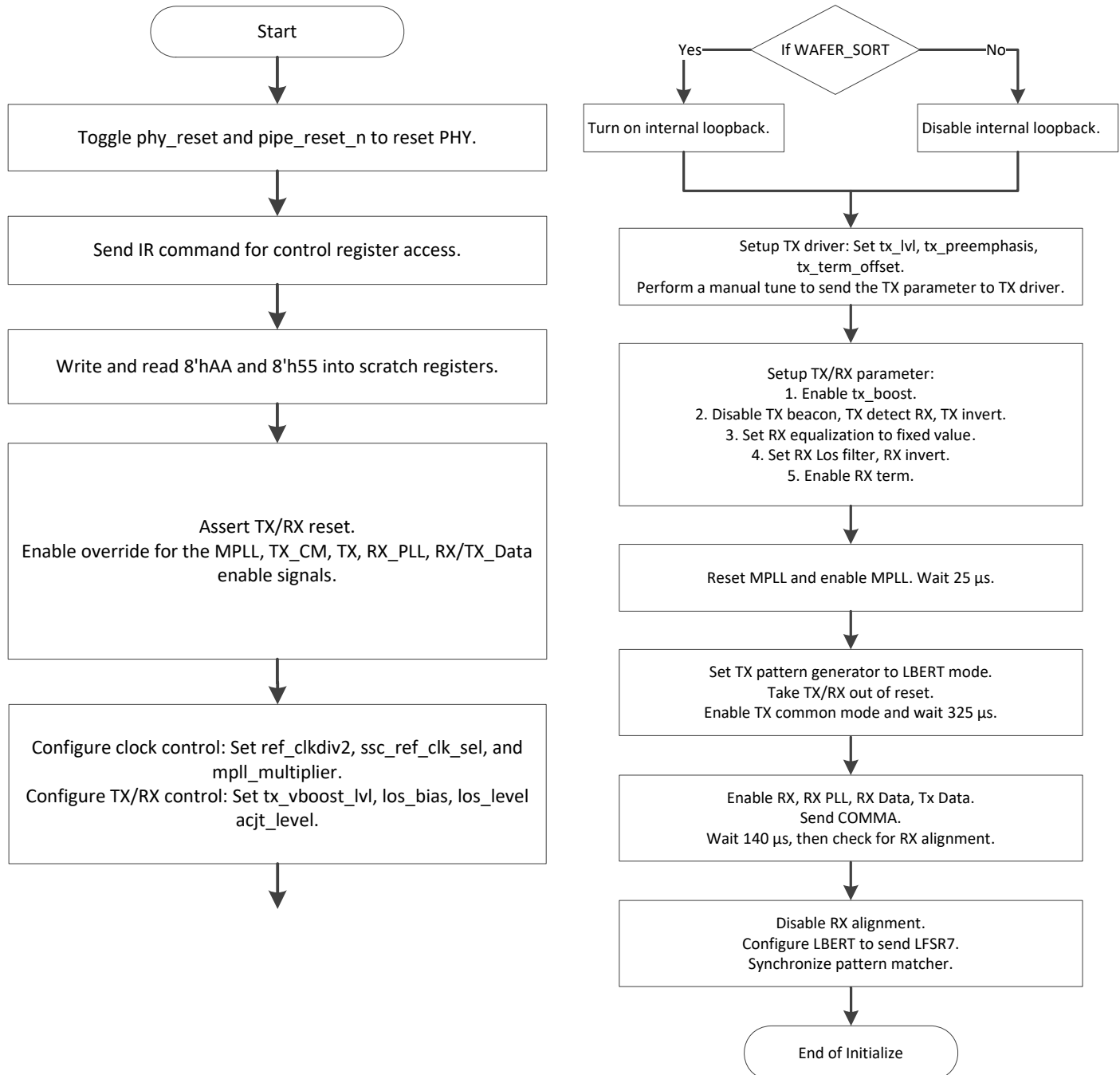


**Figure C-34 Installing Python, Step 2****Figure C-35 Installing Python, Step 3**

## C.5 Flowchart for ate\_initialize ATE Subtask

The following flowchart illustrates the ate\_initialize ATE subtask.

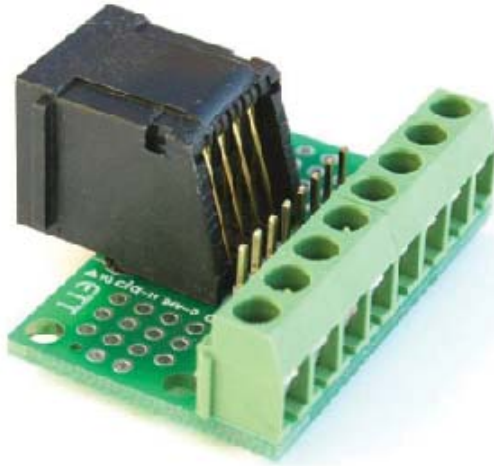
Figure C-36 ate\_initialize ATE Subtask



## C.6 RJ45 BreakOut Board Pinout

You can use the RJ45 breakout board to connect the TUMPA cable to the Synopsys USB 3.0 femtoPHY test chip through an Ethernet cable. The following figure shows the RJ45 breakout board and its pinout.

**Figure C-37 RJ45 Breakout Board**



| Pin | Name |
|-----|------|
| 1   | VCC  |
| 2   | TDO  |
| 3   | GND  |
| 4   | TRK  |
| 5   | GND  |
| 6   | TDI  |
| 7   | GND  |
| 8   | TMS  |

# D

## Application Notes

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The following application note pertains to the USB 3.0 femtoPHY and is available on SolvNet:

*USB 3.0 PHY and USB 3.0 Controller Integration Guide:* This document provides guidelines for integrating the DesignWare USB 3.0 femtoPHY with the DesignWare USB 3.0 Controller.

<https://solvnet.synopsys.com/retrieve/035804.html>.

Note that for test and debug purposes, it is recommended that you implement software control over the PIPE interface signals.

